Noetic Topos 2.0: The Marked and Modal Successor to the Conservative Noetic Site

Little Henderson

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Contents

Chapter 1. Introduction 1
1. The bookkeeping problem 1
2. From aggregation to composition 2
3. Sites and sheaves as the natural candidate 3
4. The legacy construction and its intent 5
5. The collapse problem, stated informally 5
6. The three moves of the successor 7
7. What the finite theory establishes 9
8. Reading guide 10
9. Evidence discipline 12
10. Notation and conventions 16

Chapter 2. Categories, Functors, and Presheaves: A Primer 18
1. Purpose and scope 18
2. Categories 18
3. Functors 22
4. Natural transformations 23
5. Opposite categories and contravariance 24
6. Presheaves 24
7. Representable presheaves 25
8. Subpresheaves 26
9. The Yoneda lemma 27
10. Limits and pullbacks 29
11. The small-poset laboratory 32
12. Summary 33

Chapter 3. Sieves, Grothendieck Topologies, and Sites 35
1. Purpose and scope 35
2. Sieves 35
3. Grothendieck topologies 38
4. Coverage bases and generated topologies 40
5. Matching families and subcanonicity 41
6. All Grothendieck topologies on the two-element chain 41
7. The union coverage on a powerset lattice 43
8. Summary 45

Chapter 4. Sheaves, Sheafification, and Modalities 46
1. Sections, restriction, and the idea of local data 46
2. Matching families and amalgamation 47
3. First examples on a small poset 48

4. Non-examples, and what each failure teaches 50
5. The basis criterion 50
6. The equalizer formulation, and life beyond posets 53
7. Sheafification: the universal property 55
8. The plus construction, and computing on finite sites 55
9. Reflective subcategories and idempotent monads 57
10. Modalities 58
11. Chapter summary and the road ahead 60

Chapter 5. The Legacy Noetic Site and Its Collapse 62
1. The registered category Noetica 62
2. The three cover clauses 63
3. Two elementary facts about sieves 65
4. The collapse theorem 65
5. Generation is a closure operator, and closure erases provenance 67
6. Three tempting repairs, formalized and refuted 69
7. What the v7.4 corrigendum retracts, and what it preserves 72
8. Executable corroboration 73
9. Chapter summary 74

Chapter 6. The Subcanonicity Countermodel and the Representable Repair 75
1. What subcanonicity buys 76
2. Why the legacy axioms do not derive subcanonicity 76
3. The countermodel 77
4. The failed gluing, in full 78
5. The exact logical status 80
6. The standard repair: $a_{\text{legacy}}(yX)$ 80
7. Countermodel methodology in axiomatic auditing 83

Chapter 7. The Context Lattice: Complete Combinatorics 85
1. The atom set and the lattice 85
2. The sixteen contexts 87
3. The eighty-one arrows 88
4. The transverse partitions 91
5. Symmetries of the marked lattice 92
6. The generating families 93
7. Pullback stability, case by case 95
8. Cover composition and refinement 97
9. Block completeness 98
10. The sieve census 100
11. Catalog of degenerate and boundary cases 102
12. Summary, and what has deliberately not been done 103

Chapter 8. Marked Covers and the Provenance Calculus 105
1. Why topologies forget provenance 105
2. The two-representation design 106
3. The marked-cover schema, field by field 107
4. Canonical JSON and the digest binding 109
5. The derivation calculus 110

6. Parent resolution at deserialization 111
7. The fail-closed rejection taxonomy 112
8. Operational lifecycle 114
9. The singleton coincidence, in full 114
10. Certificate check classes for the marked layer 116
11. Design alternatives, and why they were not taken 117
12. What the mechanism is, and is not 118
13. Summary 119

Chapter 9. The Finite-Context Theorems 120
1. Standing of the results and ledger discipline 120
2. The coverage-basis lemma 121
3. NT2-1: the generated topologies 122
4. Propriety: the generators contain no identities 124
5. NT2-2: subcanonicity of the context factor 124
6. NT2-3: distinct, non-identity modalities 126
7. NT2-4: the join is the singleton topology 129
8. Consolidated ledger view 132
9. What the finite theorems establish, and what they do not 133

Chapter 10. Explicit Reflectors and the Composition Theorem 134
1. Purpose and status of this chapter 134
2. Standing data 134
3. The one-partition reflector 135
4. The sheaf lemma 137
5. The universal property 138
6. The join of the two topologies 140
7. The composition theorem 142
8. Corollaries 143
9. The narrative correction: distinct but commuting 145
10. Status recapitulation 146

Chapter 11. The Executable Certificate 147
1. Why an executable layer exists 147
2. Methodology of exhaustive checking 147
3. The finite-site checker and its twenty-three checks 148
4. The modality-composition verifier 152
5. The Lean layer: scope and explicit non-scope 155
6. The certificate's own claim boundary 156

Chapter 12. The Ambient Candidate, Boundaries, and the Research Program 157
1. From the finite factor to the ambient package 157
2. Gluing boundaries: what Set-valued descent does not license 159
3. Backward compatibility as a conjectural comparison 161
4. The expressivity boundary and the cross-context obligation 162
5. Related work and novelty status 163
6. The admission gates 164
7. Conclusion 165

Appendix A. Context and Cover Tables 166
1. Naming conventions 166
2. The sixteen contexts and their complete blocks 166
3. The generating family $K_A(V)$ at every context 167
4. The generating family $K_R(V)$ at every context 168
5. Coincidence of the unmarked families 168
6. The common refinement at every context 169
7. Per-context sieve and covering-sieve counts 170

Appendix B. The Sieve Census 171
1. The underlying poset 171
2. Arrow count 171
3. Sieve and covering-sieve totals 172
4. Block-completeness table 172

Appendix C. Witness Presheaf Value Tables 174
1. Definitions and the counting rule 174
2. Value tables 174
3. Restriction maps 175
4. Gluing at Ω: the four registered computations 175
5. Gluing at the remaining contexts 176
6. Reflector values on the witnesses 177
7. Section counts at a glance 177

Appendix D. Glossary and Notation Index 178
1. Terms 178
2. Symbols 182

Bibliography 185

CHAPTER 1

Introduction

This monograph is a complete, self-contained development of the research candidate known internally as *Noetic Topos 2.0*: a marked and modal successor to the conservative Noetic site, together with the finite-context theory that the successor rests on, the collapse theorem that made a successor necessary, and the evidence discipline under which every one of these statements is held. It expands the corresponding v0.2 research paper without changing any claim, status, or boundary recorded there. The mathematics is elementary category theory — categories, presheaves, sieves, Grothendieck topologies, sheafification — applied to a bookkeeping problem that arises in the assessment of AI systems. The bookkeeping problem is real; the mathematics is standard; the specific construction is a Ring 4 research candidate that has not been independently reviewed, and this chapter explains, among other things, exactly what that sentence means and why the monograph repeats it so often.

The present chapter does six things. It describes the program context in which the construction arose and the structural (not numerical) difficulty that motivates it (Sections 1–2). It explains informally why sites and sheaves are the natural candidate formalism for that difficulty (Section 3). It recounts the legacy construction and its intent, and states the collapse problem that the legacy construction runs into — informally here, with full proofs deferred to Chapter 5 (Sections 4–5). It presents the three moves that define the successor (Section 6) and summarizes what the finite theory actually establishes (Section 7). It gives a detailed reading guide to the remaining chapters (Section 8). Finally, it states the evidence discipline — rings, ledger proof statuses, and the precise scope of the executable corroboration — and fixes the notation used throughout (Sections 9–10).

1. The bookkeeping problem

AI-assessment work in the Cognitive Architecture System (CAS) program, and in the Project Coherence effort that grew out of it, increasingly relies on heterogeneous evidence: observations of a system gathered under different prompts, different tasks, different roles, and different deployment contexts. An assessment pipeline may hold, at the same time, transcripts elicited under one prompting regime, scored behavior under another, structured probe results from a third, and audit records whose provenance passes through several hands before reaching the analyst. Each item of evidence is *local*: it was gathered under specific conditions, and it speaks, in the first instance, only about the system's behavior under those conditions.

The recurring difficulty is structural rather than numerical. It is not that the individual observations are noisy — noise is a familiar problem with familiar statistical treatments — but that the observations need not *compose*. Two local views, each internally coherent, may fail to be views of any single consistent whole. Three views may be pairwise compatible and yet jointly incompatible. A conclusion drawn from a family of local views is only as good as the compatibility of that family, and ordinary aggregation — averaging scores, concatenating

transcripts, tallying pass rates — silently assumes a compatibility that nothing in the pipeline checks.

A concrete schematic (deliberately content-free, in keeping with the claim discipline of Section 9): suppose an assessment records a system's behavior separately under context A and under context B, and each record is separately well-formed. Whether the two records can be regarded as restrictions of one description of the system — whether there exists a single global section of which both are shadows — is a further question, and it is a question with mathematical content. Sometimes the answer is yes and the gluing is unique; sometimes yes but not uniquely; sometimes no. A bookkeeping discipline for heterogeneous evidence must be able to pose this question precisely, answer it where an answer exists, and refuse to fabricate an answer where none does.

The word "bookkeeping" is chosen deliberately. Nothing in this monograph measures a system, scores a behavior, or validates an assessment method. The subject matter is the *ledger*: the formal apparatus that records which local views were gathered, under which admission rule each family of views was accepted as jointly informative, and what compatibility constraints a purported global conclusion must satisfy. The apparatus is useful exactly to the extent that it is honest — that it refuses to remember less than it was told and refuses to assert more than it can check — and the central technical drama of this monograph, the collapse theorem of Chapter 5, is precisely a case in which a proposed apparatus turned out, provably, to remember less than its designers intended.

2. From aggregation to composition

To see why the problem calls for more than statistics, consider the three questions any discipline for local-to-global reasoning must answer.

(1) **Which families of local views count?** Not every collection of observations deserves to be treated as jointly informative about a single target. A family gathered under systematically related conditions — for instance, conditions that jointly exhaust a space of contexts — has a claim to joint significance that an arbitrary grab-bag does not. The discipline must say, for each target of assessment, which families of "restrictions to sub-contexts" are admitted as *covers* of that target.

(2) **When are local data compatible?** Given an admitted family, the local data attached to its members overlap: two sub-contexts may share a common further sub-context, and the data restricted to that overlap must agree if the family is to describe one thing. The discipline must define compatibility on overlaps and must be able to check it.

(3) **What glues?** Given an admitted family with compatible local data, the discipline must say whether a global datum exists that restricts to the given local data, and whether it is unique. Existence without uniqueness, and uniqueness without existence, are both live possibilities, and both must be nameable.

Aggregation answers none of these questions; it presupposes affirmative answers to all three. Composition — in the mathematician's sense of building global objects from local ones along specified interfaces — is the activity these questions describe, and the mathematics of composition along covers is sheaf theory. The reader who has met sheaves in geometry will recognize the three questions as, respectively, the choice of a Grothendieck topology, the matching-family condition, and the sheaf (gluing) condition. The reader who has not will find all three developed from scratch, at tutorial depth, in Chapters 2–4.

2.1. Desiderata for a bookkeeping discipline. The three questions can be sharpened into desiderata against which any candidate formalism — including the one this monograph develops — can be audited. A bookkeeping discipline for heterogeneous assessment evidence should be:

(1) *Explicit about admission.* The rule by which a family of local views is admitted as jointly informative must be a registered, inspectable object, not an implicit habit of the pipeline. (In the formalism: the topology is declared, not inferred.)
(2) *Compositional.* Admitted families must behave coherently under restriction and iteration: restricting an admitted family to a sub-target yields an admitted family, and admitting families locally on an admitted family admits the composite. (The second and third topology axioms.)
(3) *Honest about gluing.* The discipline must distinguish, and be able to certify, the three outcomes: unique gluing, non-unique gluing, and no gluing. (The sheaf condition as a bijection demand, with its injectivity and surjectivity halves separately meaningful.)
(4) *Provenance-preserving.* The record of *why* each family was admitted — under which clause, on which authority, in which review state — must survive every derivation the discipline performs. (The marked-cover layer; by the collapse theorem, this cannot be delegated to the topology.)
(5) *Self-limiting.* The discipline must state the boundaries of its own guarantees — which data types it covers, which it provably does not, and which of its own components are unproven — with the same precision as its positive results. (The claim-boundary apparatus of this monograph.)

Desiderata (1)–(3) are satisfied by the standard site-and-sheaf machinery and are not project contributions. Desideratum (4) is where the legacy construction failed and where the successor's principal design work lies. Desideratum (5) is a discipline on the text itself, enforced throughout by the evidence rules of Section 9.

Two adjacent temptations should be flagged immediately, because the discipline of this monograph blocks both explicitly (the full treatment is in the seed paper's gluing-boundary section and recurs in Chapter 12). First, *probabilistic* local data do not inherit set-valued gluing by analogy: pairwise-compatible marginal distributions may admit no global joint distribution, a classical phenomenon studied by Vorob'ev [10] and, in the sheaf-theoretic language adopted here, by Abramsky and Brandenburger [11]. Second, *linear* or cellular data require a separate linear representation — stalks, inner products, weighted coboundaries — before spectral tools such as sheaf Laplacians are even defined [15]. Nothing in this monograph constructs either structure, and no set-valued result proved here licenses a probabilistic or spectral claim.

3. Sites and sheaves as the natural candidate

Category theory supplies a compact formal answer to question (1) above: a *site*. A site is a category — objects are the targets of assessment and morphisms are the ways one target restricts to, or maps into, another — together with a *Grothendieck topology*, an assignment to each object of a collection of *covering sieves*. A sieve on an object C is a set of morphisms into C closed under precomposition; intuitively, a downward-closed family of "ways of looking at C from below." The topology's three axioms (stated and motivated in Chapter 3) guarantee that the covering notion behaves like a covering notion should: the trivial cover covers, covers restrict along morphisms, and a family that covers locally on a cover covers globally.

Questions (2) and (3) are then answered uniformly by the *sheaf condition*: a presheaf — an assignment of a set of local data to each object, contravariant along morphisms — is a sheaf when, for every covering sieve, compatible families of local data over the sieve correspond bijectively to global data over the covered object. The category of sheaves on a small site is a Grothendieck topos [1, 7], and the inclusion of sheaves into presheaves has a left-exact left adjoint, *sheafification* [5]. Composing sheafification with the inclusion yields a left-exact idempotent monad on presheaves; this monograph follows the seed paper in calling such composites *modalities*, and in calling a modality *trivial* when it is naturally isomorphic to the identity.

3.1. A miniature vignette. Because the sheaf condition is the load-bearing idea of the entire monograph, we first see it informally, on a structure small enough to hold in the head — a two-point miniature of the sixteen-context lattice of Part II. Let the "contexts" be the subsets of a two-element set $\{1, 2\}$: the empty context $\varnothing$, the two singleton contexts $\{1\}$ and $\{2\}$, and the full context $\{1, 2\}$, ordered by inclusion. Declare that the family $\{\{1\} \subseteq \{1, 2\},\ \{2\} \subseteq \{1, 2\}\}$ *covers* the full context — the two singletons jointly exhaust it — and that the empty family covers the empty context.

A presheaf of "local records" assigns a set of records to each context and a restriction map to each inclusion. Consider first the presheaf G where a record over a context V is an arbitrary function from V to $\{0, 1\}$ — a bit written on each point in view — and restriction forgets the bits of points no longer in view. Take a compatible pair of local records over the cover: a bit for point 1 (a record over $\{1\}$) and a bit for point 2 (a record over $\{2\}$); compatibility on the overlap is automatic, because the overlap is the empty context, over which G has exactly one record (the empty function). The pair assembles into exactly one record over $\{1, 2\}$, and every record over $\{1, 2\}$ arises this way from exactly one pair. The gluing map is a bijection: G is a sheaf for this cover. Local bit-records compose perfectly.

Now consider the presheaf H that assigns to *every* nonempty context the same two-element set $\{\mathsf{p}, \mathsf{q}\}$ — read: a global verdict recorded redundantly at each context — with identity restrictions, and a single record over the empty context. A compatible pair over the cover is any pair of verdicts, one over $\{1\}$ and one over $\{2\}$ (again the empty overlap enforces nothing), so there are four compatible pairs; but there are only two records over $\{1, 2\}$, each of which restricts to a *matching* pair. The gluing map $2 \to 4$ is injective but not surjective: the mixed pairs (p, q) and (q, p) are locally coherent families that no global record explains. H is not a sheaf for this cover, and the failure is informative — it says that verdict-type data cannot be treated as freely localizable over these covers, on pain of manufacturing global verdicts that nothing global supports.

This tiny example already displays every phenomenon the monograph trades in: covers as admitted families, compatibility on overlaps, gluing as a bijection demand, sheafhood as a property a presheaf has *relative to a choice of covers*, and non-sheafhood as a diagnosis rather than a disaster — the associated sheaf construction of Chapter 4 repairs H canonically, and what it repairs it to is itself informative. The witness presheaves that separate the ACBE and RPM modalities in Chapter 9 are elaborations of exactly the two patterns above, and the reader may find it useful to keep the pair (G, H) in mind as the germ of that argument.

3.2. What the formalism promised. The promise this formalism held for the assessment program was twofold. First, the topology is a place to *register admission rules*: to say, once and in one place, which families of restricted observations count as covers, and to have the topos machinery enforce the compatibility and gluing disciplines downstream. Second — and this was the more ambitious hope — different *kinds* of admission rules might generate

different topologies, whose sheafification modalities would then act as distinct "lenses" on the same ambient evidence: one modality resolving evidence up to one kind of cover, another up to another kind, with the interaction of the lenses itself carrying structure. It is exactly this second promise that the legacy construction failed to deliver, for reasons that turned out to be theorems.

4. The legacy construction and its intent

The conservative Noetic Topos paper — the "legacy construction" throughout this monograph — introduced a site for the registered category $\mathcal{C}$, written Noetica in the source artifacts: a small concrete category of finite sets indexed by the forty distinguished CAS indices, with admissibility-relation generating maps and pullback closure. On $\mathcal{C}$ it defined a coverage J_{legacy} generated by three clauses:

(O) *jointly-surjective observational families*: families of maps into a target whose images jointly exhaust it;

(A) *designated ACBE descent maps*, including, by reflexivity, the descent map at equal contexts;

(R) *the RPM counit* $\varepsilon_X : R(X) \to X$, where the registered RPM data supply a unit $\eta_X : X \to R(X)$ with $\varepsilon_X \circ \eta_X = \mathrm{id}_X$ — a section-retraction pair, and deliberately *not* an adjunction, in keeping with the program-wide correction that prerequisite satisfaction is one-directional.

Each clause encodes a defensible bookkeeping intuition, and we record the intuitions, because the successor preserves them even as it discards their legacy formalization. Clause (O) says that a family of observation channels which jointly exhausts a target is entitled to speak for the target as a whole — the ordinary local-to-global entitlement. Clause (A) was meant to encode a descent discipline along designated context relationships: evidence gathered at related contexts should descend to, and be answerable at, a common context. Clause (R) was meant to encode a prerequisite discipline: the counit's passage from a prerequisite-enriched representation back to the base object was to count as an evidence-bearing cover, so that satisfaction established at the enriched level would be creditable at the base. The intuitions survive in the successor as the two formal coordinates of the context lattice; their legacy formalizations, as the next section explains, could not.

The legacy paper proved a conditional topos-existence theorem: when $\mathcal{C}$ is small, the category of sheaves $\mathrm{Sh}(\mathcal{C}, J_{\text{legacy}})$ is a Grothendieck topos. That theorem is valid and is not disputed anywhere in this monograph. The *intent* attached to the construction, however, went further: the ACBE clause (A) and the RPM clause (R) were meant to be remembered by the topology as distinguishable contributions, so that later work could build one modal operator from the (A)-generated covers and a different modal operator from the (R)-generated covers — two separate lenses on evidence, with the observational clause (O) supplying the common substrate. The names ACBE and RPM travel through this monograph as formal labels for these two intended lenses; the reader needs no expansion of the acronyms, and, as Section 9 makes precise, the labels carry no empirical interpretation in any theorem below.

5. The collapse problem, stated informally

The intent was unrealizable, and the reason is elementary. Two facts about sieves, each provable in three lines (they are proved, with care, in Chapter 3), do all the damage:

(1) A sieve containing the identity of its target is the *maximal* sieve: it contains every morphism into the target, because every morphism factors through the identity.

(2) A sieve containing a *split epimorphism* — a map with a one-sided inverse on the right — contains the identity, by closing under precomposition with the splitting; hence it is maximal by fact (1).

Now look at the legacy clauses. Clause (A) includes, by reflexivity, the descent map at equal contexts, which is an identity; every sieve it generates is maximal by fact (1). Clause (R) includes the counit ε_X, which is split by the unit η_X since $\varepsilon_X \circ \eta_X = \mathrm{id}_X$; every sieve it generates is maximal by fact (2). But maximal sieves belong to *every* Grothendieck topology — that is the first topology axiom — so neither clause contributes anything that the observational clause (O) did not already provide. The topology generated by all three clauses equals the topology generated by (O) alone, and any modality built from the (A)- or (R)-generated topology is trivial. Chapter 5 states this as the Legacy Collapse proposition and proves it; the registered finite certificate independently re-executes both mechanisms as the checks `legacy_acbe_identity_forces_maximal_sieve` and `legacy_rpm_split_epi_forces_identity`.

Three remarks fix the correct reading of this result.

First, the collapse is *not* a defect of the topos machinery, and it does not invalidate the legacy topos-existence theorem. What collapses is the intended interpretive structure. The result is best read as a theorem about which information a Grothendieck topology is *capable* of retaining: sieve generation is a closure operator, closure operators forget how their output was reached, and a topology therefore cannot remember *why* a family was admitted — only *that* the sieve it generates is covering. Provenance is exactly the kind of information a topology discards by design.

Second, the collapse cannot be escaped by relabeling. Any successor clause that again contains an identity, or again contains a split epimorphism onto its target, generates only maximal sieves and collapses the same way. A successor must *change the generators*, not rename them.

Third, a separate and independent gap accompanies the collapse: subcanonicity of the legacy site — the property that representable presheaves are sheaves, which is what licenses treating a site object as its own evidence-carrier inside the topos — is *not derivable* from the registered axioms. Chapter 6 exhibits a countermodel: a model of the axioms (finite sets of size at most two, with injections) in which a jointly-surjective family exists whose gluing map for a representable fails to be surjective. The consequence is a standing rule, not a repair: until an exact revised site is proved subcanonical, the distinguished legacy object is represented in the sheaf topos by its sheafification $a_{\mathrm{legacy}}(yX)$, never by silently treating yX as a sheaf [6].

5.1. Why a topology cannot remember: closure and provenance. We dwell for a moment on the mechanism behind the collapse, because it explains the shape of the entire successor design and not merely the failure of one construction. Passing from a generating family of morphisms to the sieve it generates, and from a collection of generating sieves to the topology they generate, are both instances of *closure*: monotone, extensive, idempotent operations that add everything the axioms force and record nothing about the order or origin of the forcing. Closure operators are forgetful by design — that is what makes them canonical. The closure of a set of premises under logical consequence does not remember which consequences came from which premises; the subgroup generated by a set of elements does not remember which generator produced which product; and the topology generated by a set of covers does not remember which clause admitted which cover. Asking the generated topology to distinguish an (A)-cover from an (R)-cover is asking a closure operator to be injective on histories, which no closure operator is.

Seen this way, the legacy failure decomposes into a general fact and a specific misfortune. The general fact: even with well-chosen generators, the topology would remember only the *join* of the admitted covers, not their labels; any design that needs labels needs a second structure. The specific misfortune: the legacy generators were not merely unlabeled but degenerate — each contained an identity or a split epimorphism, so each generated only the maximal sieve and contributed nothing at all, not even unlabeled content. The successor therefore needs two distinct repairs, and Moves 1 and 2 below are exactly those repairs: new generators that contribute genuine nonmaximal covers (curing the degeneracy), and an external provenance record that survives closure (curing the forgetfulness). Neither repair substitutes for the other. Nonmaximal covers without external marks would still be anonymous after closure — the singleton-context example of Section 6.2 exhibits this concretely — and marks without nonmaximal covers would be labels on nothing.

5.2. The two negative results are independent. The collapse (Chapter 5) and the subcanonicity gap (Chapter 6) are logically independent defects, and the monograph keeps them separate because their repairs are separate. The collapse concerns clauses (A) and (R) and is repaired by the successor's new generators; it would persist even if the legacy site were proved subcanonical. The subcanonicity gap concerns clause (O) — the jointly-surjective coverage itself — and is not repaired by the successor at all: the successor's subcanonicity theorem (Chapter 9) covers only the new context factor L, and the monograph states explicitly, twice, that it neither repairs nor bears on the legacy gap. What the successor does about the legacy gap is disciplinary rather than mathematical: it adopts the representation rule that bare legacy site objects are sheafified before use whenever J_{legacy} is not known subcanonical. A reader auditing the construction should verify that no argument in Part II quietly assumes legacy representables are sheaves; the countermodel of Chapter 6 is the standing reason such an assumption would be unearned.

6. The three moves of the successor

The successor candidate — Noetic Topos 2.0 — responds to the collapse with three moves. Each is developed fully in Part II of this monograph (Chapters 7–12); this section states them and their rationale.

6.1. Move 1: change the generators. Since no relabeling of the legacy clauses can produce nonmaximal covers, the successor introduces a new, separate, finite *context* dimension on which nontrivial coverage is possible, and leaves the legacy category untouched. The context factor is the sixteen-element powerset lattice

$$L = \mathcal{P}(\Omega), \qquad \Omega = \{0,1\} \times \{0,1\},$$

ordered by inclusion, with $U \to V$ if and only if $U \subseteq V$. The four atoms of Ω carry two coordinates; the first coordinate is the formal ACBE mark and the second the formal RPM mark. Two *transverse partitions* of Ω,

$$A_i = \{(i,0),(i,1)\}, \qquad R_j = \{(0,j),(1,j)\}, \qquad i,j \in \{0,1\},$$

generate two coverages: at a nonempty context V, the ACBE generating family is $\{V \cap A_0 \to V,\ V \cap A_1 \to V\}$ and the RPM generating family is the analogue with the R_j; the empty context is covered by the null (empty) family. The decisive structural point is local to the global context: at Ω the pieces of both generating families are *proper* subsets ($A_0, A_1 \subsetneq \Omega$ and $R_0, R_1 \subsetneq \Omega$), so neither global family contains an identity, and — a poset having no nontrivial split epimorphisms (Remark 2.17) — neither contains a split epimorphism onto its

target. The collapse mechanism therefore has nothing to grip where it mattered. At a context V contained in a single block the corresponding family degenerates to $\{V \to V,\ \varnothing \to V\}$ and does generate the maximal sieve; that degeneracy is harmless for the escape theorem, and it is exactly what makes the external mark record necessary at singleton contexts (Section 6.2). The transversality identity $A_i \cap R_j = \{(i, j)\}$ — each ACBE block meets each RPM block in exactly one atom — powers essentially every theorem of the finite theory. Chapter 7 develops the lattice, and Chapter 9 proves that the generated topologies J_A and J_R are genuine Grothendieck topologies, are subcanonical, and support distinct, non-identity sheafification modalities.

6.2. Move 2: externalize provenance. The collapse theorem says a topology cannot remember why a cover was admitted; the successor stops asking it to. Provenance is stored *outside* the topology, in immutable, hash-chained *marked cover* records that travel in parallel with the unmarked arrow families the topology consumes. A marked cover binds, under canonical JSON serialization and SHA-256 digesting: a derivation kind (generator, pullback, or common refinement), the target, the exact member arrows, a mark record (cover kind, source authority reference, assumptions, evidence references, review state, provenance chain), and the sorted set of parent-cover digests. The integrity rules are fail-closed: unknown or duplicate keys are rejected, derived records require a parent resolver that rehashes every parent digest and recomputes the child, and a common refinement lacking exactly one ACBE and one RPM generating parent is rejected rather than treated as evidence.

The division of labor is strict and is stated as an invariant of the design: *the topology consumes the unmarked family; governance and audit code consume the marked record.* The necessity of the parallel record is not hypothetical — it is exhibited inside the finite model itself. At a singleton context $\{(i, j)\}$, the bare ACBE and RPM families coincide as arrow lists, so the topology literally cannot distinguish them; the marked records carry different cover kinds and different digests, and the certificate confirms both facts (`singleton_unmarked_families_coincide`, `singleton_marked_covers_distinct`). Provenance survives exactly where the topology forgets. Chapter 8 develops the mechanism, with the claim boundary — repeated there and binding here — that the mark mechanism is an external serialization and audit invariant, not a proof of an internal topos object, not a colored-site construction from the literature, and not yet the subject of a dedicated novelty review.

6.3. Move 3: name the ambient package honestly. The third move is a naming decision with mathematical content. It would be natural — and it would be wrong — to define the successor as the category of sheaves for the *join* of all the topologies in play, the "fully resolved" subtopos. Wrong, because on that subtopos the ACBE and RPM reflectors provably restrict to the identity (every jointly resolved object already satisfies both sheaf conditions), which would erase the modal structure a second time — an ironic repetition of the collapse by over-resolution rather than by bad generators. The successor is therefore defined as the *ambient* package: the presheaf topos $\mathcal{E}_{\text{amb}} = \mathrm{PSh}(\mathcal{C} \times L)$, equipped with three topology refinements (observational in the $\mathcal{C}$ coordinate, J_A and J_R in the L coordinate), their three sheafification modalities M_{obs}, M_A, M_R, the resolved subtopos $\mathcal{E}_{\text{res}}$ for the joined topology, and the external mark record. The underlying topos candidate is $\mathrm{NoeTop}_2 = \mathcal{E}_{\text{amb}}$; the full research object is the structured package

$$\mathrm{NoeSystem}_2 = \big(\mathrm{NoeTop}_2,\ M_{\text{obs}},\ M_A,\ M_R,\ \mathcal{E}_{\text{res}},\ \textsc{external mark record}\big),$$

which is a package, not itself a topos or category, with the provenance record as external audit data rather than an internal topos object. The intended discipline: raw and partially resolved evidence lives in the ambient topos, acted on by the modalities; only jointly resolved objects enter $\mathcal{E}_{\text{res}}$. Chapter 12 develops the ambient program and its open obligations, including the explicit boundary that the v0.1 executable schema is deliberately narrower than the ambient construction: typed observational covers in the $\mathcal{C}$ coordinate, general arrows of $\mathcal{C} \times L$, and full product-site covers remain OPEN.

7. What the finite theory establishes

On the context factor L alone — sixteen objects, small enough to check exhaustively — the theory is complete, and Chapters 9–10 prove all of it. In summary, with ledger identifiers attached as recorded in the registered theorem ledger:

(1) *Generated topologies* (THM-NT2-MARKED-COVERAGE): the ACBE and RPM generating families form pullback-stable coverage bases and generate Grothendieck topologies J_A and J_R on L.

(2) *Proper global covers* (THM-NT2-NONTRIVIAL-GLOBAL-COVERS): at the global context Ω both generating families consist of proper pieces, and the generated sieves are nonmaximal — the exact point at which the successor escapes the collapse.

(3) *Subcanonicity* (THM-NT2-SUBCANONICAL-CONTEXT-FACTOR): J_A and J_R are subcanonical on L. This concerns L only and neither repairs nor bears on the legacy subcanonicity gap.

(4) *Distinct, non-identity modalities* (THM-NT2-DISTINCT-MODALITIES): explicit finite witness presheaves F_A and F_R — bit assignments on complete blocks — separate the two sheaf subcategories: F_A is a J_A-sheaf but not a J_R-sheaf, and symmetrically. Hence the reflective subcategories are distinct and neither sheafification modality is trivial. The scope is $\mathrm{PSh}(L, \mathbf{Set})$; the ambient lift proposed in the v0.1 specification contained an error, the corrected witness remains unverified, and ambient distinctness is OPEN.

(5) *The join* (THM-NT2-JOIN-DISCRETE-CONTEXT): the common refinement of the two global covers is the four-singleton cover, and $J_A \vee J_R$ is the topology generated by the singleton partition — equivalently, the ordinary union topology on the finite powerset lattice.

(6) *Explicit reflectors* (registered within THM-NT2-MODALITY-COMPOSITION; see Boundary 10.33): a product-over-blocks construction realizes each associated-sheaf reflector concretely, verified directly against the universal property rather than through the general plus construction. This item alone carries no executable corroboration — the universal property of the reflector and the sheafhood of its values have no registered check in either verifier (Chapter 11), and their machine verification is admission gate 2.

(7) *The composition theorem* (THM-NT2-MODALITY-COMPOSITION, v0.2): the two modalities satisfy unit-compatible canonical natural isomorphisms $M_A M_R \cong M_{\text{atom}} \cong M_R M_A$, and either composite is naturally isomorphic to sheafification for the joined topology.

(8) *Identity on the resolved subcategory* (THM-NT2-RESOLVED-MODALITIES-IDENTITY): every sheaf for the join is separately a sheaf for each factor topology, so both reflectors restrict to the identity on the resolved subcategory — the formal content of Move 3.

We state one consequence separately, because it corrects a live narrative risk rather than fulfilling a hope.

Adverse Finding 1.1 (Distinct but commuting). The finite-context modalities are distinct and non-identity, yet they commute up to canonical natural isomorphism. Earlier informal program materials hoped for interacting, order-sensitive modal structure; distinctness does not imply noncommutation, no current artifact licenses a noncommutation claim, and any successor construction seeking genuinely order-sensitive modalities must change the site again rather than reinterpret this one.

Beyond the finite theory, the monograph carries exactly one conjecture and a register of open obligations. The conjecture (thm-nt2-product-site-backward-compatibility, status proof required) is the product-site round trip: after atomic resolution of the context factor, an equivalence $\mathcal{E}_{\mathrm{res}} \simeq \mathcal{E}^4_{\mathrm{legacy}}$ under which a neutral-channel embedding is fully faithful and evaluation at the neutral channel is naturally isomorphic to the identity. Two round trips are required and neither is complete: the categorical comparison, and the strict serialization requirement that registered identifiers and canonical payload hashes survive migration bit-for-bit — a categorical natural isomorphism is not a serialization receipt. This monograph claims no backward compatibility. And even a proved round trip would expose an expressivity boundary, stated in the seed paper and developed in Chapter 12: the resolved product is four *independent* legacy channels with no remaining overlap, transition law, or cross-atom coupling, so the sixteen-context construction is a finite separation and provenance reference model, not a substantively rich final semantics for contextual cognition. The registered obligation obl-nt2-cross-context-semantics demands a canonical decision between naming the ambient modal package as principal semantics and adding typed coupling with proved coherence; v0.1 licenses neither branch.

8. Reading guide

The monograph is organized in two parts around a hinge. Part I (Chapters 2–6) is background and diagnosis: the standard category theory needed to read everything else, and the two negative results about the legacy construction. Part II (Chapters 7–12) is the successor: the finite context factor, the provenance layer, the finite theorems, the reflector calculus, the executable certificate, and the ambient program. Four appendices collect exhaustive finite data. In more detail:

Chapter 2 (Categories, functors, and presheaves: a primer).: A tutorial-depth development of categories, functors, natural transformations, opposite categories, presheaves, representables, the Yoneda lemma (with a full proof in the poset case, which is the case the finite theory lives in), and limits with emphasis on pullbacks — including the fact, used constantly, that pullbacks in a poset are meets and in a powerset lattice are intersections. Everything in the chapter is standard material [8, 9, 2, 1], presented as such, with many worked examples on two-, three-, and four-element posets chosen to rehearse in miniature the sixteen-element lattice of Part II.

Chapter 3 (Sieves, Grothendieck topologies, and sites).: Sieves and their identification with subfunctors of representables; pullback sieves; the three topology axioms with intuition; coverage bases and the generated topology; the two elementary maximality facts that drive the collapse; a complete enumeration of all Grothendieck topologies on the two-element chain; and the union coverage on a powerset lattice. Again standard material [1, 4, 7], at tutorial depth.

Chapter 4 (Sheaves, sheafification, and modalities).: Matching families, the sheaf condition, sheafification and its exactness [5], subcanonicity and representable sheaves [6], and the definition of modality (left-exact idempotent monad from sheafification-then-inclusion) with the triviality criterion.

Chapter 5 (The legacy construction and the collapse).: The registered legacy site in full; the Legacy Collapse proposition with proof; what the collapse does and does not invalidate; the corrigendum reading.

Chapter 6 (The countermodel).: Non-derivability of legacy subcanonicity: the injections-on-small-sets countermodel, worked in complete detail, and the resulting representation rule for legacy objects.

Chapter 7 (The context lattice).: The lattice L, its combinatorics (sixteen contexts, eighty-one inclusion arrows), the transverse partitions, the generating families, and the transversality identity.

Chapter 8 (Marked covers).: The provenance layer: record schemas, canonical serialization, hash chaining, derivation kinds, the fail-closed integrity rules, the singleton-context necessity example, and the claim boundary on the mechanism's mathematical status.

Chapter 9 (The finite theorems).: Full proofs of the coverage-basis lemma, the generated-topology theorem, properness of global covers, subcanonicity, the witness-presheaf distinctness theorem, and the join identification.

Chapter 10 (Reflectors and composition).: The explicit product-over-blocks reflector; its sheafhood; the universal-property verification; the composition theorem with its unit-compatibility and naturality checks; identity on the resolved subcategory; the distinct-but-commuting finding.

Chapter 11 (The executable certificate).: What the finite-site checker and the modality-composition verifier actually execute, item by item; the formal (Lean) layer and its limits; and the certificate's own claim boundary, quoted and adopted.

Chapter 12 (The ambient program).: The ambient package $\mathrm{NoeSystem}_2$; the gluing boundary for probabilistic and linear data; the backward-compatibility conjecture and the expressivity boundary; related work and novelty status; the full register of open obligations and admission gates.

Appendix A (Context tables).: The sixteen contexts of L with their partition pieces and inclusion data.

Appendix B (Sieve census).: The enumeration underlying the certificate's sieve counts.

Appendix C (Witness tables).: The witness presheaves F_A and F_R tabulated on all sixteen contexts, with the gluing computations displayed.

Appendix D (Glossary).: Project terms, ledger identifiers, and status vocabulary.

The logical dependencies among the chapters are summarized below; a chapter presupposes the chapters listed against it and nothing else beyond Chapter 1.

Chapter	Depends on	Character
2	—	standard background
3	2	standard background
4	2, 3	standard background
5	3	project diagnosis (negative)
6	3, 4	project diagnosis (negative)
7	2, 3	project construction
8	7	project construction (external layer)
9	4, 7	project theorems (finite, proved)
10	9	project theorems (finite, proved)
11	9, 10	corroboration record
12	5–10	program, conjecture, obligations
A–D	as cited	exhaustive finite data

Prerequisites and pedagogy. The monograph presupposes comfort with naive set theory — sets, functions, images, unions, intersections, partitions — and with reading and writing elementary proofs; a reader wanting a refresher on proof technique may consult a standard introduction such as Hammack's *Book of Proof* [16]. No prior category theory is assumed: Chapters 2–4 build everything used later, and they build it on posets wherever posets suffice, because the finite theory of Part II lives entirely on a poset and the poset case strips away every distraction. The tutorial chapters follow three habits throughout. First, every abstract definition is followed immediately by its instantiation on a two-, three-, or four-element poset, and the same small posets recur so that the reader accumulates a portable laboratory. Second, enumerations are actually carried out — all sieves on a small object, all topologies on the two-element chain — rather than gestured at, because the exhaustive habit is precisely the habit the certificate of Chapter 11 mechanizes at scale. Third, exercises are interleaved where working a computation by hand is more instructive than reading it; solutions are not provided, but every exercise is within reach of the worked examples that precede it, and none is needed for the logical development.

Suggested paths. A reader new to category theory should read Chapters 2–4 in order, then 5–6, then Part II linearly; the small-poset examples of Chapters 2–3 are designed so that every later construction has already been met in miniature. A reader fluent in topos theory can skim Chapters 2–4 for notation, read Chapter 5 and Chapter 6 for the negative results (both are short), and then work through Chapters 7–12, consulting the appendices as the exhaustive data are cited. A reader whose interest is the evidence discipline itself — what is proved, what is corroborated, what is open — should read Section 9 below, Chapter 11, and Chapter 12, in that order, before anything else.

9. Evidence discipline

Nothing in this monograph is peer-reviewed, and the monograph is written so that this fact is impossible to lose. This section states the discipline once, in full; individual chapters repeat the locally relevant boundaries in *Claim Boundary* environments at the point of use.

9.1. Artifact status and rings. The governing artifact status, inherited verbatim from the v0.2 seed paper and its source basis, is: DRAFT / GATE 1 RESEARCH CANDIDATE / RING 4 / NOT LICENSED / HELD. Ring 4 is the program's outermost evidence ring: material that may be developed, executed, and internally corroborated, but that licenses no external claim — no empirical claim, no causal claim, no safety claim, no product claim — and that

may leave the ring only through the explicit admission gates listed in Chapter 12. Those gates include, among others: independent category-theory review of every proposition and theorem; proof-assistant formalization of the finite site, topologies, witnesses, reflectors, join, and natural isomorphisms; the typed observational-cover construction; the product-site lift; the backward-compatibility conjecture with both round trips; the cross-context-semantics decision; separate probability-gluing theorems; a dedicated novelty search for the marked-cover mechanism; registry, authorship, and IP reconciliation; and external referee review with explicit confirmation that no empirical or product wording was licensed. Drafting this monograph promotes no claim and closes no gate.

9.2. Source basis. Every project fact, status, hash, and count in this monograph comes from the registered source artifacts and from nowhere else:

- the formal specification `NOETIC_TOPOS_2_SPEC_v0.1.md`, with SHA-256 digest `156A5D8C5320F36E81E3B46473BA8A038B35672DCD28839A404D96A3DB1A763A`;
- the theorem ledger v0.1 (digest abbreviated `52E00065...` in the seed paper, full value in the registered ledger);
- the proof-obligation ledger v0.1 (abbreviated `FD567E38...`);
- the finite model certificate `FINITE_MODEL_CERTIFICATE_v0.1.json`, model identifier NT2-CONTEXT-2X2-TRANSVERSE-PARTITIONS;
- the composition proof `MODALITY_COMPOSITION_PROOF_v0.1.md` and its certificate;
- the publication delta at base commit `a1cd990109e7561dfd03949bf0e1653461c7ae44`;
- the reference implementation under `src/noetic_topos2/`.

Where this monograph states a number — sixteen contexts, eighty-one arrows, and so on — the number is either proved in the text from the definitions or attributed to the certificate; where a detail is unspecified in the sources, the monograph says so rather than supplying one.

9.3. Ledger proof statuses. Every theorem-shaped statement in this monograph carries the proof status recorded in the registered theorem ledger, and the status vocabulary is fixed as follows. INFORMAL PROOF PRESENT means a written mathematical proof exists in the source artifacts and is reproduced or expanded here; all finite-context theorems of Chapters 9–10 carry this status, together with exhaustive executable corroboration. CONJECTURE / PROOF REQUIRED means precisely what it says; the backward-compatibility statement is the only such item. OPEN marks constructions specified but not built or not proved (the typed observational covers, the ambient lift of distinctness, the product-site lift of the composition theorem, the cross-context semantics decision). NOVELTY UNASSESSED is carried by all eight theorem-ledger records: no dedicated literature search has been performed, and this monograph accordingly never calls any project construction a new theorem in the scholarly sense. Standard results are cited to the standard literature [1, 7, 4, 5, 6] and presented as standard; the boundary between textbook material and project candidate is marked in every chapter.

9.4. What exhaustive finite corroboration does and does not mean. The finite theory of Part II is small enough to check by exhaustive computation, and the project does so with two independent executable layers, described fully in Chapter 11. The finite-site checker verifies, over all four atoms and sixteen contexts: properness and joint exhaustiveness of the global covers, pullback stability of every generating cover, sheafhood of all sixteen

representables, functoriality and the four sheaf/non-sheaf verdicts for the witness presheaves, the transverse refinement identity, the singleton coincidence-and-distinctness pair for marked covers, marked-cover round-trip and parent-provenance binding, and both legacy collapse mechanisms — twenty-three registered checks, all passing. The modality-composition verifier computes exact finite Grothendieck topologies by enumerating all 298 sieves over the sixteen contexts, saturating maximality, pullback stability, and local character to a fixed point; it reports 168 covering sieves for both the joined and singleton presentations, verifies the Grothendieck axioms independently on both computed topologies, and checks the composite decompositions, both canonical natural isomorphisms, inverse laws, restriction naturality, commutation naturality, and unit compatibility across 18 executable fixtures and 3,012 restriction squares. At the bound commit, the finite-site suite (8 tests), the modality-composition suite (6 tests), and the pinned Lean Gate 1 verifier pass; the Lean scope covers four-atom exhaustiveness, proper global pieces, transverse refinement, and witness-map lemmas, and does *not* formalize the sixteen presheaves, sheaf conditions, topology generation, reflector universal properties, the composition theorem, the product-site comparison, or the legacy round trip.

The two executable layers differ in method: the finite-site checker (described in the sources as dependency-free) works from the coverage bases and checks basis-level conditions directly, while the composition verifier constructs the full topologies by saturating all sieves under maximality, pullback stability, and local character to a fixed point and then re-verifies the Grothendieck axioms independently on the computed result. The methodological difference means the two layers do not simply re-run one encoding of "topology" twice; the deterministic run at the bound commit passes both.

What this corroboration *means*: for a statement quantified over a finite structure, an exhaustive check of every instance is a genuine and strong form of evidence — there is no sampling, no approximation, and no untested case within the model checked. It caught, during development, exactly the kinds of slips (a nonfunctorial witness candidate, an unstated empty-context convention) that informal proofs are prone to.

What it does *not* mean, in the certificate's own words, adopted here as governing: exhaustive execution over one sixteen-object site "corroborates the finite proofs; it is not proof-assistant verification, external peer review, empirical validation, or a product-claim license." Corroboration is relative to the implementation's encoding of the definitions; a shared misreading of a definition by prover and programmer would survive it. That is why proof-assistant formalization and independent review remain listed as open admission gates rather than as formalities.

9.5. Formal marks carry no empirical interpretation.

Claim Boundary 1.2. The coordinate names ACBE and RPM, wherever they appear in Part II, are formal labels preserved for controlled migration vocabulary. They carry no empirical interpretation until a separate operationalization and validation record exists. No theorem in this monograph attaches meaning to them beyond their combinatorics, and no theorem here establishes that either coordinate detects a cognitive, causal, safety, or alignment property.

9.6. How to read a claim in this monograph. Because the status vocabulary carries real weight, two worked parses may help calibrate the reader.

Parse 1: a finite theorem. The distinctness theorem of Chapter 9 (ledger THM-NT2-DISTINCT-MODALITIES) asserts that two explicitly constructed presheaves separate the two

sheaf subcategories of $\mathrm{PSh}(L, \mathbf{Set})$. Its full epistemic position is: a written informal proof exists (reproduced and expanded in Chapter 9); every step quantifies over a sixteen-object structure and has been exhaustively executed by the certificate (the checks `acbe_witness_functorial`, `acbe_witness_is_acbe_sheaf` true, `acbe_witness_is_rpm_sheaf` false, and their RPM-side counterparts); the statement's scope is $\mathrm{PSh}(L, \mathbf{Set})$ and *not* the ambient category, because the v0.1 ambient-lift witness was found to be erroneous and the corrected ambient statement is OPEN; the ledger record includes the ambient-lift sentence, and the publication layer — followed by this monograph — corrects it; the result is NOVELTY UNASSESSED and has no independent review. Every clause of that position is part of what the theorem *is*, here.

Parse 2: the conjecture. The backward-compatibility statement (ledger THM-NT2-PRODUCT-SITE-BACKWARD-COMPATIBILITY) is written in theorem-like language — an equivalence, a fully faithful embedding, a natural isomorphism — but carries status PROOF REQUIRED. In this monograph such a statement licenses exactly nothing: no chapter may use it as a premise, no summary may report it as a property of the construction, and the correct citation form is "conjectured, proof required." The same reading applies to every OPEN item: the ambient lift, the typed observational covers, the product-site lift of the composition theorem, and the cross-context-semantics decision are specifications of future work, not results.

The general rule the parses instantiate: in this monograph, a claim is the conjunction of its mathematical statement and its recorded status, and quoting the former without the latter is a misquotation.

9.7. Relation to the seed artifacts. The monograph's textual relationship to its sources is expansion, not revision. The v0.2 research paper (which itself superseded a v0.1 white paper) fixed the claim set: the collapse diagnosis, the countermodel, the finite-context theorems and composition law, the marked-cover mechanism, the ambient package, the conjecture, and the open-obligation register. This monograph adds tutorial background (Chapters 2–4), fully expanded proofs where the paper compressed them, worked small-poset laboratories, exhaustive appendix tables, and extended discussion of the evidence discipline — and it adds no new mathematical claim, promotes no status, and closes no obligation. Where the monograph's expanded prose and a source artifact could be read to differ, the source artifact governs; the statuses, hashes, and counts of Section 9.2 are the fixed points of that ordering. Readers who need the compressed, single-artifact form of the material should read the v0.2 paper; readers who need the executable ground truth should read the certificate and the implementation under `src/noetic_topos2/`; this monograph sits between them as the didactic layer.

9.8. Writing rules.

RULE 1.3 (Claim discipline for this monograph). (i) Standard mathematics is presented at tutorial depth, cited to the standard literature, and never presented as project novelty. (ii) Project constructions are presented with their ledger statuses attached and are never called novel; all carry NOVELTY UNASSESSED. (iii) No sentence asserts an empirical, causal, safety, or product property of any AI system. (iv) Hashes, counts, and identifiers appear only as recorded in the source artifacts; unspecified details are declared unspecified. (v) Claim boundaries travel with the material they bound: each chapter restates the boundaries relevant to its content.

10. Notation and conventions

NOTATION 1.4. Categories are written in calligraphic or roman letters; $\mathcal{C}$ is reserved for the registered legacy category (Noetica in the source artifacts). **Set** is the category of sets and functions in a fixed Grothendieck universe; smallness assumptions are stated where used. $\mathrm{Ob}(\mathcal{C})$ is the class of objects; $\mathrm{Hom}_{\mathcal{C}}(A, B)$ the set of morphisms from A to B; id_A the identity. Composition is written $g \circ f$ for $f : A \to B$ followed by $g : B \to C$, applied right to left. $\mathrm{PSh}(\mathcal{C})$ denotes the category of presheaves $\mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$, and $\mathrm{Sh}(\mathcal{C}, J)$ the full subcategory of sheaves for a topology J. The representable presheaf of an object C is written $yC = \mathrm{Hom}_{\mathcal{C}}(-, C)$.

NOTATION 1.5. A poset $(P, \leq)$ is regarded as a category with a unique morphism $U \to V$ exactly when $U \leq V$; Chapter 2 verifies that this is a category and that all diagrams in it commute. In the powerset lattice $\mathcal{P}(X)$ the order is inclusion, meets are intersections, and joins are unions. The successor's context lattice is $L = \mathcal{P}(\Omega)$ with $\Omega = \{0, 1\} \times \{0, 1\}$; its atoms are written (i, j), its ACBE blocks A_i, its RPM blocks R_j, and its generated topologies J_A, J_R, with join $J_A \vee J_R$. The seed artifacts' internal label J_{atom} for the join is retained where the ledger is quoted, with the standing caution that it does not assert the standard category-theoretic atomic topology; $J_{\mathrm{singleton}}$ is the preferred publication-layer name.

NOTATION 1.6. Modalities are composites $M = ia$ of sheafification a followed by inclusion i; M_{obs}, M_A, M_R are the observational, ACBE, and RPM modalities of the ambient construction. The ambient and resolved categories are $\mathcal{E}_{\mathrm{amb}} = \mathrm{PSh}(\mathcal{C} \times L)$ and $\mathcal{E}_{\mathrm{res}} = \mathrm{Sh}(\mathcal{C} \times L, J_{\mathrm{res}})$ with $J_{\mathrm{res}} = J_{\mathrm{obs}} \vee J_A \vee J_R$; $\mathcal{E}_{\mathrm{legacy}} = \mathrm{Sh}(\mathcal{C}, J_{\mathrm{legacy}})$. Ledger identifiers are set in small capitals (THM-NT2-MARKED-COVERAGE); executable check names and file names are set in typewriter face with escaped underscores (`finite_site.py`). Statuses are set in small capitals: OPEN, PROOF REQUIRED, INFORMAL PROOF PRESENT, NOVELTY UNASSESSED, NOT LICENSED.

NOTATION 1.7 (Environment conventions). The monograph uses a fixed repertoire of numbered environments, and their names are part of the claim discipline. *Theorem*, *Proposition*, *Lemma*, and *Corollary* are reserved for statements with written proofs in this text; when such a statement corresponds to a ledger record, the ledger identifier is displayed with the statement. *Conjecture* marks the backward-compatibility statement and nothing else. *Definition*, *Construction*, *Example*, and *Exercise* have their usual meanings; *Notation* fixes symbols. *Remark* is expository and carries no claim weight. *Claim Boundary* (the `boundary` environment) is normative: it restricts what the surrounding material may be taken to assert, and it binds every reading of that material. *Finding* records a corrective result — something the mathematics settled against a prior informal expectation — and *Rule* records a standing discipline adopted by the text. *Design argument* (the `designargument` environment) is an unnumbered block that draws a design or policy consequence from an already-proved result; it is deliberately not a proof, asserts no new mathematics, and carries no tombstone. *Axiom*, *Algorithm*, *Protocol*, *Invariant*, *Check Class*, and *Threat* are numbered alongside the theorem-class environments and name what they say. Proofs end with the usual tombstone; proof sketches are labeled as sketches and are used only for standard results whose full proofs are cited.

NOTATION 1.8. The monograph source is ASCII; all mathematical symbols are produced by LaTeX commands. Numbered environments (theorem, proposition, lemma, definition, example, remark, boundary, and the rest) are shared across the monograph with per-chapter numbering; labels are prefixed by chapter slug. Tables use booktabs rules. Displayed equations are numbered only when referenced.

The following summary table collects the recurring symbols for reference.

Symbol	Meaning
$\mathcal{C}$	registered legacy category (Noetica)
J_{legacy}	legacy coverage, clauses (O), (A), (R)
Ω	the four-atom set $\{0,1\} \times \{0,1\}$
L	context lattice $\mathcal{P}(\Omega)$, sixteen objects
A_i, R_j	ACBE and RPM partition blocks
J_A, J_R	generated context topologies
$J_A \vee J_R$	joined (singleton) topology
F_A, F_R	witness presheaves for distinctness
M_A, M_R, M_{obs}	sheafification modalities
$\mathcal{E}_{\text{amb}}$, $\mathcal{E}_{\text{res}}$	ambient and resolved categories
NoeTop_2, NoeSystem_2	topos candidate; full research package
yC	representable presheaf of C
m_C	maximal sieve on C

REMARK 1.9. A last word on tone. The monograph's subject is a repair, and the repair is presented together with the failure it repairs, the certificate that corroborates it, and the list of everything it does not yet establish. This is deliberate. In the program's experience, the value of a bookkeeping formalism is inseparable from the honesty of its own bookkeeping: a construction that overstated its status would fail the very discipline it proposes. Preserving the boundaries with the same precision as the theorems is part of the result.

CHAPTER 2

Categories, Functors, and Presheaves: A Primer

1. Purpose and scope

This chapter develops, from nothing, the standard category theory used in the rest of the monograph: categories, functors, natural transformations, opposite categories, presheaves, representable presheaves, the Yoneda lemma, and limits with particular attention to pullbacks. Everything in this chapter is textbook material [8, 9, 2, 1], presented as such; none of it is a project contribution, and the chapter's only originality is in the choice and arrangement of examples. Readers fluent in the subject should skim for notation — in particular the poset conventions of Section 2.1 and the small posets introduced in Section 11, which recur throughout Part II — and move on.

The pedagogical strategy is dictated by the destination. The finite theory of Part II lives entirely on a sixteen-element powerset lattice, viewed as a category; every presheaf that matters there is a functor out of a finite poset, every pullback that matters is an intersection of subsets, and every application of the Yoneda lemma is an application of its poset case. This chapter therefore works the poset case of everything in full, on posets small enough that all objects can be enumerated by hand: the two-element chain, the three-element chain, a three-element "vee," and the four-element powerset lattice of a two-element set. The general definitions are always given — the legacy category $\mathcal{C}$ of Chapter 5 is not a poset, and the countermodel of Chapter 6 lives in a category of finite sets and injections — but the general theory is cited rather than redeveloped where the poset case suffices for Part II. A reader wanting a refresher on elementary proof technique before starting may consult Hammack [16]; the proofs below use nothing beyond induction, case analysis, and careful bookkeeping with sets and functions.

2. Categories

Definition 2.1 (Category). A *category* $\mathcal{C}$ consists of the following data:

(1) a class $\mathrm{Ob}(\mathcal{C})$ of *objects*;
(2) for each ordered pair of objects (A, B), a set $\mathrm{Hom}_{\mathcal{C}}(A, B)$ of *morphisms* from A to B (written $f : A \to B$; A is the *domain* of f and B its *codomain*);
(3) for each triple of objects (A, B, C), a *composition* operation

$$\circ : \mathrm{Hom}_{\mathcal{C}}(B, C) \times \mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{C}}(A, C), \qquad (g, f) \mapsto g \circ f;$$

(4) for each object A, an *identity* morphism $\mathrm{id}_A \in \mathrm{Hom}_{\mathcal{C}}(A, A)$;

subject to two axioms:

(C1) *Associativity*: $h \circ (g \circ f) = (h \circ g) \circ f$ whenever the composites are defined;
(C2) *Unitality*: $\mathrm{id}_B \circ f = f = f \circ \mathrm{id}_A$ for every $f : A \to B$.

We assume morphism sets with distinct domain-codomain pairs are disjoint, so that every morphism has a well-defined domain and codomain. A category is *small* when $\mathrm{Ob}(\mathcal{C})$ is a

set (not a proper class) and *locally small* when each $\mathrm{Hom}_{\mathcal{C}}(A, B)$ is a set, as Definition 2.1 already requires. Size issues play one role in this monograph: topos-existence statements require the site to be small in a chosen universe, and the finite lattice L of Part II is not merely small but finite, which is why exhaustive checking is possible at all.

Example 2.2 (The category of sets). **Set** has all sets (in a fixed universe) as objects and all functions as morphisms, with ordinary composition of functions and the ordinary identity functions. Axioms (C1) and (C2) are the familiar facts that composition of functions is associative and that identity functions are neutral. **Set** is locally small but not small.

Example 2.3 (Finite sets). **FinSet** has finite sets as objects and all functions between them as morphisms; it is a *full subcategory* of **Set**, meaning its morphism sets are exactly those of **Set** between the retained objects. Variants with restricted morphisms also matter later: the category of finite sets and *injections* (same objects, only injective functions) is a category, because identities are injective and composites of injections are injections. Chapter 6's countermodel lives in the full subcategory of that category on sets of size at most two. Note that restricting morphisms can change categorical properties drastically — which is the point of the countermodel.

2.1. Posets as categories.

Definition 2.4 (Poset). A *partially ordered set* (poset) is a set P with a relation $\leq$ that is reflexive ($U \leq U$), transitive ($U \leq V$ and $V \leq W$ imply $U \leq W$), and antisymmetric ($U \leq V$ and $V \leq U$ imply $U = V$).

Proposition 2.5 (A poset is a category). *Let $(P, \leq)$ be a poset. Define a category, also written P, with $\mathrm{Ob}(P) = P$ and*

$$\mathrm{Hom}_P(U, V) = \begin{cases} \{\, \iota_{U,V} \,\} & \text{if } U \leq V, \\ \varnothing & \text{otherwise,} \end{cases}$$

where $\iota_{U,V}$ is a formal symbol ("the" morphism from U to V). Composition is $\iota_{V,W} \circ \iota_{U,V} = \iota_{U,W}$ and the identity on U is $\iota_{U,U}$. Then P is a category.

Proof. Composition is well defined: if $U \leq V$ and $V \leq W$ then $U \leq W$ by transitivity, so $\iota_{U,W}$ exists. Identities exist by reflexivity. Associativity (C1): both $h \circ (g \circ f)$ and $(h \circ g) \circ f$, when defined, are morphisms $U \to Z$ for the same pair (U, Z), and $\mathrm{Hom}_P(U, Z)$ has at most one element, so they are equal. Unitality (C2) holds for the same reason: $\mathrm{id}_V \circ \iota_{U,V}$ and $\iota_{U,V}$ are both elements of the at-most-one-element set $\mathrm{Hom}_P(U, V)$. □

We isolate the at-most-one-morphism property just used, because it is invoked silently on almost every page of Part II.

Proposition 2.6 (All diagrams in a poset commute). *In a poset regarded as a category, any two morphisms with the same domain and the same codomain are equal. Consequently every diagram of morphisms commutes: any two composites with the same source and target agree.*

Proof. Immediate from $|\mathrm{Hom}_P(U, V)| \leq 1$: parallel morphisms lie in a common set with at most one element. A composite of morphisms $U \to \cdots \to V$ is a morphism $U \to V$, so any two such composites are parallel, hence equal. □

Example 2.7 (The standard small posets). Four posets recur throughout this monograph.

(1) The *two-element chain* $[2] = \{0, 1\}$ with $0 \leq 1$. As a category it has two objects and three morphisms: id_0, id_1, and $f : 0 \to 1$.
(2) The *three-element chain* $[3] = \{0, 1, 2\}$ with $0 \leq 1 \leq 2$. Six morphisms: three identities and the arrows $0 \to 1$, $1 \to 2$, $0 \to 2$, with $(1 \to 2) \circ (0 \to 1) = (0 \to 2)$ — the only nontrivial composition, forced by Proposition 2.6.
(3) The *vee* $\Lambda = \{x, y, t\}$ with $x \leq t$, $y \leq t$, and x, y incomparable. Five morphisms: three identities, $x \to t$, $y \to t$. There is no morphism between x and y in either direction.
(4) The *square* $B = \mathcal{P}(\{1, 2\})$, the powerset of a two-element set ordered by inclusion: objects $\varnothing$, $\{1\}$, $\{2\}$, $\{1, 2\}$. Counting ordered pairs $U \subseteq V$: $\varnothing$ is below all four objects, $\{1\}$ below itself and $\{1, 2\}$, $\{2\}$ likewise, and $\{1, 2\}$ below itself only — $4 + 2 + 2 + 1 = 9$ morphisms.

PROPOSITION 2.8 (Counting inclusions in a powerset). *For a finite set X, the number of ordered pairs (U, V) with $U \subseteq V \subseteq X$ — that is, the number of morphisms of the category $\mathcal{P}(X)$ — is $3^{|X|}$.*

PROOF. A pair $U \subseteq V$ is determined by assigning to each element of X one of three mutually exclusive statuses: in U (hence in V), in V but not U, or outside V. Distinct assignments give distinct pairs and every pair arises, so the count is $3^{|X|}$. □

For $|X| = 2$ this gives the 9 morphisms of B just counted by hand; for $|X| = 4$ it gives $3^4 = 81$, which is the arrow count of the context lattice $L = \mathcal{P}(\{0, 1\} \times \{0, 1\})$ recorded by the finite certificate and used constantly in Part II. The agreement between this one-line combinatorial argument and the certificate's exhaustive enumeration is a small instance of the corroboration discipline of Chapter 11.

2.2. Monoids as one-object categories.

DEFINITION 2.9 (Monoid). A *monoid* is a set M with an associative binary operation $(m, m') \mapsto mm'$ and a two-sided unit $e \in M$: $em = m = me$.

PROPOSITION 2.10 (Monoids are one-object categories). *For a monoid M, define a category $\mathbf{B}M$ with a single object $*$, $\mathrm{Hom}(*, *) = M$, composition $m' \circ m = m'm$ (multiplication), and $\mathrm{id}_* = e$. Then $\mathbf{B}M$ is a category. Conversely, if $\mathcal{C}$ is a category with exactly one object $*$, then $\mathrm{Hom}_{\mathcal{C}}(*, *)$ with composition as multiplication and id_* as unit is a monoid, and the two constructions are mutually inverse.*

PROOF. For $\mathbf{B}M$: all composites are defined (there is one object), and (C1), (C2) are exactly associativity and unitality of M. Conversely, in a one-object category the composition operation is a total binary operation on the single morphism set; (C1) makes it associative and (C2) makes id_* a two-sided unit. Applying the first construction to the monoid so obtained returns the original category, and applying the second to $\mathbf{B}M$ returns M. □

Posets and monoids are the two degenerate extremes of the notion of category: a poset has many objects but at most one morphism between any two, so all information is in *which* morphisms exist; a monoid has one object and many morphisms, so all information is in *how* morphisms compose. General categories mix both kinds of information. The finite theory of Part II lives at the poset extreme; the legacy category $\mathcal{C}$ of Chapter 5 does not, which is why the general definitions are needed at all.

2.3. Isomorphisms, monomorphisms, epimorphisms.

DEFINITION 2.11. Let $f : A \to B$ be a morphism in a category $\mathcal{C}$.

(1) f is an *isomorphism* if there exists $g : B \to A$ with $g \circ f = \mathrm{id}_A$ and $f \circ g = \mathrm{id}_B$; such g is unique when it exists and is written f^{-1}.
(2) f is a *monomorphism* (mono) if for all parallel $h, k : Z \to A$, $f \circ h = f \circ k$ implies $h = k$.
(3) f is an *epimorphism* (epi) if for all parallel $h, k : B \to Z$, $h \circ f = k \circ f$ implies $h = k$.

PROPOSITION 2.12 (Monos and epis in **Set**). *In* **Set***, a function is a monomorphism if and only if it is injective, and an epimorphism if and only if it is surjective.*

PROOF. *Mono ⇔ injective.* If $f : A \to B$ is injective and $f \circ h = f \circ k$ for $h, k : Z \to A$, then for each $z \in Z$, $f(h(z)) = f(k(z))$, so $h(z) = k(z)$; thus $h = k$. Conversely, suppose f is a mono and $f(a) = f(a')$. Let $Z = \{*\}$ be a one-element set and define $h, k : Z \to A$ by $h(*) = a$, $k(*) = a'$. Then $f \circ h = f \circ k$, so $h = k$, so $a = a'$.

Epi ⇔ surjective. If f is surjective and $h \circ f = k \circ f$, then every $b \in B$ is $f(a)$ for some a, so $h(b) = h(f(a)) = k(f(a)) = k(b)$; thus $h = k$. Conversely, suppose f is not surjective; choose $b_0 \in B$ with $b_0 \notin \mathrm{im} f$. Define $h, k : B \to \{0, 1\}$ by h constantly 0, and $k(b) = 1$ if $b = b_0$, else $k(b) = 0$. Then $h \neq k$ but $h \circ f = k \circ f$, since f never hits b_0. So f is not an epi. □

PROPOSITION 2.13 (Isomorphisms in a poset are identities). *In a poset regarded as a category, the only isomorphisms are the identity morphisms.*

PROOF. Suppose $f : U \to V$ has an inverse $g : V \to U$. The existence of f and g means $U \leq V$ and $V \leq U$, so $U = V$ by antisymmetry. Then $f \in \mathrm{Hom}_P(U, U) = \{\mathrm{id}_U\}$, so $f = \mathrm{id}_U$. □

This innocuous fact has a consequence used in Section 9: in a poset, isomorphic representables have equal representing objects — there is no "isomorphism slack."

2.4. Split epimorphisms and section-retraction pairs. The next definition is elementary, but it is the single most consequential definition of the chapter for this monograph's story: the legacy collapse of Chapter 5 turns on it.

DEFINITION 2.14 (Section, retraction, split epi). A morphism $\varepsilon : B \to A$ is a *split epimorphism* if there exists $\eta : A \to B$ with $\varepsilon \circ \eta = \mathrm{id}_A$. One then calls η a *section* of ε, calls ε a *retraction* of η, and calls (η, ε) a *section-retraction pair*. Nothing is assumed about the other composite $\eta \circ \varepsilon : B \to B$, which in general is not id_B.

PROPOSITION 2.15. *Every split epimorphism is an epimorphism, and every section is a monomorphism.*

PROOF. Let $\varepsilon \circ \eta = \mathrm{id}_A$. If $h \circ \varepsilon = k \circ \varepsilon$, compose on the right with η: $h = h \circ \mathrm{id}_A = h \circ \varepsilon \circ \eta = k \circ \varepsilon \circ \eta = k$. Dually, if $\eta \circ h = \eta \circ k$, compose on the left with ε to get $h = k$. □

EXAMPLE 2.16. In **Set**, every surjection $\varepsilon : B \to A$ with A nonempty (or $B = A = \varnothing$) is split: choose, for each $a \in A$, an element $\eta(a) \in \varepsilon^{-1}(a)$. The composite $\eta \circ \varepsilon$ sends each b to the chosen representative of its fiber — generally not b itself. This is the shape to keep in mind: passing from B down to A and back up loses exactly the within-fiber information. The registered RPM data of the legacy construction have this shape: $\eta_X : X \to R(X)$, $\varepsilon_X : R(X) \to X$, $\varepsilon_X \circ \eta_X = \mathrm{id}_X$, with the composite the other way explicitly *not* required

to be the identity — a section-retraction pair, not an adjoint equivalence, encoding the program-wide rule that prerequisite satisfaction is one-directional.

REMARK 2.17. In a poset, if $\varepsilon : V \to U$ is a split epimorphism with section $\eta : U \to V$, then $U \leq V$ and $V \leq U$, so $U = V$ and both maps are id_U (Proposition 2.13). Posets harbor no nontrivial section-retraction pairs — one reason the successor's context factor, a poset, is structurally immune to the split-epi collapse mechanism of Chapter 5.

EXERCISE 2.18. Show that in a poset regarded as a category, *every* morphism is both a monomorphism and an epimorphism. (Hint: Proposition 2.6 makes the required implications vacuous.) Conclude that "mono and epi" does not imply "iso" in general categories.

3. Functors

DEFINITION 2.19 (Functor). Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A *functor* $F : \mathcal{C} \to \mathcal{D}$ assigns to each object $A \in \mathrm{Ob}(\mathcal{C})$ an object $F(A) \in \mathrm{Ob}(\mathcal{D})$ and to each morphism $f : A \to B$ a morphism $F(f) : F(A) \to F(B)$, such that

(F1) $F(g \circ f) = F(g) \circ F(f)$ whenever $g \circ f$ is defined, and
(F2) $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$ for every object A.

Functors compose in the evident way (apply one, then the other), and the identity assignment is a functor; small categories and functors between them themselves form a category [2].

EXAMPLE 2.20. (i) The inclusion **FinSet** $\to$ **Set** is a functor. (ii) For a fixed set S, the assignment $A \mapsto A \times S$, $f \mapsto f \times \mathrm{id}_S$ is a functor **Set** $\to$ **Set**. (iii) For any object C of a locally small category $\mathcal{C}$, the *covariant hom functor* $\mathrm{Hom}_{\mathcal{C}}(C, -) : \mathcal{C} \to$ **Set** sends B to $\mathrm{Hom}_{\mathcal{C}}(C, B)$ and $g : B \to B'$ to the map "postcompose with g." Functoriality is associativity and unitality of composition in $\mathcal{C}$.

PROPOSITION 2.21 (Functors between posets are monotone maps). *Let P and Q be posets regarded as categories. A functor $F : P \to Q$ is exactly the same data as a monotone (order-preserving) function $F : P \to Q$: a function with $U \leq V \Rightarrow F(U) \leq F(V)$.*

PROOF. Given a functor F, its object part is a function $P \to Q$. If $U \leq V$, the morphism $\iota_{U,V}$ exists, and $F(\iota_{U,V})$ is a morphism $F(U) \to F(V)$ in Q, which by definition of Q as a category means $F(U) \leq F(V)$. So the object part is monotone. Conversely, a monotone function F extends to morphisms in exactly one way (send $\iota_{U,V}$ to $\iota_{F(U),F(V)}$, which exists by monotonicity), and (F1), (F2) hold automatically by Proposition 2.6 applied in Q: all the required equations are between parallel morphisms in a poset. The two constructions are mutually inverse. □

EXAMPLE 2.22. The functors $[2] \to [2]$ are the monotone self-maps of the two-element chain: the constant map at 0, the constant map at 1, and the identity. The non-monotone swap $0 \mapsto 1, 1 \mapsto 0$ is not a functor: it would have to send the morphism $0 \to 1$ to a morphism $1 \to 0$, and there is none.

PROPOSITION 2.23 (Functors between monoids are homomorphisms). *For monoids M and N, functors $\mathbf{B}M \to \mathbf{B}N$ correspond exactly to monoid homomorphisms $M \to N$ (functions preserving multiplication and unit).*

PROOF. A functor sends $*$ to $*$ (no choice) and each morphism $m \in M$ to a morphism $F(m) \in N$; (F1) says $F(m'm) = F(m')F(m)$ and (F2) says $F(e_M) = e_N$ — precisely the homomorphism conditions. Conversely any homomorphism defines a functor this way. □

Definition 2.24 (Faithful, full). A functor $F : \mathcal{C} \to \mathcal{D}$ is *faithful* if each map $\mathrm{Hom}_{\mathcal{C}}(A, B) \to \mathrm{Hom}_{\mathcal{D}}(FA, FB)$ is injective, *full* if each is surjective, and *fully faithful* if both.

4. Natural transformations

Functors compare categories; natural transformations compare functors. The definition is short, but its content is the commuting square, and the examples below are chosen to show that the square has real force.

Definition 2.25 (Natural transformation). Let $F, G : \mathcal{C} \to \mathcal{D}$ be functors. A *natural transformation* $\alpha : F \Rightarrow G$ assigns to each object $A \in \mathrm{Ob}(\mathcal{C})$ a morphism $\alpha_A : F(A) \to G(A)$ in $\mathcal{D}$ (the *component* at A) such that for every morphism $f : A \to B$ in $\mathcal{C}$,

$$G(f) \circ \alpha_A = \alpha_B \circ F(f) \qquad \text{(naturality at } f\text{)}.$$

If every component is an isomorphism, α is a *natural isomorphism*, and F and G are *naturally isomorphic*, written $F \cong G$.

Natural transformations compose componentwise ($(\beta \circ \alpha)_A = \beta_A \circ \alpha_A$; naturality of the composite follows by pasting the two squares), and the identity transformation has components $\mathrm{id}_{F(A)}$. Hence:

Definition 2.26 (Functor category). For categories $\mathcal{C}$ (small) and $\mathcal{D}$, the *functor category* $[\mathcal{C}, \mathcal{D}]$ has functors $\mathcal{C} \to \mathcal{D}$ as objects and natural transformations as morphisms. Its isomorphisms are exactly the natural isomorphisms.

Example 2.27 (Natural transformations into a poset are pointwise inequalities). Let $\mathcal{C}$ be any small category and Q a poset. For functors $F, G : \mathcal{C} \to Q$, a natural transformation $F \Rightarrow G$ exists if and only if $F(A) \leq G(A)$ for every object A, and is then unique. Indeed, a component α_A is a morphism $F(A) \to G(A)$ in Q, which exists iff $F(A) \leq G(A)$ and is then unique; the naturality squares hold automatically by Proposition 2.6 applied in Q. So the functor category $[\mathcal{C}, Q]$ is itself a poset: functors ordered pointwise. This is the first hint of a theme that recurs with presheaves: when the target or source collapses to a poset, natural-transformation data collapse to order data.

Example 2.28 (Intertwiners between monoid homomorphisms). Let $f, g : M \to N$ be monoid homomorphisms, regarded as functors $\mathbf{B}M \to \mathbf{B}N$ (Proposition 2.23). A natural transformation $f \Rightarrow g$ has a single component: an element $n \in N = \mathrm{Hom}(*, *)$. Naturality at $m \in M$ reads $g(m)\, n = n\, f(m)$ — the element n *intertwines* the two homomorphisms. So natural transformations here are not automatic at all: they exist only when a genuine algebraic condition is solvable. For instance, with $M = N = (\mathbb{N}, +)$ and f, g the homomorphisms "multiply by 2" and "multiply by 3," naturality demands $3m + n = n + 2m$ for all m, i.e. $m = 0$ for all m, which fails: there is no natural transformation from f to g. The commuting square has teeth.

Example 2.29 (Morphisms of presheaves on the two-element chain). Anticipating Section 6: a contravariant **Set**-valued functor on $[2]$ amounts to a function $r : S_1 \to S_0$ (Section 6 makes this precise), and a natural transformation between two such, $r : S_1 \to S_0$ and $r' : S_1' \to S_0'$, is a pair of functions $\alpha_1 : S_1 \to S_1'$, $\alpha_0 : S_0 \to S_0'$ with $r' \circ \alpha_1 = \alpha_0 \circ r$ — a commuting square of sets. Everything in Part II about maps of presheaves is this picture with more objects.

5. Opposite categories and contravariance

Definition 2.30 (Opposite category). The *opposite category* $\mathcal{C}^{\mathrm{op}}$ of a category $\mathcal{C}$ has the same objects, and $\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A,B) = \mathrm{Hom}_{\mathcal{C}}(B,A)$, with composition inherited from $\mathcal{C}$ in reversed order and the same identities. Clearly $(\mathcal{C}^{\mathrm{op}})^{\mathrm{op}} = \mathcal{C}$.

Example 2.31. For a poset $(P,\leq)$, the opposite category is the poset $(P,\geq)$: all arrows reverse. For the powerset lattice $\mathcal{P}(X)$ ordered by inclusion, the opposite is $\mathcal{P}(X)$ ordered by reverse inclusion — which is again a powerset lattice via complementation, though we never need that fact.

Definition 2.32 (Contravariant functor). A *contravariant functor* from $\mathcal{C}$ to $\mathcal{D}$ is a functor $\mathcal{C}^{\mathrm{op}} \to \mathcal{D}$. Concretely: it assigns to each object A an object $F(A)$, and to each morphism $f : A \to B$ of $\mathcal{C}$ a morphism $F(f) : F(B) \to F(A)$ *in the reverse direction*, with $F(g \circ f) = F(f) \circ F(g)$ and $F(\mathrm{id}_A) = \mathrm{id}_{F(A)}$.

The duality principle — every theorem about categories yields a dual theorem by applying it to opposite categories — is used lightly in this monograph, mainly through the mono/epi and section/retraction dualities already visible in Proposition 2.15. What we need constantly is the contravariant direction itself: presheaves, the subject of the next section, are contravariant by definition, and the directional bookkeeping (restriction goes *down*) must become second nature.

6. Presheaves

Definition 2.33 (Presheaf). Let $\mathcal{C}$ be a small category. A *presheaf* on $\mathcal{C}$ is a functor $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$. Unwinding: F assigns to each object C a set $F(C)$ (the *sections over* C) and to each morphism $f : D \to C$ a function $F(f) : F(C) \to F(D)$ (the *restriction along* f), such that restriction along an identity is the identity and restriction along a composite is the composite of the restrictions in reverse order. The category of presheaves is $\mathrm{PSh}(\mathcal{C}) = [\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$, with natural transformations as morphisms.

When $\mathcal{C}$ is a poset and $U \leq V$, we write the restriction of a section $s \in F(V)$ along the unique morphism $U \to V$ as $s|_U \in F(U)$. The functor laws then read: $s|_V = s$ for $s \in F(V)$, and $(s|_V)|_U = s|_U$ whenever $U \leq V \leq W$ and $s \in F(W)$ — restriction is transitive and unital, exactly the algebra of "forgetting part of the picture."

Example 2.34 (Presheaves on the two-element chain). A presheaf F on $[2]$ consists of two sets $F(1)$, $F(0)$ and one restriction function $F(1) \to F(0)$ (along $0 \leq 1$); the functor laws add nothing further, since the only composites in $[2]$ involve identities. So $\mathrm{PSh}([2])$ is the category whose objects are functions between sets and whose morphisms are commuting squares (Example 2.29). This tiny observation generalizes: a presheaf on a poset is a system of sets wired together by restriction functions along the order, one function per covering relation, with commutativity along paths for free in chains but a real condition once the poset has diamonds.

Example 2.35 (Presheaves on the vee). A presheaf on Λ ($x \leq t \geq y$) is a *span*: three sets and two functions $F(t) \to F(x)$, $F(t) \to F(y)$ with common domain. There are no composable nonidentity pairs, hence no commutativity conditions.

Example 2.36 (Presheaves on the square). A presheaf on $B = \mathcal{P}(\{1,2\})$ consists of four sets $F(\{1,2\})$, $F(\{1\})$, $F(\{2\})$, $F(\varnothing)$ and five restriction functions (one per nonidentity inclusion), subject to *one* nontrivial commutativity condition: the two paths from $F(\{1,2\})$

to $F(\varnothing)$ — via $F(\{1\})$ and via $F(\{2\})$ — must agree, because both equal restriction along $\varnothing \subseteq \{1,2\}$. This is Proposition 2.6 biting for the first time: in the source poset the two paths are the same morphism, so the presheaf must send them to the same function.

Example 2.37 (The bit presheaf). On B, define $G(V) = \{0,1\}^V$, the set of all functions $V \to \{0,1\}$, with restriction $s|_U$ the literal restriction of the function s to the subset U. Then $G(\varnothing)$ is a singleton (the empty function), $|G(\{1\})| = |G(\{2\})| = 2$, and $|G(\{1,2\})| = 4$. Functoriality is immediate: restricting a function to V then to $U \subseteq V$ is restricting it to U. This presheaf — "a bit written on each point in view" — is the germ of the witness presheaves of Chapter 9, and the reader met it informally in the vignette of Chapter 1.

Example 2.38 (The constant-verdict presheaf). On B, define $H(V) = \{\mathsf{p}, \mathsf{q}\}$ for all nonempty V and $H(\varnothing) = \{*\}$, with identity restrictions among nonempty contexts and the unique map to $H(\varnothing)$. Functoriality is a short check (all composites of identities and unique-to-singleton maps agree). This is the second presheaf of the Chapter 1 vignette; its failure to be a sheaf for the union cover of $\{1,2\}$ is worked in Chapter 4's terms once the sheaf condition is available.

Example 2.39 (Constant, terminal, and initial presheaves). For any set S, the *constant presheaf* ΔS assigns S to every object and id_S to every morphism. Two special cases recur. The *terminal presheaf* $\mathbf{1} = \Delta\{*\}$ has exactly one section everywhere; every presheaf admits exactly one morphism to it (each component must be the unique map to a singleton), so it is terminal in $\mathrm{PSh}(\mathcal{C})$. The *initial presheaf* $\Delta\varnothing$ has no sections anywhere and admits exactly one morphism to every presheaf (all components empty functions). Beware a poset-specific subtlety used with force in Chapter 10: the terminal presheaf on B has one section over $\varnothing$, but a general presheaf may have zero, two, or fifty sections over the bottom context — $F(\varnothing)$ is data, not convention, and constructions that need a singleton there (as the explicit reflector of Chapter 10 does) must impose it explicitly rather than assume it.

Example 2.40 (Presheaves on a discrete poset). Let P be the discrete two-element poset (two objects, no relation besides reflexivity). A presheaf on P is just a pair of sets with no connecting data, and a morphism is a pair of functions: $\mathrm{PSh}(P) \cong \mathbf{Set} \times \mathbf{Set}$. Discreteness kills all interaction — there are no restrictions to be compatible with. This degenerate case is the shape of the *resolved* world of Part II: after the joined topology discretizes the context factor, the resolved subcategory behaves like a product of independent copies, one per atom, with no remaining interaction — the expressivity boundary discussed in Chapters 1 and 12. Interaction lives in the order; flatten the order and only juxtaposition remains.

Exercise 2.41. Count the presheaves on [2] whose section sets are subsets of a fixed two-element set $\{a,b\}$. (There are four choices for each of $F(1), F(0)$ from $\{\varnothing, \{a\}, \{b\}, \{a,b\}\}$ and then $|F(0)|^{|F(1)|}$ choices of restriction; summing over the sixteen pairs, with the convention that there is exactly one function $\varnothing \to \varnothing$, gives $4 + 2 \cdot 4 + 6 = 18$. Carrying the sum out by hand — rather than reading the total off — is a useful five-minute drill in what a presheaf *is*.)

7. Representable presheaves

Definition 2.42 (Representable presheaf). For an object C of a small category $\mathcal{C}$, the *representable presheaf* yC is

$$yC = \mathrm{Hom}_{\mathcal{C}}(-, C) : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set},$$

sending $D \mapsto \mathrm{Hom}_{\mathcal{C}}(D, C)$ and a morphism $h : D' \to D$ to the precomposition map $f \mapsto f \circ h$. A presheaf is *representable* if it is naturally isomorphic to some yC. The assignment $C \mapsto yC$ extends to a functor $y : \mathcal{C} \to \mathrm{PSh}(\mathcal{C})$ (the *Yoneda embedding*): a morphism $g : C \to C'$ induces the natural transformation $yg : yC \Rightarrow yC'$ with components "postcompose with g."

Functoriality of yC is associativity of composition; naturality of yg is associativity again. The representable yC is the category's own record of "all ways of mapping into C" — and sieves, the central objects of Chapter 3, will be exactly the subfunctors of representables.

PROPOSITION 2.43 (Representables on a poset). *Let P be a poset and $W \in P$. Then for every $V \in P$,*

$$yW(V) = \begin{cases} \{*\} & \text{if } V \leq W, \\ \varnothing & \text{otherwise,} \end{cases}$$

and all restriction maps are the unique possible functions. Thus yW is the "indicator presheaf" of the principal down-set $\downarrow W = \{V : V \leq W\}$: it has exactly one section over each context below W and none elsewhere.

PROOF. $yW(V) = \mathrm{Hom}_P(V, W)$, which is a singleton if $V \leq W$ and empty otherwise, by definition of the poset category. Restriction along $U \leq V$ maps $\mathrm{Hom}_P(V, W) \to \mathrm{Hom}_P(U, W)$; when both are singletons it is the unique function between singletons, when the source is empty it is the empty function, and the case source-nonempty, target-empty cannot occur since $U \leq V \leq W$ implies $U \leq W$. Functoriality holds because all diagrams of functions between sets with at most one element commute. □

EXAMPLE 2.44. On $B = \mathcal{P}(\{1, 2\})$, the four representables have section counts (over $\varnothing, \{1\}, \{2\}, \{1, 2\}$ in that order): $y\varnothing : (1, 0, 0, 0)$; $y\{1\} : (1, 1, 0, 0)$; $y\{2\} : (1, 0, 1, 0)$; $y\{1, 2\} : (1, 1, 1, 1)$. Each is the indicator of a down-set, as Proposition 2.43 says. Note $y\{1, 2\}$ is the *terminal* presheaf on B (one section everywhere) — in general, on a poset with a top element, the representable of the top is terminal.

8. Subpresheaves

Chapter 3's central definition — the sieve — will be introduced twice: once elementarily, as a precomposition-closed set of morphisms, and once structurally, as a subfunctor of a representable. The structural half needs the notion of subpresheaf, which we set up now and practice on the small posets.

DEFINITION 2.45 (Subpresheaf). Let F be a presheaf on a small category $\mathcal{C}$. A *subpresheaf* (or *subfunctor*) $S \subseteq F$ is a choice of subset $S(C) \subseteq F(C)$ for each object C such that restrictions preserve membership: for every morphism $f : D \to C$ and every $s \in S(C)$, one has $F(f)(s) \in S(D)$. Equivalently, S is itself a presheaf (with the restricted restriction maps) and the inclusions $S(C) \subseteq F(C)$ form a natural transformation with injective components.

The closure condition is the whole content: a subpresheaf is not an arbitrary family of subsets but a family stable under restriction — "once a section is in, all its shadows are in." On a poset the condition reads: if $s \in S(V)$ and $U \leq V$ then $s|_U \in S(U)$.

EXAMPLE 2.46 (Subpresheaves of a representable on the vee). Work on Λ ($x, y \leq t$) and take $F = yt$, which has exactly one section over each of x, y, t (Section 11.3). A subpresheaf $S \subseteq yt$ chooses, for each object, either the empty set or the full singleton, subject to restriction-stability. If $S(t) = \{*\}$, then restricting the section of $yt(t)$ along $x \leq t$ and $y \leq t$ forces $S(x) = S(y) = \{*\}$: the only option is $S = yt$ itself. If $S(t) = \varnothing$, the choices

at x and y are free and independent, since no nonidentity morphism has domain above x or y. That gives four subpresheaves: $(\varnothing,\varnothing,\varnothing)$, $(\{*\},\varnothing,\varnothing)$, $(\varnothing,\{*\},\varnothing)$, $(\{*\},\{*\},\varnothing)$, writing values over (x,y,t). Total: five subpresheaves of yt. Chapter 3 will recognize these five as exactly the five *sieves* on the object t of Λ, and the forcing argument just made — "a full top forces full shadows" — as the elementary statement that a sieve containing the identity is maximal.

EXAMPLE 2.47 (Subpresheaves of the bit presheaf). On B, let G be the bit presheaf (Example 2.37) and define $S(V) \subseteq G(V)$ to consist of the *constant* functions $V \to \{0,1\}$ (and the empty function over $\varnothing$). Then S is a subpresheaf: a restriction of a constant function is constant. Section counts: $(1,2,2,2)$ over $(\varnothing,\{1\},\{2\},\{1,2\})$. By contrast, the family $T(V) = \{\text{functions summing to 1 over } V\}$ (interpreting the sum of the empty function as 0) is *not* a subpresheaf: the function on $\{1,2\}$ with values $1,0$ sums to 1, but its restriction to $\{2\}$ is the zero function, which sums to 0 and so fails membership. Checking restriction-stability is a real check, and Chapter 11's certificate performs precisely this kind of check — mechanically, for every candidate — under the name functoriality validation.

EXERCISE 2.48. Enumerate all subpresheaves of $y1$ on the chain $[2]$. (There are three; compare them with the three sieves on the object 1 found in Chapter 3.)

9. The Yoneda lemma

The Yoneda lemma says that mapping out of a representable is the same as evaluating: natural transformations $yC \Rightarrow F$ correspond to elements of $F(C)$. It is the fundamental bookkeeping theorem of presheaf categories, and in Part II it is used chiefly in two ways: through the identification of sieves with subfunctors of representables (Chapter 3), and through the sheaf condition's formulation as a bijection $F(C) \cong \mathrm{Nat}(S,F)$ for covering sieves S (Chapter 4), which is a Yoneda-style statement with the sieve in place of the full representable.

THEOREM 2.49 (Yoneda lemma; statement). *Let $\mathcal{C}$ be a small category, $C \in \mathrm{Ob}(\mathcal{C})$, and $F \in \mathrm{PSh}(\mathcal{C})$. The map*

$$\Phi_{C,F} : \mathrm{Nat}(yC, F) \longrightarrow F(C), \qquad \alpha \longmapsto \alpha_C(\mathrm{id}_C),$$

is a bijection, natural in both C and F. In particular (taking $F = yC'$) the Yoneda embedding $y : \mathcal{C} \to \mathrm{PSh}(\mathcal{C})$ is fully faithful. See [**9**, **8**, **2**] *for the general proof.*

We prove the poset case in complete detail; it is the case Part II uses, and its proof displays the mechanism — a natural family is determined by its value at the "generic" point — without the notational overhead of the general case.

THEOREM 2.50 (Yoneda lemma for posets; full proof). *Let P be a poset, $W \in P$, and $F \in \mathrm{PSh}(P)$. Then there is a bijection*

$$\mathrm{Nat}(yW, F) \;\cong\; F(W),$$

given by evaluating at W; it is natural in W and in F.

PROOF. Recall (Proposition 2.43) that $yW(V)$ is a singleton $\{*\}$ for $V \le W$ and empty otherwise.

The map. Given a natural transformation $\alpha : yW \Rightarrow F$, its component at W is a function $\alpha_W : yW(W) \to F(W)$ with $yW(W) = \{*\}$ (as $W \le W$); define $\Phi(\alpha) = \alpha_W(*) \in F(W)$.

Φ *is injective.* Suppose $\Phi(\alpha) = \Phi(\beta)$; we show $\alpha = \beta$ componentwise. At $V \not\leq W$ both components are the unique (empty) function $\varnothing \to F(V)$. At $V \leq W$, apply naturality of α to the morphism $V \to W$: the square

$$F(V \leq W)\big(\alpha_W(*)\big) = \alpha_V\big(*|_V\big) = \alpha_V(*)$$

(the restriction map of yW between the singletons $yW(W)$ and $yW(V)$ sends $*$ to $*$). Hence $\alpha_V(*) = \Phi(\alpha)|_V$ is determined by $\Phi(\alpha)$; the same computation for β gives $\beta_V(*) = \Phi(\beta)|_V$. Since $\Phi(\alpha) = \Phi(\beta)$, all components agree, so $\alpha = \beta$.

Φ *is surjective.* Given $s \in F(W)$, define a candidate transformation α^s by

$$\alpha^s_V(*) = s|_V \quad \text{for } V \leq W,$$

and the empty function at $V \not\leq W$. Naturality must be checked at each morphism $U \leq V$ of P. If $V \leq W$ (hence $U \leq W$): going one way around the square, $* \in yW(V)$ maps to $\alpha^s_V(*) = s|_V$ and then restricts to $(s|_V)|_U = s|_U$ (functoriality of F); going the other way, $*$ restricts to $* \in yW(U)$ and then maps to $\alpha^s_U(*) = s|_U$. Equal. If $V \not\leq W$, the source $yW(V)$ is empty and the square commutes vacuously. So α^s is natural, and $\Phi(\alpha^s) = s|_W = s$.

Naturality in F. Let $\tau : F \Rightarrow G$ be a morphism of presheaves. We must show that for every $\alpha \in \mathrm{Nat}(yW, F)$, $\Phi(\tau \circ \alpha) = \tau_W(\Phi(\alpha))$; and indeed $(\tau \circ \alpha)_W(*) = \tau_W(\alpha_W(*))$ by definition of composition of natural transformations.

Naturality in W. Let $W \leq W'$, inducing $yW \Rightarrow yW'$ and, by precomposition, a map $\mathrm{Nat}(yW', F) \to \mathrm{Nat}(yW, F)$. Chasing the point: $\alpha' \in \mathrm{Nat}(yW', F)$ precomposes to the transformation whose value at W on $*$ is $\alpha'_W(*)$, and the injectivity computation above (applied to α' and the morphism $W \leq W'$) gives $\alpha'_W(*) = \alpha'_{W'}(*)|_W = \Phi(\alpha')|_W$. So the square comparing $\mathrm{Nat}(y(-), F)$ with $F(-)$ along $W \leq W'$ commutes: both routes send α' to $\Phi(\alpha')|_W$. □

Corollary 2.51 (Yoneda embedding on a poset). *For a poset P and $U, W \in P$:*

$$\mathrm{Nat}(yU, yW) \;\cong\; yW(U) \;=\; \begin{cases} \{*\} & U \leq W, \\ \varnothing & \textit{otherwise.} \end{cases}$$

Hence the Yoneda embedding $y : P \to \mathrm{PSh}(P)$ is fully faithful — it is an order-embedding of P into the presheaf category — and $yU \cong yW$ implies $U = W$.

Proof. The displayed bijection is Theorem 2.50 with $F = yW$. Full faithfulness is the statement that $\mathrm{Hom}_P(U, W) \to \mathrm{Nat}(yU, yW)$ is a bijection, and both sides are singletons or both empty according to whether $U \leq W$. If $yU \cong yW$, then in particular $\mathrm{Nat}(yU, yW)$ and $\mathrm{Nat}(yW, yU)$ are both nonempty, so $U \leq W$ and $W \leq U$, whence $U = W$ by antisymmetry. □

Example 2.52 (Yoneda verified by hand on the two-element chain). Let F be any presheaf on $[2]$, i.e. a function $r : F(1) \to F(0)$. Compute $\mathrm{Nat}(y0, F)$ directly: the representable $y0$ has $y0(0) = \{*\}$ and $y0(1) = \varnothing$. A natural transformation $y0 \Rightarrow F$ is a choice of $\alpha_0(*) \in F(0)$ and the empty function at 1; the only naturality square (along $0 \leq 1$) has empty upper-left corner, hence commutes. So $\mathrm{Nat}(y0, F) \cong F(0)$: one transformation per element of $F(0)$, as the lemma predicts. Now $\mathrm{Nat}(y1, F)$: $y1$ has a single section over each object. A transformation is a pair $\alpha_1(*) = s \in F(1)$, $\alpha_0(*) = u \in F(0)$, and the naturality square along $0 \leq 1$ forces $u = r(s)$. So the data reduce to the free choice of $s \in F(1)$: $\mathrm{Nat}(y1, F) \cong F(1)$. Again as predicted — and the mechanism is visible: the component at the representing object is free, and naturality propagates it downward.

Exercise 2.53. On B, take $F = G$ the bit presheaf of Example 2.37. Verify by direct enumeration that $\mathrm{Nat}(y\{1\}, G)$ has exactly two elements and that they correspond to the two elements of $G(\{1\}) = \{0,1\}^{\{1\}}$ as in Theorem 2.50.

Example 2.54 (Counting all morphisms between two presheaves). The Yoneda lemma computes $\mathrm{Nat}(yW, F)$ in one stroke; for non-representable sources one counts squares directly, and it is instructive to do so once. On [2], let F be the presheaf with $F(1) = \{s, s'\}$, $F(0) = \{u\}$ (restriction the constant map) and let F' be the presheaf with $F'(1) = \{v\}$, $F'(0) = \{w, w'\}$, restriction $v \mapsto w$. A morphism $\alpha : F \Rightarrow F'$ is a pair of functions $\alpha_1 : \{s, s'\} \to \{v\}$ and $\alpha_0 : \{u\} \to \{w, w'\}$ making the square commute: $\alpha_0(u)$ must equal the restriction of $\alpha_1(s)$, which is w. So α_1 is forced (constant at v), α_0 is forced ($u \mapsto w$), and $\mathrm{Nat}(F, F')$ has exactly one element — even though there are 2 functions $\{u\} \to \{w, w'\}$, the square eliminates one. In the other direction, $\mathrm{Nat}(F', F)$: $\alpha_1 : \{v\} \to \{s, s'\}$ is a free binary choice, and $\alpha_0 : \{w, w'\} \to \{u\}$ is forced; the square commutes either way (both routes land in the singleton $\{u\}$), so there are exactly 2 morphisms. Naturality is a genuine constraint whose bite depends on direction — a five-minute computation worth repeating on any pair of small presheaves the reader invents.

Remark 2.55 (Covariant representables). Everything in this section dualizes: the covariant functor $\mathrm{Hom}_{\mathcal{C}}(C, -)$ is a "co-presheaf," and the dual Yoneda lemma identifies $\mathrm{Nat}(\mathrm{Hom}_{\mathcal{C}}(C, -), F) \cong F(C)$ for covariant F. This monograph works exclusively with the contravariant side — evidence restricts downward — and we will not use the dual statement, but the reader consulting [2] or [9] should be aware that both conventions circulate and that translation is mechanical.

Remark 2.56 (On the general proof). The general Yoneda proof is the same two-step dance with id_C playing the role of the generic point $*$: injectivity because naturality forces $\alpha_D(f) = F(f)(\alpha_C(\mathrm{id}_C))$ for every $f : D \to C$, surjectivity because that formula, read as a definition, is natural. The poset proof above *is* the general proof specialized; nothing was simplified away except the need to track which morphism one is restricting along, since in a poset there is never more than one. See [9, Ch. 2] or [8, Ch. 8].

10. Limits and pullbacks

Limits formalize "the universal object mapping compatibly into a diagram." Part II needs three instances — terminal objects, products, and above all pullbacks — plus the fact that in a poset all of them reduce to order-theoretic infima. We give the general definitions and then specialize.

Definition 2.57 (Diagram, cone, limit). Let $\mathcal{J}$ be a small category (the *shape*) and $D : \mathcal{J} \to \mathcal{C}$ a functor (a *diagram* of shape $\mathcal{J}$ in $\mathcal{C}$). A *cone* over D with vertex $Z \in \mathrm{Ob}(\mathcal{C})$ is a family of morphisms $(p_j : Z \to D(j))_{j \in \mathrm{Ob}(\mathcal{J})}$ such that for every morphism $u : j \to j'$ of $\mathcal{J}$, $D(u) \circ p_j = p_{j'}$. A *limit* of D is a cone $(\lim D, (\pi_j))$ with the universal property: for every cone $(Z, (p_j))$ there is a unique morphism $u : Z \to \lim D$ with $\pi_j \circ u = p_j$ for all j.

Proposition 2.58 (Limits are unique up to unique isomorphism). *Any two limits of the same diagram are isomorphic by a unique isomorphism commuting with the cone maps.*

Proof. Let $(P, (\pi_j))$ and $(P', (\pi'_j))$ both be limits. The cone $(P', (\pi'_j))$ factors through $(P, (\pi_j))$ by a unique $u : P' \to P$, and symmetrically $(P, (\pi_j))$ factors by a unique $v : P \to P'$. The composite $u \circ v : P \to P$ is a factorization of the cone $(P, (\pi_j))$ through itself; but id_P is another such factorization, and uniqueness in the universal property forces $u \circ v = \mathrm{id}_P$. Symmetrically $v \circ u = \mathrm{id}_{P'}$. □

EXAMPLE 2.59 (Terminal object, product). The limit of the empty diagram is a *terminal object* 1: an object with exactly one morphism from every object. In **Set**, any singleton is terminal. The limit of a discrete two-object diagram (A, B) is the *product* $A \times B$ with its projections; in **Set**, the cartesian product. In a poset, a terminal object is a top element, and a product of U, V is a *meet* (greatest lower bound) $U \wedge V$: the universal property of the product says exactly "$Z \leq U$ and $Z \leq V$ iff $Z \leq U \wedge V$."

DEFINITION 2.60 (Pullback). A *cospan* in $\mathcal{C}$ is a diagram $A \xrightarrow{f} W \xleftarrow{g} B$. A *pullback* of the cospan is a limit of it: an object P with morphisms $p : P \to A$, $q : P \to B$ satisfying $f \circ p = g \circ q$, such that for any Z with $a : Z \to A$, $b : Z \to B$ and $f \circ a = g \circ b$ there is a unique $u : Z \to P$ with $p \circ u = a$ and $q \circ u = b$. (The cone leg to W is determined as $f \circ p$, so one omits it.) Notation: $P = A \times_W B$.

PROPOSITION 2.61 (Pullbacks in **Set**). *In* **Set**, *every cospan* $A \xrightarrow{f} W \xleftarrow{g} B$ *has a pullback:*

$$A \times_W B = \{(a, b) \in A \times B \mid f(a) = g(b)\},$$

with the coordinate projections.

PROOF. The projections satisfy $f \circ p = g \circ q$ by definition of the set. Given a competing cone (Z, a, b) with $f \circ a = g \circ b$, define $u : Z \to A \times_W B$ by $u(z) = (a(z), b(z))$; this lands in the pullback set precisely because $f(a(z)) = g(b(z))$, and it is the unique function whose postcompositions with the projections are a and b, since a function into a subset of a product is determined by its coordinates. □

EXAMPLE 2.62 (Pullback of subset inclusions is intersection). Let $A, B \subseteq W$ be subsets with inclusion maps $A \hookrightarrow W \hookleftarrow B$. The pullback set of Proposition 2.61 is $\{(a, b) \mid a = b \in A \cap B\}$, which the map $(a, b) \mapsto a$ bijects with $A \cap B$; under this bijection the projections become the inclusions of $A \cap B$ into A and B. So in **Set**, "pullback of monos" computes intersection — the prototype of the next proposition, which is the version Part II actually uses.

PROPOSITION 2.63 (Pullbacks in a poset are meets). *Let P be a poset and $U \to W \leftarrow V$ a cospan in P (i.e. $U \leq W$ and $V \leq W$). A pullback of this cospan exists if and only if the meet $U \wedge V$ exists in P, and in that case the pullback is $U \wedge V$ with the unique morphisms to U and V.*

PROOF. A cone over the cospan with vertex Z is a pair of morphisms $Z \to U$, $Z \to V$ whose composites to W agree; the agreement is automatic (Proposition 2.6), so a cone is exactly an element Z with $Z \leq U$ and $Z \leq V$ — a lower bound of $\{U, V\}$. The universal property of the pullback says: P^* is a lower bound such that every lower bound Z admits a (unique, automatic) morphism $Z \to P^*$, i.e. every lower bound satisfies $Z \leq P^*$. That is precisely the definition of the greatest lower bound. So pullbacks of the cospan and meets of $\{U, V\}$ are the same data. Uniqueness of morphisms is automatic throughout, again by Proposition 2.6. □

COROLLARY 2.64 (Pullbacks in a powerset lattice are intersections). *In $\mathcal{P}(X)$ ordered by inclusion, every cospan $U \subseteq W \supseteq V$ has a pullback, namely $U \cap V$.*

PROOF. $\mathcal{P}(X)$ has all binary meets, and the meet of U and V is $U \cap V$: it is a lower bound ($U \cap V \subseteq U$, $U \cap V \subseteq V$), and any Z with $Z \subseteq U$ and $Z \subseteq V$ satisfies $Z \subseteq U \cap V$. Apply Proposition 2.63. □

This corollary is used on nearly every page of Part II: the context lattice $L = \mathcal{P}(\Omega)$ has all pullbacks, they are intersections, and the pullback of a covering piece $V \cap B \to V$ along $U \subseteq V$ is $(V \cap B) \cap U = U \cap B$ — the one-line computation at the heart of the coverage-stability lemma of Chapter 9.

Example 2.65 (A poset without a pullback). In the vee Λ ($x \leq t \geq y$), the cospan $x \to t \leftarrow y$ has no pullback: a pullback would be a meet $x \wedge y$, i.e. a greatest lower bound of $\{x, y\}$, and $\{x, y\}$ has *no* lower bound at all in Λ (neither x nor y is below the other, and t is above both, not below). Existence of pullbacks is a genuine property of a poset, not a formality — powerset lattices have it; arbitrary posets need not. The registered legacy category is required to have pullback closure by its axioms; the context lattice has pullbacks by Corollary 2.64; the vee shows the requirement is not vacuous.

Exercise 2.66. In $B = \mathcal{P}(\{1, 2\})$, compute the pullback of the cospan $\{1\} \to \{1, 2\} \leftarrow \{2\}$ and verify explicitly that every lower bound of $\{\{1\}, \{2\}\}$ factors through it. (Answer: $\varnothing$; the only lower bound is $\varnothing$ itself.)

10.1. Equalizers and general limits. For completeness we record the remaining finite-limit shape and the standard reduction.

Definition 2.67 (Equalizer). Given parallel morphisms $f, g : A \to B$, an *equalizer* is a morphism $e : E \to A$ with $f \circ e = g \circ e$, universal among such: any $h : Z \to A$ with $f \circ h = g \circ h$ factors uniquely through e. In **Set**, $E = \{a \in A \mid f(a) = g(a)\}$ with the inclusion. In a poset, any parallel pair is equal (Proposition 2.6), so every equalizer is an identity — equalizers are invisible in the poset world, which is one more simplification Part II enjoys.

Remark 2.68 (Limits from products and equalizers). A category with all small products and equalizers has all small limits, by the standard construction of a limit as an equalizer of two maps between products [**2**, Ch. V]; **Set** therefore has all small limits. We do not use the general construction — every limit needed in this monograph is a terminal object, a binary product, or a pullback, built by hand — but it is the reason one may speak of "the" limit apparatus of **Set** without further inventory.

10.2. Limits of presheaves are computed pointwise. One structural fact about presheaf categories is needed in Chapter 10, where an explicit reflector is built out of products of section sets: limits of presheaves are computed objectwise.

Proposition 2.69 (Pointwise limits). *Let $\mathcal{C}$ be small and let $D : \mathcal{J} \to \mathrm{PSh}(\mathcal{C})$ be a finite (or small) diagram of presheaves. Define, for each object C of $\mathcal{C}$, $(\lim D)(C) = \lim_j \big(D(j)(C)\big)$, the limit in* **Set** *of the evaluated diagram, with restriction maps induced by the universal property. Then $\lim D$, so defined, is a presheaf and is a limit of D in $\mathrm{PSh}(\mathcal{C})$.*

Proof for binary products, which is the case used. Let F, G be presheaves and define $(F \times G)(C) = F(C) \times G(C)$ with restriction $(s, t)|_f = (F(f)(s), G(f)(t))$ for $f : D \to C$. Functoriality is componentwise. The projections π_F, π_G are natural transformations (their naturality squares are the definitions of the restriction just given). Given a presheaf Z with natural transformations $\alpha : Z \Rightarrow F$, $\beta : Z \Rightarrow G$, define $u_C(z) = (\alpha_C(z), \beta_C(z))$; each u_C is the unique function through which α_C, β_C factor (Proposition 2.61's product analogue in **Set**), and the family (u_C) is natural because its components are built from natural components. Uniqueness of u as a natural transformation follows from uniqueness of each component. The same argument, with the **Set**-limit of the evaluated diagram replacing the cartesian product, proves the general statement; see [**2**, Ch. V] or [**9**, Ch. 3]. □

Example 2.70. On B, the product of the bit presheaf G (counts $(1,2,2,4)$) and the constant-verdict presheaf H (counts $(1,2,2,2)$) has section counts $(1,4,4,8)$, each restriction acting coordinatewise. No new phenomena: the product presheaf is just both records kept side by side — which is exactly how Chapter 10's product-over-blocks reflector should be read, one factor per partition block.

10.3. Equality versus natural isomorphism. A final calibration, load-bearing for Chapter 10. Two functors can be naturally isomorphic without being equal, and the discipline of this monograph keeps the distinction visible. A representative example: on B, the presheaf $G \times H$ of Example 2.70 and the presheaf $H \times G$ are not literally the same functor — their sections are ordered pairs with opposite component order — but the swap maps $(s,t) \mapsto (t,s)$ assemble into a natural isomorphism between them. All structural claims in this monograph that identify two constructed functors are claims of natural isomorphism with named comparison maps, never claims of literal equality; the composition theorem of Chapter 10 states this explicitly for its nested-product flattenings, since a nested tuple $((a,b),(c,d))$ and a flat tuple indexed by four atoms are different set-theoretic objects that a chosen implementation must convert between. The certificate of Chapter 11 checks the conversion maps — both directions, inverse laws, naturality — rather than pretending the objects coincide. Acquire the habit of asking "equal, or isomorphic via which map?" now, on examples this small, where the answer is always computable by hand.

11. The small-poset laboratory

This closing section consolidates the running examples into a single reference, tabulating for each of the four standard posets the data that Chapters 3 and 9 will call on. Everything below is a restatement or short extension of computations already made; the point of gathering them is that Part II's sixteen-object arguments are these arguments with a larger index set, and a reader who owns the small cases owns the method.

11.1. The two-element chain. Objects $0 \le 1$; morphisms $\mathrm{id}_0, \mathrm{id}_1, f : 0 \to 1$. (For an n-element chain the morphism count is the number of pairs $i \le j$, namely $n(n+1)/2$ — here 3; the $3^{|X|}$ formula of Proposition 2.8 is specific to powerset lattices.) A presheaf is a function $F(1) \to F(0)$; a morphism of presheaves is a commuting square. Representables: $y0 = (1,0)$ and $y1 = (1,1)$, writing section counts over $(0,1)$. Yoneda: verified by hand in Example 2.52. Pullbacks: every cospan in a chain has a pullback, since any two elements of a chain are comparable and the meet is the smaller; in particular $0 \to 1 \leftarrow 0$ pulls back to 0.

11.2. The three-element chain. Objects $0 \le 1 \le 2$; six morphisms; the unique nontrivial composition $(1 \to 2) \circ (0 \to 1) = (0 \to 2)$. A presheaf is a composable pair of functions $F(2) \to F(1) \to F(0)$; the functor law adds the requirement that the composite equal the restriction along $0 \le 2$, which is a definition rather than a constraint (there is nothing else for it to be). Representables, section counts over $(0,1,2)$: $y0 = (1,0,0)$, $y1 = (1,1,0)$, $y2 = (1,1,1)$. The chain illustrates transitivity of restriction in its simplest nontrivial form and will reappear in Chapter 3 as the first example with a three-layer sieve structure.

11.3. The vee. Objects $x, y \le t$; five morphisms; no nontrivial compositions. A presheaf is a span $F(x) \leftarrow F(t) \to F(y)$. Representables, section counts over (x,y,t): $yx = (1,0,0)$, $yy = (0,1,0)$, $yt = (1,1,1)$. No meet of $\{x,y\}$, hence no pullback of the cospan $x \to t \leftarrow y$ (Example 2.65). The vee is the minimal illustration that "the overlap of two views" need not

exist as an object — in the powerset lattices of Part II it always does, and the contrast is what the pullback-closure axioms are about.

11.4. The square. $B = \mathcal{P}(\{1,2\})$: four objects, nine morphisms (Proposition 2.8), one nontrivial commutativity condition per presheaf (Example 2.36). Representables tabulated in Example 2.44; the top representable is terminal. All pullbacks exist and are intersections (Corollary 2.64); the generic instance $\{1\} \cap \{2\} = \varnothing$ is Exercise 2.66. Two working presheaves live here: the bit presheaf G (Example 2.37), with section counts $(1,2,2,4)$ over $(\varnothing, \{1\}, \{2\}, \{1,2\})$, and the constant-verdict presheaf H (Example 2.38), with counts $(1,2,2,2)$. Chapter 4 shows: for the covering family $\{\{1\},\{2\}\}$ of $\{1,2\}$, the presheaf G glues perfectly while H does not, and the numbers 4 versus 2×2 already hint at why.

The square is the $|X| = 2$ member of the family $\mathcal{P}(X)$; the context lattice L of Part II is the $|X| = 4$ member, with $X = \Omega = \{0,1\} \times \{0,1\}$ carrying its two coordinates. Everything tabulated here scales: 16 objects, 81 morphisms, representables as down-set indicators, pullbacks as intersections, and presheaves as commuting cubes of restriction functions. The scaling adds bookkeeping, not ideas — which is precisely why the finite certificate of Chapter 11 can check the scaled statements exhaustively, and why this chapter has spent its pages on the small cases where the ideas are visible.

11.5. Comparison table.

	[2]	[3]	Λ	$B = \mathcal{P}(\{1,2\})$
Objects	2	3	3	4
Morphisms	3	6	5	9
Top element	yes	yes	yes	yes
All binary meets	yes	yes	no	yes
All pullbacks	yes	yes	no	yes
Presheaf shape	function	pair	span	square
Commutativity conditions	0	0	0	1

The context lattice $L = \mathcal{P}(\Omega)$ of Part II extends the last column: 16 objects, 81 morphisms (Proposition 2.8), top element Ω, all meets as intersections, all pullbacks, and presheaves as commuting assemblies of restriction functions over the inclusion order.

11.6. Where each tool is used later. For orientation, the primer's results map onto Part II as follows. The poset dictionary and commutation (Propositions 2.5 and 2.6) underlie every diagram chase on L. The split-epi apparatus (Section 2.4) is consumed by the collapse theorem of Chapter 5; the poset immunity remark (Remark 2.17) explains why the successor's site is out of its reach. Subpresheaves (Section 8) become sieves in Chapter 3. Representables-as-indicators (Proposition 2.43) and the poset Yoneda lemma (Theorem 2.50) power the subcanonicity theorem of Chapter 9 and the sheaf condition's formulation in Chapter 4. Pullbacks-as-intersections (Corollary 2.64) is the engine of the coverage-stability lemma in Chapter 9. Pointwise limits (Proposition 2.69) legitimize the product-over-blocks reflector of Chapter 10, and the equality-versus-isomorphism calibration (Section 10.3) governs how that chapter's composition theorem may be stated. The injections category of Example 2.3 reappears as the countermodel of Chapter 6. Nothing in the primer is idle.

12. Summary

The chapter's yield, in the order Part II consumes it: categories and the poset dictionary (objects are contexts, morphisms are inclusions, all diagrams commute); functors and natural

transformations, with the functor category $[\mathcal{C}^{\mathrm{op}}, \mathbf{Set}]$ of presheaves as the ambient world; restriction algebra on posets; representables as down-set indicators (Proposition 2.43); the Yoneda lemma with full proof in the poset case (Theorem 2.50) and its corollary that the Yoneda embedding of a poset is an order-embedding; limits, with pullbacks in posets identified as meets (Proposition 2.63) and in powerset lattices as intersections (Corollary 2.64); and the two special morphism classes — isomorphisms (none nontrivial in posets) and section-retraction pairs (none nontrivial in posets, ubiquitous in **Set**) — whose interaction with sieves opens Chapter 3. All of it is standard [**8**, **9**, **2**, **1**]; the next chapter begins to aim it.

CHAPTER 3

Sieves, Grothendieck Topologies, and Sites

1. Purpose and scope

This chapter develops the covering apparatus: sieves, their identification with subfunctors of representables, pullback sieves, Grothendieck topologies and sites, coverage bases (pretopologies) and the topology they generate, and subcanonicity. As with Chapter 2, everything here is standard material, presented at tutorial depth and cited to the standard sources — Mac Lane–Moerdijk [1], the Stacks Project [4, 5, 6], and Johnstone [7] — and none of it is a project contribution. The chapter's role in the monograph is to put on the table, with complete proofs, the small collection of elementary facts on which everything in Parts I and II turns: that sieves on a poset are down-closed sets of elements; that a sieve containing an identity, or a split epimorphism, is maximal (the two facts that drive the legacy collapse of Chapter 5); that a pretopology generates a topology (the passage that legitimizes the successor's generating families in Chapter 9); and that the union coverage on a powerset lattice is a subcanonical Grothendieck topology (the ambient standard fact of which Chapter 9's join theorem identifies the project-specific instance).

Two complete worked enumerations anchor the abstractions: all sieves on each object of the four small posets of Chapter 2, and all Grothendieck topologies on the two-element chain. The second enumeration is deliberately exhaustive — there are exactly four — because exhaustive enumeration over a small structure is the method of the finite certificate of Chapter 11, and seeing the method by hand on the smallest nontrivial site calibrates what the certificate's larger enumerations do and do not establish.

2. Sieves

2.1. Definition and first examples.

Definition 3.1 (Sieve). Let $\mathcal{C}$ be a category and $C \in \mathrm{Ob}(\mathcal{C})$. A *sieve* on C is a set S of morphisms, all with codomain C, that is closed under precomposition: if $f : D \to C$ belongs to S and $h : E \to D$ is any morphism of $\mathcal{C}$, then $f \circ h$ belongs to S.

The mental picture: a sieve on C is a family of "ways of approaching C" that is closed downward — any way of approaching C that factors through an admitted way is itself admitted. The empty set is a sieve (the closure condition is vacuous), and the set of *all* morphisms into C is a sieve; the latter is important enough to name.

Definition 3.2 (Maximal sieve). The *maximal sieve* on C, written m_C, is the set of all morphisms of $\mathcal{C}$ with codomain C. It contains id_C, and it is the largest sieve on C.

Definition 3.3 (Generated sieve). For a family $\mathcal{F}$ of morphisms with codomain C, the *sieve generated by* $\mathcal{F}$, written $\langle \mathcal{F} \rangle$, is

$$\langle \mathcal{F} \rangle = \{\, f \circ h \mid f \in \mathcal{F},\ h \text{ a morphism with } \mathrm{cod}(h) = \mathrm{dom}(f) \,\}.$$

Proposition 3.4. *$\langle \mathcal{F} \rangle$ is the smallest sieve on C containing $\mathcal{F}$.*

PROOF. It contains $\mathcal{F}$, since $f = f \circ \mathrm{id}$. It is a sieve: a precomposition $(f \circ h) \circ h' = f \circ (h \circ h')$ is again of the required form, by associativity. And any sieve T containing $\mathcal{F}$ contains every $f \circ h$ with $f \in \mathcal{F}$, by T's own closure; hence $\langle \mathcal{F} \rangle \subseteq T$. □

2.2. Sieves on a poset. On a poset the notion collapses to something completely concrete. Recall that $\downarrow V = \{U : U \leq V\}$ denotes the principal down-set of V, and call a subset $D \subseteq \downarrow V$ *down-closed* if $U' \leq U \in D$ implies $U' \in D$.

PROPOSITION 3.5 (Sieves on a poset are down-closed sets). *Let P be a poset and $V \in P$. The assignment $S \mapsto \{U \in P \mid \iota_{U,V} \in S\}$ is a bijection between sieves on V (in the sense of Definition 3.1) and down-closed subsets of $\downarrow V$. Under this bijection the maximal sieve corresponds to $\downarrow V$ itself, the empty sieve to $\varnothing$, and the sieve generated by a family $\{U_i \leq V\}_{i \in I}$ to the union of principal down-sets $\bigcup_{i \in I} \downarrow U_i$.*

PROOF. A morphism with codomain V is exactly a $\iota_{U,V}$ with $U \leq V$, and distinct U give distinct morphisms, so a set of morphisms into V is exactly a subset of $\downarrow V$. Precomposition closure translates as: if $U \in S$ (meaning $\iota_{U,V} \in S$) and $U' \leq U$, then $\iota_{U,V} \circ \iota_{U',U} = \iota_{U',V} \in S$, i.e. $U' \in S$ — precisely down-closedness. The three identifications follow by unwinding: m_V collects all $U \leq V$; the generated sieve of Definition 3.3 collects all U' factoring through some U_i, i.e. $\bigcup_i \downarrow U_i$. □

From here on, when working on a poset we identify a sieve with its down-closed set of elements and write $U \in S$ without comment.

EXAMPLE 3.6 (All sieves on the small posets). Using Proposition 3.5, sieves on an object V are down-closed subsets of $\downarrow V$, and we can enumerate them completely for the four posets of Chapter 2.

(1) *Two-element chain* $[2]$ $(0 \leq 1)$. On 0: $\downarrow 0 = \{0\}$, whose down-closed subsets are $\varnothing$ and $\{0\}$ — two sieves, the empty and the maximal. On 1: $\downarrow 1 = \{0, 1\}$; down-closed subsets are $\varnothing$, $\{0\}$, and $\{0, 1\}$ — three sieves. (Note $\{1\}$ alone is *not* down-closed: $0 \leq 1$ would force $0 \in$.) Total over both objects: 5.

(2) *Three-element chain* $[3]$. On 2: down-closed subsets of $\{0, 1, 2\}$ are $\varnothing, \{0\}, \{0, 1\}, \{0, 1, 2\}$ — four sieves. On 1: three. On 0: two. Total: 9.

(3) *The vee* Λ $(x, y \leq t)$. On t: $\downarrow t = \{x, y, t\}$; down-closed subsets: $\varnothing$, $\{x\}$, $\{y\}$, $\{x, y\}$, $\{x, y, t\}$ — five sieves. (Any set containing t must contain x and y.) On x: two. On y: two. Total: 9. These five sieves on t are exactly the five subpresheaves of yt enumerated in Chapter 2 (Example 2.46) — the coincidence is Proposition 3.8 below.

(4) *The square* $B = \mathcal{P}(\{1, 2\})$. On the top $\{1, 2\}$: down-closed subsets of the whole four-element lattice. Any such set containing the top is everything; the sets omitting the top are the down-closed subsets of the three-element sub-poset $\{\varnothing, \{1\}, \{2\}\}$, namely $\varnothing$, $\{\varnothing\}$, $\{\varnothing, \{1\}\}$, $\{\varnothing, \{2\}\}$, $\{\varnothing, \{1\}, \{2\}\}$ — five of them — so there are six sieves on the top. On $\{1\}$: $\downarrow\{1\} = \{\varnothing, \{1\}\}$ is a two-element chain, giving three sieves; likewise on $\{2\}$. On $\varnothing$: two. Total: $6 + 3 + 3 + 2 = 14$.

The same computation scales to the sixteen-element context lattice $L = \mathcal{P}(\Omega)$ of Part II; the total there is 298 sieves over the sixteen contexts, a count derived in closed form in Chapter 7 (Lemma 7.46 and Theorem 7.47), tabulated in Appendix B, and independently corroborated by the exhaustive enumeration in the registered certificate.

EXERCISE 3.7. Show that on an n-element chain the number of sieves on the top element is $n+1$, and the total number of sieves over all objects is $2+3+\cdots+(n+1) = \frac{(n+1)(n+2)}{2} - 1$. Verify the formula against the enumerations above for $n = 2, 3$.

2.3. Sieves as subfunctors of representables.

PROPOSITION 3.8 (Sieve–subfunctor correspondence). *Let $\mathcal{C}$ be a small category and $C \in \mathrm{Ob}(\mathcal{C})$. For a sieve S on C, define*

$$\widetilde{S}(D) = \{\, f \in \mathrm{Hom}_{\mathcal{C}}(D, C) \mid f \in S \,\} \subseteq yC(D),$$

with restriction maps inherited from yC (precomposition). Then $S \mapsto \widetilde{S}$ is a bijection between sieves on C and subpresheaves of the representable yC.

PROOF. First, $\widetilde{S}$ is a subpresheaf precisely because S is a sieve: the restriction of yC along $h : E \to D$ sends $f \mapsto f \circ h$, and the requirement that this map carry $\widetilde{S}(D)$ into $\widetilde{S}(E)$ (Definition 2.45) says exactly that $f \in S$ implies $f \circ h \in S$ — the closure condition of Definition 3.1. So the assignment is well defined, and conversely, any subpresheaf $T \subseteq yC$ yields the set $S_T = \bigcup_D T(D)$ of morphisms into C, which is precomposition-closed by the same translation read backwards.

The two assignments are mutually inverse: starting from a sieve S, the set $S_{\widetilde{S}}$ collects the elements of all the $\widetilde{S}(D)$, which is S itself, since a morphism into C lies in exactly one $\mathrm{Hom}_{\mathcal{C}}(D, C)$ (namely for D its domain); starting from a subpresheaf T, the subpresheaf $\widetilde{S_T}$ has $\widetilde{S_T}(D) = S_T \cap \mathrm{Hom}_{\mathcal{C}}(D, C) = T(D)$. □

This correspondence is what lets sieves interact with presheaves: a *matching family* for a sieve, defined below (Definition 3.22), is exactly a natural transformation $\widetilde{S} \Rightarrow F$, and the sheaf condition of Chapter 4 compares such transformations with sections over C in Yoneda style. On a poset, the correspondence specializes to the coincidence already observed: sieves on V = down-closed subsets of $\downarrow V$ = subpresheaves of the down-set indicator yV (Proposition 2.43).

2.4. Pullback sieves.

DEFINITION 3.9 (Pullback sieve). Let S be a sieve on C and $g : D \to C$ a morphism. The *pullback sieve* g^*S on D is

$$g^*S = \{\, h : E \to D \mid g \circ h \in S \,\}.$$

g^*S is a sieve: if $g \circ h \in S$ and k is composable, then $g \circ (h \circ k) = (g \circ h) \circ k \in S$ by closure of S. Intuitively, g^*S is the sieve S "as seen from D through g": an approach to D is admitted iff its composite into C was admitted.

LEMMA 3.10 (Membership via pullback). *For any sieve S on C and morphism $g : D \to C$:*

1. *$g^*m_C = m_D$;*
2. *$g \in S$ if and only if $g^*S = m_D$;*
3. *if $S \subseteq T$ then $g^*S \subseteq g^*T$.*

PROOF. (1) Every h into D satisfies $g \circ h \in m_C$. (2) If $g \in S$, then for every $h : E \to D$ we have $g \circ h \in S$ by closure, so $h \in g^*S$; thus $g^*S = m_D$. Conversely if $g^*S = m_D$ then $\mathrm{id}_D \in g^*S$, which says $g \circ \mathrm{id}_D = g \in S$. (3) is immediate from the definition. □

Example 3.11 (Pullback sieves on a poset). On a poset, with S a down-closed subset of $\downarrow V$ and $U \leq V$, the pullback along $U \to V$ is

$$(\iota_{U,V})^* S = \{\, W \leq U \mid W \in S \,\} = S \cap \downarrow U :$$

the part of S lying below U. For instance, on $B = \mathcal{P}(\{1,2\})$, let S be the sieve on the top generated by $\{1\}$, i.e. $S = \downarrow\{1\} = \{\varnothing, \{1\}\}$. Its pullback along $\{2\} \subseteq \{1,2\}$ is $S \cap \downarrow\{2\} = \{\varnothing\}$: from the vantage of $\{2\}$, only the empty overlap of the admitted approach survives. Its pullback along $\{1\} \subseteq \{1,2\}$ is $\{\varnothing, \{1\}\} = m_{\{1\}}$, the maximal sieve — consistent with Lemma 3.10(2), since $\{1\} \in S$.

3. Grothendieck topologies

3.1. The axioms.

Definition 3.12 (Grothendieck topology; site). A *Grothendieck topology* J on a category $\mathcal{C}$ assigns to each object C a collection $J(C)$ of sieves on C, called *covering sieves*, such that:

(J1) (*Maximality*) $m_C \in J(C)$ for every C;
(J2) (*Stability*) if $S \in J(C)$ and $g : D \to C$ is any morphism, then $g^*S \in J(D)$;
(J3) (*Transitivity, or local character*) if $S \in J(C)$ and T is any sieve on C such that $f^*T \in J(D)$ for every $f : D \to C$ in S, then $T \in J(C)$.

A *site* is a pair $(\mathcal{C}, J)$ of a category and a topology on it. See [1, Ch. III] and [3].

Each axiom encodes one piece of covering common sense, and we take them one at a time, because the successor construction of Part II is engineered directly against them.

J1, maximality. The family of all ways into C covers C. This is the trivial admission: an object is always covered by everything mapping into it, including its own identity. J1 is harmless — but it is also the axiom through which the legacy collapse operates: since maximal sieves are *automatically* covering in every topology, a clause whose generated sieves are all maximal contributes literally nothing to a topology. Being maximal is the covering-theoretic way of being redundant.

J2, stability. Covers restrict. If S covers C and we change vantage to D along $g : D \to C$, the restricted family g^*S must cover D. In bookkeeping terms: admission of a family of local views is not allowed to depend on staying at the original target; any sub-target inherits an admitted family. On a poset, stability says: if the down-closed set S covers V, then $S \cap \downarrow U$ covers every $U \leq V$. This is the axiom that forces the successor's generating families to be pullback-stable, which is the content of the coverage-basis lemma of Chapter 9 and is where the intersection formula $(V \cap B) \cap U = U \cap B$ does its work.

J3, transitivity. Covers compose hierarchically. If S covers C, and T is a sieve that covers *locally on* S — pulled back along each member of S, it covers that member's domain — then T covers C outright. This is the axiom that lets one refine covers: cover C by pieces, cover each piece by smaller pieces, and conclude that the assembly of smaller pieces covers C. In Part II it is precisely J3 that makes the *common refinement* of the ACBE and RPM covers — the four-singleton family — covering for the joined topology, which is the join theorem of Chapter 9.

3.2. First consequences.

Lemma 3.13 (Monotonicity). *Let J be a Grothendieck topology, $S \in J(C)$, and T a sieve on C with $S \subseteq T$. Then $T \in J(C)$.*

PROOF. We verify the hypothesis of (J3) for T against the covering sieve S. Let $f : D \to C$ be a member of S. Then $f \in T$ as well, so by Lemma 3.10(2), $f^*T = m_D$, which is covering by (J1). Thus $f^*T \in J(D)$ for every $f \in S$, and (J3) yields $T \in J(C)$. □

LEMMA 3.14 (Intersection stability). *If $S, T \in J(C)$ then $S \cap T \in J(C)$.*

PROOF. $S \cap T$ is a sieve (both closure conditions hold). First observe that pullback commutes with intersection: $h \in f^*(S \cap T)$ iff $f \circ h \in S$ and $f \circ h \in T$, i.e. $f^*(S \cap T) = f^*S \cap f^*T$. Now let $f : D \to C$ be any member of S. Then $f \in S$ gives $f^*S = m_D$ (Lemma 3.10(2)), so $f^*(S \cap T) = f^*T$, which is covering by (J2) applied to T. Thus the hypothesis of (J3) holds for $S \cap T$ against S, and $S \cap T \in J(C)$. □

Together, Lemmas 3.13 and 3.14 say each $J(C)$ is a *filter* of sieves: upward closed and closed under finite intersection — a useful sanity check when enumerating topologies, as we do in Section 6.

3.3. The two decisive elementary facts. The next two lemmas are, mathematically, the least sophisticated results in this monograph; they are also the ones with the largest downstream consequences, since Chapter 5's collapse theorem is nothing but their application to the legacy clauses. We therefore prove them with full care.

LEMMA 3.15 (A sieve containing an identity is maximal). *Let S be a sieve on C with $\mathrm{id}_C \in S$. Then $S = m_C$.*

PROOF. Let $f : D \to C$ be any morphism with codomain C. Since $\mathrm{id}_C \in S$ and S is closed under precomposition (Definition 3.1), the composite $\mathrm{id}_C \circ f$ belongs to S. But $\mathrm{id}_C \circ f = f$ by unitality (C2). Hence every morphism into C belongs to S, i.e. $S = m_C$. □

LEMMA 3.16 (A sieve containing a split epimorphism is maximal). *Let S be a sieve on C containing a split epimorphism $\varepsilon : B \to C$, say with section $\eta : C \to B$, $\varepsilon \circ \eta = \mathrm{id}_C$. Then $S = m_C$.*

PROOF. By precomposition closure applied to $\varepsilon \in S$ and the morphism $\eta : C \to B$, the composite $\varepsilon \circ \eta$ belongs to S. By hypothesis $\varepsilon \circ \eta = \mathrm{id}_C$, so $\mathrm{id}_C \in S$, and Lemma 3.15 gives $S = m_C$. □

REMARK 3.17. Combined with the observation under (J1) above — maximal sieves are covering in *every* topology, so they never enlarge one — these lemmas give a general design constraint on sites: a covering clause whose generating families contain an identity, or a split epimorphism onto the target, is vacuous. It adds no sieve that was not already covering, and, since topology generation is a closure operator, it leaves no trace of having been imposed. Chapter 5 applies this constraint verbatim: the legacy ACBE clause contains an identity (the reflexive descent map at equal contexts) and the legacy RPM clause contains a split epimorphism (the counit ε_X, split by η_X), so both are vacuous, and the legacy topology collapses to its observational clause. Note also Remark 2.17: in a poset the only split epimorphisms are identities, so on the successor's poset site the second mechanism cannot even be formulated nontrivially — a structural immunity, not a lucky accident.

EXERCISE 3.18. Show more generally that a sieve containing any isomorphism is maximal. (Every isomorphism is a split epimorphism.)

4. Coverage bases and generated topologies

Topologies are rarely specified sieve by sieve; one specifies a *basis* of covering families and closes under the axioms. The successor construction of Part II is specified exactly this way — two-element partition families at each context — so we develop the passage carefully.

Definition 3.19 (Pretopology / coverage basis). Let $\mathcal{C}$ be a category with pullbacks. A *pretopology* (or *coverage basis*) K assigns to each object C a collection $K(C)$ of families of morphisms $\{f_i : C_i \to C\}_{i\in I}$, called *generating covers*, such that:

- (B1) (*Isomorphisms*) every family consisting of a single isomorphism $\{f : C' \xrightarrow{\cong} C\}$ — in particular the identity family $\{\mathrm{id}_C\}$ — is in $K(C)$;
- (B2) (*Stability*) if $\{f_i : C_i \to C\}_i \in K(C)$ and $g : D \to C$ is any morphism, then the pulled-back family $\{\,\pi_i : C_i \times_C D \to D\,\}_i$ is in $K(D)$;
- (B3) (*Composition*) if $\{f_i : C_i \to C\}_i \in K(C)$ and for each i a family $\{g_{ij} : C_{ij} \to C_i\}_j \in K(C_i)$ is given, then the composite family $\{f_i \circ g_{ij}\}_{i,j}$ is in $K(C)$.

See [4] and [1, Ch. III] for variants; weaker "coverage" axioms suffice to generate a topology, but the pretopology form is the one the successor's families satisfy and is simplest to state.

Theorem 3.20 (Generated topology). *Let K be a pretopology on a category $\mathcal{C}$ with pullbacks. Define, for each object C,*

$$J_K(C) = \{\, S \text{ a sieve on } C \mid S \supseteq R \text{ for some } R \in K(C)\,\},$$

where $S \supseteq R$ means S contains every member of the family R. Then J_K is a Grothendieck topology — the topology generated by K *— and it is the smallest topology in which every generating cover generates a covering sieve.*

Proof sketch, with the key steps in full. *(J1).* The identity family $\{\mathrm{id}_C\}$ lies in $K(C)$ by (B1), and m_C contains it; so $m_C \in J_K(C)$.

(J2). Let $S \in J_K(C)$ via $R = \{f_i : C_i \to C\} \subseteq S$, and let $g : D \to C$. By (B2) the pulled-back family $R' = \{\pi_i : C_i \times_C D \to D\}$ is in $K(D)$. We claim $R' \subseteq g^*S$: each π_i satisfies $g \circ \pi_i = f_i \circ \pi_i'$ (the pullback square), and $f_i \circ \pi_i' \in S$ because $f_i \in S$ and S is a sieve; hence $g \circ \pi_i \in S$, i.e. $\pi_i \in g^*S$. So $g^*S \supseteq R' \in K(D)$, giving $g^*S \in J_K(D)$.

(J3). Let $S \in J_K(C)$ via $R = \{f_i\} \subseteq S$, and let T be a sieve with $f^*T \in J_K(\mathrm{dom} f)$ for every $f \in S$. In particular, for each i, $f_i^*T \supseteq R_i$ for some $R_i = \{g_{ij} : C_{ij} \to C_i\} \in K(C_i)$. Form the composite family $\{f_i \circ g_{ij}\}_{i,j}$, which lies in $K(C)$ by (B3). Each of its members lies in T: indeed $g_{ij} \in f_i^*T$ means precisely $f_i \circ g_{ij} \in T$. Hence $T \supseteq \{f_i \circ g_{ij}\}_{i,j} \in K(C)$, so $T \in J_K(C)$.

Minimality. If J' is any topology in which each generating cover $R \in K(C)$ generates a covering sieve $\langle R\rangle \in J'(C)$, then every $S \supseteq R$ satisfies $S \supseteq \langle R\rangle$ (Proposition 3.4) and hence $S \in J'(C)$ by monotonicity (Lemma 3.13, which used only the axioms). So $J_K \subseteq J'$ objectwise. Conversely each $\langle R\rangle$ itself contains R, so $\langle R\rangle \in J_K(C)$: the generated topology does make every basis cover covering. Full details in the general (coverage) setting are in [4]; the poset case, where pullbacks are intersections and all of the above becomes finite bookkeeping, is rechecked concretely in Chapter 9 for the successor's families. □

Remark 3.21 (Sheaf condition on a basis). The companion fact, stated here and used from Chapter 4 onward: a presheaf is a sheaf for the generated topology J_K if and only if it satisfies the gluing condition for the basis families themselves [4, 1]. This is what allows Chapter 9 to verify subcanonicity and sheafhood by checking only the two-piece partition families rather than every covering sieve of J_A and of J_R separately (the figure 168 quoted in

Remark 3.35 counts the covering sieves of the *join* $J_A \vee J_R$; each factor is strictly coarser and has strictly fewer); the certificate of Chapter 11, by contrast, computes the full topologies and rechecks the equivalence exhaustively — a deliberate redundancy between basis-level proof and topology-level computation.

5. Matching families and subcanonicity

The full theory of sheaves belongs to Chapter 4, but subcanonicity is a property *of topologies* and is most naturally introduced here. We give the minimal forward definitions.

Definition 3.22 (Matching family; gluing). Let F be a presheaf on $\mathcal{C}$ and S a sieve on C. A *matching family* for S with values in F is an assignment, to each $f : D \to C$ in S, of an element $x_f \in F(D)$, such that

$$F(h)(x_f) = x_{f \circ h} \quad \text{for every } h : E \to D.$$

Equivalently (Proposition 3.8), it is a natural transformation $\widetilde{S} \Rightarrow F$. Every section $s \in F(C)$ induces a matching family by restriction: $x_f = F(f)(s)$. The presheaf F is a *sheaf* for a topology J if for every C and every $S \in J(C)$ this canonical map from $F(C)$ to matching families is a bijection; it is *separated* if the map is always injective.

Definition 3.23 (Subcanonical topology). A topology J on $\mathcal{C}$ is *subcanonical* if every representable presheaf yC is a J-sheaf. The *canonical topology* is the largest subcanonical topology; every topology contained in a subcanonical one is subcanonical [**6**, **7**].

Subcanonicity is the license to regard site objects as evidence objects: it says the topology's compatibility demands never manufacture or destroy maps into a fixed target, so C and yC can be identified inside the sheaf category. When it fails, the identification is unearned — the exact situation of the legacy site, where subcanonicity is not derivable from the registered axioms (Chapter 6), and the reason the program's representation rule sheafifies legacy objects before use. On the successor's context factor, by contrast, subcanonicity is a theorem (Chapter 9), and Section 7 below proves the standard prototype of that theorem in its natural generality.

Example 3.24 (A sheaf check on the square, by hand). On B, let $S = \{\varnothing, \{1\}, \{2\}\}$ be the sieve on the top generated by the two singletons, and let G be the bit presheaf (Example 2.37). A matching family assigns $x_{\{1\}} \in G(\{1\})$, $x_{\{2\}} \in G(\{2\})$, $x_\varnothing \in G(\varnothing)$ with all restrictions compatible; since $G(\varnothing)$ is a singleton, the only constraints are $x_{\{1\}}|_\varnothing = x_\varnothing = x_{\{2\}}|_\varnothing$, which hold automatically. So matching families correspond to free pairs $(x_{\{1\}}, x_{\{2\}})$ — there are $2 \times 2 = 4$ — and the canonical map from $G(\{1,2\})$, which has 4 elements, sends a bit-function on $\{1,2\}$ to its two restrictions. This map is a bijection (a function on $\{1,2\}$ is exactly a pair of values), so G satisfies the sheaf condition for this sieve. The constant-verdict presheaf H (Example 2.38) does not: matching families again form a 2×2 set, but $H(\{1,2\})$ has only 2 elements, and the canonical map hits only the diagonal. The count 4 versus 2 is the germ of the witness argument of Chapter 9, where the same arithmetic ($2^2 = 4$ sections against a singleton matching-family set, and conversely) separates the ACBE and RPM topologies.

6. All Grothendieck topologies on the two-element chain

We now carry out, in full, the smallest interesting classification: every Grothendieck topology on the poset $[2]$ $(0 \le 1)$. The answer is that there are exactly four, and the proof method — identify the free boolean choices, derive forcing implications among them, then

verify each consistent assignment — is the hand-scale version of the fixed-point saturation the certificate of Chapter 11 performs on the sixteen-object lattice.

Recall the sieve inventory (Example 3.6): on 0, the empty sieve $\varnothing_0$ and the maximal $m_0 = \{0\}$; on 1, the empty sieve $\varnothing_1$, the proper sieve $S_f = \{0\}$ (the down-set generated by 0, corresponding to the morphism $f : 0 \to 1$), and the maximal $m_1 = \{0, 1\}$.

By (J1), any topology contains m_0 and m_1. Three boolean choices remain:

$$a = [\varnothing_0 \in J(0)], \qquad b = [S_f \in J(1)], \qquad c = [\varnothing_1 \in J(1)].$$

LEMMA 3.25 (Forcing implications). *In any Grothendieck topology J on* [2]*:*

(1) *if $\varnothing_C \in J(C)$ for an object C, then* every *sieve on C is in $J(C)$;*
(2) $c \Rightarrow a$*;*
(3) $c \Rightarrow b$*;*
(4) $(a \wedge b) \Rightarrow c$*.*

PROOF. (1) Apply (J3) with $S = \varnothing_C$: the hypothesis "for every $f \in S$, f^*T is covering" quantifies over the empty set and is vacuously true for every sieve T; hence every T is covering. (2) If $\varnothing_1 \in J(1)$, then by (J2) its pullback along $f : 0 \to 1$ is covering; and $f^*\varnothing_1 = \{h \mid f \circ h \in \varnothing_1\} = \varnothing_0$. So a holds. (3) Immediate from (1) applied at $C = 1$: if c, then all sieves on 1 are covering, in particular S_f. (4) Assume a and b. Apply (J3) with $S = S_f \in J(1)$ and $T = \varnothing_1$. The members of S_f, as morphisms, are exactly f (the unique morphism $0 \to 1$; its precompositions with id_0 give f again). For this member, $f^*\varnothing_1 = \varnothing_0$, which is covering by a. So the hypothesis of (J3) holds, and $\varnothing_1 = T \in J(1)$, i.e. c. □

THEOREM 3.26 (Classification). *The poset* [2] *carries exactly four Grothendieck topologies:*

(1) $J_{\min}$*:* $(a, b, c) = (0, 0, 0)$ — $J(0) = \{m_0\}$, $J(1) = \{m_1\}$*;*
(2) J_{dense}*:* $(0, 1, 0)$ — $J(0) = \{m_0\}$, $J(1) = \{m_1, S_f\}$*;*
(3) J_{closed}*:* $(1, 0, 0)$ — $J(0) = \{m_0, \varnothing_0\}$, $J(1) = \{m_1\}$*;*
(4) $J_{\max}$*:* $(1, 1, 1)$ — *every sieve on every object is covering.*

PROOF. *Only these four are possible.* The eight boolean assignments (a, b, c) are cut down by Lemma 3.25: $(0, 0, 1)$, $(0, 1, 1)$ violate (2); $(1, 0, 1)$ violates (3); $(1, 1, 0)$ violates (4). The remaining four are those listed.

Each of the four is a topology. (J1) holds in all four by construction. We check (J2) and (J3) case by case; throughout, pullback along an identity is the identity operation, and pullback of a maximal sieve is maximal (Lemma 3.10(1)), so only nonmaximal sieves and the morphism f need attention.

$J_{\min}$: (J2): only maximal sieves are covering, and their pullbacks are maximal. (J3): let $S = m_1$, and T with $g^*T \in J$ for all $g \in m_1$; taking $g = \mathrm{id}_1$ gives $T = \mathrm{id}_1^*T \in J(1)$, so the conclusion holds trivially; the same argument at 0, and with $S = m_0$. Valid.

J_{dense}: (J2): S_f pulled back along id_1 is $S_f \in J(1)$; along $f : 0 \to 1$ it is $f^*S_f = \{h \mid f \circ h \in S_f\} \ni \mathrm{id}_0$ (since $f \in S_f$), hence $= m_0 \in J(0)$ (Lemma 3.10(2)). (J3): suppose $S \in J(1)$ and T has all pullbacks covering. If $\mathrm{id}_1 \in S$ (i.e. $S = m_1$), then taking $g = \mathrm{id}_1$ gives $T \in J(1)$ directly. If $S = S_f$, the members are f and its precompositions (all equal to f), and the hypothesis gives $f^*T \in J(0) = \{m_0\}$, so $\mathrm{id}_0 \in f^*T$, i.e. $f \in T$. A sieve on 1 containing f is either S_f or m_1, and both are in $J(1)$; so T is covering. At the object 0 the argument of $J_{\min}$ applies. Valid.

J_{closed}: (J2): the only nonmaximal covering sieve is $\varnothing_0$ on 0; morphisms into 0 are id_0 only, and $\mathrm{id}_0^*\varnothing_0 = \varnothing_0 \in J(0)$. (J3) at 0: covered by Lemma 3.25(1)'s reasoning — $\varnothing_0$ covering makes every sieve on 0 required to be covering, and indeed both sieves on 0 *are*

covering in J_{closed}, so no contradiction; for $S = m_0$ the identity argument applies. (J3) at 1: the only covering sieve is m_1; for $S = m_1$ and any T with all pullbacks covering, $g = \mathrm{id}_1$ gives $T \in J(1)$ as before — note the hypothesis for $T = S_f$ would require $\mathrm{id}_1^* S_f = S_f \in J(1)$, which fails, so nothing forces S_f or $\varnothing_1$ in. Valid.

$J_{\max}$: (J2) and (J3) hold trivially since every sieve is covering. Valid. □

REMARK 3.27 (Reading the four). The four topologies illustrate the design space in miniature. $J_{\min}$ admits only trivial covers: presheaves and sheaves coincide. J_{dense} admits the proper sieve S_f: it declares that "0 covers 1," and its sheaves are exactly the presheaves whose restriction $F(1) \to F(0)$ is a bijection (matching families for S_f are single elements of $F(0)$, since the compatibility conditions are trivial; the gluing map is the restriction). J_{closed} admits the empty sieve on 0: its sheaves must have $F(0)$ a singleton (the empty sieve has exactly one matching family — the empty one — so the sheaf condition forces $|F(0)| = 1$), with $F(1)$ arbitrary. $J_{\max}$ forces both conditions: only the terminal presheaf survives. The correspondence of these four with the four subtoposes of the presheaf topos on [2] is an instance of the standard bijection between Grothendieck topologies and Lawvere–Tierney topologies [1, Ch. V] [7]; we cite it for orientation and do not use it. For the monograph's purposes the important specimen is J_{dense}: a *proper* generating morphism producing a genuinely nontrivial topology whose sheaf condition has real content — the two-object preview of what the successor's proper partition pieces do on L, and a first concrete instance of a nontrivial modality (the sheafification for J_{dense} replaces $F(1)$ by $F(0)$).

EXERCISE 3.28. Classify all Grothendieck topologies on the *discrete* two-object poset (no relation between the objects). (Each object independently either admits only its maximal sieve or admits both its sieves; there are no cross-object constraints, so there are four topologies here as well — but for a structurally opposite reason. Compare Example 2.40.)

EXERCISE 3.29. Carry out the same program on the three-element chain [3]: enumerate the boolean choices (one per nonmaximal sieve per object), derive the forcing implications from (J2) and (J3), and count the topologies. The bookkeeping is heavier; the method is identical.

7. The union coverage on a powerset lattice

The final section develops the standard example that Part II's context factor instantiates: the union coverage on a powerset lattice $\mathcal{P}(X)$, for a finite set X. This is the "geometric" covering notion — a family of subsets covers V when it actually covers V, elementwise — and it is the topology that the successor's joined topology $J_A \vee J_R$ turns out to equal on $L = \mathcal{P}(\Omega)$ (the join theorem of Chapter 9, ledger THM-NT2-JOIN-DISCRETE-CONTEXT). Proving its properties here, in the general finite-powerset setting and with standard citations, keeps Chapter 9 free to focus on what is project-specific (the two coarser partition topologies and their join) and keeps the boundary between textbook fact and project candidate sharp.

DEFINITION 3.30 (Union coverage). On the poset $\mathcal{P}(X)$, let $K_\cup(V)$ consist of all families $\{V_i \subseteq V\}_{i \in I}$ of subsets of V (each $V_i \to V$ the inclusion) such that $\bigcup_{i \in I} V_i = V$. Note the empty family qualifies at $V = \varnothing$ (an empty union is empty), so the empty context is covered by the *null family*.

PROPOSITION 3.31. *$K_\cup$ is a pretopology on $\mathcal{P}(X)$.*

PROOF. Pullbacks exist and are intersections (Corollary 2.64). (B1): the singleton family $\{V \subseteq V\}$ has union V. (B2): pulling $\{V_i\}_i$ back along $U \subseteq V$ gives the family $\{V_i \cap U\}_i$,

and $\bigcup_i (V_i \cap U) = \left(\bigcup_i V_i\right) \cap U = V \cap U = U$ by distributivity of intersection over union; so the pulled-back family is in $K_\cup(U)$. (B3): if $\bigcup_i V_i = V$ and for each i, $\bigcup_j W_{ij} = V_i$, then $\bigcup_{i,j} W_{ij} = \bigcup_i V_i = V$. All three axioms hold. □

DEFINITION 3.32. The *union topology* $J_\cup$ on $\mathcal{P}(X)$ is the topology generated by $K_\cup$ (Theorem 3.20).

PROPOSITION 3.33 (Characterization of covering sieves). *Let S be a sieve on $V \in \mathcal{P}(X)$, identified with a down-closed set of subsets of V. The following are equivalent:*

(1) $S \in J_\cup(V)$;
(2) $\bigcup_{W \in S} W = V$;
(3) $\{x\} \in S$ *for every* $x \in V$.

In particular, a sieve on $\varnothing$ is covering iff it is either of the two sieves on $\varnothing$ — both the empty sieve and the maximal sieve satisfy (2) vacuously — so the empty context is covered even by the empty sieve.

PROOF. (1) ⇒ (2): if S contains a family $\{V_i\}$ with $\bigcup_i V_i = V$, then the union of *all* members of S contains $\bigcup_i V_i = V$, and is contained in V since every member is a subset of V.

(2) ⇒ (3): let $x \in V$. By (2), $x \in W$ for some $W \in S$. Then $\{x\} \subseteq W$, and S is down-closed, so $\{x\} \in S$.

(3) ⇒ (1): the family of singletons $\{\{x\} \subseteq V\}_{x \in V}$ has union V, hence lies in $K_\cup(V)$; by (3) it is contained in S, so S is covering by definition of the generated topology. (For $V = \varnothing$ this is the null family, and every sieve on $\varnothing$ contains it trivially.) □

Condition (3) exhibits $J_\cup$ as generated by the *singleton partition* at each context — the finite powerset lattice's atomic decomposition. This is exactly the form in which the joined topology appears in Chapter 9: the common refinement of the ACBE and RPM partitions is the singleton partition (transversality), and the join theorem identifies $J_A \vee J_R$ with the $J_\cup$ of the sixteen-element lattice. The seed artifacts' internal label J_{atom} for that join is a name for this topology; as Chapter 1 noted, the publication layer prefers $J_{\text{singleton}}$, and neither label asserts the standard category-theoretic "atomic topology," which is a different notion.

THEOREM 3.34 (The union topology is subcanonical). *Every representable presheaf on $\mathcal{P}(X)$ is a sheaf for $J_\cup$. Hence $J_\cup$ is subcanonical, as is every topology contained in it.*

PROOF. Let yW be a representable (Proposition 2.43: $yW(V)$ is a singleton if $V \subseteq W$, else empty). By Remark 3.21 it suffices to check the gluing condition for basis families. Let $\{V_i \subseteq V\}_i \in K_\cup(V)$, so $\bigcup_i V_i = V$.

Case $V \subseteq W$. There is exactly one section over V, and its restrictions form the unique candidate matching family; each $V_i \subseteq V \subseteq W$, so each $yW(V_i)$ is a singleton and there is exactly one matching family. Bijection: one section, one family.

Case $V \not\subseteq W$. There are no sections over V, so we must show there is no matching family either. Suppose one existed: then $yW(V_i) \neq \varnothing$ for every i, i.e. $V_i \subseteq W$ for every i; but then $V = \bigcup_i V_i \subseteq W$, contradiction. So both sides are empty, and the gluing map is (vacuously) a bijection.

The null family at $\varnothing$. A matching family over the empty family is the empty assignment (exactly one), and $yW(\varnothing)$ is a singleton since $\varnothing \subseteq W$ always. Bijection again.

The final clause holds because a sheaf for a finer topology is a sheaf for any coarser one — the coarser topology's covering sieves are a subset of the finer's, so the sheaf condition

is a weaker demand. See [6] for the general context of representable sheaves and [7] for the canonical topology on a frame, of which this is the finite powerset instance. □

REMARK 3.35 (What Chapter 9 adds). Everything in this section is standard. What is project-specific, and deferred to Chapter 9, is the *pair of coarser topologies*: the ACBE topology J_A and RPM topology J_R generated by the two transverse two-block partitions of Ω, each strictly coarser than $J_\cup$, each subcanonical (by the final clause above, once contained in $J_\cup$ — though Chapter 9 proves subcanonicity directly from the basis as well), each supporting a nontrivial modality, with join exactly $J_\cup$. The union topology is the ambient standard object; the successor's contribution, whatever its eventual assessment, lies in the choice and analysis of the coarser pair, and the certificate's counts (298 sieves over the sixteen contexts, of which 168 cover for the join, computed identically from both the join and singleton presentations) quantify the sixteen-object instance exhaustively.

EXERCISE 3.36. On $B = \mathcal{P}(\{1,2\})$, enumerate $J_\cup$ completely: for each of the four objects, list the covering sieves among the sieve inventory of Example 3.6. (Answers: on $\{1,2\}$, the covering sieves are $\{\varnothing, \{1\}, \{2\}\}$ and the maximal sieve — the two sieves containing both singletons; on $\{1\}$: the maximal sieve and $\{\varnothing, \{1\}\}$ coincide, and the covering sieves are exactly those containing $\{1\}$, i.e. the maximal one; on $\{2\}$: likewise; on $\varnothing$: both sieves. Check each against Proposition 3.33(3), and count: $2 + 1 + 1 + 2 = 6$ covering sieves out of 14.)

8. Summary

The chapter's toolkit, as Part II will draw on it: sieves as down-closed sets on a poset (Proposition 3.5) and as subfunctors of representables (Proposition 3.8); pullback sieves and the membership criterion (Lemma 3.10); the topology axioms with their covering common sense, and the filter properties (Lemmas 3.13–3.14); the two decisive maximality facts (Lemmas 3.15 and 3.16), which Chapter 5 aims at the legacy clauses; pretopologies and the generated topology (Theorem 3.20), which Chapter 9 aims at the successor's partition families; matching families and subcanonicity (Definitions 3.22 and 3.23); the complete classification of topologies on [2] (Theorem 3.26), a hand-scale rehearsal of exhaustive site analysis; and the union coverage on a finite powerset lattice with its covering-sieve characterization and subcanonicity (Propositions 3.31–3.33, Theorem 3.34), the standard fact of which the successor's join theorem is the marked, two-coordinate instance. The whole toolkit is textbook material [1, 4, 5, 6, 7]; Chapter 4 completes the standard background with sheaves and sheafification, and Chapter 5 begins the project-specific story.

CHAPTER 4

Sheaves, Sheafification, and Modalities

This chapter completes the background half of the monograph. Chapter 2 fixed the categorical vocabulary; Chapter 3 introduced sites: categories equipped with a Grothendieck topology that declares which sieves count as covers. The present chapter explains what covers are *for*. A topology is interesting only because it selects a class of presheaves—the sheaves—whose sections are determined by compatible local data, and because the passage from an arbitrary presheaf to its best sheaf approximation—sheafification—is a functor with exceptionally good properties. Composing sheafification with the inclusion of sheaves back into presheaves yields a left-exact idempotent monad, which this monograph calls a *modality*. Modalities are the objects whose existence, distinctness, triviality, and composition laws the remainder of the monograph interrogates: Chapter 5 proves that the legacy cover clauses generate only trivial modalities, and the later constructive chapters build a finite site on which two genuinely distinct nontrivial modalities exist and commute.

Everything in this chapter is standard material. The definitions and theorems are drawn from the textbook and reference literature [**1**, **7**, **5**, **4**], and no novelty whatsoever is claimed for them; they are cited as background throughout. What is specific to this monograph is the selection and the emphasis: we dwell on exactly the points—matching families on sieves containing identities, the basis criterion, the behavior of sheafification on finite posets, and the criterion for a modality to be trivial—that carry the weight of the collapse analysis in Chapter 5 and of the finite constructions later. Examples are chosen on small finite posets, deliberately miniature versions of the sixteen-context lattice used by the successor construction, so that every verification in this chapter can be redone by hand in a few minutes and every later exhaustive machine check has a hand-checked prototype here.

NOTATION 4.1. Throughout this chapter $\mathcal{C}$ denotes a small category, $\mathrm{PSh}(\mathcal{C})$ the category of presheaves $F : \mathcal{C}^{\mathrm{op}} \to \mathbf{Set}$, and $yC = \mathrm{Hom}_{\mathcal{C}}(-, C)$ the representable presheaf at $C \in \mathrm{Ob}(\mathcal{C})$. For $x \in F(C)$ and $f : D \to C$ we write $x|_f$ for $F(f)(x) \in F(D)$, and when $\mathcal{C}$ is a poset and $U \leq V$ we write $x|_U$ for the restriction of $x \in F(V)$ along the unique arrow $U \to V$. Sieves, pullback sieves g^*S, the maximal sieve m_C, and Grothendieck topologies J are as in Chapter 3. A sieve on C is identified with a subfunctor of yC.

1. Sections, restriction, and the idea of local data

A presheaf is a system of "data assignments": to every object C a set $F(C)$ of *sections over* C, and to every arrow $f : D \to C$ a *restriction* operation $F(C) \to F(D)$, contravariantly and compatibly with composition. Nothing in the presheaf axioms relates sections over an object to sections over the things that map into it, beyond the bare existence of restriction. A presheaf may have many sections over C that all restrict to the same local data, or local data that assemble compatibly yet come from no section over C at all. Both phenomena will be exhibited on a four-element poset in Section 4.

The sheaf condition is precisely the statement that neither pathology occurs over covers. Informally: *a section over C is exactly a compatible family of sections over any cover of C.* The slogan has two halves. Uniqueness—two sections over C agreeing on a cover are equal—is called *separation*. Existence—every compatible family of local sections glues to a section over C—is the properly "sheafy" half. The definition below makes "compatible family" precise for an arbitrary sieve; the version for a generating family of arrows, which is what one actually checks in practice, follows in Section 5.

Why should an evidence-integration program care? The program's ambition, recorded in the governing specification, is to treat locally gathered observations as sections of presheaves over a site of contexts, and to let the topology encode which families of contexts are jointly informative. Under that reading, the sheaf condition is a *composability discipline*: it says which systems of local observations determine a global one and insists the determination be unique. The mathematics of this chapter is entirely neutral about any such interpretation—the theorems are about sets and arrows—but the choice of what to prove is driven by it, and the claim boundaries of the program (Ring 4, not licensed, no empirical interpretation) apply to every interpretive gloss in this paragraph.

2. Matching families and amalgamation

Definition 4.2 (Matching family). Let $F \in \mathrm{PSh}(\mathcal{C})$, let $C \in \mathrm{Ob}(\mathcal{C})$, and let S be a sieve on C. A *matching family* for S with values in F is a function assigning to every arrow $f : D \to C$ belonging to S an element $x_f \in F(D)$, such that for every arrow $h : E \to D$,

$$x_{f \circ h} \;=\; x_f|_h \qquad \text{in } F(E).$$

Note that $f \circ h \in S$ automatically, because S is closed under precomposition, so the condition is well posed. We write $\mathrm{Match}(S, F)$ for the set of matching families.

The compatibility equation quantifies over *all* arrows h, not merely over some chosen overlaps. This is the right level of generality for sieves; when one works with a generating family and the category has pullbacks (or the poset has meets), compatibility reduces to agreement on pairwise overlaps, as we prove in Lemma 4.6 below.

Remark 4.3 (Matching families are natural transformations). Regard the sieve S as a subfunctor of yC: $S(D) \subseteq \mathrm{Hom}(D, C)$ is the set of members of S with domain D. Then a matching family for S with values in F is exactly a natural transformation $S \Rightarrow F$: the component at D sends $f \in S(D)$ to x_f, and the compatibility equation is exactly naturality. Hence

$$\mathrm{Match}(S, F) \;=\; \mathrm{Hom}_{\mathrm{PSh}(\mathcal{C})}(S, F),$$

and matching families inherit all the functoriality one expects: they can be restricted along inclusions of sieves $S' \subseteq S$, pulled back along arrows $g : D \to C$ (giving a matching family for g^*S), and pushed forward along maps of presheaves $F \to G$. These three operations are used constantly below and in the plus construction of Section 8.

Definition 4.4 (Amalgamation; separated; sheaf). Let S be a sieve on C and $(x_f)_{f \in S}$ a matching family for S with values in F. An *amalgamation* of the family is an element $x \in F(C)$ with $x|_f = x_f$ for every $f \in S$. Every $x \in F(C)$ determines a canonical matching family $(x|_f)_{f \in S}$; amalgamation asks for the converse. Define the *gluing map*

$$\mathrm{glue}_{S,F} \;:\; F(C) \longrightarrow \mathrm{Match}(S, F), \qquad x \longmapsto (x|_f)_{f \in S}.$$

Given a Grothendieck topology J on $\mathcal{C}$, a presheaf F is

(1) *separated* for J if $\mathrm{glue}_{S,F}$ is injective for every C and every $S \in J(C)$;
(2) a *sheaf* for J if $\mathrm{glue}_{S,F}$ is bijective for every C and every $S \in J(C)$.

We write $\mathrm{Sh}(\mathcal{C}, J) \subseteq \mathrm{PSh}(\mathcal{C})$ for the full subcategory of sheaves. A topology is *subcanonical* if every representable presheaf is a sheaf [**1, 6**].

Under the identification of Remark 4.3, the gluing map is composition with the inclusion $S \hookrightarrow yC$ followed by the Yoneda identification $\mathrm{Hom}_{\mathrm{PSh}}(yC, F) \cong F(C)$; the sheaf condition says exactly that the inclusion of every covering sieve into its representable becomes invisible to F. This one-line reformulation is the source of both the triviality criterion of Proposition 4.37 and the collapse diagnosis of Chapter 5.

Example 4.5 (Maximal sieves are free). For every presheaf F and every object C, the gluing map for the maximal sieve m_C is a bijection. Indeed a matching family $(x_f)_{f \in m_C}$ is determined by its value at the identity: $x_f = x_{\mathrm{id}_C \circ f} = x_{\mathrm{id}_C}|_f$, so the assignment $(x_f) \mapsto x_{\mathrm{id}_C}$ is inverse to $\mathrm{glue}_{m_C,F}$. Consequently axiom (J1)—every maximal sieve covers—imposes no condition at all on presheaves. A topology whose covering sieves are *only* the maximal ones therefore has every presheaf as a sheaf. This innocuous computation is the engine of the legacy collapse: a cover clause that generates only maximal sieves is a cover clause that constrains nothing (Proposition 4.37 and Chapter 5).

Lemma 4.6 (Compatibility via meets on a poset). *Let $\mathcal{C}$ be a poset with binary meets, $V \in \mathcal{C}$, and let $\{U_i \leq V\}_{i \in I}$ be a family generating the sieve $S = \{W \mid W \leq U_i \text{ for some } i\}$. Then a matching family for S with values in F is the same data as a tuple $(x_i \in F(U_i))_{i \in I}$ satisfying*

$$x_i|_{U_i \wedge U_j} \;=\; x_j|_{U_i \wedge U_j} \qquad \text{for all } i, j \in I.$$

(Standard; see [**1**, Chap. III]*.)*

Proof. Given a matching family $(x_W)_{W \in S}$, set $x_i = x_{U_i}$; compatibility at $U_i \wedge U_j \leq U_i, U_j$ follows from the matching condition applied to the two factorizations of $U_i \wedge U_j \to V$. Conversely, given a compatible tuple, define $x_W = x_i|_W$ for any i with $W \leq U_i$; this is well defined because if also $W \leq U_j$ then $W \leq U_i \wedge U_j$ and the two candidate values agree by compatibility after restriction. The matching condition for (x_W) is then immediate, since restriction is functorial and every arrow of a poset is a unique inequality. The two constructions are mutually inverse. □

Lemma 4.6 is why finite poset examples are so pleasant: a matching family is just a compatible tuple, and both the matching-family set and the section set are finite sets one can list. All the finite sites in this monograph—the miniature examples below, and the sixteen-context lattice of the successor—are posets with meets (intersections), so the lemma applies verbatim throughout.

3. First examples on a small poset

Fix the four-element poset

$$L_2 \;=\; \mathcal{P}(\{a, b\}) \;=\; \{\varnothing,\ \{a\},\ \{b\},\ \{a, b\}\},$$

ordered by inclusion. It has meets (intersection) and joins (union). Equip it with the coverage whose generating data are: the two-element family $\{\{a\} \to \{a, b\},\ \{b\} \to \{a, b\}\}$ covering the top; the identity family at each object; and the *null cover*—the empty family—covering $\varnothing$. Write J_2 for the generated topology. (That these data generate a topology is an instance of the basis machinery of Section 5; the reader may also verify (J1)–(J3) directly, a worthwhile

finger exercise.) The poset L_2 is the two-atom miniature of the four-atom context lattice $L = \mathcal{P}(\{0,1\} \times \{0,1\})$ on which the successor construction lives, and the covering pattern—partition the top into its atoms, null-cover the bottom—is the miniature of the partition coverages there.

Before any example we fix one point. The null cover at $\varnothing$ is a genuine covering sieve (the empty sieve), and the sheaf condition for it reads: $F(\varnothing) \to \mathrm{Match}(\varnothing\text{-sieve}, F)$ is a bijection. A matching family for the empty sieve is a function defined on no arrows; there is exactly one such function. Hence:

LEMMA 4.7 (The null cover forces a singleton). *If the empty sieve covers an object Z, then every sheaf F has $F(Z)$ a one-element set. (Standard; the two-line proof is self-contained.)*

PROOF. $\mathrm{Match}(\varnothing, F)$ is a singleton (the empty family, vacuously matching), and the gluing map $F(Z) \to \mathrm{Match}(\varnothing, F)$ must be a bijection. □

EXAMPLE 4.8 (The function presheaf is a sheaf). Fix a set X and let $F(V) = \mathbf{Set}(V, X)$, the set of all functions from the subset $V \subseteq \{a, b\}$ to X, with restriction of functions as restriction. Then F is a J_2-sheaf. At the null cover: $F(\varnothing) = \mathbf{Set}(\varnothing, X)$ is a singleton (the empty function), as Lemma 4.7 requires. At the top cover: by Lemma 4.6 a matching family is a pair $(s \in \mathbf{Set}(\{a\}, X),\ t \in \mathbf{Set}(\{b\}, X))$ agreeing on $\{a\} \cap \{b\} = \varnothing$—a vacuous condition since $F(\varnothing)$ is a singleton—so Match $\cong X \times X$; and $F(\{a, b\}) = \mathbf{Set}(\{a, b\}, X) \cong X \times X$ by evaluation at a and b. The gluing map is exactly this identification, hence bijective. Identity covers are free by Example 4.5. This is the paradigm sheaf: sections are literally functions, and functions are determined by, and glued from, their values on the pieces of a partition.

EXAMPLE 4.9 (Representables; subcanonicity of J_2). For $W \in L_2$ the representable yW has $yW(V)$ a singleton if $V \subseteq W$ and empty otherwise. Every representable is a J_2-sheaf, so J_2 is subcanonical. The check at the top cover splits into cases. If $W = \{a, b\}$: sections over $\{a\}$, $\{b\}$, and the top are all singletons, matching families are singletons, gluing is a bijection of singletons. If $W = \{a\}$: then $yW(\{b\}) = \varnothing$, so there is *no* matching family at all for the top cover (a matching family must supply an element over every member of the sieve, including $\{b\} \to \{a, b\}$), and also $yW(\{a, b\}) = \varnothing$; the gluing map is the unique map $\varnothing \to \varnothing$, a bijection. The case $W = \{b\}$ is symmetric, and $W = \varnothing$ likewise gives $\varnothing \to \varnothing$ at the top and singletons at the null cover ($\varnothing \subseteq W$ always). The general shape of this argument—if every piece of a partition of V lies inside W then V itself lies inside W, because V is the union of the pieces—reappears as the subcanonicity theorem for the full context lattice in the successor development, and stands in instructive contrast to Chapter 6, where a jointly-surjective coverage on a category with *too few* morphisms fails subcanonicity.

EXAMPLE 4.10 (A top-only side condition: separated but not a sheaf). Let $X = \{0, 1\}$ and let $G'(V) \subseteq \mathbf{Set}(V, X)$ consist of all functions when $V \neq \{a, b\}$, and of those functions that are *not* constantly 1 when $V = \{a, b\}$. Restriction preserves membership, since no condition is imposed below the top, so G' is a subpresheaf of the sheaf $\mathbf{Set}(-, X)$ of Example 4.8 and is therefore separated (injectivity of gluing is inherited by subpresheaves). It is not a sheaf: $s(a) = 1$ and $t(b) = 1$ are both allowed locally, they match vacuously over $\{a\} \cap \{b\} = \varnothing$, but any amalgamation would be the constant function 1 on $\{a, b\}$, which G' excludes. This is the standard failure mode of "global side conditions": a property imposed at the top but not stable under gluing from pieces destroys sheafhood without harming separation. Imposing the same condition at *every* context would not even define a presheaf—the restriction to $\{a\}$ of the function with values $1, 0$ on $\{a, b\}$ is constantly 1—which is a reminder that restriction-stability is the first thing to check.

4. Non-examples, and what each failure teaches

Non-examples are where the sheaf condition earns its keep. We record three on L_2, each isolating one failure mode, and summarize the verdicts in Table 1.

Example 4.11 (The constant presheaf fails the null cover). Fix a set X with at least two elements and let $\Delta_X(V) = X$ for *every* V, all restrictions the identity. At the top cover, matching families are pairs $(x, y) \in X \times X$ with $x|_\varnothing = y|_\varnothing$, i.e. $x = y$ since restrictions are identities; so Match $\cong X$, and gluing $X \to X$ is the identity—fine. But at the null cover, Lemma 4.7 demands $|\Delta_X(\varnothing)| = 1$, whereas $|X| \geq 2$. So Δ_X is not a J_2-sheaf, and the failure is entirely concentrated at the empty context. The lesson: the empty cover is not a degenerate afterthought; it normalizes the value at the initial context, and every finite construction later in the monograph (the reflector formula in particular, whose empty product clause sets the value at $\varnothing$ to a singleton) must and does handle it explicitly.

Example 4.12 (Separated but not a sheaf: missing gluings). Let X have at least two elements. Define $H(\varnothing) = \{*\}$, $H(\{a\}) = H(\{b\}) = H(\{a, b\}) = X$, with the restrictions between nonempty contexts the identity on X and the restrictions to $\varnothing$ the unique map to the singleton. At the top cover, a matching family is any pair $(x, y) \in X \times X$: the compatibility condition lives in $H(\varnothing) = \{*\}$ and is vacuous. An amalgamation of (x, y) is an element $z \in X$ with $z = x$ and $z = y$; it exists if and only if $x = y$ and is then unique. So glue is injective but not surjective: H is separated, not a sheaf. The presheaf "thinks" a section over the top is a single element of X, but the cover sees two independent local choices. Its sheafification, computed in Example 4.29 below, repairs this by *creating* the missing sections: the associated sheaf has $X \times X$ over the top. The witness presheaves that separate the two modalities in the successor's finite theory are exactly presheaves of this character: sheaf-like along one partition, gluing-deficient along the other.

Example 4.13 (Not separated: redundant global sections). Define $K(\{a, b\}) = \{u, v\}$ with $u \neq v$, and $K(V) = \{*\}$ for all proper $V \subseteq \{a, b\}$, with the only possible restrictions. At the top cover there is exactly one matching family (all values are forced into singletons), and *both* u and v are amalgamations of it: gluing is surjective but not injective. K is not separated. The presheaf carries a distinction at the top—u versus v—that no cover can see; sheafification will identify u and v. In audit language: K records a difference with no observational support on the covering contexts, and passage to sheaves deletes exactly that difference. Whether such deletion is desirable is a modeling decision, not a mathematical one; the mathematics only says that *if* one demands the sheaf discipline, unsupported distinctions cannot survive.

Exercise 4.14. L_2 has four objects; list all sieves on each object (there are six on the top, three on each singleton, two on $\varnothing$) and determine exactly which collections of sieves form Grothendieck topologies on L_2. Verify that J_2 is the smallest topology in which the atom family covers the top and the empty sieve covers $\varnothing$. This is the two-atom warm-up for the 298-sieve census that the finite certificate performs on the sixteen-context lattice.

5. The basis criterion

Definition 4.4 quantifies over all covering sieves of a topology, which even on the sixteen-object successor lattice means every covering sieve of the topology in question—168 of them for the joined topology $J_A \vee J_R$ (Corollary 9.24), and a strictly smaller collection for each of the coarser J_A and J_R, not separately enumerated in this monograph. In practice one

Presheaf	Defined by	Null cover	Top cover	Verdict
$F = \mathbf{Set}(-, X)$	functions into X	singleton	bijective	sheaf
yW	representable	singleton	bijective	sheaf
G'	top-only side condition	singleton	not surjective	separated only
Δ_X, $\lvert X\rvert \geq 2$	constant	fails	bijective	not a sheaf
H, $\lvert X\rvert \geq 2$	identity restrictions	singleton	not surjective	separated only
K	doubled top	singleton	not injective	not separated

TABLE 1. Verdicts for the presheaves of Sections 3–4 on the poset L_2 with topology J_2.

never checks them all directly; one checks a *generating* family of covers and appeals to the basis criterion. This section states the criterion, proves it in the finite-poset generality used throughout the monograph, and cites the general statement.

DEFINITION 4.15 (Coverage basis). A *coverage basis* (or pretopology-style basis) on $\mathcal{C}$ assigns to each object C a set $K(C)$ of families of arrows with codomain C, such that:

(1) (Identity) the singleton family $\{\mathrm{id}_C\}$ belongs to $K(C)$;
(2) (Stability) for every $\{f_i : C_i \to C\} \in K(C)$ and every arrow $g : D \to C$, there is a family $\{h_j : D_j \to D\} \in K(D)$ such that every composite $g \circ h_j$ factors through some f_i;
(3) (Composition) if $\{f_i : C_i \to C\} \in K(C)$ and for each i a family $\{g_{ik} : C_{ik} \to C_i\} \in K(C_i)$ is given, then the sieve generated by the composite family $\{f_i \circ g_{ik}\}_{i,k}$ contains the sieve generated by some family in $K(C)$.

This is the coverage (refinement) form of the axioms; Definition 3.19 gave the stricter pretopology form, which presupposes pullbacks and states stability as literal pullback of families. The successor's partition families satisfy both forms; the J_2 data of Section 3 satisfy the coverage form only, since the pullback $\{\{a\}, \varnothing\}$ of the atom family is not itself a declared generating family at $\{a\}$. The arguments below use only the coverage form.

The *generated topology* J_K is the smallest Grothendieck topology in which, for every C and every family in $K(C)$, the sieve generated by that family is covering. Smallest makes sense because the collection of topologies containing a given set of sieves is closed under intersection; this closure point is developed at length in Chapter 5, where it becomes the provenance-erasure argument. See [4] and [1, Chap. III] for the standard treatment.

THEOREM 4.16 (Basis criterion; general case cited, finite-poset case proved here). *Let K be a coverage basis on $\mathcal{C}$ and J_K the generated topology. A presheaf F is a J_K-sheaf if and only if for every object C and every generating family $\{f_i : C_i \to C\} \in K(C)$, the gluing map from $F(C)$ to matching families for the sieve generated by $\{f_i\}$ is a bijection. The general statement, for an arbitrary small category with a coverage basis, is standard background* [4, 1] *and is not reproved here. The proof below establishes the case of finite posets with meets in which K is moreover* strongly pullback-stable*: for every $\{f_i : C_i \to C\} \in K(C)$ and every $U \leq C$, the pulled-back family generates a sieve on U that is itself basis-generated or maximal. That is the only case used in this monograph, and the hypothesis has to be verified per basis rather than assumed.*

The forward implication is trivial (generated sieves are covering). The substantive converse is standard [4, 1]; we prove it in the setting of finite posets with meets, which covers every site analyzed computationally in this monograph and exposes the mechanism—saturation—in its executable form.

ALGORITHM 4.17 (Saturation of a basis on a finite poset). Input: a finite poset P with meets and a coverage basis K. Initialize $\mathcal{S}$ as the set of all pairs (V, S) where S is either a maximal sieve or the sieve generated by a member of $K(V)$. Repeat until no change:

(1) (Maximality) for every $(V, S) \in \mathcal{S}$ with $S = m_V$, keep; add (V, m_V) for every V;
(2) (Pullback) for every $(V, S) \in \mathcal{S}$ and every $U \leq V$, add (U, U^*S) where $U^*S = \{W \leq U \mid W \in S\}$;
(3) (Local character) for every V and sieve T on V: if some $(V, S) \in \mathcal{S}$ has, for every $W \in S$, the pair $(W, W^*T) \in \mathcal{S}$, add (V, T).

Output the fixed point $\mathcal{S}^\infty$.

LEMMA 4.18. *On a finite poset, Algorithm 4.17 terminates and its fixed point is exactly the generated topology J_K.*

PROOF. Termination: the set of (object, sieve) pairs is finite and $\mathcal{S}$ grows monotonically. The fixed point satisfies (J1) by step (1), (J2) by step (2), and (J3) by step (3), so it is a topology containing the generated sieves; conversely each step only adds sieves that every topology containing the generators must contain—(J1), (J2), (J3) are exactly the three closure rules—so by induction on the stage at which a pair enters $\mathcal{S}$, the fixed point is contained in every such topology. Hence it equals the smallest one. □

This is not merely an existence device: the successor's modality-composition verifier literally runs this saturation on the sixteen-context lattice, enumerating all 298 sieves and reporting 168 covering ones, and then re-verifies (J1)–(J3) on the computed fixed point as an independent check. Algorithm 4.17 is the hand-sized description of that computation.

PROOF OF THEOREM 4.16 FOR FINITE POSETS WITH MEETS. Suppose F glues bijectively along every basis-generated sieve; we show F glues bijectively along every $S \in J_K(V)$, by induction over the stages of Algorithm 4.17. The induction hypothesis is carried in pullback-closed form: what is proved for a pair (V, S) is that F glues bijectively along U^*S for *every* $U \leq V$, the case $U = V$ included. The strengthening is what discharges the pullback step, and it is needed, because the strong stability hypothesis is a hypothesis about generating families only: a sieve that enters at the local-character step need not contain any basis-generated sieve, so that hypothesis says nothing about its pullbacks. Stage 0: basis-generated sieves hold by hypothesis; maximal sieves hold by Example 4.5.

Stage 0 in pullback-closed form: let S be generated by a family $\{V_i \to V\} \in K(V)$ and let $U \leq V$. On a poset, $U^*S = \{W \leq U \mid W \in S\}$, and on a poset with meets this is precisely the sieve generated by the pulled-back family $\{U \wedge V_i \to U\}$. By the strong stability hypothesis of Theorem 4.16 that family generates a basis-generated or maximal sieve at U, so U^*S is again basis-generated or maximal and the base case applies to it directly; and if $S = m_V$ then $U^*S = m_U$. That hypothesis must be checked per basis, not assumed of "the poset setting": for the successor's partition bases it is Lemma 9.2 (intersection with U carries blocks to blocks); for the J_2 basis of Section 3 the pullback of the atom family along $\{a\} \subseteq \{a, b\}$ is $\{\{a\}, \varnothing\}$, which is *not* a declared generating family at $\{a\}$, but it generates $m_{\{a\}}$, so the hypothesis holds there as well. (This is also the point where a general-category proof needs more care; the general statement is [4].)

Pullback step: a pair (U, U^*S) added at step (2) from (V, S) needs no argument of its own, since $U'^*(U^*S) = U'^*S$ for every $U' \leq U$: the pullback-closed statement for (V, S) already contains the pullback-closed statement for (U, U^*S). This is the step at which the unstrengthened induction would appeal to strong stability outside its scope.

Local-character step: suppose T is a sieve on V, $S \in J_K(V)$ satisfies the bijective-gluing conclusion (inductively, together with all its pullbacks), and for every $W \in S$ the pullback W^*T does too. Injectivity of gluing along T: if $x, y \in F(V)$ agree on all of T, then for each $W \in S$, the restrictions $x|_W, y|_W$ agree on W^*T, which glues bijectively over W; hence $x|_W = y|_W$ for all $W \in S$, and gluing along S (injective) gives $x = y$. Surjectivity: let $(x_W)_{W \in T}$ be a matching family for T. For each $W \in S$, the family restricts to a matching family for W^*T, which glues to a unique $z_W \in F(W)$. The z_W form a matching family for S: for $W' \leq W$ in S, both $z_W|_{W'}$ and $z_{W'}$ glue the restricted family on W'^*T, and uniqueness of gluing along W'^*T forces them equal. Gluing along S yields $z \in F(V)$; one checks $z|_W = x_W$ for $W \in T$ by comparing both sides on the covering sieve W^*S of W and using injectivity there (each $W'' \in W^*S$ lies in both S-land and T-land, where the values agree by construction). Hence gluing along T is bijective. The pullback-closed form for (V, T) follows by rerunning the same argument at each $U \leq V$ with U^*S in place of S: gluing along U^*S and along all of its pullbacks is available from the pullback-closed hypothesis for (V, S), and each $W \in U^*S$ lies in S with $W^*(U^*T) = W^*T$, so the pullback-closed hypothesis for (W, W^*T) supplies the rest. This completes the induction. □

Remark 4.19. The practical import: to certify sheafhood on the successor lattice one checks each presheaf against the generating partition families only—two families per nonempty context plus the null cover—rather than against every covering sieve of the topology concerned (168 for the join $J_A \vee J_R$, fewer for each of J_A and J_R). The finite certificate does exactly this for the sixteen representables and the two witness presheaves, and Theorem 4.16 is the license. Conversely, when *refuting* sheafhood (as for the witness presheaves against the transverse topology, or for the representable in the countermodel of Chapter 6), a single failed generating family suffices, since generated sieves are covering.

6. The equalizer formulation, and life beyond posets

The matching-family formulation of the sheaf condition is the right one for sieves; for a generating family of arrows there is an older and equally useful formulation as an equalizer, which we record both for completeness and because it is the form in which gluing computations are usually organized in the literature [**1**, **4**].

Proposition 4.20 (Equalizer form). *Let $\mathcal{C}$ have pullbacks, let $\{f_i : C_i \to C\}_{i \in I}$ be a family of arrows generating the sieve S, and let $F \in \mathrm{PSh}(\mathcal{C})$. Then the gluing map for S is a bijection if and only if the diagram*

$$F(C) \longrightarrow \prod_{i \in I} F(C_i) \rightrightarrows \prod_{i,j \in I} F(C_i \times_C C_j)$$

is an equalizer of sets, where the first map is restriction along the f_i and the parallel pair restricts along the two pullback projections $C_i \times_C C_j \to C_i$ and $C_i \times_C C_j \to C_j$.

Proof sketch. An element of the equalizer is a tuple (x_i) with $x_i|_{\pi_1} = x_j|_{\pi_2}$ on every pullback. Given such a tuple, define $x_f = x_i|_g$ for any factorization $f = f_i \circ g$ of a member $f \in S$; the pullback compatibility makes this independent of the chosen factorization (two factorizations through f_i and f_j induce a map to $C_i \times_C C_j$ through which both restrictions

agree), and the resulting assignment is a matching family. Conversely a matching family restricts to an equalizer element on the generators. The two constructions are mutually inverse; details are in [**1**, Chap. III]. □

On a poset with meets, $C_i \times_C C_j$ is the meet $U_i \wedge U_j$ and the proposition specializes to Lemma 4.6. Two warnings attach to the general case, and both matter later. First, if $\mathcal{C}$ lacks the relevant pullbacks, the equalizer form is not available and one must either use the sieve form directly or check compatibility over *all* pairs of factorizations, as in Definition 4.2; the sieve form is the primitive notion, the equalizer form a convenience. Second, even when pullbacks exist, a family can have empty or degenerate pullbacks that make the compatibility condition vacuous while gluing still fails—this is exactly the geometry of the countermodel in Chapter 6, where two point inclusions into a two-element set have empty overlap, so every pair of local sections is "compatible," yet no global section exists. Vacuous compatibility plus failed gluing is the signature of a cover that is surjective on points but not effective on morphisms, and the reader should hold the L_2 intuition (Example 4.8) and that failure side by side: the function presheaf glues because functions out of a union are determined by functions on the pieces; a representable in an injection-only category may not glue because the would-be glued map is not an arrow of the category at all.

Example 4.21 (A non-poset site in miniature). Let $\mathcal{C}$ be the category with two objects D, C, two parallel arrows $u, v : D \to C$, and identities. Let S be the sieve on C generated by $\{u, v\}$; its members are u, v (and nothing else, as there is nothing to precompose with except identities). A matching family for S valued in F is a pair (x_u, x_v) with $x_u, x_v \in F(D)$ and *no* compatibility condition, since there are no nonidentity arrows into D. The gluing map sends $x \in F(C)$ to $(x|_u, x|_v)$. For $F = yD$: $yD(C) = \varnothing$ (no arrows $C \to D$) while $yD(D) = \{\mathrm{id}_D\}$, so the matching-family set is a singleton and the gluing map $\varnothing \to \{*\}$ is not surjective—the representable yD fails gluing for any topology declaring S covering. This tiny computation is the structural germ of the subcanonicity countermodel of Chapter 6: nothing about it is exotic, and its entire content is that the category supplies no arrow to serve as the glued section. It also illustrates why Definition 4.2 quantifies over all precompositions: on posets that quantification collapses to meets, but in general the sieve's own arrow structure is the compatibility data.

Remark 4.22 (Separated presheaves as an intermediate reflection). Between $\mathrm{PSh}(\mathcal{C})$ and $\mathrm{Sh}(\mathcal{C}, J)$ sits the full subcategory of separated presheaves. It too is reflective: the first plus construction (Section 8) composed with an image factorization reflects a presheaf onto its separated quotient, identifying sections that agree on some cover. The two-step structure of sheafification—first kill unsupported distinctions, then adjoin missing gluings—is thus a composite of two reflections, and the two steps are conceptually independent: Example 4.13 needs only the first, Example 4.12 only the second. In audit terms the two steps have different characters: separation *deletes* data (a quotient), gluing *creates* data (new sections from compatible families), and a provenance-conscious pipeline must track both, since neither is injective on information. This is one of the structural reasons the successor stores provenance outside the topology; the inside-the-topos operations are irreversibly lossy by design.

Remark 4.23 (The canonical topology). For every small category $\mathcal{C}$ there is a largest subcanonical topology, the *canonical topology*, whose covering sieves are the *universally effective-epimorphic* sieves: those S on C such that every representable glues uniquely along S and along every pullback g^*S, $g : D \to C$ [**1, 7**]. Subcanonicity of a given topology J is exactly the containment of J in the canonical topology. This gives the right frame for two later results:

the successor's partition topologies on the context lattice are subcanonical (they sit inside the canonical topology, as the representable checks show), while the legacy jointly-surjective coverage on the countermodel category of Chapter 6 is *not* contained in the canonical topology of that category—joint surjectivity on points simply does not imply effective epimorphy of the family. "Subcanonical" is thus not a decoration but a containment statement with content, and it can fail.

7. Sheafification: the universal property

Not every presheaf is a sheaf, but every presheaf has a best sheaf approximation, characterized by a universal property, constructed by the plus construction, and enjoying left exactness. We take these in order.

Definition 4.24 (Associated sheaf). Let $(\mathcal{C}, J)$ be a site and $F \in \mathrm{PSh}(\mathcal{C})$. An *associated sheaf* (or *sheafification*) of F is a sheaf aF together with a map $\eta_F : F \to aF$ such that for every sheaf G, composition with η_F is a bijection

$$\mathrm{Hom}_{\mathrm{PSh}}(aF,\, G) \xrightarrow{\;\cong\;} \mathrm{Hom}_{\mathrm{PSh}}(F,\, G).$$

Proposition 4.25. *An associated sheaf, if it exists, is unique up to unique isomorphism compatible with the units; and the assignment $F \mapsto aF$ extends uniquely to a functor $a : \mathrm{PSh}(\mathcal{C}) \to \mathrm{Sh}(\mathcal{C}, J)$ left adjoint to the inclusion $i : \mathrm{Sh}(\mathcal{C}, J) \hookrightarrow \mathrm{PSh}(\mathcal{C})$.*

Proof. Both statements are formal consequences of the universal property, by the standard argument for reflections [2, 9]: uniqueness because two reflections of the same object corepresent the same functor on the subcategory; functoriality because $\eta_{F'} \circ \alpha : F \to aF'$ extends uniquely across η_F for any $\alpha : F \to F'$; adjointness because the defining bijection is exactly the adjunction isomorphism, and its naturality follows from uniqueness of extensions. □

Theorem 4.26 (Existence and exactness; background). *For every small site $(\mathcal{C}, J)$ the associated-sheaf functor $a : \mathrm{PSh}(\mathcal{C}) \to \mathrm{Sh}(\mathcal{C}, J)$ exists, and it is* left exact*: it preserves all finite limits. Consequently $\mathrm{Sh}(\mathcal{C}, J)$ is a reflective subcategory of $\mathrm{PSh}(\mathcal{C})$ whose reflector preserves finite limits, and $\mathrm{Sh}(\mathcal{C}, J)$ is a Grothendieck topos.*

This is textbook material [1, Chap. III, §§5–6 and Chap. V, §3], [5], [7]; we present the construction in the next section both because the monograph's finite computations imitate it and because seeing it makes the left-exactness statement believable rather than oracular. Left exactness matters for the program for a concrete reason: a left-exact reflector preserves monomorphisms, finite products, and equalizers, hence preserves the shape of relational and subobject structure when evidence objects are pushed from the presheaf level into sheaves. A right adjoint that merely existed, without exactness, would permit the reflection to distort exactly the finite-limit structure that admissibility relations and typed comparisons live in.

8. The plus construction, and computing on finite sites

Construction 4.27 (The plus construction). For $F \in \mathrm{PSh}(\mathcal{C})$ and $C \in \mathrm{Ob}(\mathcal{C})$, the covering sieves of C form a directed poset under *refinement*: S' refines S when $S' \subseteq S$, and any two covering sieves have a common covering refinement $S \cap S'$ (an instance of (J2) plus (J3)). Restriction of matching families along inclusions gives a directed system, and one defines

$$F^{+}(C) \;=\; \operatorname*{colim}_{S \in J(C)} \mathrm{Match}(S, F),$$

the colimit over ever-finer covering sieves. An element of $F^+(C)$ is an equivalence class of matching families, two families being identified when they agree on a common covering refinement. Restriction along $g : D \to C$ sends the class of $(S, (x_f))$ to the class of $(g^*S, (x_{g\circ h})_h)$; the unit $F \to F^+$ sends $x \in F(C)$ to the class of the canonical family $(x|_f)_{f\in m_C}$.

THEOREM 4.28 (Background; [5, 1]). *F^+ is a separated presheaf for every F; if F is separated, F^+ is a sheaf. Hence $aF = F^{++}$, with unit the composite $F \to F^+ \to F^{++}$, satisfies the universal property of Definition 4.24. Moreover a is left exact: each application of $(-)^+$ is a filtered colimit of finite-limit-preserving operations* $\mathrm{Match}(S, -)$*, and filtered colimits commute with finite limits in* **Set**.

Two applications of $(-)^+$ are needed in general: the first pass merges sections that a cover cannot distinguish (repairing separation), the second pass adjoins gluings for compatible families (repairing existence). On particular presheaves one pass may happen to suffice, but no general theorem says so, and the successor's proof artifacts are deliberately structured so as never to assume it.

EXAMPLE 4.29 (A worked sheafification on L_2). Take H from Example 4.12 (H separated, identity restrictions between nonempty contexts, $|X| \geq 2$). We compute H^+ for J_2. On L_2 every object has a *minimum* covering sieve: the intersection of all covering sieves is covering (finitely many sieves, and covering sieves are closed under pairwise intersection), so the directed colimit collapses to its terminal stage:

$$F^+(V) \;\cong\; \mathrm{Match}(S_{\min}(V),\, F) \qquad \text{on any finite site.}$$

Here $S_{\min}(\{a, b\})$ is the sieve generated by the two atoms, $S_{\min}$ at each singleton is the maximal sieve, and $S_{\min}(\varnothing)$ is the empty sieve. Therefore $H^+(\{a, b\}) = \mathrm{Match}(S_{\min}, H) \cong X \times X$ (Example 4.12), H^+ agrees with H on singletons, and $H^+(\varnothing)$ is a singleton. One checks directly that H^+ is already a J_2-sheaf (it is isomorphic to the function presheaf $\mathbf{Set}(-, X)$ of Example 4.8), so here a single pass sufficed—because H was separated to begin with, exactly as Theorem 4.28 predicts. Redoing the computation for the non-separated K of Example 4.13 shows the first pass collapsing u and v to a single class, after which the second pass has nothing left to add.

REMARK 4.30 (Why direct constructions can beat the plus construction in finite settings). On a finite site the plus construction is entirely computable, as Example 4.29 shows, but it is rarely the best way to *certify* a sheafification, for three reasons. First, bookkeeping: the colimit ranges over all covering sieves of the topology in question (168 of them for the joined topology on the successor lattice), and although the minimum-sieve collapse tames this, an implementation still owes a proof that the collapse is legitimate on the exact site at hand. Second, the one-pass question: whether F^+ is already a sheaf depends on separation hypotheses that must be re-established for every input, and an implementation that silently assumes one pass is wrong on non-separated inputs. Third, and decisively for audit: the universal property, not the construction, is the specification. A closed-form candidate reflector—such as the product-over-blocks formula the successor uses on its partition coverages, which assigns to each context the product of the presheaf's values on the pieces—can be verified *directly* against Definition 4.24: prove the candidate lands in sheaves, prove composition with the unit is a bijection onto $\mathrm{Hom}_{\mathrm{PSh}}(F, G)$ for every sheaf G, and conclude by uniqueness (Proposition 4.25) that the candidate *is* sheafification, with no plus-construction bookkeeping anywhere in the trusted path. The successor's modality-composition proof takes exactly this route, and its stated reason is the one just given: the direct proof avoids assuming

that a single generic plus construction sheafifies an arbitrary presheaf. In finite settings the direct route is simultaneously more honest and easier to check exhaustively: a formula evaluated at sixteen contexts, against a universal property checked on finitely many fixtures, is a better certificate substrate than a two-pass colimit whose intermediate objects are themselves quotients of function spaces.

Exercise 4.31. Compute Δ_X^+ and Δ_X^{++} on (L_2, J_2) for $|X| = 2$. Check that Δ_X^+ has one section over $\varnothing$, X over each atom, and only X over the top—the matching condition through $\Delta_X(\varnothing)$ forces the two atom values to agree—so that Δ_X^+ is separated but not yet a sheaf; then check that the second pass enlarges the top to $X \times X$, so that $a\Delta_X = \Delta_X^{++} \cong \mathbf{Set}(-, X)$, the function presheaf of Example 4.8. Say which pass deletes sections and which creates them, and where each happens.

9. Reflective subcategories and idempotent monads

Sheafification exhibits $\mathrm{Sh}(\mathcal{C}, J)$ as a reflective subcategory of $\mathrm{PSh}(\mathcal{C})$. This section isolates the abstract shape, because the monograph's later chapters manipulate several reflections at once and their composites, and the abstract calculus is what makes those manipulations safe.

Definition 4.32. A full subcategory $\mathcal{B} \subseteq \mathcal{A}$ is *reflective* if the inclusion $i : \mathcal{B} \to \mathcal{A}$ has a left adjoint $a : \mathcal{A} \to \mathcal{B}$, the *reflector*. The composite $M = ia : \mathcal{A} \to \mathcal{A}$, together with the unit $\eta : \mathrm{Id} \Rightarrow M$ and multiplication induced by the adjunction counit, is a monad on $\mathcal{A}$ [**2**, **9**, **8**].

Proposition 4.33 (Idempotency). *For a reflective subcategory as above, the monad $M = ia$ is* idempotent*: the multiplication $\mu : MM \Rightarrow M$ is a natural isomorphism; equivalently, $M\eta_F$ and η_{MF} are equal isomorphisms $MF \to MMF$ for every F.*

Proof. Because $\mathcal{B}$ is full, the counit of the adjunction is an isomorphism $ai \cong \mathrm{Id}_{\mathcal{B}}$ (the reflection of an object already in $\mathcal{B}$ is itself, by the universal property applied to the identity). Whiskering, $MM = iaia \cong ia = M$, and the triangle identities identify this isomorphism with μ; the two unit maps into MMF agree because both are the unique extension of η_F across itself. □

Idempotency is what lets one speak of the subcategory and the monad interchangeably: $\mathcal{B}$ is recovered inside $\mathcal{A}$ as the objects for which the unit is an isomorphism, and M is determined up to canonical isomorphism by its image. The next definition packages the objects side.

Definition 4.34 (Local objects). Let $M = ia$ arise from a reflective subcategory. An object $F \in \mathcal{A}$ is *M-local* (or just *local* when M is understood) if $\eta_F : F \to MF$ is an isomorphism.

Proposition 4.35. *For $M = ia$ with a sheafification for a topology J on $\mathcal{C}$, an object of $\mathrm{PSh}(\mathcal{C})$ is M-local if and only if it is a J-sheaf. In general, the M-local objects are exactly the objects isomorphic to objects of the reflective subcategory, and M restricted to local objects is naturally isomorphic to the identity.*

Proof. If F is a sheaf then η_F is an isomorphism by the universal property (extend id_F across η_F; the two composites are identities by uniqueness of extensions). Conversely if η_F is an isomorphism then F is isomorphic to the sheaf iaF, and the sheaf condition—bijectivity of a class of gluing maps—transfers across isomorphism. The general statements are proved the same way, using only fullness and the universal property. □

10. Modalities

Definition 4.36 (Modality). A *modality* on a presheaf category $\mathrm{PSh}(\mathcal{C})$ is a left-exact idempotent monad M on $\mathrm{PSh}(\mathcal{C})$ arising as $M = ia$ for a reflective subcategory whose reflector a preserves finite limits. A modality is *trivial* if it is naturally isomorphic to the identity functor. Two modalities are *distinct* if they are not naturally isomorphic; equivalently (by Proposition 4.35) if their subcategories of local objects differ.

The word "modality" is borrowed deliberately. In topos theory, left-exact reflective subcategories of a presheaf topos correspond to Lawvere–Tierney topologies on it, which for $\mathrm{PSh}(\mathcal{C})$ correspond in turn to Grothendieck topologies on $\mathcal{C}$; the closure operators j on the subobject classifier that encode them behave like modal operators on the internal logic ("it is locally the case that"), and the sheaf condition says exactly that a predicate or object is insensitive to the modal blur. See [1, Chap. V] and [7] for this correspondence; in the formal-logic and type-theoretic literature the same abstract shape—a left-exact idempotent monad and its local objects—also travels under the name modality. This monograph uses the term exclusively in the sense of Definition 4.36, and always for modalities of sheafification type: $M_J = i_J a_J$ for a Grothendieck topology J on the site in question. No claim is made here about modal logics, and none is needed: every later theorem about "modalities" is a theorem about sheafification monads.

The program-level motivation, stated once and bounded: the governing specification intends distinct modalities to serve as distinct *resolution disciplines* on evidence presheaves—one reflector enforcing compatibility along one family of covers, another along a transverse family—so that partially resolved objects (local for one modality but not the other) are representable states of an audit pipeline. For that intent to be realizable at all, the modalities in play must be provably distinct and provably nontrivial, and their composites must be understood. Those are precisely the questions the finite theory answers, and the questions the legacy site fails, which is why the criterion below is the hinge of the whole monograph.

Proposition 4.37 (Triviality criterion). *Let J be a Grothendieck topology on a small category $\mathcal{C}$ and $M_J = i_J a_J$ the associated modality on* $\mathrm{PSh}(\mathcal{C})$. *The following are equivalent:*

1. *M_J is trivial (naturally isomorphic to the identity);*
2. *every presheaf on $\mathcal{C}$ is a J-sheaf;*
3. *every covering sieve of J is maximal: $J(C) = \{m_C\}$ for all C.*

This equivalence is standard; see [1, Chap. V] *and* [7]. *The proof is given in full because Chapter 5 applies the criterion directly to the legacy cover clauses.*

Proof. (3)⇒(2): Example 4.5—gluing along a maximal sieve is bijective for every presheaf. (2)⇒(1): if every presheaf is a sheaf then every unit $\eta_F : F \to M_J F$ is an isomorphism (Proposition 4.35), and these assemble to a natural isomorphism $\mathrm{Id} \cong M_J$. (1)⇒(3): we prove the contrapositive with an explicit witness. Suppose some $S_0 \in J(C)$ is not maximal. Consider the presheaf of sieves

$$\Omega(D) \;=\; \{\,\text{sieves on } D\,\}, \qquad \Omega(g : E \to D) \;=\; g^*(-),$$

the subobject classifier of $\mathrm{PSh}(\mathcal{C})$; functoriality is the identity $(g \circ h)^* = h^* g^*$ on sieves. Define a matching family for S_0 with values in Ω by assigning to every $f : D \to C$ in S_0 the maximal sieve $m_D \in \Omega(D)$. This matches: $h^* m_D = m_E$ for every $h : E \to D$. An amalgamation is a sieve T on C with $f^* T = m_D$ for all $f \in S_0$; since $f^* T = m_D$ if and only if $\mathrm{id}_D \in f^* T$ if and only if $f \in T$, the amalgamations are exactly the sieves $T \supseteq S_0$. There are at least two: S_0 itself and m_C, distinct because S_0 is not maximal. So Ω is not even separated for J, hence

not a sheaf; if M_J were naturally isomorphic to the identity, $\Omega \cong M_J\Omega$ would be isomorphic to a sheaf and hence a sheaf, a contradiction. □

REMARK 4.38. Return to the proof of (1)⇒(3), because it is the cleanest possible statement of what a nonmaximal cover *does*: it forces the presheaf of sieves to carry unsupported global distinctions (many sieves over C restrict to the maximal sieve everywhere on the cover), and sheafification must therefore act nontrivially on Ω. Conversely, a topology all of whose covers are maximal never forces anything. The relevance to the program is immediate and is spelled out in Chapter 5: the legacy ACBE and RPM clauses each generate only maximal sieves, so by Proposition 4.37 the modalities built from either clause alone are trivial—naturally isomorphic to the identity, local on every presheaf, incapable of supporting any distinction between resolution states. The collapse theorem is, in this exact sense, an instance of the triviality criterion, and the successor's entire design problem is to manufacture covers that are honestly nonmaximal.

10.1. Closed sieves and the classifier viewpoint. The subobject classifier Ω of $\mathrm{PSh}(\mathcal{C})$, which appeared as the witness in Proposition 4.37, supports a second description of modalities, recorded here even though the monograph's proofs never depend on it. Given a topology J, call a sieve T on C *J-closed* if every arrow that lies in T locally already lies in T: whenever $f : D \to C$ has $f^*T \in J(D)$, then $f \in T$. Closing a sieve (adding all such arrows) defines an operator $j : \Omega \to \Omega$, natural in C, which is inflationary, idempotent, and meet-preserving; such an operator on the classifier is a *Lawvere–Tierney topology*, and on a presheaf topos these correspond exactly to Grothendieck topologies on the base category [**1**, Chap. V], [**7**]. The J-sheaves are then the objects orthogonal to the class of *dense* monomorphisms (those whose closure is everything), and sheafification is the universal way of inverting dense monos. Three consequences of this viewpoint are used later, and we state them as background [**1**, **7**]:

(1) The covering sieves of J are exactly the sieves whose closure is maximal; in particular a topology is determined by its closed sieves, and the presheaf Ω_j of closed sieves is the subobject classifier of $\mathrm{Sh}(\mathcal{C}, J)$.
(2) A modality of sheafification type inverts the inclusions $S \hookrightarrow yC$ for covering S; this is the precise sense in which "the topology cannot see the difference between a covering sieve and the whole representable," and it is the abstract root of the provenance-erasure discussion in Chapter 5.
(3) Meets of topologies correspond to pointwise meets of the associated closure operators, and the join of two topologies is the topology generated by their union—the operation the successor computes explicitly when it identifies the join of its two partition topologies with the singleton-partition topology.

For the finite sites of this monograph, every one of these statements is also directly checkable by enumeration, and the certificate chapters do so; the classifier viewpoint is offered as orientation, not as a load-bearing dependency.

EXERCISE 4.39. On (L_2, J_2), compute the closure of each of the six sieves on the top object and verify: (i) the closed sieves on the top are exactly the maximal sieve and the sieves generated by $\{a\}$, by $\{b\}$, and by the bottom arrow $\varnothing \to \{a, b\}$ (note the *empty* sieve is not closed, since the null cover makes $\varnothing$ covered and so forces $\varnothing$ into every closure); (ii) the covering sieves are exactly those whose closure is maximal; (iii) Ω_j at the top has exactly four elements, matching the sheaf condition's verdict on Exercise 4.42.

REMARK 4.40 (Distinctness, composition, and what is *not* claimed). Two further points of the abstract calculus recur later. First, distinctness of modalities is witnessed by objects: $M_J \neq M_{J'}$ (not naturally isomorphic) as soon as some presheaf is local for one and not the other, since reflectors fix exactly their local objects. This is how the successor's two finite modalities are separated: an explicit presheaf that satisfies one partition's gluing and fails the other's. Second, composites of modalities are subtle in general—the composite of two left-exact idempotent monads need not be idempotent, and when it is, identifying it requires a theorem, not a reflex. On the successor's finite lattice the composite of the two partition modalities is, by a proved and exhaustively corroborated canonical natural isomorphism, the modality of the joined topology, and in particular the two modalities *commute*; distinctness does not imply noncommutation, and no artifact of the program licenses a noncommutation claim. Both facts are proved in the finite-theory chapters; they are previewed here only so that the reader does not import an expectation of order-sensitivity that the mathematics does not support.

CLAIM BOUNDARY 4.41. Status discipline for this chapter: every numbered statement above is either (i) standard background, cited to [**1**], [**2**], [**7**], [**5**], [**4**], [**9**], or [**8**], with no novelty claimed; or (ii) an elementary finite verification (the L_2 examples, Algorithm 4.17 and its lemma, and the finite-poset case of Theorem 4.16) offered at tutorial level. Forward references to properties of the successor construction (distinct nontrivial commuting modalities, the 298-sieve and 168-cover census, the direct reflector) describe results whose registered status is INFORMAL PROOF PRESENT with exhaustive executable corroboration; they are not peer-reviewed theorems, and nothing in this chapter promotes them. No statement in this chapter carries any empirical, causal, safety, or product interpretation.

11. Chapter summary and the road ahead

The chapter assembled the four-piece toolkit the rest of the monograph runs on. (1) *The sheaf condition*: sections over an object biject with matching families over each covering sieve; separation is the injective half, gluing the surjective half; on posets with meets, matching families are compatible tuples (Lemma 4.6); the null cover normalizes the initial context (Lemma 4.7). The examples and non-examples of Sections 3–4 exhibit each failure mode in the smallest possible habitat. (2) *The basis criterion*: sheafhood needs checking only on generating families (Theorem 4.16), and on finite posets the generated topology is the fixed point of an explicit saturation (Algorithm 4.17)—the exact computation the finite certificate later performs. (3) *Sheafification*: a left-exact reflection characterized by its universal property (Definition 4.24, Theorem 4.26), constructible by the two-pass plus construction (Construction 4.27) but, on finite sites, better certified by closed-form candidates verified directly against the universal property (Remark 4.30). (4) *Modalities*: left-exact idempotent monads of sheafification type, with local objects the sheaves (Proposition 4.35), and with a sharp triviality criterion—trivial exactly when every cover is maximal (Proposition 4.37).

The criterion is the pivot on which the monograph now turns. Chapter 5 takes the registered legacy site, whose ACBE and RPM cover clauses were intended to seed two distinct modalities, and shows by two one-line sieve computations that every sieve either clause generates is maximal. By Proposition 4.37, the intended modalities are trivial; by the closure-operator analysis there, no relabeling, sub-sieve surgery, or object-adjunction can help; and by Chapter 6, the same axioms also fail to deliver subcanonicity, so even the representables must be handled through their associated sheaves. Only after both negative results are fully absorbed does the successor construction begin.

EXERCISE 4.42. Compute Ω explicitly on L_2 (six sieves on the top, three on each singleton, two on the bottom; compare Exercise 4.14), identify the matching family used in the proof of Proposition 4.37 for the atom cover of the top, and list its amalgamations. Then compute $a\Omega$ for J_2 and verify that the result classifies J_2-closed sieves only.

EXERCISE 4.43. On L_2, let J_a be generated by the single family $\{\{a\} \to \{a, b\}\}$ (and identities and the null cover), and J_2 as above. Determine which representables are sheaves for J_a, whether $J_a \subseteq J_2$, and whether the two modalities M_{J_a} and M_{J_2} are distinct, exhibiting a separating presheaf if so. (This exercise previews the witness-presheaf technique of the finite theory.)

EXERCISE 4.44. Using the description $F^+(C) = \mathrm{Match}(S_{\min}(C), F)$ valid on finite sites (Example 4.29), verify by direct computation that $(-)^+$ preserves finite products and equalizers on (L_2, J_2), and exhibit the natural isomorphism $a(F \times G) \cong aF \times aG$ for $F = G = \Delta_X$.

CHAPTER 5

The Legacy Noetic Site and Its Collapse

This chapter is the conservative correction at the heart of the monograph. It takes the registered legacy site—the category Noetica with its three cover clauses—exactly as the conservative source states it, and proves that two of the three clauses contribute nothing to the topology they were intended to enrich. The ACBE clause admits, by its own reflexivity provision, an identity arrow into its covering sieves; the RPM clause admits a split epimorphism whose section drags an identity in by sieve closure. A sieve containing an identity is maximal, maximal sieves belong to every Grothendieck topology, and so each clause generates only covers that every topology already has. The combined topology collapses to the jointly-surjective observational coverage alone, and the modalities the two clauses were meant to seed are trivial in the precise sense of Chapter 4.

The result is negative but not destructive, and stating its exact perimeter is as important as proving it. The conditional topos-existence theorem of the conservative source survives untouched: the legacy site is still a site, its sheaf category is still a Grothendieck topos, and nothing in this chapter casts doubt on that. What falls is the *interpretive superstructure*: the inference that the ACBE and RPM clauses, as registered, could ever support two distinct nontrivial modal operators. We prove the collapse in fully expanded form (Section 4), diagnose it structurally—sieve generation is a closure operator, and closure operators erase provenance (Section 5)—and then do something the terse research paper could not afford: we take the three most tempting repairs seriously, formalize each, and refute each (Section 6). Each repair either regenerates maximal sieves or changes nothing; the only exit is to change the cover generators themselves, which is not a repair of the legacy clauses but the replacement the successor chapters carry out. The chapter closes with the exact scope of the v7.4 corrigendum: what it retracts, what it preserves, and what discipline it imposes on all downstream text (Section 7).

Claim Boundary 5.1. Status discipline. The mathematical content of this chapter consists of elementary sieve computations whose registered status is INFORMAL PROOF PRESENT, with the two collapse mechanisms additionally corroborated by named executable checks in the registered finite certificate (Section 8). Independent category-theory review of these arguments remains an open obligation; nothing here is peer-reviewed, and nothing here is promoted by appearing in this monograph. The chapter makes no empirical, causal, safety, or product claim; the words "ACBE" and "RPM" name registered clause labels and formal structure only.

1. The registered category Noetica

Notation 5.2. Throughout this chapter, $\mathcal{C}$ denotes the registered legacy category Noetica, and J_{legacy} denotes the coverage defined by the conservative source's three clauses, made precise below. We write J_{js} for the topology generated by the jointly-surjective observational clause alone.

The conservative source registers Noetica as a small concrete category of finite sets indexed by the forty distinguished CAS indices, whose morphisms are generated by admissibility-relation maps and closed under pullback. Three features of this registration matter for the present chapter, and only these three.

First, *concreteness and finiteness*: the objects are finite sets, so "jointly surjective" has its ordinary elementwise meaning for a family of maps into a common codomain, and the observational cover clause is a set-level condition. Second, *smallness*: Noetica is small in the chosen universe, which is the hypothesis under which the background machinery of Chapter 4—sheafification, the sheaf topos, the modality calculus—applies at all. Third, *the registered structural maps*: among the arrows of Noetica the source designates certain families as cover-generating, and it is the exact arrows so designated, not their intended meanings, that the topology machinery consumes. The precise object class, the full admissibility generator list, and the index combinatorics are registry data recorded in the frozen source artifacts and their hashes (the governing specification is registered under SHA-256 `156A5D8C5320F36E81E3B46473BA8A038B35672DCD28839A404D96A3DB1A763A`); none of those details enter the collapse argument, which uses only the three clause statements quoted in the next section. Where a detail of the registration is not restated here, it is because the argument does not depend on it, and this chapter deliberately refrains from paraphrasing registry content it does not use.

Claim Boundary 5.3. The collapse theorem is a schema-level result: it holds for *every* category and every designation of clause data satisfying the three quoted clauses, because its proof manipulates only the clause structure (a reflexivity provision and a section-retraction identity). In particular it does not depend on which model of the registered axioms one takes—a point that matters again in Chapter 6, where a different schema-level question (subcanonicity) is settled by exhibiting a particular model.

2. The three cover clauses

The conservative source admits a sieve as covering when it arises from any of three clauses. We state them as registered axioms, in the notation of this monograph, and then record the intent attached to each—keeping statement and intent rigorously apart, because the entire content of this chapter is that the statements do not support the intent.

Axiom 5.4 (Clause (O): observational covers). *For each object X of $\mathcal{C}$, every family $\{f_i : U_i \to X\}_{i \in I}$ of arrows of $\mathcal{C}$ that is jointly surjective—every element of the finite set underlying X lies in the image of some f_i—generates a covering sieve on X.*

Axiom 5.5 (Clause (A): ACBE descent covers). *The source designates a class of* ACBE descent maps *between contexts, and declares the sieve generated by the designated descent data at X to be covering. The designation is reflexive: at equal contexts the designated descent map is the identity, so the generating family at X contains* id_X.

Axiom 5.6 (Clause (R): RPM counit covers). *The RPM structure supplies, for each relevant object X, a pair of arrows*

$$\eta_X : X \longrightarrow R(X), \qquad \varepsilon_X : R(X) \longrightarrow X, \qquad \varepsilon_X \circ \eta_X = \mathrm{id}_X,$$

a section-retraction pair*: η_X is a section of the split epimorphism ε_X. The clause declares the sieve generated by $\{\varepsilon_X\}$ to be covering on X.*

Remark 5.7 (Section-retraction, not adjunction). The RPM data are registered as a section-retraction pair and expressly *not* as an adjunction. The identity $\varepsilon_X \circ \eta_X = \mathrm{id}_X$

holds; the other composite $\eta_X \circ \varepsilon_X$ is *not* required to be the identity of $R(X)$, and in the intended reading it typically is not: re-embedding after realization may lose information, and prerequisite satisfaction is one-directional. This asymmetry is a registered correction in its own right elsewhere in the program, and the collapse argument respects it scrupulously: the proof below uses only the one identity that is actually postulated, $\varepsilon_X \circ \eta_X = \mathrm{id}_X$, and at no point assumes $\eta_X \circ \varepsilon_X = \mathrm{id}_{R(X)}$, invertibility of either map, or any adjunction triangle. The collapse is not an artifact of over-reading the RPM structure; it follows from the weakest identity the structure carries.

Remark 5.8 (The recorded intent). The conservative source attached interpretations to the clauses: clause (O) was to encode which families of observations jointly determine a context; clauses (A) and (R) were to seed, downstream, two distinct modal operators—two "lenses" on evidence presheaves, an ACBE-descent lens and an RPM-resolution lens—by the mechanism of Chapter 4: each clause generates a topology, each topology a sheafification modality. The acronyms are registered clause names; their operational interpretation carries no weight in what follows and is quarantined by the program's claim boundaries. What matters here is only the mathematical shape: (A) puts an identity among its generators by reflexivity, and (R) puts a split epimorphism there. Those two shapes decide everything.

2.1. Why the clauses were seductive. Before refuting the construction we explain why competent drafting produced it, because the error has a recognizable genealogy and naming it inoculates against repetition. Both offending provisions are natural—indeed mandatory—in *adjacent* formalisms, and the legacy source imported them into the one formalism where they are fatal.

Reflexivity first. In the theory of descent data, and everywhere equivalence relations or groupoids organize "being related across contexts," reflexivity is a harmless and structurally required axiom: an object is related to itself, descent along the identity is trivial descent, and omitting reflexivity would break composition. A designer specifying "ACBE descent maps, including descent of every context to itself" is following that healthy instinct. The instinct fails here because a cover clause does not feed a groupoid; it feeds a sieve, and a sieve is a *downward-closed class of arrows into a fixed target*, where an identity member is not a harmless degenerate case but a universal solvent (Lemma 5.9). Descent data and covering sieves are both "families of maps with conditions," and the type distinction between them is exactly what the reflexivity provision missed.

The counit second. In the RPM reading, $\varepsilon_X : R(X) \to X$ is the realization of prerequisites—an arrow along which structured resources map onto the target—and declaring it a cover expresses the reasonable intent "X is covered by its resolved form." In formalisms where covers are single structured morphisms with extra properties (fibrations, effective epimorphisms taken singly, projections of resolutions), such a declaration can carry content. Under sieve generation it cannot, whenever the realization admits a section—and the RPM axioms themselves supply the section η_X, since realization-after-embedding being the identity is the whole point of the section-retraction discipline. The very axiom that makes the RPM data well behaved (Remark 5.7) is the axiom that trivializes the cover (Lemma 5.10). There is a genuine lesson in irony here: the legacy construction did not fail by sloppiness in the RPM axioms—it failed because the RPM axioms are strong enough to defeat their own intended topological use.

3. Two elementary facts about sieves

The collapse rests on two observations so short that their consequences were easy to overlook. We isolate them, prove them, and add the one-step generalization to isomorphisms for completeness.

LEMMA 5.9 (Identities force maximality). *Let S be a sieve on an object X of any category. If $\mathrm{id}_X \in S$, then $S = m_X$, the maximal sieve.*

PROOF. Let $f : D \to X$ be any arrow with codomain X. Then $f = \mathrm{id}_X \circ f$, and S is closed under precomposition with arbitrary arrows, so $f \in S$. Hence S contains every arrow into X. □

LEMMA 5.10 (Split epimorphisms force maximality). *Let S be a sieve on X and let $\varepsilon : E \to X$ be a split epimorphism with section $\eta : X \to E$, i.e. $\varepsilon \circ \eta = \mathrm{id}_X$. If $\varepsilon \in S$, then $S = m_X$.*

PROOF. S is closed under precomposition, so $\varepsilon \circ \eta \in S$; but $\varepsilon \circ \eta = \mathrm{id}_X$, so $\mathrm{id}_X \in S$ and Lemma 5.9 applies. □

LEMMA 5.11 (Isomorphisms force maximality). *If a sieve S on X contains any isomorphism $g : D \to X$, then $S = m_X$.*

PROOF. An isomorphism is in particular a split epimorphism, with section g^{-1}; apply Lemma 5.10. (Directly: $g \circ g^{-1} = \mathrm{id}_X \in S$.) □

REMARK 5.12. Pause on *why* these lemmas are inescapable. A sieve is by definition a precomposition-closed class: the definition is what makes the sheaf condition well posed (Definition 4.2 quantifies over precompositions) and what makes pullback of covers (axiom (J2)) meaningful. Precomposition closure is therefore not an optional convenience one might weaken to save the legacy clauses; it is constitutive of the notion of cover in a Grothendieck topology. And precomposition closure is exactly the rule that converts "contains ε_X" into "contains id_X" the moment a section exists. The legacy defect is thus not a subtle interaction of axioms but a direct consequence of the definition of sieve meeting the definition of section.

4. The collapse theorem

We first fix what "the topology generated by the clauses" means, so that the collapse statement is exact.

DEFINITION 5.13 (Generated topology). Let G be a class of pairs (X, S) with S a sieve on X. The collection of all sieves on all objects is a Grothendieck topology (the largest one), and the intersection of any family of Grothendieck topologies is a Grothendieck topology. Hence there is a smallest Grothendieck topology containing G; we call it the topology *generated* by G and write $\mathrm{Gen}(G)$. The legacy topology is

$$J_{\text{legacy}} \;=\; \mathrm{Gen}\big(G_{\mathrm{O}} \cup G_{\mathrm{A}} \cup G_{\mathrm{R}}\big),$$

where G_{O}, G_{A}, G_{R} are the classes of (object, generated sieve) pairs supplied by Axioms 5.4, 5.5, 5.6 respectively, and $J_{\mathrm{js}} = \mathrm{Gen}(G_{\mathrm{O}})$.

THEOREM 5.14 (Legacy collapse). *Under the registered axioms:*

(1) *every sieve in G_{A} is maximal;*
(2) *every sieve in G_{R} is maximal;*

(3) *consequently* $J_{\text{legacy}} = J_{\text{js}}$*: the topology generated by all three clauses equals the topology generated by the observational clause alone, and the generated topology retains no ACBE or RPM provenance.*

PROOF. (1) Fix an object X at which clause (A) applies, and let $S_{\text{A}}(X)$ be the sieve generated by the designated ACBE descent family at X. By the reflexivity provision of Axiom 5.5, the designated family contains the reflexive descent map at equal contexts, which is id_X. A generated sieve contains its generators, so $\text{id}_X \in S_{\text{A}}(X)$, and Lemma 5.9 gives $S_{\text{A}}(X) = m_X$.

(2) Fix an object X at which clause (R) applies, and let $S_{\text{R}}(X)$ be the sieve generated by $\{\varepsilon_X\}$. Then $\varepsilon_X \in S_{\text{R}}(X)$, and ε_X is a split epimorphism with section η_X by the registered identity $\varepsilon_X \circ \eta_X = \text{id}_X$ of Axiom 5.6. By Lemma 5.10, $S_{\text{R}}(X) = m_X$. Explicitly and in full: $S_{\text{R}}(X)$ is closed under precomposition, so it contains $\varepsilon_X \circ \eta_X$; that composite *is* id_X; a sieve containing id_X contains $f = \text{id}_X \circ f$ for every $f : D \to X$; hence $S_{\text{R}}(X)$ is all of m_X.

(3) We show the two generated topologies coincide by showing they have the same family of upper bounds. Let J be any Grothendieck topology on $\mathcal{C}$ containing G_{O}. By axiom (J1), J contains every maximal sieve; by parts (1) and (2), G_{A} and G_{R} consist of maximal sieves; hence J contains $G_{\text{O}} \cup G_{\text{A}} \cup G_{\text{R}}$. Conversely any topology containing the union contains G_{O}. So a topology contains G_{O} if and only if it contains the union, the two intersections defining Gen range over the same family, and $J_{\text{legacy}} = J_{\text{js}}$. For the final phrase of (3): the equality says that the covering sieves of the legacy site are exactly the jointly-surjective ones; no covering sieve, and hence no sheaf condition, no sheaf, and no sheafification value, differs from what clause (O) alone determines. Any structure downstream of the topology—the sheaf topos, the modality, the internal logic—is therefore blind to whether clauses (A) and (R) were ever stated. □

The presheaf-level restatement is given in full because it shows the collapse operating not as an abstract maximality fact but as a concrete gluing computation: *every* presheaf satisfies the ACBE and RPM sheaf conditions automatically, with the RPM section itself manufacturing the amalgamation.

PROPOSITION 5.15 (Clause-wise gluing is automatic). *Let F be any presheaf on $\mathcal{C}$.*

(1) *For the ACBE generating family at X (which contains id_X), the gluing map of F is a bijection.*
(2) *For the RPM generating family $\{\varepsilon_X\}$ at X, the gluing map of F is a bijection; explicitly, a matching datum $x \in F(R(X))$ compatible over ε_X has the unique amalgamation $y = x|_{\eta_X} \in F(X)$.*

PROOF. By the standard reduction (Chapter 4), matching families for a generating family biject with matching families for the sieve it generates, so part (1) is Example 4.5 applied to $S_{\text{A}}(X) = m_X$: a matching family on the maximal sieve is determined by its value at id_X, and gluing is the corresponding bijection.

Part (2) can be quoted the same way, but we also carry out the computation directly, because it exhibits the mechanism. A matching datum for $\{\varepsilon_X\}$ is an element $x \in F(R(X))$ satisfying the compatibility condition: whenever $h, h' : E \to R(X)$ obey $\varepsilon_X \circ h = \varepsilon_X \circ h'$, then $x|_h = x|_{h'}$. Set $y := x|_{\eta_X} \in F(X)$. We claim $y|_{\varepsilon_X} = x$. By contravariant functoriality, $y|_{\varepsilon_X} = x|_{\eta_X \circ \varepsilon_X}$ (restriction along η_X then along ε_X is restriction along the composite $\eta_X \circ \varepsilon_X : R(X) \to R(X)$). Now apply the compatibility condition with $h = \eta_X \circ \varepsilon_X$ and $h' = \text{id}_{R(X)}$: indeed $\varepsilon_X \circ (\eta_X \circ \varepsilon_X) = (\varepsilon_X \circ \eta_X) \circ \varepsilon_X = \varepsilon_X = \varepsilon_X \circ \text{id}_{R(X)}$, so $x|_{\eta_X \circ \varepsilon_X} = x|_{\text{id}_{R(X)}} = x$. Hence

$y|_{\varepsilon_X} = x$: y is an amalgamation. For uniqueness, suppose $y' \in F(X)$ also has $y'|_{\varepsilon_X} = x$. Then

$$y' \;=\; y'|_{\mathrm{id}_X} \;=\; y'|_{\varepsilon_X \circ \eta_X} \;=\; \big(y'|_{\varepsilon_X}\big)\big|_{\eta_X} \;=\; x|_{\eta_X} \;=\; y.$$

So gluing along the RPM family is a bijection for every presheaf F: the sheaf condition it imposes is empty. Note where the two halves came from: *existence* of the amalgamation used the compatibility condition once, at the pair $(\eta_X \circ \varepsilon_X,\ \mathrm{id}_{R(X)})$; *uniqueness* used only the section identity. The very map η_X that was meant to express one-directional prerequisite structure is the map that trivializes the cover. □

Corollary 5.16 (Triviality of the intended modalities). *Let $J_{\mathrm{A}}^{\mathrm{leg}} = \mathrm{Gen}(G_{\mathrm{A}})$ and $J_{\mathrm{R}}^{\mathrm{leg}} = \mathrm{Gen}(G_{\mathrm{R}})$ be the topologies generated by clause (A) alone and clause (R) alone. Then every covering sieve of either topology is maximal; every presheaf on $\mathcal{C}$ is a sheaf for both; both sheafification modalities are trivial (naturally isomorphic to the identity of* $\mathrm{PSh}(\mathcal{C})$*); and the two modalities are naturally isomorphic to each other. Moreover, relative to the full legacy site, adjoining clauses (A) and (R) to clause (O) changes neither the sheaf category, nor the sheafification functor, nor the resulting topos:*

$$\mathrm{Sh}(\mathcal{C}, J_{\mathrm{legacy}}) \;=\; \mathrm{Sh}(\mathcal{C}, J_{\mathrm{js}}) \quad \textit{with equal, not merely equivalent, subcategories of } \mathrm{PSh}(\mathcal{C}).$$

Proof. The generators of $J_{\mathrm{A}}^{\mathrm{leg}}$ and $J_{\mathrm{R}}^{\mathrm{leg}}$ are maximal sieves (Theorem 5.14(1),(2)), and the collection consisting of exactly the maximal sieves on each object is already a Grothendieck topology (axiom (J1) holds by construction; pullback of a maximal sieve along g contains id, hence is maximal, giving (J2); and (J3) adds nothing new, since if T has f^*T maximal for all f in a maximal sieve, take $f = \mathrm{id}$ to get T maximal). Being the smallest Grothendieck topology on $\mathcal{C}$, it is therefore the generated one. Proposition 4.37 then gives triviality of both modalities, and trivial modalities are naturally isomorphic to the identity, hence to each other. The final statement is Theorem 5.14(3): equal topologies have identical sheaf conditions, hence identical sheaf subcategories, and sheafification is determined up to canonical isomorphism by its subcategory (Proposition 4.25). □

Remark 5.17 (What the collapse is not). Three misreadings should be blocked immediately. The collapse is *not* a contradiction in the legacy axioms: the axioms are consistent, the site is a site, and the topos exists. It is *not* a defect of topos theory or an exotic boundary case: Lemmas 5.9–5.10 are among the first exercises one can state about sieves, and the theorem is their immediate application. And it is *not* a statement that ACBE or RPM structure is meaningless: the descent maps and the section-retraction pair remain perfectly good arrows and identities of $\mathcal{C}$, available to any construction that consumes arrows. What collapses is specifically the attempt to make the *topology* remember them, because they were fed to it in a form (containing an identity; containing a split epi) that generation immediately saturates. The failure is a type error, in the following sense: cover generation is the wrong interface for provenance, and the next section makes that precise.

5. Generation is a closure operator, and closure erases provenance

The collapse theorem shows two particular clauses dissolving into the topology. The deeper point is that *any* clause information beyond the bare arrow families must dissolve, because of what kind of operation topology generation is. This section formalizes that claim; it is the mathematical justification for the successor's architectural decision to carry provenance in parallel records outside the topology, rather than trying again with cleverer generators.

Consider the complete lattice $\mathcal{L}$ whose elements are classes G of (object, sieve) pairs on $\mathcal{C}$, ordered by inclusion. Grothendieck topologies are particular elements of $\mathcal{L}$, and, as noted in Definition 5.13, they are closed under arbitrary intersection and include the top element. The map $\mathrm{Gen} : \mathcal{L} \to \mathcal{L}$ sending G to the smallest topology containing it is therefore the closure operator associated with a closure system, and it enjoys the three defining properties:

PROPOSITION 5.18 (Generation is a closure operator). *For all $G, G' \in \mathcal{L}$:*

(1) *(Extensive)* $G \subseteq \mathrm{Gen}(G)$*;*
(2) *(Monotone)* $G \subseteq G'$ *implies* $\mathrm{Gen}(G) \subseteq \mathrm{Gen}(G')$*;*
(3) *(Idempotent)* $\mathrm{Gen}(\mathrm{Gen}(G)) = \mathrm{Gen}(G)$*;*

and the fixed points of Gen *are exactly the Grothendieck topologies.*

PROOF. All four statements are the standard facts about the closure operator of a closure system: $\mathrm{Gen}(G)$ is the intersection of all topologies containing G; that intersection contains G (extensivity); a larger G is contained in fewer topologies, so its intersection is larger (monotonicity); $\mathrm{Gen}(G)$ is itself a topology, so the topologies containing it are the topologies containing G that contain it, and the intersection is unchanged (idempotency); and G is a fixed point iff G is a topology, since topologies are exactly the elements containing themselves as their own closure. □

Now observe what the *input* to Gen is. A clause system, as registered, is richer than an element of $\mathcal{L}$: it is a family of *labeled* designations—this family is admitted *as an observational cover*, this one *as ACBE descent*, this one *as the RPM counit*, each with its authority, its assumptions, its intended downstream use. Formally, a clause system is a set Λ of labels together with a function $u : \Lambda \to \mathcal{L}$ assigning to each label the (object, sieve) pairs it contributes. The topology the site carries is

$$J \;=\; \mathrm{Gen}\Big(\bigcup_{\lambda \in \Lambda} u(\lambda)\Big).$$

Two erasures happen in this formula, in order. The union forgets the *partition* of the generators by label: two clause systems with the same underlying pairs but different labelings have the same union. Then Gen forgets the *generators themselves*: it returns a fixed point, and a fixed point carries no record of which of its elements were inputs and which arose from saturation. Both erasures are structural, not accidental:

ADVERSE FINDING 5.19 (Provenance erasure). The passage from a labeled clause system to its Grothendieck topology factors as

$$\text{labeled clauses} \xrightarrow{\text{forget labels}} \text{generator classes in } \mathcal{L} \xrightarrow{\mathrm{Gen}} \text{topologies},$$

and each factor is non-injective in the relevant way: the first identifies all labelings of the same generators; the second identifies all generator classes with the same closure. Consequently no construction that consumes only the topology—sheaves, sheafification, the modality, the internal logic, the topos—can distinguish two clause systems whose generators have equal closure, and *a fortiori* none can recover which clause admitted a given cover. A Grothendieck topology is constitutionally incapable of remembering why a family was admitted.

EXAMPLE 5.20. Non-injectivity of Gen is easy to witness even without the collapse: adding to any generator class all maximal sieves, or any sieve already in its closure, changes nothing. The legacy site is the pointed instance: G_{O} and $G_{\mathrm{O}} \cup G_{\mathrm{A}} \cup G_{\mathrm{R}}$ are different generator classes—materially different, in that the latter records the program's two interpretive

clauses—with identical closure (Theorem 5.14). The interpretive investment in clauses (A) and (R) was placed entirely in the part of the input that the closure operator is guaranteed to forget.

REMARK 5.21 (The design consequence, stated once). Finding 5.19 converts the collapse from an accident into a specification. If cover provenance—which authority admitted a family, under which assumptions, with which review state—is to survive, it must be stored in a structure that is *not* downstream of Gen: a parallel record bound to the exact member arrows, versioned and hash-chained, consumed by governance code while the topology consumes the bare family. That is precisely the marked-cover architecture the successor adopts, developed in the marked-covers chapter of this monograph; nothing in the present chapter depends on it, but the reader should carry forward the exact division of labor it implements: *the topology is for compatibility semantics; the mark record is for provenance; neither can do the other's job.*

6. Three tempting repairs, formalized and refuted

Faced with Theorem 5.14, three repairs suggest themselves so naturally that each was, at one time or another, informally proposed within the program. Each keeps the legacy site "as intended" while adjusting the offending detail. We formalize all three and show that each either still yields maximal sieves or changes nothing. The section is deliberately thorough: the refutations, not the collapse itself, are what justify the successor's decision to abandon the legacy generators rather than continue repairing them.

6.1. Repair 1: relabel the generators. *Proposal.* Keep the category and the designated arrows, but rename or re-partition the clauses: introduce new clause labels (say ACBE-prime and RPM-prime), or redistribute the same designated arrows among differently named clauses, or designate the same arrows "with different intended semantics," expecting the resulting topologies to differ and the modalities to separate.

PROPOSITION 5.22 (Generation is label-blind). *Let (Λ, u) and (Λ', u') be two labeled clause systems with the same underlying generators: $\bigcup_\lambda u(\lambda) = \bigcup_{\lambda'} u'(\lambda')$. Then they generate the same topology, the same sheaf category, and canonically isomorphic modalities. In particular, any relabeling of the legacy clauses that preserves the designated arrows yields exactly J_{js} again; and any relabeled clause whose designated family still contains the reflexive descent map, or still contains a split epimorphism onto its target, still generates only maximal sieves.*

PROOF. The first sentence is immediate from the factorization in Finding 5.19: the topology depends on the clause system only through the union of its generators. The second sentence is Theorem 5.14 applied verbatim: the collapse proof used no property of the labels, only membership of id_X (via reflexivity) or of a split epimorphism (via the section identity) in the generating family, and those memberships are properties of the arrows, which the relabeling preserves. □

REMARK 5.23. The refutation is short precisely because the proposal misidentifies where the defect lives. The names "ACBE" and "RPM" never reached the mathematics; only the arrow families did, and they are unchanged. A variant of the proposal designates *different* arrows under the old names—but then either the new families still satisfy the reflexivity and section-retraction provisions that the registered axioms impose on those clauses (collapse again, by the same two lemmas), or the provisions have been dropped, in which case the axioms have been changed and the proposal is no longer a relabeling but a new site, to be analyzed on its own terms. In either reading, relabeling per se accomplishes nothing.

6.2. Repair 2: restrict to sub-sieves. *Proposal.* The full generated sieves are maximal only because closure drags in the identity; so declare covering, instead, a proper *subcollection* of the generated sieve—the RPM arrows "without" the identity they generate, or the ACBE family "without" its reflexive member—and expect nonmaximal covers.

PROPOSITION 5.24 (Sub-sieve restriction is unstable or vacuous). *Fix X and the RPM data of Axiom 5.6.*

(1) *Any sieve on X containing ε_X is maximal; hence there is no proper sub-*sieve *of m_X containing ε_X, and any covering sieve that retains the clause's designated arrow is maximal.*

(2) *The precomposition closure of $\{\varepsilon_X\}$ with id_X deleted,*

$$S^- \;=\; \{\varepsilon_X \circ h \mid h \text{ an arrow into } R(X)\} \setminus \{\mathrm{id}_X\},$$

is not a sieve: it contains ε_X but not $\varepsilon_X \circ \eta_X$. Declaring it covering is not admissible data for a Grothendieck topology, and applying Gen *to it restores id_X and yields m_X again.*

(3) *If instead the clause is weakened to designate a family omitting ε_X (respectively, omitting the reflexive ACBE map at every context where it appears), then at every object where the omission leaves the clause with* no designation at all*, or leaves a family whose generated sieve already lies in J_{js}, the clause contributes no covering sieve beyond J_{js}: the modification changes nothing. The remaining reading—retaining the clause but declaring the* empty family *covering at a nonempty X—is not a weakening at all: the empty sieve is then covering, the hypothesis of (J3) is vacuous over it, so every sieve on X becomes covering and every sheaf F is forced to $|F(X)| = 1$.*

Consequently every variant of the proposal either regenerates maximal sieves, trivializes the topology at X, or reduces the clause to a vacuous appendix of clause (O).

PROOF. (1) is Lemma 5.10. (2): membership of ε_X and closure under precomposition would force $\varepsilon_X \circ \eta_X = \mathrm{id}_X \in S^-$, contradicting the deletion; so S^- fails the sieve axiom at exactly the composite the repair tried to excise. A Grothendieck topology's covers are sieves by definition, so S^- cannot be declared covering as it stands; the two available completions are to close it (restoring id_X, hence maximality by Lemma 5.9) or to shrink it until it is precomposition-closed without ε_X—which is case (3). For (3): a clause that no longer designates ε_X (or the reflexive map) no longer contains the arrows that defined it; whatever family it now designates either generates sieves already covering in J_{js} (no change to the topology, by idempotency of Gen), or generates genuinely new covers—but then the clause is a *new* cover axiom about *other* arrows, whose justification, pullback-stability, and consistency with the registered axioms must be established from scratch. Within the registered clause data, no such family is specified; the registered content of clauses (A) and (R) beyond their identity and counit members is exactly the designation recorded in the source, and the source attaches the reflexivity and counit provisions to it. Deleting those provisions deletes the clause's registered content. The one remaining reading is disposed of directly: if the clause is retained but designates the empty family at a nonempty X, the empty sieve is covering, so the hypothesis of (J3) is vacuous over it and every sieve on X becomes covering, while the sheaf condition against the empty sieve forces $|F(X)| = 1$ for every sheaf F. □

REMARK 5.25 (The pretopology variant). A more sophisticated version of the proposal keeps families, not sieves, as the covering data—"use $\{\varepsilon_X\}$ as a pretopology cover and never close it." This changes nothing either, and the reason is structural: the *sheaf condition* for

a family is equivalent to the sheaf condition for the sieve it generates (matching families biject, as used in Proposition 5.15), so even a formalism that never materializes the closed sieve imposes exactly the closed sieve's gluing requirement—which, by Proposition 5.15(2), every presheaf satisfies automatically. The collapse is visible at the level of what the covers *demand*, not merely at the level of how they are stored: a cover containing a split epimorphism demands nothing, however the bookkeeping is arranged. There is no formalism-shopping exit.

6.3. Repair 3: add new objects without changing covers. *Proposal.* Embed $\mathcal{C}$ into a larger category $\mathcal{C}'$—adjoin auxiliary context objects, resolution stages, or interpolants—while keeping the three clauses as they are, expecting the richer ambient structure to make the ACBE/RPM sieves nonmaximal ("there will be more arrows, so the old sieves will no longer be everything") or otherwise to reactivate the lost distinction.

PROPOSITION 5.26 (Object adjunction preserves the collapse). *Let $\iota : \mathcal{C} \to \mathcal{C}'$ be a full embedding into any category $\mathcal{C}'$, and suppose the clause data are transported along ι unchanged: the designated ACBE families (with reflexivity) and RPM data (with $\varepsilon_X \circ \eta_X = \mathrm{id}_X$) generate covers on the images $\iota(X)$, and no new cover designations are made. Then:*

1. *the sieve generated in $\mathcal{C}'$ by the transported ACBE family at $\iota(X)$ is the maximal sieve on $\iota(X)$ in $\mathcal{C}'$;*
2. *likewise for the transported RPM family;*
3. *objects of $\mathcal{C}'$ outside the image receive no covering sieves from the clauses at all (only what (J1) and any transported (O)-covers give them);*
4. *hence the generated topology on $\mathcal{C}'$ bears the same relation to its observational part as before: the (A)- and (R)-clauses contribute only maximal sieves, and the collapse statement holds verbatim with $\mathcal{C}$ replaced by $\mathcal{C}'$.*

PROOF. (1) and (2): the two collapse lemmas are absolute, not relative. The proof that a sieve containing $\mathrm{id}_{\iota(X)}$ contains every arrow into $\iota(X)$ quantifies over the arrows of the *ambient* category and uses only precomposition closure; more arrows in $\mathcal{C}'$ means more members forced into the sieve, never fewer. Concretely: functors preserve identities and composites, so the transported reflexive member is $\mathrm{id}_{\iota(X)}$ and the transported RPM identity reads $\iota(\varepsilon_X) \circ \iota(\eta_X) = \mathrm{id}_{\iota(X)}$; Lemmas 5.9 and 5.10 then apply in $\mathcal{C}'$ exactly as in $\mathcal{C}$. The intuition behind the proposal—that enlarging the category "dilutes" a sieve—has the logic backwards: maximality of a sieve generated by an identity is stable under arbitrary enlargement of the ambient arrow supply, because the identity swallows whatever arrows the enlargement provides. (3) is immediate: a cover designation at $\iota(X)$ is data at $\iota(X)$; new objects received no designations by hypothesis. (4) restates Theorem 5.14(3) in $\mathcal{C}'$, whose proof repeats word for word. □

REMARK 5.27. The proposal has a second reading: adjoin new objects *and* give them new covers. That, of course, can produce nontrivial modalities—it is exactly what the successor does, adjoining a finite context factor whose partition covers are proper by construction. But it falls outside the repair's own terms ("without changing covers"), and honesty requires saying so plainly: once new cover generators are introduced, the legacy clauses are not being repaired; they are being retired as topology inputs, surviving only as labels in the parallel provenance layer. The refutation of Repair 3 is thus also the derivation of the successor's design from first principles: since object adjunction alone changes nothing and generator surgery within the clauses is unstable or vacuous (Propositions 5.24–5.26), any successor with genuinely distinct modalities must introduce new proper generators somewhere, and the collapse lemmas dictate the design rule they must obey.

Repair	Formalization	Mechanism of failure	Verdict
Relabel generators	Prop. 5.22	generation is label-blind	changes nothing
Restrict to sub-sieves	Prop. 5.24	closure restores id; else clause vacuous	maximal or nothing
Keep bare families	Rem. 5.25	family condition equals sieve condition	changes nothing
Add objects, same covers	Prop. 5.26	maximality is absolute under embedding	changes nothing
New objects, new covers	Rem. 5.27	—	not a repair; successor route

TABLE 1. The tempting repairs and their fates. Every route that keeps the registered generators yields maximal sieves or leaves the topology untouched; the only escape changes the generators, which is the successor construction, not a repair.

INVARIANT 5.28 (Design rule for nontrivial cover generators). A generating family intended to produce a nonmaximal covering sieve on X must contain no identity of X, no isomorphism onto X, and no split epimorphism onto X; on a poset, equivalently, every generating member must be a proper subobject of X. This invariant is extracted directly from Lemmas 5.9–5.11 and governs the successor's partition coverages, whose global cover pieces are proper subcontexts by construction and which therefore escape the collapse at exactly this point.

7. What the v7.4 corrigendum retracts, and what it preserves

The v7.4 corrigendum is the registered document through which the program absorbed Theorem 5.14. Because corrigenda are easily over-read—both by critics, as a concession of more than was conceded, and by proponents, as conceding less—this section states its exact scope in both directions.

7.1. Retracted. First, the corrigendum retracts the inference that the legacy ACBE and RPM clauses generate distinct nontrivial modalities. By Corollary 5.16 the clause-generated topologies have only maximal covers, both associated modalities are naturally isomorphic to the identity, and every statement in earlier informal material that spoke of "the ACBE lens" or "the RPM lens" as operators acting nontrivially on legacy evidence presheaves is withdrawn without residue. There is no weaker salvage—no "partially distinct," no "distinct on a subclass of presheaves": a trivial modality fixes every presheaf up to canonical isomorphism, and two trivial modalities are isomorphic to each other.

Second, and consequentially, the corrigendum retracts the inference that the legacy topology retains ACBE or RPM provenance in any form. This is Finding 5.19 applied to the registered clauses: the topology equals J_{js}, and nothing downstream of the topology can testify that clauses (A) and (R) were ever part of the site description.

Third—developed in full in Chapter 6 and only registered here—the corrigendum retracts any silent treatment of representable presheaves as sheaves for the legacy topology. Subcanonicity of $(\mathcal{C}, J_{\mathrm{legacy}})$ is not derivable from the registered axioms, so every use of a bare yX as an object of the sheaf topos is replaced by its associated sheaf.

7.2. Preserved. First, *conditional topos existence stands.* The legacy site is a genuine site: J_{legacy} is a Grothendieck topology (a collapsed one, but the collapse is an equality of topologies, not a failure of the axioms (J1)–(J3)), and for a small site the sheaf category is a Grothendieck topos by the standard background theorem (Theorem 4.26; [**1**, **7**]). The conditional statement—*if* the registered site data are well formed and small, *then* $\mathrm{Sh}(\mathcal{C}, J_{\text{legacy}})$ is a Grothendieck topos—was true before the collapse was noticed and is true after. The corrigendum takes care to permit this statement while blocking the modal inference; the two have entirely different evidentiary bases.

Second, the *observational coverage survives with its content intact.* $J_{\text{legacy}} = J_{\text{js}}$ means the legacy sheaf topos is exactly the topos of jointly-surjective descent: everything the legacy construction validly proved about compatibility of observations over jointly-surjective families remains proved, and the legacy topos remains the reference object for the successor's backward-compatibility program (whose comparison statement is registered as a conjecture with proof required, and is not advanced by this chapter).

Third, the *ACBE and RPM structures survive as structure.* The descent maps are still arrows of $\mathcal{C}$; the section-retraction identity $\varepsilon_X \circ \eta_X = \mathrm{id}_X$ still holds; the one-directionality of Remark 5.7 is untouched. What is withdrawn is one particular consumption of this structure—as cover generators—not the structure itself. Indeed the successor re-consumes the same names in two new, carefully separated roles: as formal coordinate labels on a new finite context factor (where the mathematics never depends on the labels), and as cover-kind tags in the external marked-cover records (where provenance belongs, per Remark 5.21).

RULE 5.29 (Editorial rule derived from the corrigendum). No artifact of the program may assert or imply that the legacy ACBE or RPM clauses generate distinct or nontrivial modalities, that the legacy topology records clause provenance, or that a bare legacy representable is a sheaf. Permitted statements are: the conditional topos existence for the legacy site; the equality $J_{\text{legacy}} = J_{\text{js}}$; and the collapse theorem itself with its registered proof status. Text describing the *intent* of the legacy clauses must mark it as retracted intent, as Remark 5.8 does.

8. Executable corroboration

Both collapse mechanisms are small enough to check by machine, and the registered finite certificate does so: its check `legacy_acbe_identity_forces_maximal_sieve` constructs a sieve from a generating family containing an identity and verifies elementwise that the closure is the maximal sieve, and `legacy_rpm_split_epi_forces_identity` constructs the sieve generated by a split epimorphism with an explicit section and verifies that the identity lands in the closure, hence maximality. These checks live in the same certificate that exhaustively corroborates the successor's finite theory, and they are reported as passing in the registered certificate artifact; the certificate chapter of this monograph documents the full check roster and its claim boundary. Two remarks keep the corroboration in proportion. The checks are corroboration of Theorem 5.14's mechanisms on executable instances, not a formal verification of the theorem, whose quantification is over all registered clause instances; and they are deliberately redundant—the informal proof is three lines—because the program's evidence discipline prices redundancy at approximately zero and prices unexecuted claims high. An auditor who trusts neither the prose nor the author can rerun the certificate and watch the identity arrive in the closed sieve.

9. Chapter summary

The registered legacy site admits covers from three clauses. The ACBE clause's reflexivity provision places id_X among its generators; the RPM clause's counit is split by its registered section, and closure places $\varepsilon_X \circ \eta_X = \mathrm{id}_X$ in its sieve. By the two elementary lemmas, every clause-generated sieve is maximal (Theorem 5.14(1),(2)); maximal sieves cost nothing, so $J_{\text{legacy}} = J_{\text{js}}$ (Theorem 5.14(3)); at the presheaf level, every presheaf satisfies both clauses' gluing conditions automatically, the RPM amalgamation being manufactured by the section itself (Proposition 5.15); and the intended modalities are trivial (Corollary 5.16 via Proposition 4.37). The collapse is an instance of a structural fact: topology generation is a closure operator and erases both clause labels and generator identity (Proposition 5.18, Finding 5.19). The three tempting repairs fail on schedule: relabeling is label-blind (Proposition 5.22), sub-sieve surgery is unstable or vacuous (Proposition 5.24, Remark 5.25), and object adjunction preserves maximality absolutely (Proposition 5.26); Table 1 is the ledger. The v7.4 corrigendum retracts the modal inference, the provenance inference, and silent representable-as-sheaf usage, while preserving conditional topos existence, the observational topos, and the ACBE/RPM structure as structure (Section 7); Rule 5.29 enforces the boundary in prose, and Invariant 5.28 carries the mathematical lesson forward: nontrivial covers require proper generators—no identities, no isomorphisms, no split epimorphisms onto the target.

Two questions remain before the successor construction can begin. The first is whether the legacy topology, collapsed though it is, at least treats the legacy objects themselves well—whether representables are sheaves, so that legacy objects pass into the topos unmodified. Chapter 6 answers no: subcanonicity is not derivable from the registered axioms, by an explicit small countermodel with three objects up to isomorphism, and the standard repair routes every legacy object through its associated sheaf. The second is whether proper generators supporting genuinely distinct, nontrivial, well-understood modalities can be exhibited at all within the program's conservative constraints. The remaining chapters answer yes, on a finite context factor, with the entire finite theory proved directly and corroborated exhaustively.

Exercise 5.30. (i) Show that a sieve containing a *monomorphism that is also covering-split* in the weaker sense that some h with $\varepsilon \circ h = \mathrm{id}$ exists merely in a larger ambient category $\mathcal{C}' \supseteq \mathcal{C}$ need *not* be maximal in $\mathcal{C}$, and explain exactly which step of Lemma 5.10 fails; conclude that the collapse depends on the section being an arrow of the site itself, and verify that Axiom 5.6 does place η_X in $\mathcal{C}$. (ii) Show that on a poset the converse of Invariant 5.28 also holds: a generating family every member of which is a *proper* subobject of X can never generate the maximal sieve on X, because the generated sieve is $\bigcup_i {\downarrow}U_i$ and $X \notin {\downarrow}U_i$ whenever $U_i \subsetneq X$. Then exhibit a *non*-poset example—a category carrying a split epimorphism $\varepsilon : E \to X$ that is not an isomorphism—in which a generating family of non-identity, non-isomorphism arrows nonetheless generates m_X, and identify which hypothesis of the poset argument fails.

Exercise 5.31. On the poset L_2 of Chapter 4, compute $\mathrm{Gen}(G)$ for all 2^6 subsets G of the six sieves on the top object (holding the null cover fixed), verify idempotency and monotonicity empirically, and count how many distinct topologies arise. This is the smallest nontrivial instance of the non-injectivity in Example 5.20.

CHAPTER 6

The Subcanonicity Countermodel and the Representable Repair

Chapter 5 established that the legacy topology is exactly the jointly-surjective observational topology and that the modalities intended to ride on the ACBE and RPM clauses are trivial. One question about the legacy site remains, and it is the question that determines how legacy *objects*—as opposed to legacy covers—must be handled by any successor: is the legacy topology subcanonical? If every representable presheaf yX is a sheaf, then the objects of Noetica pass into the sheaf topos unmodified, carrying their morphisms and their identities with them, and every downstream artifact may treat "the legacy object X" and "the sheaf yX" as the same thing. If subcanonicity fails, or merely fails to be provable, that identification is illegitimate, and a repair with its own discipline is required.

This chapter shows that subcanonicity is *not derivable* from the registered axioms, by exhibiting a complete, elementary countermodel with three objects up to isomorphism: the category of sets of size at most two with injections, carrying its jointly-surjective coverage. In that model the representable of a one-element set fails the sheaf condition against the two-point cover of a two-element set, by a computation short enough to display in full and instructive enough to reward the display: the matching-family set is a singleton, the global-section set is empty, and the gluing map from the empty set to the singleton has nowhere to send nothing. We then delimit the exact logical status of the result—non-derivability, not universal non-subcanonicity—and present the standard repair: the distinguished legacy object is represented in the sheaf topos by the associated sheaf $a_{\text{legacy}}(yX)$, never by silently treating yX as a sheaf [6]. Because the countermodel is fully computable, we can go further than the repair's statement and compute exactly what the repair does in the model—the answer is drastic, and knowing it is drastic is itself valuable calibration. The chapter closes with extended remarks on countermodel methodology in axiomatic auditing, since the technique used here—small finite model, exhaustive verification, sharp scope statement—is a template the program relies on repeatedly.

Claim Boundary 6.1. Status discipline. The countermodel argument originates in the registered mathematical red-team report of Project Coherence (registered under SHA-256 `03AEC81BF9F6AA12376C2DF78E6CC410EB636589DA37D2E2534460132BE720C6`) and is restated here with full proofs; its status, like that of every finite result in this monograph, is INFORMAL PROOF PRESENT—elementary, finite, and checkable, but not peer-reviewed and not formalized in a proof assistant. The extended computations of Section 6.1 (the explicit identification of the associated sheaves in the countermodel) are tutorial-level verifications presented in this monograph with informal proofs; they are not among the registered certificate checks, and nothing promotes them by their appearance here. The chapter makes no empirical, causal, safety, or product claim.

1. What subcanonicity buys

A topology J on a small category $\mathcal{C}$ is *subcanonical* when every representable presheaf is a J-sheaf (Definition 4.4; [1, 7]). The definition is one line; its consequences are the reason anyone cares, and we list them with some precision, because each one is something the legacy construction implicitly wanted and—as this chapter shows—was not entitled to assume.

First, *the site embeds in its topos.* If J is subcanonical, the Yoneda embedding factors through sheaves,

$$y \;:\; \mathcal{C} \longrightarrow \mathrm{Sh}(\mathcal{C}, J) \subseteq \mathrm{PSh}(\mathcal{C}),$$

and remains full and faithful: for all objects X, Y,

$$\mathrm{Hom}_{\mathrm{Sh}(\mathcal{C},J)}(yX,\, yY) \;=\; \mathrm{Hom}_{\mathrm{PSh}(\mathcal{C})}(yX,\, yY) \;\cong\; \mathrm{Hom}_{\mathcal{C}}(X, Y),$$

the first equality because Sh is a full subcategory, the second by the Yoneda lemma. The objects of the site appear in the topos *as themselves*: same morphisms, no new morphisms, no identifications. For a program whose objects carry registered identities, this is the property that lets "the object X" name one thing on both sides of the site/topos boundary.

Second, *evaluation is representable.* For any sheaf F, the Yoneda lemma gives $\mathrm{Hom}(yX, F) \cong F(X)$; under subcanonicity this computes sections of sheaves by maps in the topos from an object that is itself a sheaf. Diagrammatic reasoning about sections—gluing arguments, naturality arguments, limit computations—can then be conducted entirely inside $\mathrm{Sh}(\mathcal{C}, J)$.

Third, *covers become genuine colimit statements.* A covering sieve $S \in J(X)$ with yX a sheaf exhibits yX as a colimit of the representables over S in the sheaf category; covering families become effective-epimorphic families. The topology's covers acquire an exactness meaning, not merely a bookkeeping one [1, 7].

Fourth, there is a largest subcanonical topology on any small category, the *canonical topology*, whose covers are the universally effective-epimorphic sieves (Remark 4.23). Subcanonicity of J is precisely the containment of J in the canonical topology. Asserting subcanonicity is therefore asserting a comparison between the given coverage and an intrinsic invariant of the category—and comparisons require proofs. The legacy source supplied none, and the next section explains why none was forthcoming.

2. Why the legacy axioms do not derive subcanonicity

Unfold what subcanonicity demands of the jointly-surjective coverage. Let $\{f_i : U_i \to X\}$ be a jointly surjective family and Y any object. The sheaf condition for yY along (the sieve generated by) this family says: for every family of morphisms $g_i : U_i \to Y$ compatible on all overlaps, there is a unique morphism $g : X \to Y$ with $g \circ f_i = g_i$. That is: the family is *effective-epimorphic*, and since covers pull back, it must be so universally.

Joint surjectivity is a statement about *elements*: every point of X is hit. Effective epimorphy is a statement about the *morphism supply* of the category: the function one would define piecewise—send $x = f_i(u)$ to $g_i(u)$, well defined by compatibility—must actually be an arrow of $\mathcal{C}$. In the category of all finite sets and all functions, it always is: the piecewise definition is a function, functions are arrows, and jointly surjective families are universally effective-epimorphic; the jointly-surjective coverage on that category is subcanonical. The registered axioms, however, do not say that Noetica has all functions as morphisms. They say its morphisms are generated by designated admissibility-relation maps and closed under pullback—a *restricted* morphism supply, of unspecified extent. Nothing in the axioms requires the piecewise-glued function to be admissible. The gap between elementwise covering and morphism-level gluing is exactly the gap the axioms leave open, and a derivation of

subcanonicity would have to close it for every admissible instantiation. The countermodel shows it cannot be closed: there is an instantiation—finite concrete sets, a composition-closed and pullback-closed class of structure maps, the jointly-surjective coverage—in which a glued arrow fails to exist at the very first nontrivial cover.

One more framing before the construction. The failure mode has a recognizable signature, previewed in Example 4.21: *vacuous compatibility plus missing gluing.* When cover pieces have small or empty overlaps, the compatibility condition on a would-be glued family degenerates—every family is compatible—while the gluing requirement stays fully strong. Categories with restricted morphism supply are precisely where this bites, because the compatibility condition is the only brake on the demand side, and the morphism supply is the only source on the offer side. The countermodel is engineered so that the brake is fully released and the source is empty.

3. The countermodel

Definition 6.2 (The countermodel site). Fix a countable set $\mathcal{U}$ of atoms (so that the category is small) and let $\mathcal{I}_{\leq 2}$ be the category whose objects are the subsets of $\mathcal{U}$ of size at most two, and whose morphisms are the *injections* between them, with ordinary composition of functions. Declare a sieve S on an object X to be covering when the family of its members is jointly surjective: $\bigcup_{f \in S} \mathrm{im}(f) = X$. Write J_{js} for this assignment.

Everything about $\mathcal{I}_{\leq 2}$ is finite and enumerable at each object, and it is useful to have the counts explicit. For objects D, X with $|D| = d$ and $|X| = n$ (each in $\{0, 1, 2\}$), the morphism counts are:

$$|\mathrm{Hom}(D, X)| \;=\; \begin{cases} 1 & d = 0, \\ n & d = 1, \\ n(n-1) & d = 2, \end{cases}$$

so in particular $\mathrm{Hom}(D, X) = \varnothing$ whenever $d > n$: there is no injection from a two-element set to a one-element set (pigeonhole), and this single emptiness is the engine of the entire countermodel. Composites of injections are injections; identities are injections; and pullbacks exist and are computed as in **Set**: the pullback of two injections $U \to X \leftarrow V$ is (isomorphic to) the intersection of their images, which has size at most two, so $\mathcal{I}_{\leq 2}$ is closed under the pullbacks the site machinery uses. Up to isomorphism the category has three objects—$\varnothing$, a singleton, a doubleton—and readers who prefer a skeleton may work with $0 = \varnothing$, $1 = \{1\}$, $2 = \{1, 2\}$ throughout; we keep named sets $\{a\}$, $\{b\}$, $\{a, b\}$, $\{*\}$ for readability.

Lemma 6.3 (Covering sieves contain all point inclusions). *Let S be a sieve on X in $\mathcal{I}_{\leq 2}$. Then S is jointly surjective if and only if for every $x \in X$ the point inclusion $p_x : \{x\} \hookrightarrow X$ belongs to S. (For $X = \varnothing$ every sieve, including the empty sieve, is jointly surjective.)*

Proof. ($\Leftarrow$) The images of the point inclusions are the singletons $\{x\}$, whose union is X. ($\Rightarrow$) Let $x \in X$ and choose $f : E \to X$ in S with $x \in \mathrm{im}(f)$, say $f(e) = x$. Let $k : \{x\} \to E$ be the injection sending x to e. Then $f \circ k : \{x\} \to X$ is a member of S by precomposition closure, and it is an injection from $\{x\}$ sending x to x—but any two injections $\{x\} \to X$ with the same image point are equal as functions, and $f \circ k$ has image $\{x\}$, so $f \circ k = p_x$. Hence $p_x \in S$. □

Proposition 6.4 (J_{js} is a Grothendieck topology). *The assignment J_{js} of Definition 6.2 satisfies (J1), (J2), and (J3); hence $(\mathcal{I}_{\leq 2}, J_{\mathrm{js}})$ is a site.*

PROOF. (J1) The maximal sieve on X contains id_X, whose image is X.

(J2) Let S be covering on X and $g : D \to X$ any arrow; we must show $g^*S = \{h \mid g \circ h \in S\}$ is covering on D. Let $d \in D$. Then $g(d) \in X$, so by Lemma 6.3 the point inclusion $p_{g(d)} : \{g(d)\} \to X$ lies in S. Consider the point inclusion $p_d : \{d\} \to D$; the composite $g \circ p_d : \{d\} \to X$ is an injection with image $\{g(d)\}$. As in the proof of Lemma 6.3, $g \circ p_d$ equals $p_{g(d)}$ precomposed with the bijection $\{d\} \to \{g(d)\}$, hence lies in S by closure. So $p_d \in g^*S$, for every $d \in D$; by Lemma 6.3, g^*S is covering. (If $D = \varnothing$ there is nothing to check.)

(J3) Let S be covering on X and let T be a sieve on X such that f^*T is covering on D for every $f : D \to X$ in S. Let $x \in X$; we find a member of T hitting x. By Lemma 6.3, $p_x \in S$; by hypothesis p_x^*T is covering on $\{x\}$, so it contains the point inclusion $\mathrm{id}_{\{x\}}$ (Lemma 6.3 again, at the object $\{x\}$). Membership $\mathrm{id}_{\{x\}} \in p_x^*T$ means $p_x \circ \mathrm{id} = p_x \in T$, and $\mathrm{im}(p_x) \ni x$. Hence T is jointly surjective, i.e. covering. □

REMARK 6.5 (The countermodel instantiates the registered axioms). To serve as a countermodel to *derivability*, the site must satisfy what the registered axioms demand, at the level at which a derivation would have to operate (Boundary 5.3). The checklist: the objects are finite concrete sets; the category is small; the morphisms form a designated class of structure-preserving maps (injections) containing all identities, closed under composition and under the pullbacks used by the site machinery; the observational clause (O) is instantiated by the jointly-surjective coverage just verified. Clauses (A) and (R) are instantiated degenerately but legitimately: take the ACBE designation to be the reflexive one (the identity at each context), and take the RPM data to be $R(X) = X$ with $\eta_X = \varepsilon_X = \mathrm{id}_X$, a valid section-retraction pair. By Theorem 5.14 these clauses contribute only maximal sieves in *every* model, so the total legacy topology of this model is exactly the J_{js} of Proposition 6.4. Where the registry's fuller instantiation data (the forty-index object class, the specific admissibility generators) are not reproduced here, that is because a derivation of subcanonicity from the axioms could not lean on them either—no registered axiom ties the index structure to the existence of glued morphisms, and a proof from the axioms must survive every instantiation the axioms permit. That is precisely what the next section refutes.

4. The failed gluing, in full

We now carry out the central computation with nothing suppressed. Fix two atoms $a \neq b$ and a third atom $*$, giving objects $\{a\}$, $\{b\}$, $\{a, b\}$, $\{*\}$ of $\mathcal{I}_{\leq 2}$.

LEMMA 6.6 (The cover and its sieve). *Let S_2 be the sieve on $\{a, b\}$ generated by the two point inclusions $p_a : \{a\} \hookrightarrow \{a, b\}$ and $p_b : \{b\} \hookrightarrow \{a, b\}$. Then:*

1. *S_2 is covering for J_{js};*
2. *S_2 consists exactly of the arrows into $\{a, b\}$ whose domain has size at most one—that is, for every singleton domain $\{d\} \subseteq \mathcal{U}$ the two injections $\{d\} \to \{a, b\}$ (one with image $\{a\}$, one with image $\{b\}$), together with the unique arrow from $\varnothing$. In particular S_2 is infinite, though only four of its members have domain $\{a\}$ or $\{b\}$;*
3. *S_2 is not maximal: neither $\mathrm{id}_{\{a,b\}}$ nor the swap automorphism belongs to it;*
4. *S_2 is the* minimum *covering sieve on $\{a, b\}$: it is contained in every covering sieve.*

PROOF. (1) The images $\{a\}$ and $\{b\}$ of the generators union to $\{a, b\}$.

(2) Membership in the generated sieve means factoring through p_a or p_b. Any arrow $f : D \to \{a, b\}$ with $|D| \leq 1$ has image contained in $\{a\}$ or in $\{b\}$ (its image has at most one point), hence factors through the corresponding point inclusion: for $D = \varnothing$ via the empty

injection into either singleton, and for $D = \{d\}$ via the bijection $\{d\} \to \{f(d)\}$ followed by $p_{f(d)}$. Conversely any arrow factoring through a singleton has domain of size at most one, injections being injective. So the members are exactly the arrows from objects of size at most one.

(3) The two automorphisms of $\{a, b\}$ have two-element domain, hence are not members by (2). (Equivalently: were either present, Lemma 5.11 would make S_2 maximal, and (2) shows it is not.)

(4) By Lemma 6.3 every covering sieve on $\{a, b\}$ contains p_a and p_b, hence contains the sieve they generate. □

THEOREM 6.7 (The representable $y\{*\}$ is not a sheaf). *Consider the representable presheaf* $y\{*\} = \mathrm{Hom}_{\mathcal{I}_{\leq 2}}(-, \{*\})$ *and the covering sieve* S_2 *on* $\{a, b\}$. *Then:*

1. *the matching-family set* $\mathrm{Match}(S_2,\, y\{*\})$ *has exactly one element;*
2. *the section set* $y\{*\}(\{a, b\})$ *is empty;*
3. *hence the gluing map* $y\{*\}(\{a, b\}) \to \mathrm{Match}(S_2,\, y\{*\})$ *is injective but not surjective:* $y\{*\}$ *is separated for* S_2 *but has a matching family with no amalgamation, and is therefore not a* J_{js}*-sheaf.*

Consequently $(\mathcal{I}_{\leq 2}, J_{\mathrm{js}})$ *is not subcanonical.*

PROOF. (1) A matching family assigns to each $f : D \to \{a, b\}$ in S_2 an element $x_f \in y\{*\}(D) = \mathrm{Hom}(D, \{*\})$, naturally. By Lemma 6.6(2), every member has $|D| \leq 1$, and for such D the set $\mathrm{Hom}(D, \{*\})$ is a *singleton*: the empty injection when $D = \varnothing$, the unique bijection $D \to \{*\}$ when $|D| = 1$. So there is exactly one candidate assignment—x_f is the unique element of its singleton—and no freedom anywhere. It remains to observe that the candidate is in fact matching: for any $h : E \to D$, both $x_{f \circ h}$ and $x_f|_h = x_f \circ h$ are elements of $\mathrm{Hom}(E, \{*\})$, which is again a singleton (as $|E| \leq |D| \leq 1$), so they are equal. Hence $\mathrm{Match}(S_2, y\{*\})$ is a singleton: the compatibility conditions are all vacuously satisfied by forced data. This is the "vacuous compatibility" half of the signature announced in Section 2: the pieces $\{a\}$ and $\{b\}$ overlap only over $\varnothing$, where $y\{*\}$ holds no information, so the cover imposes no constraint that could rule the family out.

(2) $y\{*\}(\{a, b\}) = \mathrm{Hom}(\{a, b\}, \{*\})$ is the set of injections from a two-element set to a one-element set, which is empty by pigeonhole. This is the "missing gluing" half: the would-be amalgamation—the function sending both a and b to $*$—exists perfectly well as a function, but it is not injective, so it is not an arrow of $\mathcal{I}_{\leq 2}$. The category's morphism supply stops exactly one arrow short of what the cover demands.

(3) The gluing map is a map $\varnothing \to \{\mathrm{pt}\}$; it is vacuously injective and not surjective. A sheaf requires bijectivity for every covering sieve; S_2 is covering by Lemma 6.6(1); so $y\{*\}$ is not a sheaf, and subcanonicity fails for the site. □

COROLLARY 6.8 (Non-derivability). *Subcanonicity of the legacy site is not derivable from the registered axioms: there exists a model of the axioms (Remark 6.5) whose jointly-surjective coverage is not subcanonical (Theorem 6.7). Hence every artifact that treated a bare legacy representable as a sheaf was relying on an unproved assumption, and the v7.4 corrigendum's retraction of that usage (Chapter 5, Section 5.7) is mandatory, not precautionary.*

REMARK 6.9 (How little was needed). Tally the ingredients, because the economy of the countermodel is itself methodologically instructive (Section 7). The argument used: one cover with two pieces whose overlap carries no data; one target object small enough that pigeonhole empties the crucial hom-set; and the injectivity constraint on morphisms, which is what prevents the set-level gluing from being an arrow. No infinite structure, no exotic

category, no interaction with the ACBE/RPM clauses (which contribute only maximal sieves and hence cannot rescue gluing: a maximal sieve's gluing condition is automatic and so adds no constraint that would exclude the offending matching family—nor any new way to satisfy it). Three objects up to isomorphism suffice. When an axiom system's intended theorem fails, it usually fails small, and the audit practice of searching small models first (Protocol 6.17) is calibrated to exactly this experience.

5. The exact logical status

Precision about what has and has not been proved is the difference between an audit finding and a rhetorical flourish. Three statements must be held apart.

First: what is proved. The registered axioms do not entail subcanonicity. This is a non-derivability result, established by the standard model-theoretic route: a structure satisfying the axioms in which the candidate theorem is false. Any purported proof of subcanonicity from the registered axioms alone is unsound—it would prove a false statement about $(\mathcal{I}_{\leq 2}, J_{\text{js}})$.

Second: what is not proved. Universal non-subcanonicity is not established. Nothing here shows that the intended, fully instantiated Noetica—with its registered forty-index object class and its actual admissibility generators—fails subcanonicity. That question is simply open, and it is a question about a particular site, answerable only by working with that site's registered data. The countermodel neither answers it nor prejudges it.

Third: the independence observation. The same axiom schema also admits models in which subcanonicity *holds*, so the statement is independent of the schema: neither it nor its negation is derivable. A witness is already in hand from Chapter 4: take the site to be a finite powerset poset with inclusion maps—inclusions are injections between finite sets, identities are present, composition and pullback (intersection) closure hold—with the jointly-surjective coverage, which for inclusions is exactly the union coverage. There, every representable is a sheaf: if every piece of a covering family lands in W, their union, which is the covered object, lands in W; this is Example 4.9 generalized, and it is the same argument that proves the successor's context-factor topologies subcanonical later in the monograph. The two models bracket the axioms from both sides, and the bracket is the correct summary: *the registered axioms underdetermine subcanonicity.* Deriving it for an exact revised site is a registered open obligation of the program, with a proof required; until that obligation closes, the repair of the next section is the only licensed handling of legacy representables.

Adverse Finding 6.10. Subcanonicity is independent of the registered legacy axiom schema: $(\mathcal{I}_{\leq 2}, J_{\text{js}})$ satisfies the axioms and refutes it; finite powerset posets with union coverage satisfy the axioms and verify it. All prior text asserting or silently using legacy subcanonicity is retracted to the conditional form: *if* a revised exact site is proved subcanonical, *then* representables may be used as sheaves there; otherwise the associated-sheaf repair applies.

6. The standard repair: $a_{\text{legacy}}(yX)$

The repair is classical and requires no invention; what it requires is discipline in use. Let a_{legacy} denote sheafification for the legacy site, and let X be a distinguished legacy object. The representable yX is a presheaf in any case; its associated sheaf

$$a_{\text{legacy}}(yX) \;\in\; \text{Sh}(\mathcal{C},\, J_{\text{legacy}}), \qquad \eta_{yX} \;:\; yX \longrightarrow a_{\text{legacy}}(yX),$$

is the canonical—and, absent a subcanonicity proof, the only licensed—avatar of X in the legacy sheaf topos [6]. Three properties govern its use.

PROPOSITION 6.11 (What the repair preserves). *For every sheaf $G \in \mathrm{Sh}(\mathcal{C}, J_{\text{legacy}})$, composition with the unit induces a bijection*

$$\mathrm{Hom}_{\mathrm{Sh}}\big(a_{\text{legacy}}(yX),\, G\big) \;\cong\; \mathrm{Hom}_{\mathrm{PSh}}(yX,\, G) \;\cong\; G(X),$$

naturally in G. That is: mapping out of *the repaired object into sheaves computes exactly evaluation at X, just as mapping out of yX does.*

PROOF. The first bijection is the universal property of sheafification (Definition 4.24); the second is the Yoneda lemma. Both are natural in G. □

So a pipeline that uses legacy objects only as *probes*—evaluating sheaves at X, forming matching data over covers of X, indexing assessments by X—loses nothing under the repair: the repaired object represents evaluation at X on the entire sheaf category. What the repair does *not* preserve is the internal structure of the object itself: its sections over other objects, its automorphisms in the topos, its distinctness from other repaired objects. The countermodel makes this concrete, and we do the computation in full because it demonstrates that the repair can be as far from innocuous as it is from optional.

6.1. The repair computed in the countermodel.

PROPOSITION 6.12 (Associated sheaves of representables in $\mathcal{I}_{\leq 2}$). *In $(\mathcal{I}_{\leq 2}, J_{\text{js}})$, write $\mathbf{Set}(-, X)$ for the presheaf $D \mapsto \mathbf{Set}(D, X)$ of* all *functions into the underlying set of X, with restriction by precomposition. Then for every object X:*

1. *$\mathbf{Set}(-, X)$ is a J_{js}-sheaf;*
2. *the evident inclusion $yX \hookrightarrow \mathbf{Set}(-, X)$ (injections are functions) exhibits $\mathbf{Set}(-, X)$ as the associated sheaf of yX, i.e. $a(yX) \cong \mathbf{Set}(-, X)$ with the inclusion as unit;*
3. *in particular $a(y\{*\}) \cong \mathbf{1}$, the terminal sheaf: the repaired singleton representable has exactly one section over every object.*

PROOF. (1) We check the sheaf condition at every object directly, using the minimum covering sieves supplied by Lemma 6.3 (every covering sieve on D contains the sieve generated by the point inclusions of D) and treating the larger covering sieves explicitly in each case. At $D = \varnothing$: the empty sieve covers, and $\mathbf{Set}(\varnothing, X)$ is a singleton (the empty function), as required (Lemma 4.7). At $|D| = 1$: by Lemma 6.3 a covering sieve contains id_D (the point inclusion of the unique point is the identity up to the bijection of singletons; precomposition closure then yields id_D itself), hence is maximal, and maximal sieves glue freely (Example 4.5). At $|D| = 2$, say $D = \{x, y\}$: a matching family for the minimum sieve $S_{\min}(D)$ assigns to each singleton-domain member an element of a $\mathbf{Set}(E, X)$ with $|E| = 1$, naturally; unwinding as in Theorem 6.7(1), the family is freely determined by its values at the two point inclusions p_x, p_y, i.e. by a pair of functions $\{x\} \to X$, $\{y\} \to X$, i.e. by an arbitrary function $D \to X$; and the gluing map sends $g \in \mathbf{Set}(D, X)$ to exactly that pair. It is therefore a bijection. Larger covering sieves on D contain additionally arrows with doubleton domain (isomorphisms onto D or injections into D from other doubletons—the latter do not exist for $|D| = 2$ except isomorphisms); a matching family's values there are forced by naturality from the point values, and the same bijection results.

(2) By Proposition 4.25 it suffices to show the inclusion $\iota : yX \to \mathbf{Set}(-, X)$ has the universal property: every map $\alpha : yX \to G$ with G a sheaf factors uniquely through ι. Both existence and uniqueness follow from the plus-construction description, but the direct route is cleaner here: the presheaf $\mathbf{Set}(-, X)$ is generated over yX by gluing, in the precise sense that every section $g \in \mathbf{Set}(D, X)$ restricts along the minimum covering sieve of D to sections

lying in the image of ι (over domains of size at most one, every function is injective or empty, hence *is* an injection—this is the pivotal smallness fact). Given $\alpha : yX \to G$, define $\bar{\alpha}_D(g)$ as the amalgamation in $G(D)$ of the matching family $\big(\alpha_E(g \circ h)\big)_{(h:E\to D)\in S_{\min}(D)}$—well defined since G is a sheaf and the family matches by naturality of α; uniqueness of amalgamations makes $\bar{\alpha}$ natural and forces it on all of $\mathbf{Set}(-, X)$, and $\bar{\alpha} \circ \iota = \alpha$ because for injective g the family already has the amalgamation $\alpha_D(g)$. Any other factorization agrees with $\bar{\alpha}$ on restrictions to the covering sieve, hence everywhere, by separatedness of G.

(3) $\mathbf{Set}(D, \{*\})$ is a singleton for every D. □

Remark 6.13 (Reading the computation). Proposition 6.12 says that in the counter-model, sheafification of representables *manufactures exactly the missing non-injective maps*: the topology demanded function-like gluing, the category supplied only injections, and the associated sheaf completes the hom-presheaf $\mathrm{Inj}(-, X)$ to the full function presheaf $\mathbf{Set}(-, X)$. Three consequences follow, each a live risk for any successor that handles legacy objects casually. First, *isomorphism classes are not reconstructible from the site's identities*: the two singleton objects $\{a\}$ and $\{b\}$ are distinct registry objects but already isomorphic in the site, so their repaired avatars are the same object of the topos, $a(y\{a\}) \cong a(y\{b\}) \cong \mathbf{1}$ (Proposition 6.12(3)). Sheafification cannot restore a distinction the site itself does not draw; in this model $X \mapsto a(yX)$ happens to reflect isomorphism (evaluating $\mathbf{Set}(-, X)$ at a singleton recovers $|X|$), but nothing in general guarantees that, which is why Invariant 6.15 keeps object identity at the registry rather than in the topos. Second, *morphisms can appear*: computing with Proposition 6.11, $\mathrm{Hom}_{\mathrm{Sh}}(a(yX), a(yY)) \cong a(yY)(X) = \mathbf{Set}(X, Y)$, which strictly contains $\mathrm{Hom}_{\mathcal{I}_{\leq 2}}(X, Y)$ whenever a non-injective function $X \to Y$ exists. The repaired image of the site sees maps the site does not have; the composite $X \mapsto a(yX)$ is faithful here but not full. Third, *units need not be surjective or bijective*: the unit $yX \to \mathbf{Set}(-, X)$ is a proper inclusion at every doubleton evaluated at doubletons. None of this is pathology; it is what sheafification is *for*—enforcing descent that the site's raw hom-structure does not satisfy. But it means that under a non-subcanonical (or not-known-subcanonical) topology, "the object X" and "the sheaf avatar of X" are different mathematical citizens, and no identification between them may be made silently.

6.2. Handling discipline for legacy objects.

Rule 6.14 (Representable repair rule). Until an exact revised legacy site is proved subcanonical, every successor artifact that places a distinguished legacy object X inside the legacy sheaf topos must use $a_{\text{legacy}}(yX)$ and must record that it has done so. Prohibited: silently treating yX as a sheaf; transferring site-level identities of objects to topos-level isomorphism classes (Remark 6.13 shows repaired objects may become isomorphic); inferring site-level morphism counts from topos-level hom-sets (the repair may add morphisms). Permitted without further proof: using $a_{\text{legacy}}(yX)$ as the representing object for evaluation at X on sheaves (Proposition 6.11); forming its covers, subobjects, and diagrams inside the sheaf topos.

Invariant 6.15 (Identity lives at the registry, not in the topos). The identity of a legacy object—its registered identifier and canonical payload hash—is site-level and serialization-level data. It must travel with the marked records of the successor architecture, never be reconstructed from topos-level isomorphism class, and never be presumed stable under sheafification. A categorical isomorphism is not an identity receipt. (This invariant restates, for objects, the division of labor that Chapter 5 established for covers: the topos computes semantics; the registry carries identity and provenance.)

REMARK 6.16 (If subcanonicity is later proved). The repair is self-liquidating. If a future exact revised site is proved subcanonical, then for that site every yX is already a sheaf, the unit $\eta_{yX} : yX \to a(yX)$ is an isomorphism (Proposition 4.35), and every use of $a_{\text{legacy}}(yX)$ made under Rule 6.14 becomes canonically isomorphic to the bare representable—no migration, no debt, no stranded artifacts. This is the reason the rule mandates the repaired form rather than waiting: the repaired form is correct in both futures, the bare form in only one. Successors should prefer constructions that remain valid when open obligations close in either direction; the repair is the type specimen of that design principle.

7. Countermodel methodology in axiomatic auditing

The countermodel of this chapter is a two-page argument, but it carried real weight in the program: it converted an implicit assumption into a named open obligation, forced a corrigendum, and imposed a handling rule on every downstream artifact. This closing section abstracts the technique, because the program's audit posture treats countermodel construction as a first-class tool, on par with proof and executable check, and the abstract lessons transfer beyond this instance.

Countermodels refute derivability, which is the auditable claim. An assertion inside a formal program rarely fails by being false in the intended model, which is usually rich and forgiving; it fails by being underivable from what was actually written down—and underivability is what a countermodel certifies. The audit question is never "do we believe subcanonicity?" but "does the registered text entail it?". Model both answers: exhibit a structure satisfying exactly the registered text where the claim fails (refuting entailment), and where possible a second structure where it holds (blocking the overcorrection to "the axioms refute it"). The bracket of Finding 6.10 is the complete deliverable; either half alone invites misuse.

Search small, and search where the demands are strongest and the supplies weakest. The heuristic that found this countermodel generalizes: identify what the candidate theorem demands (here: glued morphisms over covers) and what the axioms supply (here: an unspecified morphism class), then construct the leanest structure that honors every axiom while starving the demand. Finiteness is not a concession but a weapon—finite models are exhaustively checkable, communicable in full, and immune to the suspicion that the failure hides an error in an infinite limit. The size cap of two in $\mathcal{I}_{\leq 2}$ is exactly the point at which pigeonhole first bites; no smaller cap yields a countermodel, which is itself worth verifying as an exercise (Exercise 6.19).

A countermodel is a permanent regression test. Because the model is concrete and finite up to isomorphism, its failing computation can be executed, not merely believed: enumerate the sieve in the three-object skeleton $0 = \varnothing$, 1, 2 (Lemma 6.6), count the matching families (one), count the sections (zero), compare. The program's practice of registering such computations alongside certificates—as it does for the collapse mechanisms of Chapter 5—turns each audit finding into a machine-checkable invariant that future revisions must continue to pass or consciously retire. An axiom change that silently restores the broken derivation will trip the executable form of the countermodel before it reaches prose.

Countermodels bound repairs. The countermodel does not merely destroy; it specifies. Any future subcanonicity proof for a revised site must visibly use some feature that $(\mathcal{I}_{\leq 2}, J_{\text{js}})$ lacks—a richer morphism supply, a restricted cover class, or additional axioms—and referees of such a proof should demand to see where the countermodel is excluded. Conversely the repair adopted in the meantime (Rule 6.14) is exactly the construction the countermodel

shows to be necessary and the universal property shows to be sufficient for the mapping-out uses the program actually has. This demand–supply–repair triangle is the shape of a completed audit item.

Protocol 6.17 (Countermodel audit protocol). For a claim Σ asserted or assumed within an axiomatized artifact: (1) isolate the registered axioms A actually written, distinguishing them from intended-model folklore; (2) restate Σ as a demand on structures satisfying A; (3) attempt derivation sketches to locate the load-bearing gap; (4) search finite structures satisfying A that starve the gap, smallest first; (5) on success, verify A and refute Σ in the model with complete proofs, and where feasible register an executable form; (6) determine the complementary side—does some model satisfy Σ?—to classify Σ as refuted, derivable, or independent; (7) issue the scope statement (non-derivability versus universal failure), the corrigendum for affected text, and the interim handling rule; (8) record the obligations that would license removing the rule. Steps (5)–(7) are the deliverable; a countermodel without a scope statement and a handling rule is an incident report without a fix.

Claim Boundary 6.18. Applying Protocol 6.17 to the present chapter: Σ is legacy subcanonicity; the model refuting it is $(\mathcal{I}_{\leq 2}, J_{\mathrm{js}})$ with the degenerate but licensed clause instantiation of Remark 6.5; the model satisfying it is the powerset-poset instantiation of Section 5; the classification is *independent*; the corrigendum item is registered in v7.4; the handling rule is Rule 6.14; and the discharging obligation—a subcanonicity proof for an exact revised site—remains OPEN in the registered obligation ledger. No part of this chapter licenses any statement beyond that classification.

Exercise 6.19. Show that the size cap two is minimal: in the category of sets of size at most one with injections, every jointly-surjective sieve on every object is maximal (use Lemma 6.3 and Lemma 5.11), hence the coverage is subcanonical there. Conclude that every countermodel to subcanonicity built from injections requires an object with at least two elements, and identify exactly which step of Theorem 6.7 uses it.

Exercise 6.20. Determine all sheaves on $(\mathcal{I}_{\leq 2}, J_{\mathrm{js}})$ up to isomorphism in terms of their values at the skeleton objects $0, 1, 2$ and the induced structure maps: show a sheaf is determined by a set $F_1 = F(1)$ together with an action of the two-element automorphism group of 2 on $F(2) \cong F_1 \times F_1$ compatible with the projections, and verify your description against Proposition 6.12.

Exercise 6.21. Verify that $y\varnothing$ *is* a sheaf on $(\mathcal{I}_{\leq 2}, J_{\mathrm{js}})$: its value is a singleton at $\varnothing$ and empty elsewhere; check the null cover, the singleton covers, and the doubleton covers separately, and reconcile the doubleton case with Theorem 6.7 (why does the empty matching-family set rescue $y\varnothing$ but not $y\{*\}$?).

Exercise 6.22. Using Proposition 6.11, show that in any site the assignment $X \mapsto a(yX)$ extends to a functor $\mathcal{C} \to \mathrm{Sh}(\mathcal{C}, J)$, and that it is full and faithful if and only if J is subcanonical. Then locate the failure of fullness in the countermodel explicitly: exhibit the map $a(y\{a,b\}) \to a(y\{a,b\})$ corresponding under Remark 6.13 to the constant function and show it is not in the image of $\mathrm{Hom}_{\mathcal{I}_{\leq 2}}(\{a,b\}, \{a,b\})$.

CHAPTER 7

The Context Lattice: Complete Combinatorics

The two preceding chapters were destructive: Chapter 5 proved that the legacy ACBE and RPM cover clauses generate only maximal sieves, and Chapter 6 exhibited a model in which the surviving jointly-surjective topology fails subcanonicity. The repair announced in Chapter 1 begins here. The successor candidate does not attempt to re-engineer covers on the legacy category $\mathcal{C}$ at all. Instead it introduces a second, separate, finite index of *contexts*, arranges the ACBE and RPM marks as two transverse coordinate partitions of a four-element atom set, and generates coverage from proper partition pieces rather than from identity-containing clauses. Everything the later theorems use about that context factor—its objects, its arrows, its pullbacks, its two partitions, its generating families, their pullbacks and refinements, and its block-containment facts—is finite and exactly derivable from the definitions. This chapter derives all of it, exhaustively, before a single topology is generated.

The division of labor with the following chapters is strict. This chapter is pure combinatorics: it proves no theorem about Grothendieck topologies, sheaves, or modalities, and it consumes no ledger record. Chapter 8 builds the provenance calculus on top of the same finite data. Chapter 9 then states and proves the finite-context theorems NT2-1 through NT2-4, each with its registered ledger status. The reason for this ordering is methodological: every case analysis in Chapter 9 terminates in a finite check against a table in this chapter, so the tables must exist first, and must be complete.

Claim Boundary 7.1. Throughout this chapter the words "ACBE" and "RPM" are formal coordinate labels preserved for controlled migration vocabulary, exactly as registered in the governing specification. The first coordinate of an atom is the formal ACBE mark and the second is the formal RPM mark. These labels carry no empirical interpretation until a separate operationalization and validation record exists; no statement in this chapter, and no statement derived from this chapter later in the monograph, attaches meaning to them beyond their combinatorics. The entire monograph remains DRAFT / GATE 1 RESEARCH CANDIDATE / RING 4 / NOT LICENSED.

1. The atom set and the lattice

Definition 7.2 (Atom set). The *atom set* is the four-element product

$$\Omega \;=\; \{0,1\} \times \{0,1\} \;=\; \{(0,0),\,(0,1),\,(1,0),\,(1,1)\}.$$

We write $\omega_{ij} = (i,j)$ for the atom whose ACBE coordinate is i and whose RPM coordinate is j, so that

$$\Omega = \{\omega_{00}, \omega_{01}, \omega_{10}, \omega_{11}\}.$$

Definition 7.3 (Context lattice). The *context lattice* is the powerset

$$L \;=\; \mathcal{P}(\Omega),$$

regarded as a poset ordered by inclusion, and hence as a category: for $U, V \in L$ there is an arrow $U \to V$ precisely when $U \subseteq V$, and at most one such arrow. Elements of L are called *contexts*. The bottom element is the empty context $\varnothing$; the top element is the global context Ω.

Three elementary observations should be made explicit at the outset, because they are used silently everywhere afterwards.

First, L is a category in the usual way a poset is: the identity arrow at V is the inclusion $V \subseteq V$, and composition of $U \subseteq V$ with $V \subseteq W$ is $U \subseteq W$, which is again an arrow of L by transitivity of inclusion. Associativity and the unit laws are automatic because each hom-set has at most one element; a poset category can never fail them. This triviality is a feature: every diagram in L commutes, so the only content a diagram in L can carry is *which* objects and arrows it involves, never how they compose. All the combinatorial weight of this chapter therefore lies in counting and classifying, not in diagram chasing.

Second, L is a finite Boolean lattice. Meets are intersections, joins are unions, and every context V has a complement $V^c = \Omega \setminus V$ with $V \cap V^c = \varnothing$ and $V \cup V^c = \Omega$. Complementation is an order-reversing involution of L; we will use it in Section 2 to pair off the sixteen contexts.

Third, and most importantly for the topology to come, L has pullbacks, and they are intersections.

Proposition 7.4 (Pullbacks are intersections). *Let $U \to V$ and $W \to V$ be arrows of L, that is, $U \subseteq V$ and $W \subseteq V$. Then the pullback of $U \to V$ along $W \to V$ exists in L and is $U \cap W$, with the two inclusion arrows $U \cap W \to U$ and $U \cap W \to W$.*

Proof. The square with vertices $U \cap W$, U, W, V commutes because every diagram in a poset commutes. It remains to check the universal property. Let $Z \in L$ carry arrows $Z \to U$ and $Z \to W$, that is, $Z \subseteq U$ and $Z \subseteq W$. Then $Z \subseteq U \cap W$ by the definition of intersection, which is exactly an arrow $Z \to U \cap W$; it is compatible with the projections because compatibility is automatic, and it is unique because L has at most one arrow between any two objects. Hence $U \cap W$ satisfies the universal property of the pullback; this is the powerset instance of Proposition 2.63 (Corollary 2.64). □

Remark 7.5. More is true: L is a complete lattice, so as a category it has all small limits and colimits, computed as intersections and unions respectively. We isolate Proposition 7.4 because pullback stability of covering families—axiom (J2) in the standard presentation of Grothendieck topologies [**3**, **7**]—is the axiom whose verification in Chapter 9 reduces entirely to a computation with intersections. The single most frequently used identity in this monograph is the associativity-with-absorption computation

$$(V \cap B) \cap U \;=\; U \cap (V \cap B) \;=\; (U \cap V) \cap B \;=\; U \cap B \qquad \text{whenever } U \subseteq V,$$

which says: intersecting a piece of V down to U is the same as taking the corresponding piece of U directly. Section 7 verifies this identity case by case across all context classes; the one-line algebraic proof is the display above, since $U \subseteq V$ gives $U \cap V = U$.

Notation 7.6. For $U \subseteq V$ we write the corresponding arrow of L either as $U \to V$ or, when the ambient target matters, as $U \hookrightarrow V$. Since hom-sets are subsingletons, we freely identify an arrow with the ordered pair (U, V) of its domain and codomain. A *family at V* is a finite indexed set of arrows with common codomain V.

Proposition 7.7 (Boolean structure). *$(L, \cap, \cup, (-)^c, \varnothing, \Omega)$ is a Boolean algebra. In particular L is a Heyting algebra with implication*

$$U \Rightarrow W \;=\; U^c \cup W,$$

characterized by the adjunction $Z \cap U \subseteq W \iff Z \subseteq U \Rightarrow W$*, and negation* $\neg U = U \Rightarrow \varnothing = U^c$ *satisfies* $\neg\neg U = U$*.*

PROOF. The Boolean-algebra axioms for a powerset are the standard set identities (distributivity of intersection over union and conversely, De Morgan, complementation). For the adjunction: if $Z \cap U \subseteq W$, then any $z \in Z$ is either outside U (so $z \in U^c$) or inside U, in which case $z \in Z \cap U \subseteq W$; either way $z \in U^c \cup W$. Conversely if $Z \subseteq U^c \cup W$ then $Z \cap U \subseteq (U^c \cup W) \cap U = (U^c \cap U) \cup (W \cap U) \subseteq W$. Taking $W = \varnothing$ gives $\neg U = U^c$, and $(U^c)^c = U$ is involutivity of complement. □

REMARK 7.8 (Why record the Heyting structure). Categorically, Proposition 7.7 says the poset category L is cartesian closed with exponentials $W^U = U \Rightarrow W$, and that its internal propositional logic is classical. This needs one paragraph of care, because the sheaf categories built on L in Chapters 9 and 10 have their *own* internal logic, which is in general intuitionistic and need not inherit Booleanness from the base; no artifact of this program asserts anything about the internal logic of the successor's sheaf categories, and this monograph derives none. The proposition is used below only in its combinatorial clothing: complements exist and behave (Remark 7.11), and implication gives a compact notation for containment facts, e.g., the block-completeness predicate $A_i \subseteq V$ of Section 9 is the statement $\Omega = A_i \Rightarrow V$.

2. The sixteen contexts

Since $|\Omega| = 4$, the lattice L has $2^4 = 16$ objects. The binomial distribution of their sizes is

$$\binom{4}{0} + \binom{4}{1} + \binom{4}{2} + \binom{4}{3} + \binom{4}{4} \;=\; 1 + 4 + 6 + 4 + 1 \;=\; 16.$$

The five cardinality classes have unequal standing in the later theory, and the six doubletons split further into three structurally distinct kinds once the partitions of Section 4 are in place. We fix names now.

DEFINITION 7.9 (Context classes and standard names). The sixteen contexts are classified and named as follows.

(1) **The empty context** $\varnothing$ (one context).
(2) **Singletons** $\{\omega_{ij}\}$ for $i, j \in \{0, 1\}$ (four contexts).
(3) **Doubletons** (six contexts), subdivided into:
   (a) the two *ACBE blocks* $A_0 = \{\omega_{00}, \omega_{01}\}$ and $A_1 = \{\omega_{10}, \omega_{11}\}$, the fibers of the first coordinate;
   (b) the two *RPM blocks* $R_0 = \{\omega_{00}, \omega_{10}\}$ and $R_1 = \{\omega_{01}, \omega_{11}\}$, the fibers of the second coordinate;
   (c) the two *diagonal doubletons* $D_= = \{\omega_{00}, \omega_{11}\}$ and $D_{\neq} = \{\omega_{01}, \omega_{10}\}$, which are fibers of neither coordinate.
(4) **Triples** $T_{ij} = \Omega \setminus \{\omega_{ij}\}$ for $i, j \in \{0, 1\}$ (four contexts), each the complement of a singleton.
(5) **The global context** Ω (one context).

REMARK 7.10 (Why the doubletons split three ways). A doubleton $\{\omega, \omega'\}$ with $\omega \neq \omega'$ either agrees in the first coordinate (then it is A_0 or A_1), agrees in the second coordinate (then it is R_0 or R_1), or agrees in neither (then it is $D_=$ or $D_{\neq}$). Two distinct atoms cannot agree in both coordinates, so the three cases are exclusive and exhaustive: $2 + 2 + 2 = 6$, as required. The diagonal doubletons admit a second description that explains their names: a

subset of $\{0,1\} \times \{0,1\}$ meeting each vertical fiber exactly once is the graph of a function $\{0,1\} \to \{0,1\}$, and there are exactly two bijections: the identity, with graph $D_{=}$, and the swap, with graph $D_{\neq}$. The diagonals are thus the graphs of the two bijections of the coordinate set, a fact used again in Section 5.

Remark 7.11 (Complementation pairing). The order-reversing involution $V \mapsto V^c$ pairs the classes as follows: $\varnothing \leftrightarrow \Omega$; each singleton $\{\omega_{ij}\} \leftrightarrow$ the triple T_{ij}; and among doubletons, $A_0 \leftrightarrow A_1$, $R_0 \leftrightarrow R_1$, and $D_{=} \leftrightarrow D_{\neq}$. In particular each of the three doubleton kinds is closed under complementation—the complement of an ACBE block is the other ACBE block, and likewise for RPM blocks and diagonals. This is the first hint of the block structure developed in Section 4: the two ACBE blocks partition Ω, as do the two RPM blocks.

Table 1 lists all sixteen contexts with their class, their atoms, and their complement. It is the master index for every case analysis in this and the next two chapters.

Table 1. The sixteen contexts of $L = \mathcal{P}(\Omega)$.

Context	Class	Atoms	Complement
$\varnothing$	empty	—	Ω
$\{\omega_{00}\}$	singleton	ω_{00}	T_{00}
$\{\omega_{01}\}$	singleton	ω_{01}	T_{01}
$\{\omega_{10}\}$	singleton	ω_{10}	T_{10}
$\{\omega_{11}\}$	singleton	ω_{11}	T_{11}
A_0	ACBE block	ω_{00}, ω_{01}	A_1
A_1	ACBE block	ω_{10}, ω_{11}	A_0
R_0	RPM block	ω_{00}, ω_{10}	R_1
R_1	RPM block	ω_{01}, ω_{11}	R_0
$D_{=}$	diagonal	ω_{00}, ω_{11}	$D_{\neq}$
$D_{\neq}$	diagonal	ω_{01}, ω_{10}	$D_{=}$
T_{00}	triple	$\omega_{01}, \omega_{10}, \omega_{11}$	$\{\omega_{00}\}$
T_{01}	triple	$\omega_{00}, \omega_{10}, \omega_{11}$	$\{\omega_{01}\}$
T_{10}	triple	$\omega_{00}, \omega_{01}, \omega_{11}$	$\{\omega_{10}\}$
T_{11}	triple	$\omega_{00}, \omega_{01}, \omega_{10}$	$\{\omega_{11}\}$
Ω	global	all four	$\varnothing$

3. The eighty-one arrows

The arrow count of L is the first number the executable certificate of Chapter 11 checks. We prove it twice: once by summation over targets, and once by the atomwise argument that generalizes.

Theorem 7.12 (Arrow count). *The category L has exactly $81 = 3^4$ arrows.*

First proof: summation over targets. An arrow of L is an ordered pair (U, V) with $U \subseteq V$. Fixing V, the arrows with codomain V correspond to the subsets of V, of which there are $2^{|V|}$. Summing over the sixteen contexts, grouped by cardinality class using Table 1:

$$\underbrace{1 \cdot 2^0}_{\varnothing} + \underbrace{4 \cdot 2^1}_{\text{singletons}} + \underbrace{6 \cdot 2^2}_{\text{doubletons}} + \underbrace{4 \cdot 2^3}_{\text{triples}} + \underbrace{1 \cdot 2^4}_{\Omega} = 1 + 8 + 24 + 32 + 16 = 81.$$

□

SECOND PROOF: ATOMWISE TRICHOTOMY. A pair (U, V) with $U \subseteq V \subseteq \Omega$ is the same data as a function assigning to each atom $\omega \in \Omega$ one of exactly three states:

$$\omega \notin V \quad (\text{“neither”}), \qquad \omega \in V \setminus U \quad (\text{“}V\text{ only”}), \qquad \omega \in U \quad (\text{“both”}).$$

The constraint $U \subseteq V$ rules out precisely the fourth conceivable state ($\omega \in U \setminus V$), and imposes no coupling between distinct atoms: the state of each atom may be chosen freely and independently. Hence the arrows of L are in bijection with the functions $\Omega \to \{\text{neither}, V \text{ only}, \text{both}\}$, of which there are $3^{|\Omega|} = 3^4 = 81$. □

The atomwise proof has the advantage of computing more than a total. It exhibits the arrow set of L as itself a product of four independent three-element factors, one per atom—a structure that recurs when we count composable pairs and chains.

COROLLARY 7.13 (Identity and proper arrows). *Of the* 81 *arrows, exactly* 16 *are identities (one per context) and exactly* 65 *are proper inclusions* $U \subsetneq V$.

PROOF. Identities correspond to pairs (V, V), one per object. $81 - 16 = 65$. □

COROLLARY 7.14 (Local arrow counts). *For a context* V *with* $|V| = k$*: the number of arrows* into V *(its in-degree, counting the identity) is* 2^k*, and the number of arrows* out of V *(its out-degree, counting the identity) is* 2^{4-k}*. The in-degree and out-degree of every context multiply to* $2^4 = 16$*, and both checksums*

$$\sum_{V \in L} 2^{|V|} = 81, \qquad \sum_{V \in L} 2^{4-|V|} = 81$$

hold, the second because complementation $V \mapsto V^c$ *is a bijection of* L *exchanging the two counts.*

Table 2 records the degrees per class; the row sums re-derive Theorem 7.12 in both directions at once.

TABLE 2. Arrow degrees by context class.

Class	Count	In-degree each	Out-degree each	In-degree subtotal
empty	1	1	16	1
singleton	4	2	8	8
doubleton	6	4	4	24
triple	4	8	2	32
global	1	16	1	16
total	16	—	—	81

PROPOSITION 7.15 (Chains). *The number of composable pairs of arrows in* L*—that is, chains* $U \subseteq V \subseteq W$*—is* $4^4 = 256$*. More generally, the number of chains* $U_1 \subseteq U_2 \subseteq \cdots \subseteq U_n$ *of length* n *is* $(n+1)^4$*.*

PROOF. By the same atomwise argument as in the second proof of Theorem 7.12: a chain of n nested subsets assigns to each atom the (possibly empty) upper interval of chain positions at which it is present. Nestedness means the set of positions containing a given atom is upward closed in $\{1, \ldots, n\}$, so it is determined by its least element or by being empty:

$n+1$ states per atom, independently. For $n=2$ this gives $3^4=81$ (recovering the arrow count, an arrow being a 2-chain), and for $n=3$, composable pairs, it gives $4^4=256$. □

3.1. The arrow-class matrix. The atomwise proof localizes; the summation proof classifies. Combining them yields the complete matrix of arrow counts between cardinality classes, which the case analyses of Sections 7 and 9 implicitly traverse and which we therefore record explicitly.

PROPOSITION 7.16 (Arrows by source and target cardinality). *For $0 \le k \le n \le 4$, the number of arrows $U \to V$ with $|U|=k$ and $|V|=n$ is*

$$\binom{4}{n}\binom{n}{k},$$

the first factor choosing the target among the contexts of its class and the second choosing the source among the subsets of the target. The full matrix is given in Table 3; its row sums are the out-degree totals $\binom{4}{k}2^{4-k}$, its column sums are the in-degree totals $\binom{4}{n}2^n$, and its grand total is 81.

PROOF. The product formula is the two-step choice just described; independence of the two choices is immediate. Row sum for fixed k: $\sum_{n\ge k}\binom{4}{n}\binom{n}{k}=\binom{4}{k}\sum_{n\ge k}\binom{4-k}{n-k}=\binom{4}{k}2^{4-k}$ by the subset-of-a-subset identity $\binom{4}{n}\binom{n}{k}=\binom{4}{k}\binom{4-k}{n-k}$. Column sum for fixed n: $\sum_{k\le n}\binom{4}{n}\binom{n}{k}=\binom{4}{n}2^n$. Grand total: $\sum_n\binom{4}{n}2^n=(1+2)^4=81$ by the binomial theorem—a third proof of Theorem 7.12. □

TABLE 3. The arrow-class matrix: entries count arrows from sources of size k (rows) to targets of size n (columns).

$k\backslash n$	0	1	2	3	4	row sum
0	1	4	6	4	1	16
1	—	4	12	12	4	32
2	—	—	6	12	6	24
3	—	—	—	4	4	8
4	—	—	—	—	1	1
column sum	1	8	24	32	16	81

REMARK 7.17 (Reading the matrix). Several later case analyses are visible as rows and columns. The $k=1$ row says there are 32 arrows out of singletons, of which 12 land in doubletons—these are the arrows along which the singleton coincidence of Proposition 7.31 propagates by pullback in Chapter 8. The $n=4$ column says 16 arrows enter the global context, one from each context; these index the pullbacks of the global generating covers, which the certificate checks exhaustively. The diagonal entries $1,4,6,4,1$ are the identities, refining Corollary 7.13's count of 16.

REMARK 7.18 (The general pattern). Nothing in these counts is special to four atoms: for $|\Omega|=m$ the lattice $\mathcal{P}(\Omega)$ has 2^m objects, 3^m arrows, and $(n+1)^m$ chains of length n. The registered model fixes $m=4$ because the two binary marks generate exactly four atoms; the counting technology is uniform. The certificate quantities "16 contexts" and "81 inclusion arrows" are instances of 2^m and 3^m at $m=4$.

4. The transverse partitions

The successor's escape from the legacy collapse lives in two partitions of the atom set, one per coordinate.

Definition 7.19 (Coordinate partitions). The *ACBE partition* A and the *RPM partition* R of Ω are

$$A = \{A_0, A_1\}, \qquad A_i = \{(i,0),(i,1)\} = \{\omega_{i0}, \omega_{i1}\},$$
$$R = \{R_0, R_1\}, \qquad R_j = \{(0,j),(1,j)\} = \{\omega_{0j}, \omega_{1j}\}.$$

That is, A_i collects the atoms whose first (ACBE) coordinate is i, and R_j collects the atoms whose second (RPM) coordinate is j. We also fix the *singleton partition*

$$Q = \big\{\{\omega_{00}\}, \{\omega_{01}\}, \{\omega_{10}\}, \{\omega_{11}\}\big\},$$

whose blocks are the four singletons.

Proposition 7.20 (Partition axioms, verified). *Each of A, R, Q is a partition of Ω: its blocks are nonempty, pairwise disjoint, and jointly exhaustive.*

Proof. For A: the blocks A_0 and A_1 are visibly nonempty (each lists two atoms). They are disjoint because an atom in both would have first coordinate equal to both 0 and 1. They are exhaustive because every atom has a first coordinate, which is 0 or 1. The argument for R is identical with the second coordinate, and for Q it is the statement that every atom is exactly one of the four atoms. □

Theorem 7.21 (Transversality identity). *For all $i, j \in \{0,1\}$,*

$$A_i \cap R_j = \{\omega_{ij}\} = \{(i,j)\}.$$

Explicitly, the four cases are

$$A_0 \cap R_0 = \{\omega_{00}\}, \quad A_0 \cap R_1 = \{\omega_{01}\}, \quad A_1 \cap R_0 = \{\omega_{10}\}, \quad A_1 \cap R_1 = \{\omega_{11}\}.$$

Proof. An atom lies in $A_i \cap R_j$ if and only if its first coordinate is i and its second coordinate is j, if and only if it equals (i,j). For the case-by-case confirmation: $A_0 \cap R_0 = \{\omega_{00},\omega_{01}\} \cap \{\omega_{00},\omega_{10}\} = \{\omega_{00}\}$; $A_0 \cap R_1 = \{\omega_{00},\omega_{01}\} \cap \{\omega_{01},\omega_{11}\} = \{\omega_{01}\}$; $A_1 \cap R_0 = \{\omega_{10},\omega_{11}\} \cap \{\omega_{00},\omega_{10}\} = \{\omega_{10}\}$; $A_1 \cap R_1 = \{\omega_{10},\omega_{11}\} \cap \{\omega_{01},\omega_{11}\} = \{\omega_{11}\}$. □

The transversality identity is the single most load-bearing fact in the finite theory. It powers the four-versus-one witness computation in Theorem NT2-3, the four-singleton common refinement in Theorem NT2-4, and the flattening isomorphism of the composition theorem in Chapter 10. We therefore record its immediate structural consequences in full.

Corollary 7.22 (No containments between blocks of different partitions). *No ACBE block is contained in an RPM block, and no RPM block is contained in an ACBE block: for all i, j,*

$$A_i \not\subseteq R_j \qquad \text{and} \qquad R_j \not\subseteq A_i.$$

Proof. $|A_i| = |R_j| = 2$, so a containment either way would force $A_i = R_j$ and hence $A_i \cap R_j$ to have two elements, contradicting Theorem 7.21, by which the intersection is a singleton. □

Corollary 7.23 (Common refinement of the partitions). *The common refinement of the partitions A and R—the partition whose blocks are the nonempty intersections of an A-block with an R-block—is exactly the singleton partition Q.*

PROOF. By Theorem 7.21 every such intersection is a singleton $\{\omega_{ij}\}$, all four singletons arise, and none is empty. □

REMARK 7.24 (Transversality as a bijection statement). Theorem 7.21 can be restated: the map sending an atom to the pair (index of its A-block, index of its R-block) is a bijection $\Omega \to \{0,1\} \times \{0,1\}$. Unwound, this is nothing but the definition of Ω as a product—which is precisely the point. Transversality is a genuine constraint in general: for partitions into blocks of sizes p and q it forces $|\Omega| = pq$ and each cross-intersection to be a single atom. At $|\Omega| = 4$ with two blocks on each side, however, it is nearly automatic—there are only three partitions of a four-element set into two doubletons, and any two *distinct* ones already meet in singletons, so the sole non-transverse pair is the degenerate $A = R$. The content of the registered design decision is therefore not that transversality was achieved at four atoms, but that the two marks were placed as the two coordinates of a product, which is what makes transversality persist when the mark alphabets grow (Exercise 7.51). It is that identity, and not its four-atom instance, that makes the two generated topologies of Chapter 9 interact the way they do rather than collapse onto each other or ignore each other.

5. Symmetries of the marked lattice

Several arguments in Chapter 9 are stated for the ACBE side and concluded "symmetrically" for the RPM side. In a finite exhaustive development that phrase should be discharged by an actual symmetry, not by an appeal to analogy. We construct it.

PROPOSITION 7.25 (The coordinate swap). *Let $s : \Omega \to \Omega$ be the coordinate swap $s(i,j) = (j,i)$. Then:*

1. *s is an involutive bijection of Ω, fixing ω_{00} and ω_{11} and exchanging $\omega_{01} \leftrightarrow \omega_{10}$;*
2. *the direct-image map $S : L \to L$, $S(V) = \{s(\omega) \mid \omega \in V\}$, is an order-preserving involutive automorphism of the lattice L;*
3. *S exchanges the partitions: $S(A_i) = R_i$ and $S(R_j) = A_j$ for all i, j;*
4. *S fixes both diagonals setwise: $S(D_=) = D_=$ and $S(D_{\neq}) = D_{\neq}$;*
5. *S commutes with intersections and unions: $S(U \cap V) = S(U) \cap S(V)$ and $S(U \cup V) = S(U) \cup S(V)$.*

PROOF. (1) $s(s(i,j)) = (i,j)$, and the fixed points of s are the atoms with $i = j$. (2) Direct image along a bijection preserves and reflects inclusion, and $S \circ S = \mathrm{id}_L$ because $s \circ s = \mathrm{id}_\Omega$. (3) $S(A_i) = \{s(i,0), s(i,1)\} = \{(0,i),(1,i)\} = R_i$, and applying S again gives $S(R_i) = A_i$. (4) $S(D_=) = \{s(0,0), s(1,1)\} = \{(0,0),(1,1)\} = D_=$; $S(D_{\neq}) = \{s(0,1), s(1,0)\} = \{(1,0),(0,1)\} = D_{\neq}$. (5) Direct image along any bijection preserves all Boolean operations. □

PROPOSITION 7.26 (Coordinate flips). *Let $f_A(i,j) = (1-i,j)$ and $f_R(i,j) = (i,1-j)$. Their direct-image automorphisms $F_A^\flat, F_R^\flat$ of L satisfy: $F_A^\flat$ exchanges $A_0 \leftrightarrow A_1$ and fixes each R_j setwise; $F_R^\flat$ exchanges $R_0 \leftrightarrow R_1$ and fixes each A_i setwise; both exchange $D_= \leftrightarrow D_{\neq}$. The group generated by S, $F_A^\flat$, $F_R^\flat$ has order eight and acts on the six doubletons preserving the three kinds $\{A\text{-blocks}\}$, $\{R\text{-blocks}\}$, $\{\text{diagonals}\}$ as a system of blocks, with S exchanging the first two kinds.*

PROOF. The block computations are direct: $f_A(A_0) = \{(1,0),(1,1)\} = A_1$; $f_A(R_j) = \{(1,j),(0,j)\} = R_j$; $f_A(D_=) = \{(1,0),(0,1)\} = D_{\neq}$; and symmetrically for f_R. The three generators are involutions; f_A and f_R commute; conjugating f_A by s gives f_R. The group is therefore generated by the commuting pair (f_A, f_R), giving a Klein four-group, extended

by s which swaps them: a group of order $4 \cdot 2 = 8$, isomorphic to the dihedral group of the square. (Indeed the atoms form a four-cycle $\omega_{00}, \omega_{01}, \omega_{11}, \omega_{10}$ on which these maps act as the symmetries of a square.) The action on doubleton kinds follows from the computations above. □

REMARK 7.27 (How the symmetry is used). The swap S transports every ACBE-side construction to its RPM-side mirror: it carries the generating family $K_A(V)$ of Section 6 to $K_R(S(V))$, and (in Chapter 9) the witness presheaf F_A to F_R. Every "symmetric argument" in Chapter 9 is shorthand for conjugation by S, which is legitimate precisely because Proposition 7.25 makes S an automorphism of all the structure in sight. No appeal to informal analogy is required anywhere in the finite theory.

6. The generating families

We now define the covering data. Chapter 9 will feed these families into topology generation; here we compute them, in every context, exactly.

DEFINITION 7.28 (Generating families and the null cover). Let $P \in \{A, R, Q\}$ be one of the three fixed partitions, with block set $P = \{B_1, \ldots, B_r\}$ ($r = 2$ for A and R; $r = 4$ for Q). For a *nonempty* context $V \in L$, the *generating family* of P at V is the indexed family

$$K_P(V) \;=\; \big\{\, V \cap B \longrightarrow V \;\big|\; B \in P \,\big\},$$

indexed by the blocks of P, *retaining an empty piece* $\varnothing \to V$ whenever $V \cap B = \varnothing$. At the empty context the generator is the *null cover*: the empty family of arrows with codomain $\varnothing$, which is stipulated to cover $\varnothing$.

Three points of the definition require comment before the tables.

First, $K_P(V)$ is an *indexed* family: its members are indexed by blocks, so $K_A(V)$ always has exactly two members (possibly with repeated or empty domains) and $K_Q(V)$ always has exactly four. When we later compare families "as unmarked arrow lists" we forget the indexing and keep only the set of arrows; the distinction matters at exactly the coincidence loci computed in Proposition 7.31.

Second, empty pieces are retained deliberately. Discarding them would make the family shape depend on the context, complicating both the pullback-stability bookkeeping of Section 7 and the serialization of Chapter 8, where the exact member arrows are digest-bound. Retention is harmless for the generated topology: an arrow $\varnothing \to V$ adds to the generated sieve only arrows that factor through $\varnothing$, namely $\varnothing \to V$ itself, and (as verified in Chapter 9) the empty context is covered by the null cover in any case.

Third, the null cover at $\varnothing$ is a genuine stipulation, standard for sites built from partitions or unions [7]: the empty family has no members, every compatibility condition over it is vacuous, and the sheaf condition it induces forces sheaves to take singleton value at $\varnothing$—a fact that does real work in Chapters 9 and 10.

Table 4 now computes K_A and K_R at every context: for each V it lists the four intersections $V \cap A_0$, $V \cap A_1$, $V \cap R_0$, $V \cap R_1$. Every later "by Table 4" resolves to one row of this table.

We extract the per-class shape of the families in words; each claim reads off the corresponding rows of Table 4.

PROPOSITION 7.29 (Shape of K_A and K_R by context class). (1) ***Empty context.*** *The generator is the null family, for all three partitions.*

TABLE 4. The generating families at every context: $K_A(V)$ has members $V \cap A_0 \to V$ and $V \cap A_1 \to V$; $K_R(V)$ has members $V \cap R_0 \to V$ and $V \cap R_1 \to V$. At $V = \varnothing$ the generator is the null (empty) family.

V	$V \cap A_0$	$V \cap A_1$	$V \cap R_0$	$V \cap R_1$
$\varnothing$	(null cover: empty family)			
$\{\omega_{00}\}$	$\{\omega_{00}\}$	$\varnothing$	$\{\omega_{00}\}$	$\varnothing$
$\{\omega_{01}\}$	$\{\omega_{01}\}$	$\varnothing$	$\varnothing$	$\{\omega_{01}\}$
$\{\omega_{10}\}$	$\varnothing$	$\{\omega_{10}\}$	$\{\omega_{10}\}$	$\varnothing$
$\{\omega_{11}\}$	$\varnothing$	$\{\omega_{11}\}$	$\varnothing$	$\{\omega_{11}\}$
A_0	A_0	$\varnothing$	$\{\omega_{00}\}$	$\{\omega_{01}\}$
A_1	$\varnothing$	A_1	$\{\omega_{10}\}$	$\{\omega_{11}\}$
R_0	$\{\omega_{00}\}$	$\{\omega_{10}\}$	R_0	$\varnothing$
R_1	$\{\omega_{01}\}$	$\{\omega_{11}\}$	$\varnothing$	R_1
$D_=$	$\{\omega_{00}\}$	$\{\omega_{11}\}$	$\{\omega_{00}\}$	$\{\omega_{11}\}$
$D_\neq$	$\{\omega_{01}\}$	$\{\omega_{10}\}$	$\{\omega_{10}\}$	$\{\omega_{01}\}$
T_{00}	$\{\omega_{01}\}$	A_1	$\{\omega_{10}\}$	R_1
T_{01}	$\{\omega_{00}\}$	A_1	R_0	$\{\omega_{11}\}$
T_{10}	A_0	$\{\omega_{11}\}$	$\{\omega_{00}\}$	R_1
T_{11}	A_0	$\{\omega_{10}\}$	R_0	$\{\omega_{01}\}$
Ω	A_0	A_1	R_0	R_1

(2) ***Singletons.*** *At $V = \{\omega_{ij}\}$: $K_A(V)$ consists of the identity piece $V \to V$ (from block A_i) and one empty piece (from A_{1-i}); $K_R(V)$ likewise consists of $V \to V$ (from R_j) and one empty piece (from R_{1-j}).*

(3) ***ACBE blocks.*** *At $V = A_i$: $K_A(V) = \{A_i \to A_i,\ \varnothing \to A_i\}$ contains the identity; $K_R(V) = \{\{\omega_{i0}\} \to A_i,\ \{\omega_{i1}\} \to A_i\}$ consists of two* proper *singleton pieces partitioning V.*

(4) ***RPM blocks.*** *At $V = R_j$, symmetrically: $K_R(V)$ contains the identity and an empty piece; $K_A(V) = \{\{\omega_{0j}\} \to R_j,\ \{\omega_{1j}\} \to R_j\}$ consists of two proper singleton pieces.*

(5) ***Diagonals.*** *At $V = D_=$ or $V = D_\neq$: both $K_A(V)$ and $K_R(V)$ consist of the two singleton pieces $\{\omega\} \to V$, $\{\omega'\} \to V$ for the two atoms ω, ω' of V—both partitions cut a diagonal into its two atoms.*

(6) ***Triples.*** *At $V = T_{ij}$: $K_A(V)$ has one singleton piece $\{\omega_{i,1-j}\} \to V$ and one complete-block piece $A_{1-i} \to V$; $K_R(V)$ has one singleton piece $\{\omega_{1-i,j}\} \to V$ and one complete-block piece $R_{1-j} \to V$. All pieces are proper and nonempty.*

(7) ***Global context.*** *$K_A(\Omega) = \{A_0 \to \Omega,\ A_1 \to \Omega\}$ and $K_R(\Omega) = \{R_0 \to \Omega,\ R_1 \to \Omega\}$: two proper, disjoint, jointly exhaustive doubleton pieces each.*

PROOF. Every clause is a row (or four rows) of Table 4. For clause (6), instantiate: at $T_{00} = \{\omega_{01}, \omega_{10}, \omega_{11}\}$ the ACBE pieces are $T_{00} \cap A_0 = \{\omega_{01}\}$ (the surviving atom of A_0 after removing ω_{00}) and $T_{00} \cap A_1 = A_1$ (untouched, since the removed atom lies in A_0); the RPM pieces are $\{\omega_{10}\}$ and R_1 by the same reasoning on the second coordinate. The other three triples follow by applying the flip automorphisms of Proposition 7.26, or by direct reading of their rows. □

PROPOSITION 7.30 (Joint exhaustiveness). *For every nonempty context V and each partition $P \in \{A, R, Q\}$, the union of the domains of $K_P(V)$ is V:*

$$\bigcup_{B \in P} (V \cap B) \;=\; V \cap \bigcup_{B \in P} B \;=\; V \cap \Omega \;=\; V.$$

PROOF. Intersection distributes over union, and the blocks of a partition are jointly exhaustive by Proposition 7.20. □

PROPOSITION 7.31 (Coincidence loci of the unmarked families). *Forget the block indexing and regard $K_A(V)$ and $K_R(V)$ as bare sets of arrows into V ("unmarked families"). Then:*

(1) *at the empty context, both generators are the null family (trivially equal);*
(2) *at each singleton $\{\omega_{ij}\}$, the unmarked families coincide:*

$$K_A(\{\omega_{ij}\}) = \big\{\{\omega_{ij}\} \to \{\omega_{ij}\},\ \varnothing \to \{\omega_{ij}\}\big\} = K_R(\{\omega_{ij}\});$$

(3) *at each diagonal doubleton, the unmarked families coincide: for $D \in \{D_=, D_{\neq}\}$ with atoms ω, ω',*

$$K_A(D) = \big\{\{\omega\} \to D,\ \{\omega'\} \to D\big\} = K_R(D),$$

although for $D_{\neq}$ the block indexings differ (A_0 picks out ω_{01} where R_0 picks out ω_{10});
(4) *at the ACBE blocks, RPM blocks, triples, and the global context, the unmarked families differ.*

PROOF. All four clauses read off Table 4. For (2): the row of $\{\omega_{ij}\}$ shows one full piece and one empty piece on each side; as sets of arrows both families are $\{\mathrm{id},\ \varnothing \hookrightarrow\}$. For (3): the $D_=$ row shows ACBE pieces $\{\omega_{00}\}, \{\omega_{11}\}$ and RPM pieces $\{\omega_{00}\}, \{\omega_{11}\}$—equal as sets; the $D_{\neq}$ row shows ACBE pieces $\{\omega_{01}\}, \{\omega_{10}\}$ and RPM pieces $\{\omega_{10}\}, \{\omega_{01}\}$—equal as sets, crossed as indexed families. For (4): at A_0 the ACBE family contains the identity $A_0 \to A_0$ while the RPM family contains only singleton pieces; at T_{00} the ACBE family contains $A_1 \to T_{00}$, which the RPM family (members $\{\omega_{10}\}$, R_1) does not; at Ω the families $\{A_0, A_1\}$ and $\{R_0, R_1\}$ share no member since no A-block equals an R-block (Corollary 7.22). □

REMARK 7.32 (Forward pointer: why the coincidence loci matter). Proposition 7.31(2) is the combinatorial core of the *singleton coincidence*: at singleton contexts the topology generated from the ACBE family and the topology generated from the RPM family receive literally identical input, so no Grothendieck topology can remember, at those contexts, which partition a cover came from. Clause (3) shows the same phenomenon at the diagonals. This is exactly the information loss that Chapter 8's marked-cover records are designed to survive; the registered certificate checks the singleton case executably (`singleton_unmarked_families_coincide` alongside `singleton_marked_covers_distinct`). Clause (4) is equally important in the other direction: at Ω the two families are disjoint and both consist of proper pieces, which is where the generated topologies visibly diverge (Chapter 9, Theorem NT2-3).

7. Pullback stability, case by case

Axiom (J2) of a Grothendieck topology demands stability of covers under pullback. For the generated topologies of Chapter 9 this reduces, via the standard basis criterion [4], to a statement about the generating families themselves, which we now verify exhaustively. The general identity is one line; the value of the case analysis is that it confirms the identity in every context class, exhibits the resulting families concretely, and isolates the two degenerate boundary cases (empty pullback pieces and the empty subcontext) that the executable checker exercises separately.

PROPOSITION 7.33 (Pullback of generating families). *Let $P \in \{A, R, Q\}$, let V be a nonempty context, and let $U \subseteq V$. Pulling back the family $K_P(V)$ along $U \to V$ (in the sense of Proposition 7.4: each member $V \cap B \to V$ pulls back to $(V \cap B) \cap U \to U$) yields:*

1. *if $U \neq \varnothing$: exactly the indexed family $K_P(U)$, block by block;*
2. *if $U = \varnothing$: the family all of whose members are $\mathrm{id}_\varnothing$, and the null cover covers $\varnothing$ in any case.*

PROOF. For each block $B \in P$, the pullback of $V \cap B \to V$ along $U \to V$ is $(V \cap B) \cap U$ by Proposition 7.4, and $(V \cap B) \cap U = U \cap B$ by the identity of Remark 7.5 (using $U \subseteq V$). If $U \neq \varnothing$ this says the pulled-back family is $\{U \cap B \to U \mid B \in P\} = K_P(U)$, with matching block indices. If $U = \varnothing$ then $U \cap B = \varnothing$ for every B, each pulled-back member is $\varnothing \to \varnothing = \mathrm{id}_\varnothing$, and the covering of $\varnothing$ is supplied by the registered null cover. □

The identity $(V \cap B) \cap U = U \cap B$ never mentions which class V and U belong to; nevertheless we now walk the classes, because the finite certificate does, and because the walk surfaces exactly where empty pieces appear and disappear. Throughout, "pulling back to U" means along the inclusion $U \subseteq V$, and we describe the ACBE case; the RPM case is its image under the swap automorphism S of Proposition 7.25.

7.1. Pullbacks to the empty context. For any nonempty V, every piece of $K_A(V)$ pulls back to $\varnothing \cap A_i = \varnothing$, giving the two-member family $\{\mathrm{id}_\varnothing, \mathrm{id}_\varnothing\}$. This is not the null family (it has two members, both identities), and that is acceptable: the basis convention of Chapter 9 admits identity families at every context, and the null cover independently covers $\varnothing$. The executable checker verifies precisely this convention (pullback to the empty context yields identity arrows on the empty object).

7.2. Pullbacks to singletons. Let $U = \{\omega_{ij}\} \subseteq V$. Then $U \cap A_i = U$ and $U \cap A_{1-i} = \varnothing$, so the pulled-back family is $\{U \to U,\ \varnothing \to U\}$, which by Proposition 7.29(2) is exactly $K_A(U)$. Note what happened to the pieces of $K_A(V)$: whichever piece contained the atom ω_{ij} pulled back to the full piece U, and the other piece pulled back to $\varnothing$. For example, pulling $K_A(\Omega) = \{A_0, A_1\}$ back to $\{\omega_{10}\}$: $A_0 \cap \{\omega_{10}\} = \varnothing$ and $A_1 \cap \{\omega_{10}\} = \{\omega_{10}\}$, matching the $\{\omega_{10}\}$ row of Table 4.

7.3. Pullbacks to doubletons. There are three subcases, one per doubleton kind.

1. $U = A_i$ (an ACBE block; necessarily $V \supseteq A_i$, so $V \in \{A_i, T_{i',j'}$ (with $\omega_{i'j'} \notin A_i$), $\Omega\}$). Pulling back $K_A(V)$: the piece from block A_i pulls back to $A_i \cap A_i = A_i = U$ (the identity piece) and the piece from A_{1-i} pulls back to $A_i \cap A_{1-i} = \varnothing$. Result: $\{U \to U,\ \varnothing \to U\} = K_A(A_i)$, per Proposition 7.29(3). Concretely, from $V = \Omega$: $\{A_0 \cap A_i,\ A_1 \cap A_i\} = \{A_i, \varnothing\}$; from $V = T_{1-i,j}$ (a triple containing A_i): the complete-block piece $A_i \to T_{1-i,j}$ pulls back to A_i, the singleton piece to $\varnothing$.
2. $U = R_j$ (an RPM block). Pulling back $K_A(V)$: block A_0 contributes $R_j \cap A_0 = \{\omega_{0j}\}$ and block A_1 contributes $\{\omega_{1j}\}$ (Theorem 7.21). Result: the two-singleton family $K_A(R_j)$ of Proposition 7.29(4). This is the transversality identity acting as a pullback computation.
3. $U = D$ a diagonal. Each ACBE block meets D in one atom, so the pulled-back family is the two-singleton family $K_A(D)$ of Proposition 7.29(5).

7.4. Pullbacks to triples. Let $U = T_{ij} \subseteq V$; since $|U| = 3$, necessarily $V \in \{T_{ij}, \Omega\}$. The identity case is trivial. From $V = \Omega$: $A_0 \cap T_{ij}$ and $A_1 \cap T_{ij}$ are, by the T_{ij} row of Table 4, one singleton and one complete block—exactly $K_A(T_{ij})$ as computed in Proposition 7.29(6). For instance, pulling $K_A(\Omega)$ back to T_{00}: $A_0 \cap T_{00} = \{\omega_{01}\}$, $A_1 \cap T_{00} = A_1$.

7.5. Pullback to the global context. Only $V = \Omega$, $U = \Omega$ qualifies, and the pullback along the identity is the family itself.

REMARK 7.34 (Summary of the case walk). In every case the pulled-back indexed family is $K_P(U)$ on the nose—not merely a family generating the same sieve. Empty pieces enter exactly when U misses a block that V met, and leave exactly when U is contained in a single block, in which case the other block's piece is empty. The stability statement is therefore as strong as it could possibly be at the level of raw families, which is what makes the serialization story of Chapter 8 clean: a pullback record can name its parent and its exact member arrows, and the recomputation check of the parent-resolution algorithm can literally recompute the child members from the parent by intersection.

8. Cover composition and refinement

The second basis axiom needed in Chapter 9 is stability under composition: refining each member of a generating family by generating families of its own must land inside the sieve machinery correctly. Here we work the underlying combinatorics: same-partition refinement is absorbing, and cross-partition refinement produces singletons. Both computations are pure intersection algebra.

LEMMA 7.35 (Same-partition refinement is absorbing). *Let $P \in \{A, R\}$, let V be a nonempty context, and refine the member $V \cap B \to V$ of $K_P(V)$ by the family $K_P(V \cap B)$ (or by the null cover if $V \cap B = \varnothing$). The composite family, over all members, is*

$$\{\, V \cap B \cap B' \to V \;\mid\; B, B' \in P \,\} \;=\; \{\, V \cap B \to V \;\mid\; B \in P \,\} \;\cup\; \{\, \varnothing \to V \,\},$$

because block disjointness gives $B \cap B' = \varnothing$ for $B \neq B'$ and $B \cap B = B$. As a set of arrows, the composite is the original family together with (at most) the empty piece it already contained or a redundant empty piece.

PROOF. $V \cap B \cap B' = V \cap (B \cap B')$. If $B' = B$ this is $V \cap B$, reproducing the original member; if $B' \neq B$ it is $V \cap \varnothing = \varnothing$. The displayed identity follows. Since the arrow $\varnothing \to V$ contributes to any generated sieve only the arrows factoring through $\varnothing$—that is, itself—the composite generates the same sieve as the original family; this last sentence anticipates the sieve language of Chapter 9 but is recorded here because it is finite set algebra. □

LEMMA 7.36 (Cross-partition refinement produces singletons). *Let V be a nonempty context. Refining each member $V \cap A_i \to V$ of $K_A(V)$ by the family $K_R(V \cap A_i)$ (null cover on empty pieces) yields the composite family with nonempty members*

$$\{\, V \cap A_i \cap R_j \to V \;\mid\; i, j \in \{0,1\} \,\} \;=\; \{\, V \cap \{\omega_{ij}\} \to V \;\mid\; i, j \in \{0,1\} \,\},$$

that is, exactly the singleton subsets $\{\omega\} \to V$ for $\omega \in V$ (together with empty pieces for the atoms outside V). The same holds with the roles of A and R exchanged.

PROOF. $V \cap A_i \cap R_j = V \cap (A_i \cap R_j) = V \cap \{\omega_{ij}\}$ by Theorem 7.21. As (i, j) ranges over all four pairs, $V \cap \{\omega_{ij}\}$ is $\{\omega_{ij}\}$ when $\omega_{ij} \in V$ and $\varnothing$ otherwise. The exchanged statement follows by the swap automorphism (Proposition 7.25) or by redoing the two lines with $R_j \cap A_i$. □

EXAMPLE 7.37 (Worked composite at the global context). Start with $K_A(\Omega) = \{A_0 \to \Omega,\ A_1 \to \Omega\}$. Refine A_0 by $K_R(A_0) = \{\{\omega_{00}\} \to A_0,\ \{\omega_{01}\} \to A_0\}$ and A_1 by $K_R(A_1) = \{\{\omega_{10}\} \to A_1,\ \{\omega_{11}\} \to A_1\}$ (rows A_0, A_1 of Table 4). The composite family at Ω is

$$\big\{\{\omega_{00}\} \to \Omega,\ \{\omega_{01}\} \to \Omega,\ \{\omega_{10}\} \to \Omega,\ \{\omega_{11}\} \to \Omega\big\} = K_Q(\Omega)\,:$$

the four-singleton family, with no empty pieces since every atom lies in Ω. This is the *transverse refinement* at the global context, the combinatorial heart of Theorem NT2-4 in Chapter 9 and of the joint-refinement derivation records of Chapter 8.

EXAMPLE 7.38 (Worked composite at a triple). Start with $K_A(T_{00}) = \{\{\omega_{01}\} \to T_{00},\ A_1 \to T_{00}\}$. Refine the singleton piece by $K_R(\{\omega_{01}\}) = \{\{\omega_{01}\} \to \{\omega_{01}\},\ \varnothing \to \{\omega_{01}\}\}$ and the block piece by $K_R(A_1) = \{\{\omega_{10}\} \to A_1,\ \{\omega_{11}\} \to A_1\}$. The composite family at T_{00} has members

$$\{\omega_{01}\} \to T_{00}, \quad \varnothing \to T_{00}, \quad \{\omega_{10}\} \to T_{00}, \quad \{\omega_{11}\} \to T_{00}\,:$$

the three singletons of T_{00} plus one empty piece contributed by the atom $\omega_{00} \notin T_{00}$. Again this matches Lemma 7.36 exactly: the nonempty members are the singletons of V, and the empty members count the atoms of $\Omega \setminus V$.

EXAMPLE 7.39 (Worked absorbing composite). Start with $K_A(\Omega)$ and refine each piece by its *own* ACBE family: $K_A(A_0) = \{A_0 \to A_0,\ \varnothing \to A_0\}$ and $K_A(A_1) = \{\varnothing \to A_1,\ A_1 \to A_1\}$. The composite family at Ω is $\{A_0 \to \Omega,\ \varnothing \to \Omega,\ A_1 \to \Omega\}$: the original two pieces plus a redundant empty piece, exactly as Lemma 7.35 predicts. Nothing new is produced; same-partition refinement is idempotent up to empty pieces.

REMARK 7.40 (What these computations will become). In Chapter 9, Lemma 7.35 discharges the cover-composition axiom of the coverage basis (refining by the same partition regenerates the same sieve), and Lemma 7.36 supplies both the four-singleton common refinement of Theorem NT2-4 and the lower bound $J_{\text{atom}} \le J_A \vee J_R$ in the join computation. In Chapter 10 the same transversality algebra, applied inside a product over blocks, produces the flattening isomorphism of the composition theorem. In Chapter 8, the cross-partition composite is the mathematical content of a `COMMON_REFINEMENT` derivation record, which is required to carry exactly one ACBE and one RPM generating parent—one for each family entering Lemma 7.36.

9. Block completeness

The witness presheaves of Chapter 9 assign to a context the set of bit-functions on the *complete* blocks it contains. The relevant combinatorics is: which contexts contain which complete blocks? The answer is a small table with large consequences.

DEFINITION 7.41 (Complete blocks in a context). Let $V \in L$. An ACBE block A_i is *complete in* V when $A_i \subseteq V$; likewise an RPM block R_j is complete in V when $R_j \subseteq V$. Write

$$\alpha(V) = \#\{\, i \in \{0,1\} \mid A_i \subseteq V \,\}, \qquad \rho(V) = \#\{\, j \in \{0,1\} \mid R_j \subseteq V \,\}$$

for the number of complete ACBE and RPM blocks in V.

THEOREM 7.42 (Block-completeness census). *The values of α and ρ on the sixteen contexts are exactly as in Table 5. In summary, by class:*

(1) $\alpha(\varnothing) = \rho(\varnothing) = 0$*;*
(2) $\alpha = \rho = 0$ *on all singletons;*

(3) *on doubletons:* $\alpha(A_i) = 1$, $\rho(A_i) = 0$; $\alpha(R_j) = 0$, $\rho(R_j) = 1$; $\alpha(D) = \rho(D) = 0$ *for both diagonals;*
(4) *on triples:* $\alpha(T_{ij}) = \rho(T_{ij}) = 1$; *specifically* $A_{1-i} \subseteq T_{ij}$ *and* $R_{1-j} \subseteq T_{ij}$, *and no other block is complete;*
(5) $\alpha(\Omega) = \rho(\Omega) = 2$.

PROOF. A block has two atoms, so $A_i \subseteq V$ if and only if both $\omega_{i0} \in V$ and $\omega_{i1} \in V$. We verify each class.

(1) The empty context contains no atoms, hence no complete blocks.

(2) A singleton has one atom and a block needs two: no complete blocks.

(3) $A_i \subseteq A_i$ trivially, and $A_{1-i} \cap A_i = \varnothing$ shows $A_{1-i} \not\subseteq A_i$; also $R_j \not\subseteq A_i$ by Corollary 7.22: $\alpha(A_i) = 1$, $\rho(A_i) = 0$. Symmetrically for R_j (or apply the swap S). A diagonal contains one atom from each ACBE block and one atom from each RPM block (its two atoms differ in both coordinates), so it contains no block completely: at $D_=$, for instance, $A_0 \not\subseteq D_=$ because $\omega_{01} \notin D_=$, and $R_0 \not\subseteq D_=$ because $\omega_{10} \notin D_=$; the other three checks are identical.

(4) $T_{ij} = \Omega \setminus \{\omega_{ij}\}$. The removed atom ω_{ij} lies in A_i and in R_j, so $A_i \not\subseteq T_{ij}$ and $R_j \not\subseteq T_{ij}$; the untouched blocks A_{1-i} and R_{1-j} remain complete. Hence exactly one complete block of each kind.

(5) Ω contains everything: both ACBE blocks and both RPM blocks are complete. □

TABLE 5. Block completeness: for each context V, the complete ACBE blocks, the complete RPM blocks, and the counts $\alpha(V)$, $\rho(V)$. The last two columns anticipate Chapter 9: the witness presheaves there satisfy $|F_A(V)| = 2^{\alpha(V)}$ and $|F_R(V)| = 2^{\rho(V)}$.

V	Complete A-blocks	Complete R-blocks	α	ρ	2^α	2^ρ
$\varnothing$	none	none	0	0	1	1
$\{\omega_{00}\}$	none	none	0	0	1	1
$\{\omega_{01}\}$	none	none	0	0	1	1
$\{\omega_{10}\}$	none	none	0	0	1	1
$\{\omega_{11}\}$	none	none	0	0	1	1
A_0	A_0	none	1	0	2	1
A_1	A_1	none	1	0	2	1
R_0	none	R_0	0	1	1	2
R_1	none	R_1	0	1	1	2
$D_=$	none	none	0	0	1	1
$D_\neq$	none	none	0	0	1	1
T_{00}	A_1	R_1	1	1	2	2
T_{01}	A_1	R_0	1	1	2	2
T_{10}	A_0	R_1	1	1	2	2
T_{11}	A_0	R_0	1	1	2	2
Ω	A_0, A_1	R_0, R_1	2	2	4	4

Chapter 9 leans on two consequences of the census at its two decisive moments; we isolate them now.

COROLLARY 7.43 (Blocks never survive crossing). *No RPM block contains a complete ACBE block, and no ACBE block contains a complete RPM block:* $\alpha(R_j) = 0$ *and* $\rho(A_i) = 0$ *for all* i, j.

PROOF. Rows R_0, R_1, A_0, A_1 of Table 5; structurally this is Corollary 7.22 again, since a complete block inside a doubleton would be the whole doubleton. □

This is the fact that will make the ACBE witness presheaf collapse to a single section on each RPM half of Ω, producing the four-versus-one gluing failure of Theorem NT2-3: the global context supports $2^{\alpha(\Omega)} = 4$ sections while each RPM block supports $2^{\alpha(R_j)} = 1$.

COROLLARY 7.44 (Monotonicity and locality of completeness). *If $U \subseteq V$ then every block complete in U is complete in V; that is, α and ρ are monotone along inclusions. Moreover, completeness localizes to same-partition pieces: for any V and any i,*

$$A_k \subseteq V \cap A_i \iff k = i \text{ and } A_i \subseteq V,$$

so each complete ACBE block of V is complete in exactly one piece of $K_A(V)$, and no complete ACBE block of V is complete in any piece of the form $V \cap A_{i'}$ with $i' \neq i$.

PROOF. Monotonicity: $A_k \subseteq U \subseteq V$. Locality: $V \cap A_i \subseteq A_i$, so a block contained in it is contained in A_i, forcing $k = i$ by block disjointness ($A_k \cap A_i = \varnothing$ for $k \neq i$ and blocks are nonempty); conversely if $A_i \subseteq V$ then $A_i = V \cap A_i$. □

Corollary 7.44 is the combinatorial engine of ACBE gluing for the witness presheaf in Chapter 9: sections over the pieces of $K_A(V)$ are bit-functions on complementary index sets, which assemble uniquely over V.

10. The sieve census

Although topology generation belongs to Chapter 9, the raw material it operates on—sieves on the objects of L—is pure order combinatorics, and the executable certificate's headline count of 298 sieves is exactly derivable here. We include the derivation both for completeness and because it gives independent meaning to a number that would otherwise be an opaque implementation output. (The per-context sieve and covering-sieve counts appear in Appendix A; the registered aggregate totals are reproduced in Appendix B. This section proves the totals.)

DEFINITION 7.45 (Sieves on a poset object). A sieve on $V \in L$ is a set σ of arrows with codomain V closed under precomposition [**1**, **7**]. Since arrows into V correspond to subsets of V, a sieve on V is the same thing as a *down-closed* family $\sigma \subseteq \mathcal{P}(V)$: if $S \in \sigma$ and $T \subseteq S$ then $T \in \sigma$. In particular the collection of sieves on V depends only on $|V|$.

LEMMA 7.46 (Down-set counts for small Boolean lattices). *Write $\delta(k)$ for the number of down-closed families in $\mathcal{P}(X)$ with $|X| = k$. Then*

$$\delta(0) = 2, \qquad \delta(1) = 3, \qquad \delta(2) = 6, \qquad \delta(3) = 20, \qquad \delta(4) = 168.$$

These are the Dedekind numbers; Chapter 11 and the appendices write $M(n)$ for $\delta(n)$.

PROOF. $\delta(0)$: the families in $\mathcal{P}(\varnothing) = \{\varnothing\}$ are $\{\}$ and $\{\varnothing\}$; both are down-closed: 2.

$\delta(1)$: subsets of $\{a\}$ are $\varnothing, \{a\}$; down-closed families: $\{\}$, $\{\varnothing\}$, $\{\varnothing, \{a\}\}$: 3.

$\delta(2)$: a down-closed family in $\mathcal{P}(\{a, b\})$ either is empty, or contains $\varnothing$; if it contains the top $\{a, b\}$ it is everything (down-closure). The families: $\{\}$; $\{\varnothing\}$; $\{\varnothing, \{a\}\}$; $\{\varnothing, \{b\}\}$; $\{\varnothing, \{a\}, \{b\}\}$; all of $\mathcal{P}(\{a, b\})$: 6.

$\delta(3)$: split by the top $\{a, b, c\}$. If present, the family is all of $\mathcal{P}$: 1 family. If the family is empty: 1. Otherwise it contains $\varnothing$ and is determined by its two-element members X_2 (any subfamily of the three doubletons) and its one-element members X_1, which must contain every singleton inside a chosen doubleton and may freely contain the rest. Writing

$u = |\bigcup X_2|$, the free singletons number $3-u$, giving $\sum_{X_2} 2^{3-u}$: for $X_2 = \varnothing$, $2^3 = 8$; for the three one-doubleton choices, $u = 2$, $2^1 = 2$ each, total 6; for the three two-doubleton choices, $u = 3$, $2^0 = 1$ each, total 3; for all three doubletons, 1. Sum $8 + 6 + 3 + 1 = 18$. Total $\delta(3) = 1 + 1 + 18 = 20$.

$\delta(4)$: split by the top $\Omega' = \{a, b, c, d\}$. Top present: 1 (everything). Family empty: 1. Otherwise the family contains $\varnothing$, excludes the top, and is determined by down-closed choices at levels 3, 2, 1: a set X_3 of triples, a set X_2 of doubletons containing every doubleton inside a chosen triple, and a set X_1 of singletons containing every singleton inside a chosen doubleton, the rest free. We sum over X_3, then over the free part of X_2, weighting by $2^{\#\text{free singletons}}$.

Case $X_3 = \varnothing$. X_2 is an arbitrary set of doubletons—equivalently a graph G on the four vertices $\{a, b, c, d\}$ with the doubletons as edges—and the forced singletons are the non-isolated vertices of G. Grouping the $2^6 = 64$ graphs by their number k of non-isolated vertices: $k = 0$: the empty graph, 1 graph, weight $2^4 = 16$. $k = 2$: a single edge, 6 graphs, weight $2^2 = 4$, subtotal 24. $k = 3$: choose the three vertices (4 ways) and a graph on them with no isolated vertex; of the 8 subgraphs of a triangle, those covering all three vertices are the three two-edge paths and the full triangle, 4 each, so 16 graphs, weight 2, subtotal 32. $k = 4$: the remaining $64 - 1 - 6 - 16 = 41$ graphs, weight 1, subtotal 41. Case total $16 + 24 + 32 + 41 = 113$.

Case $|X_3| = 1$ (4 choices): the chosen triple forces its three doubletons and its three singletons; the free doubletons are the three containing the fourth vertex d, and d becomes forced exactly when at least one of them is chosen. Sum over the 2^3 subsets: the empty choice leaves d free ($2^1 = 2$); each of the 7 nonempty choices forces d ($2^0 = 1$): $2 + 7 = 9$ per triple, subtotal 36.

Case $|X_3| = 2$ (6 choices): two triples share two vertices and force $3+3-1 = 5$ doubletons and all four singletons; one doubleton is free: 2 per choice, subtotal 12.

Case $|X_3| = 3$ (4 choices): the union of doubletons inside three triples is all six; everything forced: 1 each, subtotal 4.

Case $|X_3| = 4$: everything forced, 1.

Total without the top and nonempty: $113 + 36 + 12 + 4 + 1 = 166$; adding the empty family and the full family, $\delta(4) = 166 + 2 = 168$. □

THEOREM 7.47 (The 298-sieve census). *The total number of sieves over all sixteen objects of L is*

$$\sum_{V \in L} \delta(|V|) = 1 \cdot \delta(0) + 4 \cdot \delta(1) + 6 \cdot \delta(2) + 4 \cdot \delta(3) + 1 \cdot \delta(4) = 2 + 12 + 36 + 80 + 168 = 298.$$

PROOF. Combine Definition 7.45 (sieves on V are down-closed families in $\mathcal{P}(V)$, so their number is $\delta(|V|)$), the class sizes of Section 2, and Lemma 7.46. □

REMARK 7.48 (Reconciliation with the certificate, and a numerical coincidence). The exact-topology verifier of Chapter 11 reports precisely 298 sieves enumerated over the sixteen contexts, matching Theorem 7.47; the derivation here shows the number is forced by the order structure alone. One caution for readers checking figures: the number 168 occurs twice in the registered artifacts with two different meanings. Here, $\delta(4) = 168$ is the number of sieves on the single object Ω. Separately, the certificate reports 168 *covering* sieves summed over all sixteen contexts for the joined topology—a count that belongs to Chapter 9 (where it is derived after Theorem NT2-4) and involves a different computation that happens to land on the same value. The two 168s are unrelated; conflating them would misread both artifacts.

11. Catalog of degenerate and boundary cases

Finite exhaustive verification lives or dies on its boundary cases: the configurations where a general formula degenerates and an implementation is most likely to be wrong. The combinatorics above surfaced all of them, and we catalog them in one place, both because Chapter 9's proofs must handle each explicitly and because Chapter 11's certificate checks are organized around exactly these configurations.

(1) **The empty context as target.** The generator at $\varnothing$ is the null family (Definition 7.28), not the formula family $\{\varnothing \cap B \to \varnothing\}$—the formula would produce a two-member family of identity arrows, which generates the maximal sieve, whereas the registered convention is the empty family. The two conventions differ, and the difference is semantically loaded: under the null-cover convention, sheaves are forced to take singleton value at $\varnothing$ (Chapter 9), which drives the witness computations and the reflector formula of Chapter 10. An implementation that "simplifies" the empty case to the formula family changes the sheaf category.

(2) **The empty context as pullback result.** Pulling any generating family back along $\varnothing \subseteq V$ produces identity arrows on $\varnothing$ (Proposition 7.33(2)); the basis nevertheless covers $\varnothing$ by the null family. Both facts hold simultaneously and neither replaces the other.

(3) **Empty pieces inside nonempty covers.** At every singleton context, and at the block contexts of the opposite partition, one piece of the generating family is empty (Table 4). Empty pieces are retained in the family, contribute only the arrow $\varnothing \to V$ to generated sieves, and are preserved verbatim by the serialization of Chapter 8.

(4) **Identity pieces inside generating covers.** At singletons and at own-partition blocks, one piece equals the whole context (rows $\{\omega_{ij}\}$, A_i, R_j), so the generated sieve is maximal there. Propriety of the generators is a fact about the *global* context (and, more broadly, about the nine contexts meeting both blocks of the relevant partition), not about all sixteen; Chapter 9 is precise about this quantification.

(5) **Coincidence loci.** At $\varnothing$, the four singletons, and the two diagonals, the unmarked ACBE and RPM families are equal as arrow sets (Proposition 7.31); at $D_{\neq}$ they are equal as sets but crossed as indexed families. Any code that compares families must specify whether it compares indexed families or arrow sets, and the two answers differ exactly at $D_{\neq}$.

(6) **Blocks of the two partitions never coincide.** $A_i \neq R_j$ always (Corollary 7.22), so cover-kind information can never be recovered from the block sets at block contexts—only the marked layer carries it.

(7) **The two meanings of 168.** The count of all sieves on Ω and the count of covering sieves for the joined topology across all contexts coincide numerically (Remark 7.48); any reconciliation of certificate figures must keep them apart.

Remark 7.49 (Degeneracy is where the discipline pays). None of the seven items is deep, and that is the point: each is a place where an informal argument would wave and a checker cannot. The registered proofs in Chapter 9 cite this catalog at the corresponding steps (null cover in the subcanonicity and witness proofs; empty pieces in the basis lemma; maximality loci in the propriety proposition; coincidence loci in the marked-cover chapter). The catalog is also the right reading order for the certificate's check list: most checks are one catalog item made executable.

12. Summary, and what has deliberately not been done

The chapter's yield, in one place:

(1) $L = \mathcal{P}(\Omega)$ with $\Omega = \{0,1\} \times \{0,1\}$ is a sixteen-object poset category with all pullbacks, computed as intersections (Proposition 7.4).
(2) The sixteen contexts fall into five cardinality classes—$1 + 4 + 6 + 4 + 1$—with the six doubletons splitting into two ACBE blocks, two RPM blocks, and two diagonals (Definition 7.9, Table 1).
(3) There are exactly $81 = 3^4$ arrows, by summation and by the atomwise neither/V-only/both trichotomy (Theorem 7.12); 16 identities, 65 proper inclusions, 256 composable pairs.
(4) The coordinate partitions A and R are transverse: $A_i \cap R_j = \{\omega_{ij}\}$, verified in all four cases (Theorem 7.21); their common refinement is the singleton partition (Corollary 7.23); an explicit order-8 symmetry group exchanges and preserves the structure, with the coordinate swap interchanging the ACBE and RPM sides (Section 5).
(5) The generating families $K_A(V)$, $K_R(V)$ are computed at every context (Table 4), with per-class shapes (Proposition 7.29), joint exhaustiveness (Proposition 7.30), and exact coincidence loci of the unmarked families: the empty context, the four singletons, and the two diagonals (Proposition 7.31).
(6) Pulling a generating family back along any inclusion yields the generating family of the subcontext, block by block, in every context class, with the empty-context degeneration handled by identity arrows and the null cover (Proposition 7.33 and the case walk of Section 7).
(7) Same-partition refinement is absorbing; cross-partition refinement produces exactly the singletons of the context, worked at the global context and at a triple (Section 8).
(8) The block-completeness census: α and ρ vanish on the empty context, singletons, and diagonals; equal 1 on own-blocks and on every triple; equal 2 at Ω; and cross-partition blocks never contain complete blocks (Theorem 7.42, Table 5).
(9) The sieve census: 298 sieves in total across the lattice, derived from first principles (Theorem 7.47).

Claim Boundary 7.50. Nothing in this chapter constructs a Grothendieck topology, proves a sheaf condition, or licenses any statement about modalities: those results are Chapter 9's, where each carries its registered ledger status (`INFORMAL_PROOF_PRESENT`, `IN_REVIEW`, `NOVELTY_UNASSESSED`, Ring 4). Equally, nothing here interprets the ACBE or RPM coordinate empirically; the marks are formal (Boundary 7.1). The chapter's claims are finite set-theoretic identities and counts, each derived above and each independently corroborated by the executable finite certificate discussed in Chapter 11—and, per the certificate's own claim boundary, exhaustive execution over one sixteen-object site is not proof-assistant verification, external peer review, empirical validation, or a product-claim license.

Exercise 7.51. Show that for $|\Omega| = m$ with two transverse partitions into blocks of sizes p and q (so $m = pq$ and each pair of cross-blocks meets in one atom), the analogue of Corollary 7.22 holds whenever $p, q \geq 2$, and fails when $p = 1$ or $q = 1$. Conclude that the four-atom model is the smallest in which both partitions have only proper blocks—the property Chapter 9 turns into nonmaximality of the generated sieves.

Exercise 7.52. Verify from Table 4 that the unmarked families $K_A(V)$ and $K_R(V)$ coincide exactly on the seven contexts $\varnothing$, the four singletons, and the two diagonals, and use

the block-completeness census to show these are exactly the contexts V with $\alpha(V) = \rho(V) = 0$. (This equivalence is special to the 2×2 model; formulating the right general statement is part of the exercise.)

Exercise 7.53. Show that a nonempty sieve (in the down-set reading) on any context necessarily contains the bottom arrow $\varnothing \to V$, and hence that the sieves *not* containing it are exactly the sixteen empty sieves, one per context. Using Theorem 7.47, conclude that exactly $298 - 16 = 282$ of the sieves contain their bottom arrow. Then explain why, of the sixteen empty sieves, only the one at the empty context is ever *covering* for the topologies of Chapter 9 (via the null cover), while the other fifteen never are.

CHAPTER 8

Marked Covers and the Provenance Calculus

Chapter 5 proved a negative result: the legacy ACBE and RPM cover clauses generate only maximal sieves, so the topology they define collapses onto the jointly-surjective coverage and retains no trace of which clause admitted which family. Chapter 7 then built a finite context factor whose generating families escape that collapse—the new generators contain no identities and no split epimorphisms. But a second, subtler information loss survives the repair, and no choice of generators can remove it: a Grothendieck topology, *by its nature*, forgets why a cover was admitted. Sieve generation is a closure operator; closure operators erase history. The successor candidate therefore does not ask the topology to remember provenance. It maintains provenance in a parallel data structure—the *marked cover*—whose design, serialization, derivation calculus, verification algorithm, and failure discipline are the subject of this chapter.

The chapter is deliberately positioned between the combinatorics of Chapter 7 and the theorems of Chapter 9. The marked-cover mechanism is not a theorem about topoi; it is an executable audit invariant that travels alongside the mathematics. Keeping that distinction sharp—in both directions—is itself one of the registered claim boundaries, and Section 12 states it in full.

Claim Boundary 8.1. Everything in this chapter concerns the v0.1 executable reference schema, which is deliberately narrower than the ambient research construction of Chapter 12: it represents ACBE/RPM context-factor families and their joint refinement only. Typed observational covers in the legacy $\mathcal{C}$ coordinate, general arrows of $\mathcal{C} \times L$, full product-site covers, and their serialization remain OPEN. The artifact status is DRAFT / GATE 1 RESEARCH CANDIDATE / RING 4 / NOT LICENSED; the ACBE and RPM cover-kind labels are formal marks with no empirical interpretation.

1. Why topologies forget provenance

We recapitulate the loss mechanism precisely, because the design of the marked record is dictated point-for-point by what is lost.

1.1. Sieve generation is a closure operator. Fix a small category $\mathcal{C}$ and an object V. The sieves on V form a complete lattice under inclusion, and for any set $\mathcal{F}$ of arrows into V there is a smallest sieve $\langle \mathcal{F} \rangle$ containing $\mathcal{F}$: the set of all arrows into V that factor through some member of $\mathcal{F}$ [1, 7]. The assignment $\mathcal{F} \mapsto \langle \mathcal{F} \rangle$ is a closure operator: extensive, monotone, idempotent. Three consequences matter here.

First, *many families generate one sieve.* The map $\mathcal{F} \mapsto \langle \mathcal{F} \rangle$ is far from injective: adding to $\mathcal{F}$ any arrow that already factors through a member changes nothing; deleting a member that factors through another member changes nothing; reindexing changes nothing, since a sieve is a bare set of arrows. Whatever distinguished two generating families—their indexing, their origin, their intended reading—is unrecoverable from the sieve.

Second, *a topology stores even less.* A Grothendieck topology J assigns to each object a set $J(V)$ of covering sieves satisfying the maximality, stability, and transitivity axioms [3]. The data structure is a subset of the sieve lattice at each object—nothing more. There is no slot in the definition where an annotation could live. If two different coverage clauses contribute the same sieve, the topology contains that sieve once, with no record of double parentage. And saturation under the axioms adds further covering sieves that no clause contributed directly, orphaned by construction.

Third, *the finite model realizes the loss concretely.* Proposition 7.31 (Chapter 7) computed the coincidence loci exactly: at each of the four singleton contexts, and at both diagonal doubletons, the unmarked ACBE family and the unmarked RPM family are literally the same set of arrows. At those seven contexts (including $\varnothing$, where both are the null family), no topology generated from these inputs can distinguish the ACBE contribution from the RPM contribution, because the inputs are indistinguishable before generation even begins.

Remark 8.2 (The legacy collapse was the extreme case). The Chapter 5 collapse can be re-read in this light. There, the ACBE clause contained an identity and the RPM clause a split epimorphism, so both clauses generated the *maximal* sieve—the top of the sieve lattice, which belongs to every topology by axiom (J1). Provenance was not merely forgotten; the contributed content was indistinguishable from no contribution at all. The successor's proper generators (Chapter 7) repair the collapse: at the global context both generated sieves are nonmaximal and distinct. But the singleton coincidence shows the repaired construction still forgets provenance wherever the input families coincide, and the closure-operator argument shows that some forgetting is intrinsic. The design conclusion, registered in the governing specification: *the ordinary Grothendieck topology intentionally forgets a tag after generation; the parallel provenance record must not.*

1.2. What governance needs that the topology cannot supply. The Project Coherence audit discipline requires, for every cover admitted into any registered construction: which kind of cover it is (its clause of origin); on whose authority it was admitted; under what assumptions; with what evidence or proof references; in what review state; and, for derived covers, from which parent covers it was derived, with tamper-evident linkage. None of these fields exists in a Grothendieck topology, and Section 1 shows none can be reconstructed from one. Hence the two-representation design.

2. The two-representation design

Construction 8.3 (Unmarked families and marked covers). The v0.1 reference model maintains, for each admitted cover, two representations with disjoint consumers:

(1) the *unmarked family*: the exact list of arrows consumed by topology generation—for the finite context factor, an indexed family $\{V \cap B \to V\}$ as computed in Chapter 7, or the null family at $\varnothing$;
(2) the *marked cover*: an immutable record

$$(\texttt{derivation_kind}, \texttt{target}, \texttt{exact_member_arrows}, \texttt{COVER_MARK_RECORD}, \texttt{parent_cover_sha256s})$$

carrying the same arrows together with the provenance fields of Section 3, serialized canonically and bound by a SHA-256 digest.

Invariant 8.4 (Consumer separation). The topology consumes the unmarked family and nothing else; audit, migration, and governance code consume the marked record and

nothing else. No code path derives topological structure from mark fields, and no code path reconstructs mark fields from topological structure.

The invariant has a mathematical face and an engineering face. Mathematically, it guarantees that the theorems of Chapters 9 and 10 are theorems about ordinary sites: erasing every mark leaves the sites, sheaves, and modalities untouched, so no proof secretly depends on provenance data, and independent reviewers can check the mathematics against the standard literature [**1**, **4**] without reference to the serialization layer. In the other direction, it guarantees that provenance claims never rest on topological reconstruction, which Section 1 showed to be impossible at the coincidence loci.

REMARK 8.5 (Both representations name the same arrows). The marked record's `exact_member_arrows` field repeats, verbatim, the arrow list of the unmarked family—including retained empty pieces. This redundancy is deliberate: it is what allows the parent-resolution algorithm (Section 6) to *recompute* a derived record's members from its parents and compare against the stored list, converting the mathematical determinism of pullback and refinement (Chapter 7, Sections 7.7–7.8) into an executable integrity check.

3. The marked-cover schema, field by field

We now specify the record. The schema has two layers: the outer `MARKED_COVER`, which fixes what the cover *is* and where it came from derivationally, and the inner `COVER_MARK_RECORD`, which fixes its governance metadata. In the registered notation:

`MARKED_COVER`	`COVER_MARK_RECORD`
`derivation_kind`	`cover_kind`
`target`	`source_authority_ref`
`exact_member_arrows`	`assumptions`
`COVER_MARK_RECORD`	`evidence_or_proof_refs`
`parent_cover_sha256s`	`review_state`
	`provenance_chain`

3.1. `derivation_kind`**.** One of exactly three controlled values:

- `GENERATOR` — the record describes a primitive generating cover: one of the families $K_A(V)$ or $K_R(V)$ of Chapter 7 (or the null cover), admitted directly from the registered specification rather than derived from other covers. Generator records have an empty parent set.
- `PULLBACK` — the record was derived from a single parent cover by pulling back along an inclusion $U \to V$, in the sense of Proposition 7.33. Exactly one parent digest is required.
- `COMMON_REFINEMENT` — the record was derived by the cross-partition refinement of Lemma 7.36: refining an ACBE cover by RPM covers of its pieces (equivalently, and verified equivalently, the other order). Exactly two parent digests are required, one from each family, with the multiplicity rule of Section 5.

The derivation kind is not decoration: the parent-resolution algorithm branches on it, and each kind has its own recomputation obligation and its own rejection conditions.

3.2. `target`**.** The context $V \in L$ that the cover covers, serialized as its atom set. For a `PULLBACK` record the target is the subcontext U; for a `COMMON_REFINEMENT` record it is the common target of both parents (the record is rejected if the parents' targets differ).

3.3. `exact_member_arrows`**.** The complete list of member arrows, each an ordered pair (domain, target), with empty pieces retained exactly as in Definition 7.28. "Exact" is normative: the list is neither normalized by dropping empty pieces nor deduplicated, because it must equal, arrow for arrow, what the corresponding unmarked family feeds to topology generation, and—for derived records—what recomputation from the parents produces.

3.4. `COVER_MARK_RECORD.cover_kind`**.** The clause of origin: for the v0.1 context-factor schema, `ACBE` or `RPM` for the partition generators, the joint kind for common refinements, and, for a `PULLBACK`, whatever kind its parent carries. This is the field the topology cannot carry (Section 1) and the reason the whole mechanism exists. The successor architecture reserves the kind `OBSERVATIONAL` for legacy-coordinate covers, but—per Boundary 8.1—observational cover records are OPEN and not part of the executable v0.1 schema.

3.5. `COVER_MARK_RECORD.source_authority_ref`**.** A reference identifying the registered authority under which the cover was admitted: for the v0.1 generators, the governing specification section that defines K_A and K_R; for derived records, the derivation rule invoked. The field makes "who says this is a cover" a first-class, digest-bound datum rather than an ambient assumption.

3.6. `COVER_MARK_RECORD.assumptions`**.** The explicit assumptions under which the admission is valid—for the context-factor generators, for instance, that arrows of L are inclusions, that pullbacks are intersections, and that the empty family covers the empty context. Recording assumptions in the record itself means a future change of ambient convention cannot silently re-validate old covers under new rules: the old records name the old rules.

3.7. `COVER_MARK_RECORD.evidence_or_proof_refs`**.** References to the proof artifacts and executable evidence backing the admission—for v0.1, the specification's theorem sections and the finite certificate. The field does not contain the evidence; it contains stable references. The distinction between the record and the raw evidence bytes is governed by the Evidence Packet boundary of Section 12.

3.8. `COVER_MARK_RECORD.review_state`**.** The record's position in the registered review lifecycle. The vocabulary is registry-controlled; values encountered in the ledger artifacts of this program include `IN_REVIEW` and `UNASSESSED`. The deserializer rejects any record whose review state is not in the controlled vocabulary (Section 7); in particular, a record cannot claim a review state that the registry does not define, and—because the digest binds the field—cannot have its review state altered without changing its identity.

3.9. `COVER_MARK_RECORD.provenance_chain`**.** The ordered chain of derivation steps that produced this record, each step naming its parent digest(s). For a generator the chain is empty; for a pullback it extends the parent's chain by one step naming the parent digest; for a common refinement it records both generating parents. The chain duplicates information recoverable from recursive parent resolution, and that duplication is intentional: it lets an auditor read a record's full history without resolving parents, while the resolver (Section 6) independently verifies that the chain and the parent graph agree.

3.10. `parent_cover_sha256s`**.** The set of SHA-256 digests of the immediate parent records, stored *sorted*. Sorting matters for canonicalization: a set has no intrinsic order, so a deterministic serialization must impose one, and lexicographic order on the hex digests is the registered choice. `GENERATOR` records carry the empty set; `PULLBACK` records exactly one digest; `COMMON_REFINEMENT` records exactly two.

4. Canonical JSON and the digest binding

Construction 8.6 (Canonical serialization and digest). Every marked cover is serialized to *canonical JSON*: a deterministic byte encoding in which object keys appear in a fixed sorted order, primitive values have a single admissible encoding, and no insignificant whitespace or alternative escapings are permitted—so that equal records produce equal bytes and distinct records produce distinct bytes [17]. The record's digest is the SHA-256 hash of its canonical bytes. The digest therefore binds, jointly and inseparably: the derivation kind, the target, the exact member arrows, every field of the mark record, and the sorted set of parent digests.

4.1. What canonicalization must guarantee. "Canonical" is a load-bearing word, and the schema's rejection rules (Section 7) are partly consequences of it. The serialization must have four properties, each of which closes a specific integrity hole.

(1) **Determinism.** Equal records yield byte-identical serializations, across implementations, platforms, and runs. Without determinism, a digest names an accident of serialization rather than a record, and the round-trip protocol below is unsatisfiable. Determinism is why object keys are emitted in a fixed sorted order and why the parent digest set is stored sorted: JSON objects and mathematical sets are unordered, and every unordered thing that enters the bytes must first receive a canonical order.

(2) **Injectivity on records.** Distinct records yield distinct byte strings. This holds when every field has exactly one admissible encoding—no optional whitespace, no alternative number formats, no equivalent escape sequences. Injectivity is what lets digest inequality certify record inequality in the singleton-coincidence argument of Section 9.

(3) **Unambiguous parseability.** The bytes parse back to exactly one record. Duplicate keys are the classical violation: JSON parsers disagree on whether the first or last duplicate wins, so a byte string with duplicate keys can denote different records to different consumers while carrying one digest. The schema does not adjudicate the ambiguity; it rejects the bytes (Table 1).

(4) **Schema closure.** The bytes contain nothing but the declared fields. Unknown keys are rejected rather than ignored because an ignored-but-digested field is a covert channel: two records could differ in a field no consumer reads yet carry different digests, or—worse under a sloppier digest scope—agree in digest while differing in content a future consumer might read.

Properties 1–3 above are the normative content of the JSON Canonicalization Scheme (RFC 8785) [17], which the v0.1 serializer follows; nothing in those three is claimed as new. Property 4 is not supplied by RFC 8785, which is schema-agnostic and delegates content validation to the consuming ecosystem: schema closure is a Project Coherence schema rule, added here together with the domain-specific field set of Section 3 and the sorted-digest rule for parent sets.

Remark 8.7 (Digests as names). Under these four properties, the digest is a *name*: a fixed-length, location-independent, collision-resistant identifier that any party can recompute from the record itself. All cross-record references in the schema—parent sets, provenance chains—are by digest, never by file path, database key, or human-readable label. This is what makes the derivation DAG portable: a population of records can be scattered across stores and reassembled by any resolver, because the reference graph is intrinsic to the records.

Invariant 8.8 (Immutability). A marked cover is immutable: any change to any bound field yields a different canonical byte string and hence, up to SHA-256 collision, a different

digest. There is no update operation on marked covers; there is only the creation of new records, which—if derived—name their parents by digest.

Remark 8.9 (Hash chaining and tamper evidence). Because the parent digests are themselves bound into the child's digest, the records form a hash-chained directed acyclic graph, with generators as sources: the same discipline that makes append-only logs tamper-evident. Altering any ancestor record changes its digest, which invalidates the stored parent reference in every descendant, which the parent-resolution algorithm detects as a missing or mismatched parent. Tampering anywhere in a derivation's past is thus visible at its present, provided resolution is actually run—which is why the schema makes resolution mandatory at deserialization time rather than optional at audit time.

Claim Boundary 8.10. The cover digest is the canonical *semantic-record* digest only: it certifies the integrity of the marked cover's own fields and derivational ancestry. It does not certify raw ingress evidence. An enclosing Evidence Packet must separately bind the raw bytes of any external material referenced by `evidence_or_proof_refs`; conflating the two digests would let a record vouch for bytes it never saw. This division is registered in the governing specification and restated in Section 12.

5. The derivation calculus

Derived records are not free-form: exactly two derivation rules exist in v0.1, mirroring the two combinatorial operations verified in Chapter 7. We state them below, preceded by the generator-admission rule they build on—three rules in all—each with its preconditions and the content of its recomputation obligations.

Rule 8.11 (Generator admission). A `GENERATOR` record for partition $P \in \{A, R\}$ at nonempty context V has target V, member arrows exactly $\{V \cap B \to V \mid B \in P\}$ with empty pieces retained (Definition 7.28), cover kind the partition's kind, empty parent set, and empty provenance chain. The null-cover record at $\varnothing$ is the degenerate generator with the empty member list. A generator record whose member list disagrees with the computation of Table 4 of Chapter 7 for its declared target and kind is invalid.

Rule 8.12 (Pullback derivation). Given a valid record c with target V and an inclusion $U \subseteq V$, the pullback record $c|_U$ has: derivation kind `PULLBACK`; target U; member arrows obtained by intersecting each member domain with U (Proposition 7.33), retaining empties; mark record inheriting the parent's cover kind, with the provenance chain extended by a step naming the parent digest; and parent set $\{\,\mathrm{sha256}(c)\,\}$. By Chapter 7 the members of $c|_U$ are exactly the generating family $K_P(U)$ for the partition $P \in \{A, R, Q\}$ underlying the parent (Q, the singleton partition, for a joint parent)—the executable checker verifies this as `marked_pullbacks_preserve_topology_family`—and the parent linkage as `marked_pullbacks_bind_parent_provenance`.

Rule 8.13 (Common refinement derivation). Given a valid ACBE generating record a and a valid RPM generating record r with the same target V, the common-refinement record $a \wedge r$ has: derivation kind `COMMON_REFINEMENT`; target V; member arrows the pairwise intersections $\{V \cap A_i \cap R_j \to V\}$, which by transversality (Theorem 7.21 and Lemma 7.36) are exactly the singleton pieces $V \cap \{\omega_{ij}\} \to V$; a mark record carrying the joint kind with both parent digests in the provenance chain; and parent set $\{\mathrm{sha256}(a),\ \mathrm{sha256}(r)\}$, sorted. The rule's precondition is strict: *exactly one* ACBE generating parent and *exactly one* RPM generating parent. A putative join with two ACBE parents, two RPM parents, one parent, three parents,

or parents of undeclared kind does not satisfy the rule and is rejected at resolution. At the global context the members are the four singletons with no empty pieces (Example 7.37); the checker records this as `joint_refinement_is_four_singletons` and the double parentage as `joint_refinement_retains_both_parent_digests`.

REMARK 8.14 (Why the parent multiset is load-bearing). The one-ACBE-plus-one-RPM precondition is not bureaucratic fussiness. The common refinement is the point where the two coordinate topologies interact—in Chapter 9 it computes the join $J_A \vee J_R$, and in Chapter 10 it underlies the composition theorem. A derivation record claiming joint status must therefore witness genuine two-sided parentage. Allowing, say, two ACBE parents would let a record assert jointness that the mathematics does not have (a same-partition refinement is absorbing, by Lemma 7.35, and yields no singletons); the rule makes such an assertion unserializable rather than merely discouraged.

REMARK 8.15 (The derivation graph). Under the three rules, any population of records forms a DAG whose sources are generators, whose unary edges are pullbacks, and whose binary joins are common refinements. Acyclicity is not an assumption but a checked property (Section 6): a cycle would require a record to be its own ancestor, hence its digest to depend on itself, which the resolver detects structurally rather than trusting cryptographic infeasibility alone.

6. Parent resolution at deserialization

Serialization is trivial; the discipline lives in *de*serialization. A marked record arriving from storage or transit is untrusted bytes. The v0.1 rule: a derived record may not be accepted—not logged, not displayed, not counted as evidence—until a parent resolver has reconstructed and verified its entire derivation. The certificate check `marked_derived_records_resolve` exercises exactly this path.

ALGORITHM 8.16 (Parent resolution). *Input:* a candidate record c and a resolver mapping digests to available records. *Output:* acceptance of c, or rejection with a taxonomy code (Section 7).

(1) **Syntactic validation.** Parse the canonical JSON. Reject on any unknown key, any duplicate key, any missing required field, or any field value outside its controlled vocabulary (including `derivation_kind` and `review_state`).
(2) **Digest check.** Recompute the canonical bytes of c and their SHA-256; reject if the recomputed digest disagrees with any externally supplied identity for c.
(3) **Kind arity.** Check the parent set arity against the derivation kind: zero parents for `GENERATOR`, one for `PULLBACK`, two for `COMMON_REFINEMENT`. Reject on mismatch.
(4) **Parent fetch and rehash.** For each digest d in `parent_cover_sha256s`: obtain the parent record from the resolver; reject if absent (missing parent). Serialize the fetched parent canonically, rehash, and reject unless the hash equals d (parent digest mismatch). *Every* parent is rehashed; no digest is trusted on the resolver's say-so.
(5) **Recursive resolution and acyclicity.** Resolve each parent by recursive application of this algorithm, maintaining the set of digests on the current descent path; reject if a parent's digest already occurs on the path (cycle). Recursion terminates at `GENERATOR` records, which are validated against Rule 8.11.
(6) **Recomputation.** Recompute what c ought to contain from its verified parents: for a `PULLBACK`, apply Rule 8.12 to the parent and the declared target; for a

`COMMON_REFINEMENT`, check the parents are exactly one ACBE and one RPM generating record with the common target, and apply Rule 8.13. Reject unless the recomputed member arrows equal the stored `exact_member_arrows` arrow-for-arrow and the recomputed mark (cover kind, provenance chain) equals the stored mark.

(7) **Accept.** Only a record surviving all six stages is treated as a marked cover.

REMARK 8.17 (What recomputation buys). Stage 6 is the point where Chapter 7's determinism becomes a security property. Because pullback members are *functions* of the parent and the target (intersection with U), and refinement members are *functions* of the two parents (pairwise intersection, landing on singletons by transversality), a derived record contains no degrees of freedom beyond its parents and its declared operation. The resolver therefore does not check the stored members for plausibility; it checks them for *equality* with an independent recomputation. A record that stored the right digest linkage but the wrong arrows—or the right arrows but a laundered cover kind—fails stage 6 even though every hash verifies.

PROTOCOL 8.18 (Round-trip stability). For every record: serialize, deserialize, re-serialize; the two byte strings and the two digests must be identical, and the resolved record must be field-for-field equal to the original. The executable certificate registers this as `marked_cover_roundtrip_stable`. Round-trip stability is what entitles later artifacts to treat a digest as a stable name for a record across process and storage boundaries.

7. The fail-closed rejection taxonomy

The v0.1 integrity rules are *fail-closed*: every anomaly is a rejection, and a rejected record contributes nothing—it is not evidence, not a warning-annotated cover, not a partially trusted object. The registered formulation is blunt: a join without exactly one ACBE and one RPM generating parent, or a pullback without its exact parent, is *rejected rather than treated as evidence*. Table 1 organizes the rejection classes; each row names the anomaly, the algorithm stage that detects it, and what accepting it would have corrupted.

TABLE 1. Fail-closed rejection taxonomy for marked-cover deserialization.

Rejection class	Detected at	What acceptance would corrupt
Unknown JSON key	stage 1	schema closure; smuggled fields
Duplicate JSON key	stage 1	canonical-bytes uniqueness
Missing required field	stage 1	record completeness
Uncontrolled `review_state`	stage 1	review-lifecycle integrity
Uncontrolled `derivation_kind`	stage 1	derivation calculus closure
Self-digest mismatch	stage 2	record identity
Parent arity mismatch	stage 3	derivation-rule preconditions
Missing parent	stage 4	ancestry completeness
Parent digest mismatch	stage 4	hash-chain integrity
Cycle in parent graph	stage 5	well-foundedness of derivations
Invalid generator members	stage 5	anchoring in Table 4 of Chapter 7
Wrong parent kinds for join	stage 6	one-ACBE-one-RPM rule
Member recomputation mismatch	stage 6	arrow-exactness
Mark recomputation mismatch	stage 6	provenance faithfulness

Remark 8.19 (Why fail-closed, and why here). The alternative—accept with a warning, quarantine, best-effort repair—reintroduces exactly the ambiguity the mechanism exists to remove. A provenance record that might be wrong is not a weaker provenance record; it is no provenance record, because the consumer of provenance is an audit trail whose value is its trustworthiness under adversarial review. Fail-closed rejection also composes: since acceptance requires full recursive resolution (stage 5), an anomaly anywhere in an ancestry poisons every descendant, which is the correct propagation for tamper evidence (Remark 8.9). The cost is availability—a lost parent makes its descendants unusable—and the design accepts that cost explicitly: in this artifact class, silent degradation is the failure mode being engineered against, not an acceptable fallback.

7.1. Three rejection walkthroughs. Abstract taxonomies invite the question of what a rejection actually looks like. Three miniature scenarios, each exercising a different stage of Algorithm 8.16.

Example 8.20 (A tampered ancestor). An auditor holds the three-generation ancestry of Example 8.30: generators g_A, g_R, join j, pullback $p = j|_{T_{00}}$. Suppose the stored copy of g_R is altered after the fact—say its `review_state` is edited in place to a more favorable controlled value. The edit changes g_R's canonical bytes, hence its digest. Resolving p: stage 4 fetches j's parents by the digests stored *in* j; the resolver returns the altered g_R; rehashing yields a digest different from the stored d_R; rejection (parent digest mismatch), propagating to p. Note what did *not* help the tamperer: the altered field was within the controlled vocabulary (stage 1 passes) and the member arrows were untouched (stage 6 would pass); the hash chain alone catches the edit. Note also the fail-closed cost: the honest descendants j and p are now unusable until an untampered g_R is restored—by design.

Example 8.21 (A laundered join). A record j' claims `COMMON_REFINEMENT` at Ω with parent set $\{d_A, d'_A\}$, where d_A resolves to the ACBE `GENERATOR` record for $K_A(\Omega)$ and d'_A to a validly derived `PULLBACK` of it (hence also of ACBE cover kind, but at a proper subcontext). Stage 3 passes: two parents. Stages 4–5 pass: both parents exist, rehash, and resolve. Stage 6 fails on two independent counts: the recomputation precondition requires exactly one ACBE and one RPM *generating* parent, and it requires both parents to share the record's target Ω; here the parent kinds are (ACBE, ACBE) and d'_A is neither generating nor targeted at Ω; rejection (wrong parent kinds for join). The mathematical backstop, per Remark 8.14: even setting the kind rule aside, no recomputation from these two parents yields the four singletons the record stores, so the member-recomputation check would reject it independently. The arity-and-kind rule and the member recomputation are redundant defenses, and the schema keeps both.

Example 8.22 (Duplicate-key bytes). A byte string arrives containing the key `review_state` twice, with values `IN_REVIEW` and a second, more advanced state. A lenient parser would silently pick one—and different parsers pick differently, so the "record" denoted by the bytes depends on the consumer. Stage 1 rejects the bytes outright (duplicate JSON key) before any semantic field is examined. The scenario is the canonical argument for parse-level strictness: every later stage operates on a parsed record, so ambiguity at the byte layer would poison every downstream guarantee while leaving every downstream check formally satisfied.

Threat 8.23 (What the mechanism defends against, and what it does not). Defended: post-hoc alteration of any field of any record (digest breaks); re-parenting a derivation (parent digests are bound); laundering a cover kind through a derived record (recomputation of the mark fails); asserting jointness without two-sided parentage (arity and kind checks);

fabricating members not derivable from parents (member recomputation fails); circular self-justification (acyclicity). Not defended: compromise of the registry that defines the controlled vocabularies; collisions in SHA-256 (assumed infeasible, not proved); a fabricated *generator* record that correctly imitates Rule 8.11 for a legitimate family—generators are anchored in the specification's tables, not in cryptography, so their trust derives from the source-verification artifacts, whose raw-byte binding is the Evidence Packet's job (Boundary 8.10).

8. Operational lifecycle

The schema fixes what records are; a brief lifecycle account fixes when they come to exist and who touches them. Four phases, each with its protocol obligations.

Protocol 8.24 (Creation). Generator records are created once, from the registered specification's definitions, at model-construction time—their member arrows computed from Table 4 of Chapter 7, their mark fields populated from the specification's clauses, their digests computed and recorded. Derived records are created only by executing Rule 8.12 or Rule 8.13 against already-validated parents; a derived record is never authored by hand. Creation is the only write operation in the system; there is no update and no delete, only the growth of an append-only DAG.

Protocol 8.25 (Storage and exchange). Records are stored and transmitted as their canonical bytes. Because references are by digest (Remark 8.7), storage layout is immaterial to integrity: records may be co-located, sharded, or duplicated freely, and duplication is idempotent—two copies of the same record have the same digest and are the same record. A store that loses a record loses, with it, the resolvability of all descendants (fail-closed), which is an argument for redundant storage of ancestries, not for weakening resolution.

Protocol 8.26 (Consumption). Every read path begins with Algorithm 8.16; no consumer parses fields out of unresolved bytes. Two consumer classes exist. *Audit* reads records to answer provenance queries—what kind, whose authority, which ancestry—and treats the resolved DAG as evidence about admissions. *Migration* reads records to carry covers across schema or model versions, and must preserve digests of ancestors it re-embeds (a migrated derivation cites the same parents, or explicitly re-derives with new records; it never silently rewrites old ones). Per Invariant 8.4, no consumer feeds mark data into topology generation.

Protocol 8.27 (Review-state evolution). Because records are immutable, a change of review state is a new record superseding an old one, not an edit. The v0.1 model's controlled-vocabulary check guarantees only that stored states are legal values; the workflow by which states advance (and which registry authority may advance them) is governed by the program's review process outside this schema, and nothing in this chapter asserts that any record's state has advanced beyond what the ledger of Chapter 9 reports.

9. The singleton coincidence, in full

We now work the registered example end to end: the context where the topology's forgetting is total and the marked layer's remembering is exact.

Example 8.28 (Marked covers at a singleton context). Fix the singleton context $V = \{\omega_{00}\}$ (the other three singletons are identical up to the flip automorphisms of Proposition 7.26).

Step 1: the unmarked families coincide. From the $\{\omega_{00}\}$ row of Table 4 of Chapter 7:

$$K_A(V) = \{\, V \cap A_0 \to V,\ V \cap A_1 \to V \,\} = \{\, \{\omega_{00}\} \to V,\ \varnothing \to V \,\},$$

$$K_R(V) = \{\, V \cap R_0 \to V,\ V \cap R_1 \to V \,\} = \{\, \{\omega_{00}\} \to V,\ \varnothing \to V \,\}.$$

As bare sets of arrows the two families are equal: one identity arrow and one empty-piece arrow.

Step 2: the generated sieves are equal—and maximal. Both families contain id_V, so both generate the maximal sieve on V (any arrow into V factors through id_V). This is not a defect: singleton contexts are where the partitions bottom out (each singleton lies inside a single block of each partition), and a cover of a context by itself is legitimately trivial. But it means the two topologies of Chapter 9, restricted to what they say at V, receive identical contributions from the two clauses. No downstream construction that sees only J_A and J_R can recover, at V, which clause contributed what.

Step 3: the marked records differ. The two generator records are

$$c_A = \big(\texttt{GENERATOR},\ \{\omega_{00}\},\ [\,\{\omega_{00}\} \to V,\ \varnothing \to V\,],\ m_A,\ \varnothing\big),$$
$$c_R = \big(\texttt{GENERATOR},\ \{\omega_{00}\},\ [\,\{\omega_{00}\} \to V,\ \varnothing \to V\,],\ m_R,\ \varnothing\big),$$

where the mark records m_A and m_R agree in review state and differ in `cover_kind` (`ACBE` versus `RPM`) and in `source_authority_ref` (the ACBE-family clause versus the RPM-family clause of the specification). The records differ in a bound field, so their canonical byte strings differ, so—up to SHA-256 collision—their digests differ:

$$\mathrm{sha256}(c_A) \neq \mathrm{sha256}(c_R).$$

Every downstream derivation that pulls back or refines these records inherits the distinction through its parent digests. Provenance survives exactly where the topology forgets.

Step 4: the executable corroboration. The finite certificate registers this configuration as a matched pair of checks, both passing: `singleton_unmarked_families_coincide` (step 1–2) and `singleton_marked_covers_distinct` (step 3). The pairing is the point: either check alone would be unremarkable; together they exhibit, on one concrete context, the separation of concerns that justifies the entire two-representation design.

Table 2 extends the computation to all four singleton contexts at once; the pattern is uniform, with the full piece supplied by the atom's own blocks and the empty piece by the opposite blocks.

TABLE 2. The singleton coincidence at all four singleton contexts: unmarked members of K_A and K_R (equal as arrow sets in every row), and the mark fields on which the records differ.

V	$K_A(V)$ members (from A_0, A_1)	$K_R(V)$ members (from R_0, R_1)	Records differ in
$\{\omega_{00}\}$	$V,\ \varnothing$	$V,\ \varnothing$	kind, authority, digest
$\{\omega_{01}\}$	$V,\ \varnothing$	$\varnothing,\ V$	kind, authority, digest
$\{\omega_{10}\}$	$\varnothing,\ V$	$V,\ \varnothing$	kind, authority, digest
$\{\omega_{11}\}$	$\varnothing,\ V$	$\varnothing,\ V$	kind, authority, digest

REMARK 8.29 (The diagonal coincidence). Proposition 7.31(3) showed the unmarked families also coincide at the two diagonal doubletons, where both partitions cut the context into its two atoms—there the common sieve is a genuinely proper cover, not the maximal sieve, which makes the forgetting arguably more consequential than at singletons. The marked records again differ in cover kind and digest, by the same argument as Example 8.28, step 3.

The v0.1 certificate registers the singleton case; the diagonal case follows from the same field-inequality reasoning and the Chapter 7 tables, and we record it here as a derived observation, not as a separately certificated check.

Example 8.30 (A derived record with its ancestry). To see the calculus operate end to end, derive the joint refinement at Ω and pull it back—or rather, observe the discipline that the schema imposes on that plan. Let g_A and g_R be the generator records for $K_A(\Omega)$ and $K_R(\Omega)$, with digests $d_A = \mathrm{sha256}(g_A)$ and $d_R = \mathrm{sha256}(g_R)$. Rule 8.13 produces the record

$$j \;=\; \big(\texttt{COMMON_REFINEMENT},\ \Omega,\ [\,\{\omega_{00}\} \to \Omega,\ \{\omega_{01}\} \to \Omega,\ \{\omega_{10}\} \to \Omega,\ \{\omega_{11}\} \to \Omega\,],\ m_j,\ \{d_A, d_R\}\big),$$

whose member list is forced by Example 7.37 and whose parent set is forced by the one-ACBE-one-RPM rule; the sorted order of $\{d_A, d_R\}$ in the serialization is the lexicographic order of the two hex digests, whichever that turns out to be. A resolver receiving j fetches both parents, rehashes both, validates both against Rule 8.11 (their member lists must match the Ω row of Table 4 of Chapter 7), recomputes the pairwise intersections, and compares with the stored four singletons. Now pull j back along $T_{00} \subseteq \Omega$: Rule 8.12 yields a `PULLBACK` record with target T_{00}, parent set $\{\mathrm{sha256}(j)\}$, and members the intersections of the four singletons with T_{00}: three singletons and one empty piece, exactly the family of Example 7.38. Its resolution recursively re-verifies j, and through j, both generators: a three-generation ancestry, every link rehashed, every member list recomputed.

10. Certificate check classes for the marked layer

The finite certificate of Chapter 11 registers, among its checks, eight that exercise the marked layer specifically. We enumerate them here as check classes, each tied to the section of this chapter whose discipline it makes executable; all eight pass in the registered run, and per the certificate's claim boundary, passing is exhaustive execution on this one finite model, not proof-assistant verification or external review.

Check Class 8.31 (`singleton_unmarked_families_coincide`). At each singleton context, the unmarked ACBE and RPM generating families are equal as arrow lists. This is Proposition 7.31(2) made executable, and it establishes the *premise* of the provenance problem: the check must pass for the marked layer to have anything to add (Section 9, steps 1–2).

Check Class 8.32 (`singleton_marked_covers_distinct`). At each singleton context, the marked ACBE and RPM generator records differ—in cover kind and hence in digest. The conclusion of Section 9, step 3. Paired with Check Class 8.31, it exhibits the two-representation separation on one context.

Check Class 8.33 (`marked_cover_roundtrip_stable`). Serialize–deserialize–reserialize yields byte-identical output and digest-identical identity for every record class in the model. Protocol 8.18; the executable form of canonicalization determinism (Section 4, property 1).

Check Class 8.34 (`marked_pullbacks_bind_parent_provenance`). Every pullback record's parent set contains exactly the digest of the record it was derived from, and its provenance chain extends the parent's by the corresponding step. Rule 8.12's linkage clause.

Check Class 8.35 (`marked_pullbacks_preserve_topology_family`). The member arrows of every pullback record equal the generating family, at the subcontext, of the partition $P \in \{A, R, Q\}$ underlying the parent record—the marked-layer echo of Proposition 7.33. This check ties the serialization to the mathematics: a marked pullback is not merely well-linked but combinatorially correct.

Check Class 8.36 (`joint_refinement_is_four_singletons`). The common-refinement record at the global context stores exactly the four singleton member arrows, matching Example 7.37 and the transversality identity. Rule 8.13's member clause.

Check Class 8.37 (`joint_refinement_retains_both_parent_digests`). The join record's parent set contains both generating digests—one ACBE, one RPM—and nothing else. Rule 8.13's arity-and-kind clause, whose adversarial face is Example 8.21.

Check Class 8.38 (`marked_derived_records_resolve`). Every derived record in the model passes the full parent-resolution algorithm: parents fetched, rehashed, recursively resolved, acyclicity confirmed, members and marks recomputed and matched. Algorithm 8.16 end to end.

Remark 8.39 (Division of labor among the checks). The eight checks are not redundant restatements of one property. Checks 8.31–8.32 establish the problem and the solution's minimal claim; 8.33 secures the byte layer; 8.34 and 8.37 secure the reference graph; 8.35 and 8.36 secure the arithmetic content against the Chapter 7 tables; and 8.38 secures the composition of all of the above under the resolver. A future change that broke exactly one layer—say, a canonicalization drift that preserved semantics but changed bytes—would fail exactly the corresponding check, which is what makes the list a usable regression surface rather than a ceremonial one.

11. Design alternatives, and why they were not taken

The two-representation design is not the only conceivable response to provenance loss. Three alternatives present themselves naturally, and the registered design rejects each for stated reasons; recording the rejections guards against re-litigating them.

11.1. Alternative 1: enrich the topology itself. One could try to define a "topology with remembered tags": covering sieves paired with labels, with the axioms extended label-wise. The closure argument of Section 1 shows why this fails *as a topology*: axiom (J3) saturation and superset closure manufacture covering sieves with no unique label ancestry—at the coincidence loci, the same sieve arises from differently-labeled inputs, so the label assignment is either non-functional (a sieve carries a set of labels, and the set degenerates as saturation proceeds) or arbitrary. Whatever such labeled structure is, it is not a Grothendieck topology, and every standard theorem this program relies on—the basis criterion, sheafification's existence and exactness, subcanonicity arguments [**1**, **4**, **5**]—would need re-proof for the enriched notion. The registered design refuses to pay that cost: the mathematics stays standard, the enrichment lives outside.

11.2. Alternative 2: never generate; work with indexed families forever. Since indexed families *do* distinguish K_A from K_R at every context (the block indexing differs even where the arrow sets coincide), one could refuse to pass to sieves at all. But the sheaf condition, the reflector, and the topology join are sieve-level or saturation-level notions; a families-only development either re-derives the sieve theory under another name or forfeits the theorems of Chapters 9 and 10. Moreover the coincidence at $D_{\neq}$ (Proposition 7.31(3)) shows indexed families are a fragile carrier: they distinguish the partitions only through index bookkeeping that ordinary mathematical operations (reindexing, set-image) silently destroy. Governance data deserves a governance carrier, not an indexing convention.

11.3. Alternative 3: internalize marks as topos objects. One could hope to represent marks *inside* a topos—as objects of a slice, a glued topos, or a category of descent data—so that provenance becomes categorical structure. Perhaps something along these lines is possible; the registered position is precisely that no such construction has been carried out, reviewed, or compared against the colored-site and indexed-category literature, and that claiming it without doing the work would be exactly the kind of unlicensed structural claim the corrigendum culture of this program exists to prevent. Hence Boundary 8.41: the mark record is *external* audit data, full stop, with the internalization question left open and explicitly not claimed.

Remark 8.40 (The design in one sentence). Where Alternative 1 would change the mathematics and Alternative 3 would overclaim it, the registered design changes nothing and claims nothing mathematically: the topology is the standard one on unmarked families, and provenance is an engineering invariant with cryptographic integrity—auditable, falsifiable, and outside the theorems.

12. What the mechanism is, and is not

The marked-cover layer sits at the boundary between mathematics and governance engineering, which makes precise boundary language unusually important. The following boundaries are registered in the governing specification and the corresponding paper; this section collects them.

Claim Boundary 8.41 (External audit invariant, not internal structure). The mark mechanism is an external serialization and audit invariant. It is not a proof that a marked cover is an internal object of any topos, nor a categorical semantics for evidence. The sites, topologies, sheaves, and modalities of Chapters 9 and 10 are constructed from the unmarked families alone; the marked records are audit data *about* the construction, held outside it. In the packaging of Chapter 12, the provenance record enters the research object NoeSystem_2 explicitly as an EXTERNAL MARK RECORD—a component of a structured package, not an object or morphism of the candidate topos.

Claim Boundary 8.42 (No colored-site or novelty claim). The literature contains constructions that enrich sites with additional data—colored or labeled covers, indexed categories, fibrations, stacks. The v0.1 marked-cover mechanism makes no claim of membership in, equivalence to, or improvement over any of them: its mathematical relationship to indexed categories and stacks has not received a dedicated novelty review, and the theorem-ledger records that touch the mechanism carry `NOVELTY_UNASSESSED` without exception. Until a dedicated literature search is complete, the mechanism should be cited as a Project Coherence engineering artifact, not as a contribution to the theory of sites.

Claim Boundary 8.43 (The Evidence Packet binds raw bytes; the cover digest does not). The cover digest certifies the marked record's own canonical bytes—the *semantic record*. It does not certify the raw ingress bytes of any external evidence the record references. That binding is the job of the enclosing Evidence Packet, which must separately hash raw inputs. A reviewer who finds a cover record citing, say, a source-verification artifact should expect two independent digests: the cover's semantic digest here, and the raw-byte digest in the packet. Neither substitutes for the other.

Claim Boundary 8.44 (A cover record is not a sheaf object). The v0.1 schema serializes *covers*. An application-level marked *sheaf object* would additionally need at least a

`sheaf_object_ref` and a `context_atom_or_block` field; that separate schema is registered as OPEN and is not part of v0.1. A cover record must not be presented as identifying a sheaf object, and no artifact of this monograph does so.

CLAIM BOUNDARY 8.45 (Scope of the executable schema). Restating Boundary 8.1 with the chapter's machinery in hand: the derivation calculus of Section 5 covers ACBE and RPM context-factor generators, their pairwise common refinement, and pullbacks of any of these along inclusions of L—nothing else. `OBSERVATIONAL` cover records for the legacy coordinate, records over general arrows of $\mathcal{C} \times L$, and product-site cover records are OPEN obligations (Chapter 12), and the fail-closed discipline applies to their absence too: a record claiming an unimplemented kind is rejected at stage 1 as outside the controlled vocabulary, not provisionally accepted as forward-compatible.

13. Summary

A Grothendieck topology cannot remember why a cover was admitted: sieve generation is a many-to-one closure, the topology's data structure has no annotation slot, and the finite model realizes the loss concretely at seven of its sixteen contexts, where the ACBE and RPM generating families are identical as arrow sets (Section 1). The v0.1 response is architectural, not topological: every cover exists in two representations—the unmarked family consumed by topology generation and the immutable marked record consumed by governance—with consumer separation as a standing invariant (Section 2). The marked record binds derivation kind, target, exact member arrows, the six-field mark record, and the sorted parent digests into a canonical-JSON SHA-256 identity (Sections 3–4); derivations are confined to a two-rule calculus whose outputs are recomputable functions of their parents (Section 5); deserialization runs a six-stage resolver that rehashes every ancestor, checks acyclicity, and recomputes members and marks (Section 6); and every anomaly, from an unknown JSON key to a recomputation mismatch, is a fail-closed rejection (Section 7). The singleton coincidence exhibits the whole design in one context: equal families, equal sieves, distinct records, distinct digests, with both facts executable and both passing (Section 9).

Eight certificate check classes make each layer of the discipline separately falsifiable (Section 10); the operational lifecycle confines writes to append-only creation and reads to post-resolution consumption (Section 8); and the rejected design alternatives—an enriched topology, indexed families forever, internalized marks—are recorded with their reasons so they are not silently revisited (Section 11). What the mechanism is not—internal topos structure, a colored-site contribution, a raw-evidence binding, a sheaf-object identifier, or anything beyond the context factor—is registered with equal precision (Section 12).

Chapter 9 now returns to the mathematics: the four finite-context theorems generated from the unmarked families, each stated with full proof and each carrying its ledger record—including the theorem whose title the present chapter has already explained, that the *marked* coverages generate ordinary, unmarked Grothendieck topologies.

CHAPTER 9

The Finite-Context Theorems

This chapter proves the finite-context theory: the four registered theorems NT2-1 through NT2-4, together with the propriety proposition that separates the successor's generators from the legacy collapse. Every combinatorial input was computed in Chapter 7; every provenance question was quarantined into Chapter 8; what remains is the mathematics of sites and sheaves on the sixteen-object lattice $L = \mathcal{P}(\Omega)$, carried out at full length. The associated-sheaf reflectors and the composition theorem—which require constructing sheafification explicitly—are deferred to Chapter 10; the executable corroboration of everything is Chapter 11.

1. Standing of the results and ledger discipline

Each theorem below is registered in the Noetic Topos 2.0 theorem ledger (`NOETIC_TOPOS_2_THEOREMS_v0.1.json`, schema `noetic_topos2.theorem_ledger.v0.1`, as of 2026-08-04, document status DRAFT / GATE 1 RESEARCH CANDIDATE / NOT LICENSED). This monograph reproduces, for each theorem, the ledger record's identity and status fields verbatim: its `id`, statement kind, proof status, review state, novelty status, admission ring, and claim-boundary text. Two disciplines govern the reproduction.

First, *status fidelity*. Every finite-context record carries proof status `INFORMAL_PROOF_PRESENT`, review state `IN_REVIEW`, novelty status `NOVELTY_UNASSESSED`, and admission ring 4. The proofs in this chapter are those informal proofs, expanded; writing them out at monograph length changes none of the statuses. In particular: these are not peer-reviewed theorems, they are not proof-assistant-verified beyond the bounded Lean scope described in Chapter 11, and no novelty in the scholarly sense is claimed for any of them. What entitles the word "theorem" here is the registered statement kind together with the finite, exhaustively checkable character of the claims; what forbids more is the ledger.

Second, *corroboration is not verification*. Each theorem is corroborated by the executable finite certificate (`FINITE_MODEL_CERTIFICATE_v0.1.json`, model NT2-CONTEXT-2X2-TRANSVERSE-PARTITIONS): a dependency-free checker exhausts all sixteen contexts, all sixteen representables, both witness presheaves, and every generating-cover pullback. Where a certificate check name is relevant we cite it. The certificate's own claim boundary governs every such citation: exhaustive execution over one sixteen-object finite context site "is not proof-assistant verification, external peer review, empirical validation, or a product-claim license."

NOTATION 9.1. Throughout, L, Ω, the atoms ω_{ij}, the partitions $A = \{A_0, A_1\}$, $R = \{R_0, R_1\}$, Q (singletons), the generating families $K_A(V)$, $K_R(V)$, $K_Q(V)$ with retained empty pieces, and the null cover at $\varnothing$ are as in Chapter 7 (Definitions 7.2, 7.3, 7.19, 7.28). Presheaves are ordinary small **Set**-valued functors $L^{\mathrm{op}} \to \mathbf{Set}$ in a fixed universe. For a sieve σ on V (equivalently a down-closed family of subsets of V, Definition 7.45) and a presheaf F, a *matching family* for σ is a choice of $x_S \in F(S)$ for each $S \in \sigma$ with $x_S|_T = x_T$ whenever

$T \subseteq S$; for a finite indexed family $\{S_k \to V\}$ a matching family is a choice of $x_k \in F(S_k)$ agreeing on all pairwise intersections $S_k \cap S_l$. The *gluing map* sends $x \in F(V)$ to its family of restrictions. F satisfies the sheaf condition for the family (or sieve) when the gluing map is a bijection onto matching families [1, 4].

2. The coverage-basis lemma

The passage from generating families to a Grothendieck topology runs through the standard notion of a coverage basis (pretopology): a system of families containing identity covers, stable under pullback, and stable under composition, whose generated sieve-collections then satisfy the topology axioms [4, 7]. We verify each basis axiom for each partition, in all cases.

LEMMA 9.2 (Coverage basis). *For each $P \in \{A, R, Q\}$, the system consisting of (i) the families $K_P(V)$ at all nonempty V, (ii) identity families $\{\mathrm{id}_V\}$ at all V, and (iii) the null cover at $\varnothing$, is a pullback-stable coverage basis on L: pullback along any $U \subseteq V$ carries a basis family at V to (a family generating the same sieve as) a basis family at U, and refining the members of a basis family by basis families yields a family whose generated sieve contains a basis-generated sieve.*

PROOF. We take the axioms in turn, with all cases.

Identity covers. Clause (ii) supplies $\{\mathrm{id}_V\}$ at every context by fiat. This is the basis-level source of axiom (J1) below.

Pullback stability, main case. Let $V \neq \varnothing$, $U \subseteq V$, $U \neq \varnothing$. By Proposition 7.33, the pullback of the indexed family $K_P(V)$ along $U \to V$ is, block for block, exactly the indexed family $K_P(U)$: the member $V \cap B \to V$ pulls back to $(V \cap B) \cap U = U \cap B \to U$, using $U \subseteq V$. Chapter 7's Section 7.7 walked this identity through every context class—pullbacks to singletons, to each of the three doubleton kinds, to triples—so the stability holds not merely abstractly but with the pulled-back family computed and identified in each case.

Pullback stability, empty subcontext. If $U = \varnothing$, each member of $K_P(V)$ pulls back to $\varnothing \cap B = \varnothing$, i.e., to $\mathrm{id}_\varnothing$, giving a family of identity arrows on $\varnothing$; independently, clause (iii) covers $\varnothing$ by the null family. Either way $\varnothing$ is covered, and the generated-sieve comparison below is unaffected.

Pullback of identity and null covers. An identity family pulls back to an identity family. The null cover sits at the bottom object, along which the only pullback is via $\mathrm{id}_\varnothing$, fixing it.

Composition, same partition. Refine each member $V \cap B \to V$ of $K_P(V)$ by its own family $K_P(V \cap B)$ (null cover if the piece is empty). By Lemma 7.35, block disjointness collapses the composite: $V \cap B \cap B'$ is $V \cap B$ when $B' = B$ and $\varnothing$ otherwise, so the composite family is the original together with redundant empty pieces. An empty piece $\varnothing \to V$ contributes to a generated sieve only the arrows factoring through $\varnothing$—that is, the single arrow $\varnothing \to V$ itself—and the sieve generated by $K_P(V)$ already contains $\varnothing \to V$, since $\varnothing$ factors through any member (or, when all members are empty, is a member). Hence the composite generates *the same* sieve as the original family, which is stability under composition in the strongest form.

Composition, with identity and null covers. Refining by identity families changes nothing. Refining an empty piece by the null cover contributes no arrows, and the generated sieve again contains everything it contained before. □

REMARK 9.3. The lemma's proof delivered more than the axioms require: pulled-back families are equal (not merely cofinal) to the target families, and same-partition composites

generate equal (not merely finer) sieves. This exactness is what the marked-cover recomputation checks of Chapter 8 rely on, and it is why the certificate can verify pullback stability as a family-level identity rather than a sieve-level inclusion.

Before using the basis, we record why checking sheaf conditions on basis families suffices for the generated topology—the reduction invoked constantly below, made explicit once.

LEMMA 9.4 (Basis check suffices). *Let J be the topology generated by a coverage basis on L as in Lemma 9.2, and let F be a presheaf. If the gluing map of F is a bijection for every basis family (including the null cover), then F is a J-sheaf: its gluing map is a bijection for every J-covering sieve.*

PROOF SKETCH. This is the standard equivalence between the sheaf condition for a coverage and for its generated topology [4, 7]; we indicate the shape of the argument on our finite poset. The covering sieves of J are reachable from basis-generated sieves by three closure steps, and the sheaf condition propagates along each. (i) *Generated sieves.* For the sieve $\langle K\rangle$ generated by a basis family K, matching families for $\langle K\rangle$ restrict to matching families for K, and conversely a matching family for K extends uniquely to $\langle K\rangle$: an arrow $T \to V$ of $\langle K\rangle$ factors through some member S_k, and one defines $x_T = x_{S_k}|_T$, well-defined because two factorizations $T \subseteq S_k$, $T \subseteq S_l$ give $T \subseteq S_k \cap S_l$, where the family's compatibility applies; identity covers make this trivial and the null cover has no arrows to extend over. Hence bijectivity for K gives bijectivity for $\langle K\rangle$. (ii) *Supersets.* Let $\sigma \subseteq \sigma'$ be sieves on V with σ covering, and suppose the gluing map of F is bijective for σ and for the pullback $g^*\sigma$ at every $g \in \sigma'$. A matching family for σ' restricts to one for σ and so glues to a unique candidate section; the candidate restricts correctly on all of σ' because the two sides agree after pullback along the covering sieve $g^*\sigma$ of each $g \in \sigma'$, where gluing is injective. (iii) *Local character.* Sieves admitted by (J3) are handled by the same two-step restriction-extension argument applied along the locally-covering data. On a finite poset every such propagation is a finite computation, and the exact-topology verifier of Chapter 11 checks the end result—sheafhood against every generating cover at all sixteen contexts, for each of J_A and J_R—directly, so the reduction is corroborated as well as cited. (The 168 covering sieves counted in Corollary 9.24 belong to the join $J_A \vee J_R$, not to either factor.) □

3. NT2-1: the generated topologies

THEOREM 9.5 (NT2-1: transverse marked context coverages generate Grothendieck topologies). *The families K_A, with identity families and the null cover, generate a Grothendieck topology J_A on L; likewise K_R generates J_R, and K_Q generates a topology J_{atom}. Explicitly, for each $P \in \{A, R, Q\}$ the collection of sieves is*

$$J_P(V) \;=\; \{\, \sigma \text{ a sieve on } V \;\mid\; \sigma \supseteq \langle K_P(V)\rangle \,\} \quad (V \neq \varnothing),$$

with $J_P(\varnothing)$ every sieve on $\varnothing$; and this collection satisfies the three Grothendieck axioms: (J1) the maximal sieve covers; (J2) covering sieves pull back to covering sieves; (J3) a sieve locally covered by a covering sieve covers.

PROOF. This is the standard passage from a coverage basis to a topology [4], instantiated on Lemma 9.2; we spell out how each axiom is discharged. The displayed description is Theorem 3.20 applied to that basis: a sieve covers when it contains some basis family at V; at nonempty V the only other basis family is the identity family, whose sieve is maximal and so already contains $\langle K_P(V)\rangle$, leaving the single condition $\sigma \supseteq \langle K_P(V)\rangle$; and at $\varnothing$ the null cover is the empty family, which every sieve contains.

(J1) The identity family $\{\mathrm{id}_V\}$ is a basis family; the sieve it generates is the maximal sieve m_V (every arrow into V factors through id_V), so $m_V \in J_P(V)$ for every V. At $\varnothing$, additionally, the null cover generates the empty sieve, which is also covering.

(J2) It suffices to check stability on basis-generated sieves and then close up. Let σ be generated by a basis family at V and let $U \subseteq V$. The pullback sieve $\iota^*\sigma$ along $\iota : U \to V$ consists of the subsets $T \subseteq U$ with $T \in \sigma$ (in the down-set reading of sieves on a poset). By Lemma 9.2, the pullback of the generating family is the basis family at U (or identities on $\varnothing$), and the sieve generated by the pulled-back family is contained in $\iota^*\sigma$: each pulled-back member $U \cap B$ satisfies $U \cap B \subseteq V \cap B \in \sigma$, hence $U \cap B \in \sigma$ by down-closure, hence lies in $\iota^*\sigma$. So $\iota^*\sigma$ contains a basis-generated covering sieve and is covering. For a general covering sieve $\sigma' \supseteq \sigma$, monotonicity of pullback gives $\iota^*\sigma' \supseteq \iota^*\sigma$, and supersets of covering sieves are covering by construction of J_P.

(J3) Suppose $\sigma \in J_P(V)$ and τ is a sieve on V with $f^*\tau \in J_P(U)$ for every $f : U \to V$ in σ. We must show $\tau \in J_P(V)$. At $V = \varnothing$ there is nothing to prove, since every sieve on $\varnothing$ covers; so let V be nonempty, where the displayed description reads $\tau \in J_P(V)$ as $\tau \supseteq \langle K_P(V)\rangle$. It therefore suffices to put each member $V \cap B \to V$ of $K_P(V)$ into τ, and each such member lies in σ—because $\sigma \supseteq \langle K_P(V)\rangle$—so the hypothesis applies to it. Suppose first $V \cap B \neq \varnothing$, and write ι for its inclusion into V. Block disjointness gives $(V \cap B) \cap B' = V \cap B$ when $B' = B$ and $\varnothing$ otherwise (Lemma 7.35), so $K_P(V \cap B)$ contains the identity of $V \cap B$ and $\langle K_P(V \cap B)\rangle$ is the maximal sieve there. The hypothesis makes $\iota^*\tau$ covering on $V \cap B$, hence a superset of that maximal sieve, hence equal to it; membership of $\mathrm{id}_{V\cap B}$ in $\iota^*\tau$ says exactly $V \cap B \in \tau$. Suppose next $V \cap B = \varnothing$: the member is then $\varnothing \to V$, which belongs to τ by down-closure as soon as τ is nonempty—and τ is nonempty, because $V \neq \varnothing$ forces at least one block of P to meet V, and that piece was just placed in τ. Every member of $K_P(V)$ therefore lies in τ, so $\tau \supseteq \langle K_P(V)\rangle$ and τ covers. □

Remark 9.6 (Ledger record). Theorem 9.5 is registered as:

Ledger id	`THM-NT2-MARKED-COVERAGE` (version 0.1)
Statement kind	`THEOREM`
Proof status	`INFORMAL_PROOF_PRESENT`
Review state	`IN_REVIEW`
Migration disposition	`IN_REVIEW`
Novelty status	`NOVELTY_UNASSESSED`
Admission ring	4

The registered statement also covers the provenance half of the construction: "Parallel immutable mark records preserve cover kind, source authority, evidence references, and pullback ancestry even when the generated unmarked families coincide"—that half is Chapter 8's, and the record's claim boundary keeps the two halves apart, verbatim: *"The topology theorem is mathematical; the hash-stable provenance behavior is an executable serialization invariant. Neither is evidence that either marking measures an empirical property."*

Claim Boundary 9.7. Per the record's claim boundary just quoted: nothing in Theorem 9.5 or its proof interprets the ACBE or RPM labels empirically. The theorem is about two pullback-stable partition bases on a finite poset; the names of the partitions do no mathematical work.

4. Propriety: the generators contain no identities

The legacy collapse (Chapter 5) turned on a single mechanism: a sieve containing an identity, or a split epimorphism composable to an identity, is maximal. The successor's claim to have escaped is therefore exactly the claim that its global generators are *proper*, and it calls for a careful proof rather than an inspection.

PROPOSITION 9.8 (NT2 global covers are proper). *At the global context, both pieces of* $K_A(\Omega) = \{A_0 \to \Omega,\ A_1 \to \Omega\}$ *and both pieces of* $K_R(\Omega) = \{R_0 \to \Omega,\ R_1 \to \Omega\}$ *are proper subcontexts of* Ω*, and the sieves these families generate are nonmaximal.*

PROOF. *Properness.* $|A_i| = |R_j| = 2 < 4 = |\Omega|$, so each piece is a proper subset (rows A_0, A_1, R_0, R_1 of Table 7.1).

Nonmaximality, by the factoring argument. Let σ be the sieve on Ω generated by $K_A(\Omega)$. By definition of generation, an arrow $U \to \Omega$ lies in σ if and only if it factors through some member, i.e., $U \subseteq A_0$ or $U \subseteq A_1$. Suppose toward a contradiction that σ were maximal; then $\mathrm{id}_\Omega \in \sigma$, so $\Omega \subseteq A_0$ or $\Omega \subseteq A_1$. But properness gives $A_i \subsetneq \Omega$, so neither containment holds: no proper subset of a set contains it. Hence $\mathrm{id}_\Omega \notin \sigma$ and σ is nonmaximal. The identical argument applies to $K_R(\Omega)$ with R_j in place of A_i. (Concretely, σ misses more than the identity: it contains none of the arrows from Ω, the four triples, the two RPM blocks, or the two diagonals, since none of those is contained in a single ACBE block—the RPM blocks and the diagonals each meet both ACBE blocks by Remark 7.10, and anything of size ≥ 3 exceeds a block's size. Its seven members are $\varnothing$, the four singletons, and A_0, A_1.) □

REMARK 9.9 (The exact point of escape). Compare mechanism against mechanism. The legacy ACBE clause contained the reflexive descent map id_X; any sieve containing an identity is maximal, since every f factors as $\mathrm{id} \circ f$. The legacy RPM clause contained the counit ε_X of a section-retraction pair, and sieve closure under precomposition then contains $\varepsilon_X \circ \eta_X = \mathrm{id}_X$: maximal again. The successor's generators contain no identity (both pieces are proper) and no split epimorphism onto the target (in a poset, every arrow is monic, and an arrow $U \to \Omega$ admits a section only if $\Omega \subseteq U$, forcing $U = \Omega$). Both legacy failure mechanisms are thus structurally unavailable—and the executable certificate closes the loop by re-verifying the legacy mechanisms themselves as sanity checks (`legacy_acbe_identity_forces_maximal_sieve`, `legacy_rpm_split_epi_forces_identity`) alongside the propriety of the new generators.

REMARK 9.10 (Ledger record). Proposition 9.8 is registered as:

Ledger id	`THM-NT2-NONTRIVIAL-GLOBAL-COVERS` (version 0.1)
Statement kind	`THEOREM`
Proof status	`INFORMAL_PROOF_PRESENT`
Review state	`IN_REVIEW`
Migration disposition	`IN_REVIEW`
Novelty status	`NOVELTY_UNASSESSED`
Admission ring	4

Claim boundary, verbatim: *"Nonmaximality repairs the inherited formal collapse; it does not establish semantic correctness of the labels."*

5. NT2-2: subcanonicity of the context factor

A topology is subcanonical when every representable presheaf is a sheaf [6]. On the poset L, the representable $yW = \mathrm{Hom}_L(-, W)$ has a transparent description: $yW(V)$ is

a one-element set if $V \subseteq W$ and empty otherwise, with the (unique possible) restriction maps. Subcanonicity is the statement that these sixteen truth-valued presheaves already glue correctly along the new covers—a basic sanity property, since a site whose own representables fail its sheaf condition supports no Yoneda-style reasoning inside its sheaf category.

THEOREM 9.11 (NT2-2: both context topologies are subcanonical). *Every representable presheaf on L is a J_A-sheaf and a J_R-sheaf. Hence J_A and J_R are subcanonical.*

PROOF. By the basis criterion it suffices to verify the sheaf condition for the generating families and the null cover [4]; supersets and saturations add no obligations beyond those closed under by the topology construction. Fix $W \in L$ and write $y = yW$. Fix $P \in \{A, R\}$.

Null cover at $\varnothing$. The matching-family set over the empty family is a singleton (there is exactly one empty family of sections, and its compatibility condition is vacuous). The sheaf condition thus requires $y(\varnothing)$ to be a singleton, and it is: $\varnothing \subseteq W$ always, so $y(\varnothing)$ has exactly one element.

Generating cover $K_P(V)$ at nonempty V. A matching family assigns to each block $B \in P$ a section over the piece $V \cap B$, compatibly on overlaps. Since y takes only empty or singleton values, both injectivity and the compatibility bookkeeping simplify, but we track them fully.

Injectivity of the gluing map. $y(V)$ has at most one element, so any map out of it is injective.

Surjectivity. Suppose a matching family exists; we must produce a section over V, i.e., show $V \subseteq W$. For each block B with $V \cap B \neq \varnothing$: the family provides an element of $y(V \cap B)$, so $V \cap B \subseteq W$. For blocks with $V \cap B = \varnothing$: trivially $V \cap B = \varnothing \subseteq W$. Hence every piece is contained in W, and by joint exhaustiveness (Proposition 7.30)

$$V = \bigcup_{B \in P} (V \cap B) \subseteq W.$$

So $y(V)$ is nonempty; its unique element restricts to the unique element of each $y(V \cap B)$, necessarily matching the given family since those sets are singletons.

Conversely, if $y(V)$ is nonempty ($V \subseteq W$), its element restricts to a matching family, and we have shown every matching family arises this way, uniquely. The gluing map is therefore a bijection for every generator, for every representable, and both generated topologies are subcanonical. □

The certificate corroborates the theorem as `representables_are_sheaves`: all sixteen representables checked against every generating cover of both partitions.

REMARK 9.12 (Ledger record).

Ledger id	`THM-NT2-SUBCANONICAL-CONTEXT-FACTOR` (version 0.1)
Statement kind	`THEOREM`
Proof status	`INFORMAL_PROOF_PRESENT`
Review state	`IN_REVIEW`
Migration disposition	`IN_REVIEW`
Novelty status	`NOVELTY_UNASSESSED`
Admission ring	4

Claim boundary, verbatim: *"This applies to the context factor L only. It does not make the inherited jointly-surjective topology on Noetica subcanonical."*

CLAIM BOUNDARY 9.13 (What NT2-2 does not repair). We expand the boundary beyond the ledger sentence, because the temptation it blocks is real. Chapter 6 exhibited a model of

the registered legacy axioms in which the jointly-surjective topology on $\mathcal{C}$ is *not* subcanonical: subcanonicity of $(\mathcal{C}, J_{\text{legacy}})$ is not derivable. Theorem 9.11 concerns the context factor L alone; the two results coexist without tension, and the second neither repairs nor bears on the first. The operative consequence for the whole program: a distinguished legacy object X is represented in the legacy sheaf topos by its sheafification $a_{\text{legacy}}(yX)$, never by silently treating yX as a sheaf [6]. No amount of good behavior on the finite factor changes that rule.

6. NT2-3: distinct, non-identity modalities

The heart of the finite theory: the two topologies are not merely well-defined and subcanonical but *different*, and different in the strong sense that their sheafification modalities are distinct and neither is trivial. Distinctness is witnessed, not deduced: we construct an explicit presheaf that satisfies the ACBE gluing condition and fails the RPM one, and symmetrically.

6.1. The witness presheaves.

Definition 9.14 (Witness presheaves). Recall from Definition 7.41 the complete-block index sets and their counts $\alpha(V) = \#\{i \mid A_i \subseteq V\}$, $\rho(V) = \#\{j \mid R_j \subseteq V\}$. Define $F_A : L^{\text{op}} \to \mathbf{Set}$ by

$$F_A(V) \;=\; \big\{\, \text{functions } b : \{\, i \in \{0,1\} \mid A_i \subseteq V \,\} \to \{0,1\} \,\big\},$$

the set of *bit assignments to the complete ACBE blocks contained in* V. For $U \subseteq V$, the restriction map $F_A(V) \to F_A(U)$ sends b to its restriction along the inclusion of index sets $\{i \mid A_i \subseteq U\} \subseteq \{i \mid A_i \subseteq V\}$ (Corollary 7.44): it *forgets* the bits of blocks no longer completely contained. Define F_R symmetrically, with RPM blocks in place of ACBE blocks; equivalently, F_R is the transport of F_A along the swap automorphism S of Proposition 7.25.

By Table 7.5, the cardinalities are

$$|F_A(V)| = 2^{\alpha(V)} : \qquad \begin{array}{ll} 1 \text{ on } \varnothing,\ \text{singletons},\ R\text{-blocks},\ \text{diagonals}; & \\ 2 \text{ on } A\text{-blocks and triples}; & 4 \text{ on } \Omega, \end{array}$$

and dually $|F_R(V)| = 2^{\rho(V)}$.

6.2. Functoriality first.

Lemma 9.15 (F_A and F_R are presheaves). *The restriction maps of Definition 9.14 satisfy the identity and composition laws, so F_A and F_R are functors $L^{\text{op}} \to \mathbf{Set}$.*

Proof. Identity: restricting along $V \subseteq V$ restricts b along the identity inclusion of index sets, giving b. Composition: for $T \subseteq U \subseteq V$, the index sets satisfy $\{i \mid A_i \subseteq T\} \subseteq \{i \mid A_i \subseteq U\} \subseteq \{i \mid A_i \subseteq V\}$, and restriction of functions along a composite inclusion equals the composite of restrictions. The same argument applies to F_R. □

Remark 9.16 (The functoriality-checked-first discipline). The lemma looks unmissable, and that is the danger: a *proposed* witness that assigns section sets per context but garbles restriction maps can appear to "fail gluing" for reasons that are actually failures to be a presheaf at all, silently invalidating a distinctness argument. The finite implementation therefore validates functoriality of any proposed witness across all 81 restriction maps

and all 256 composable pairs $T \subseteq U \subseteq V$—the two counts of Theorem 7.12 and Proposition 7.15 respectively—*before* evaluating any sheaf condition, so that a nonfunctorial assignment can never pass as a witness. The certificate registers `acbe_witness_functorial` and `rpm_witness_functorial` as separate, prior checks to the four sheaf verdicts. The monograph observes the same discipline: Lemma 9.15 precedes Theorem 9.17.

6.3. The theorem.

THEOREM 9.17 (NT2-3: distinct, non-identity modalities). *F_A is a J_A-sheaf but not a J_R-sheaf, and F_R is a J_R-sheaf but not a J_A-sheaf. Consequently:*

(1) *the reflective subcategories* $\mathrm{Sh}(L, J_A)$ *and* $\mathrm{Sh}(L, J_R)$ *of* $\mathrm{PSh}(L, \mathbf{Set})$ *are distinct;*
(2) *the sheafification modalities M_A and M_R (sheafify, then include) are distinct;*
(3) *neither M_A nor M_R is naturally isomorphic to the identity on* $\mathrm{PSh}(L, \mathbf{Set})$.

PROOF. We prove four assertions about the witnesses, then derive the consequences.

(a) F_A satisfies the null-cover condition. $F_A(\varnothing)$ is the set of functions from the empty index set to $\{0, 1\}$, which is a singleton (the empty function), as the null cover requires.

(b) F_A satisfies ACBE gluing at every nonempty V. Fix the cover $K_A(V) = \{V \cap A_0 \to V,\ V \cap A_1 \to V\}$. First, localize the index sets. By Corollary 7.44, $A_k \subseteq V \cap A_i$ iff $k = i$ and $A_i \subseteq V$: each complete block of V is complete in exactly one piece (its own), and a piece contains no other complete blocks. Hence the index set of V is the disjoint union of the index sets of the two pieces:

$$\{\, i \mid A_i \subseteq V \,\} \;=\; \{\, i \mid A_i \subseteq V \cap A_0 \,\} \;\sqcup\; \{\, i \mid A_i \subseteq V \cap A_1 \,\}.$$

Second, dispose of overlaps. The two pieces intersect in $(V \cap A_0) \cap (V \cap A_1) = V \cap (A_0 \cap A_1) = \varnothing$, where $F_A(\varnothing)$ is a singleton by (a); the compatibility condition of a matching family over $K_A(V)$ is therefore automatic. A matching family is thus exactly a free pair: a bit assignment on the complete blocks of $V \cap A_0$ and one on those of $V \cap A_1$. By the disjoint-union decomposition, such pairs correspond exactly to bit assignments on the complete blocks of V; the correspondence is implemented by the gluing map (restrict a $b \in F_A(V)$ to each piece, i.e., forget the other piece's indices), which is therefore a bijection. This holds uniformly in V—including the degenerate cases: when a piece is empty, its section set is the singleton and its index set is empty, contributing nothing to either side; when V has no complete blocks at all (singletons, RPM blocks, diagonals: Table 7.5), both sides are singletons.

(c) F_A fails RPM gluing at Ω: the four-versus-one computation. Consider the cover $K_R(\Omega) = \{R_0 \to \Omega,\ R_1 \to \Omega\}$. Count both sides of the gluing map.

Global sections: $\alpha(\Omega) = 2$, so $F_A(\Omega) = \{0, 1\}^{\{0,1\}}$ has exactly $2^2 = 4$ elements: the four bit-pairs $(b(0), b(1)) \in \{00, 01, 10, 11\}$.

Matching families: each RPM block contains no complete ACBE block—$A_i \subseteq R_j$ would contradict transversality, since $A_i \cap R_j = \{\omega_{ij}\}$ is a singleton while $|A_i| = 2$ (Corollary 7.43)—so $\alpha(R_0) = \alpha(R_1) = 0$ and $F_A(R_0)$, $F_A(R_1)$ are singletons. The overlap is $R_0 \cap R_1 = \varnothing$ with $F_A(\varnothing)$ a singleton, so compatibility is automatic, and the matching-family set is a product of two singletons: exactly 1 element.

The verdict: the gluing map is a function from a four-element set to a one-element set. It is surjective but drastically non-injective: all four global sections restrict to the same (unique) matching family, because restriction to either RPM block forgets both bits. The sheaf condition requires a bijection; $4 \neq 1$. Hence F_A is not a J_R-sheaf.

(d) The symmetric facts for F_R. Transport along the swap automorphism S (Proposition 7.25, Remark 7.27): S exchanges the partitions and carries K_A-covers to K_R-covers, and

$F_R = F_A \circ S$ by construction, so (b) transports to "F_R satisfies RPM gluing everywhere" and (c) transports to "F_R fails ACBE gluing at Ω," with the same 4-versus-1 count against the cover $K_A(\Omega)$.

Consequences. (1) $F_A \in \mathrm{Sh}(L, J_A) \setminus \mathrm{Sh}(L, J_R)$ and $F_R \in \mathrm{Sh}(L, J_R) \setminus \mathrm{Sh}(L, J_A)$: each subcategory contains an object the other excludes, so they are distinct (indeed incomparable). (2) Reflectors onto distinct reflective subcategories are distinct. (3) Suppose $M_A \cong \mathrm{id}$. Sheafhood is invariant under isomorphism of presheaves, and $M_A F$ is always a J_A-sheaf; applying the assumed isomorphism at $F = F_R$ would make F_R isomorphic to the J_A-sheaf $M_A F_R$, hence itself a J_A-sheaf, contradicting (d). So $M_A \ncong \mathrm{id}$; symmetrically $M_R \ncong \mathrm{id}$. (More pointedly: a reflector fixes, up to its unit isomorphism, exactly the objects of its subcategory, and each witness lies outside the other's subcategory.) □

The certificate records the four verdicts exactly as proved: `acbe_witness_is_acbe_sheaf` true, `acbe_witness_is_rpm_sheaf` false, `rpm_witness_is_rpm_sheaf` true, `rpm_witness_is_acbe_sheaf` false.

Remark 9.18 (Where else the RPM gluing of F_A fails). The proof used the failure at Ω, and one failure suffices. But the census of Table 7.5 locates the full failure set, and it is instructive: against $K_R(A_i) = \{\{\omega_{i0}\} \to A_i,\ \{\omega_{i1}\} \to A_i\}$, the context A_i has $|F_A(A_i)| = 2$ while both singleton pieces and the empty overlap carry singletons, so the gluing map is 2-to-1: RPM gluing of F_A already fails at both ACBE blocks. It also fails at every triple. At T_{ij} one has $|F_A(T_{ij})| = 2$, because A_{1-i} is complete in T_{ij}, while the two RPM pieces $T_{ij} \cap R_j = \{\omega_{1-i,j}\}$ and $T_{ij} \cap R_{1-j} = R_{1-j}$ both carry singleton section sets and meet in $\varnothing$, so the matching-family set is a singleton and the gluing map is again 2-to-1. Indeed, by transversality no RPM piece of any context contains a complete ACBE block—$V \cap R_j \subseteq R_j$ and $\alpha(R_j) = 0$, hence $\alpha(V \cap R_j) = 0$, by Corollary 7.43 together with monotonicity, Corollary 7.44; see also Appendix C, Finding C.2—so the RPM matching-family set of F_A is a singleton at *every* context: the gluing map is a bijection exactly where $\alpha(V) = 0$—at $\varnothing$, the four singletons, the two RPM blocks, and the two diagonals—and fails injectivity exactly where $\alpha(V) \geq 1$. The witness is thus not a boundary-case curiosity: it separates the topologies at seven of the sixteen contexts—the two ACBE blocks, the four triples, and Ω—with the global context giving the widest gap, 4 versus 1. Appendix C tabulates the pattern context by context.

Remark 9.19 (Ledger record).

Ledger id	`THM-NT2-DISTINCT-MODALITIES` (version 0.1)
Statement kind	`THEOREM`
Proof status	`INFORMAL_PROOF_PRESENT`
Review state	`IN_REVIEW`
Migration disposition	`IN_REVIEW`
Novelty status	`NOVELTY_UNASSESSED`
Admission ring	4

Claim boundary, verbatim: *"No commutation or noncommutation claim is made in this version."* (The commutation question is settled—affirmatively—by the composition theorem of Chapter 10, registered as a separate record at version 0.2; the NT2-3 record deliberately does not anticipate it.)

6.4. The ambient-lift caveat. Faithful reproduction of the ledger requires reporting a discrepancy between the registered statement and what this monograph asserts. The ledger

statement for `THM-NT2-DISTINCT-MODALITIES` ends with the sentence: *"External product with a representable yC_0 lifts the witnesses to the ambient successor topos PSh(C x L), so neither ambient modality is the identity."* The publication layer has since identified that sentence as unproved as stated, and this monograph, following the registered correction, does *not* assert it. The theorem proved above is scoped to $\mathrm{PSh}(L, \mathbf{Set})$.

Claim Boundary 9.20 (The external-product witness is false in general). The proposed lift takes a witness F on L and a representable $y\mathcal{C}_0$ on $\mathcal{C}$ and forms the external product presheaf on $\mathcal{C} \times L$, with value $\mathrm{Hom}_{\mathcal{C}}(c, \mathcal{C}_0) \times F(V)$ at (c, V). Two obstructions:

(1) *Failure at the null cover.* In any reasonable lifted topology, the null cover of the L coordinate covers $(c, \varnothing)$, forcing sheaves to take singleton value there. But the external product's value at $(c, \varnothing)$ is $\mathrm{Hom}_{\mathcal{C}}(c, \mathcal{C}_0) \times F(\varnothing) \cong \mathrm{Hom}_{\mathcal{C}}(c, \mathcal{C}_0)$, which need not be a singleton for a general $\mathcal{C}$ and $\mathcal{C}_0$. The candidate witness can thus fail to be a sheaf for the topology on the side where it was supposed to *pass*, collapsing the intended contrast.

(2) *Duplicated representable coordinates.* Even away from the null cover, gluing computations in the $\mathcal{C}$ coordinate can be obstructed by the representable factor when covers interact with it nontrivially; the finite counting argument of Theorem 9.17 does not survive the product unexamined.

A terminal $\mathcal{C}$-presheaf in place of $y\mathcal{C}_0$ is a candidate repair—its value is a singleton everywhere, removing obstruction (1)—but it becomes a proof only after the lifted topology on $\mathcal{C} \times L$ is actually defined, $\mathcal{C}$ is assumed nonempty, and naturality is checked in both coordinates. None of that is part of v0.1. *Ambient distinctness of the modalities therefore remains* OPEN; the proved result is Theorem 9.17 on $\mathrm{PSh}(L, \mathbf{Set})$, and no artifact of this monograph asserts the ambient statement.

7. NT2-4: the join is the singleton topology

The two topologies interact through their join—the coarsest topology refining both. On the finite lattice the join is computed exactly, and it is exactly what transversality suggests: the topology of the four-singleton partition.

7.1. The transverse refinement.

Lemma 9.21 (Common refinement is the four-singleton cover). *At the global context, refining the cover $K_A(\Omega)$ by the covers $K_R(A_i)$ of its pieces (in either order of partitions) yields the family whose members are the four singletons:*

$$\{\, A_i \cap R_j \to \Omega \,\}_{i,j} \;=\; \{\, \{\omega_{ij}\} \to \Omega \,\}_{i,j} \;=\; K_Q(\Omega).$$

More generally, at any nonempty V the cross-partition composite has nonempty members exactly the singletons of V (with one empty piece per atom of $\Omega \setminus V$), and at $V = \varnothing$ the composite is the null cover.

Proof. The global case is Example 7.37; the general case is Lemma 7.36, both resting on the transversality identity $A_i \cap R_j = \{\omega_{ij}\}$ (Theorem 7.21). □

7.2. The join computation. Order topologies on L by $J \leq J'$ iff every J-covering sieve is J'-covering; the join $J_A \vee J_R$ is the smallest topology above both. Write J_{atom} for the topology generated by the singleton partition Q with the null cover, as in Theorem 9.5.

Lemma 9.22 (Join identification). $J_A \vee J_R = J_{\mathrm{atom}}$.

PROOF. Both inequalities are proved directly.

(≤): J_{atom} contains both J_A and J_R, hence their join. Since the join is the smallest upper bound, it suffices to show $J_A \leq J_{\text{atom}}$ and $J_R \leq J_{\text{atom}}$, and by the basis criterion it suffices to show every generating K_A- or K_R-sieve is J_{atom}-covering. Fix nonempty V and consider the sieve σ generated by $K_A(V)$. Every singleton $\{\omega\} \subseteq V$ lies inside exactly one ACBE piece: $\omega \in A_i$ for exactly one i, so $\{\omega\} \subseteq V \cap A_i$, and by down-closure $\{\omega\} \in \sigma$. Thus σ contains *every* singleton of V, hence contains the sieve generated by the singleton family $K_Q(V)$. Since $\sigma \supseteq \langle K_Q(V)\rangle$ and $\langle K_Q(V)\rangle$ is J_{atom}-covering by construction, monotonicity of covering sieves (Lemma 3.13) gives $\sigma \in J_{\text{atom}}(V)$. The same argument verbatim handles $K_R(V)$, and the null covers of all three bases agree at $\varnothing$. Hence $J_A, J_R \leq J_{\text{atom}}$ and so $J_A \vee J_R \leq J_{\text{atom}}$.

(≥): any topology containing both partial topologies covers by singletons. Let J be any Grothendieck topology with $J \geq J_A$ and $J \geq J_R$; we show every K_Q-generated sieve is J-covering, whence $J_{\text{atom}} \leq J$ and in particular $J_{\text{atom}} \leq J_A \vee J_R$. Fix nonempty V. Cover V by its ACBE pieces: the sieve $\langle K_A(V)\rangle$ is J-covering since $J \geq J_A$. Let $\tau = \langle K_Q(V)\rangle$ be the singleton-generated sieve on V; we verify the hypothesis of (J3) for τ against the covering sieve $\langle K_A(V)\rangle$. A generic generator of $\langle K_A(V)\rangle$ is a piece $V \cap A_i \to V$ (arrows factoring through pieces reduce to the pieces by (J2) and monotonicity). Pull τ back to $V \cap A_i$: the result contains every singleton of $V \cap A_i$ (a singleton of the piece is a singleton of V, hence in τ; its pullback membership is immediate). If $V \cap A_i = \varnothing$, the pullback contains the null cover's content and the empty context is J-covered because $J \geq J_A$ includes the null cover. If $V \cap A_i \neq \varnothing$, the sieve on $V \cap A_i$ generated by its singletons contains the sieve generated by $K_R(V \cap A_i)$'s refinement—concretely, by transversality (Lemma 9.21 applied at $V \cap A_i$), the RPM pieces of $V \cap A_i$ are

$$(V \cap A_i) \cap R_j = V \cap \{\omega_{ij}\},$$

which are exactly singletons of $V \cap A_i$ or empty. So the pullback of τ to the piece contains the K_R-generated covering sieve of the piece, hence is J-covering (supersets again). The hypothesis of (J3) holds for every arrow of the covering sieve $\langle K_A(V)\rangle$, so $\tau \in J(V)$. At $V = \varnothing$ the singleton basis prescribes the null cover, which is J-covering by assumption. Hence $J_{\text{atom}} \leq J$.

Together: $J_A \vee J_R = J_{\text{atom}}$. □

7.3. The theorem, and the union-topology reading.

THEOREM 9.23 (NT2-4: the join resolves all four atoms). *The common refinement of the ACBE and RPM global covers is the four-singleton cover, and*

$$J_A \vee J_R \;=\; J_{\text{atom}},$$

the topology generated by the singleton partition with the null cover. Moreover J_{atom} admits the following exact description: a sieve σ on V is covering if and only if it contains every singleton of V—equivalently, if and only if the union of its members is V. The joined context coverage is thus the ordinary union coverage on the finite powerset lattice.

PROOF. The refinement computation is Lemma 9.21; the join identification is Lemma 9.22. For the exact description: if σ contains every singleton of V then $\sigma \supseteq \langle K_Q(V)\rangle$ is covering. Conversely let $\sigma \in J_{\text{atom}}(V)$ and fix $\omega \in V$; pulling back along $\{\omega\} \to V$ (axiom (J2)) gives a covering sieve on $\{\omega\}$. Covering sieves on a singleton context contain the identity: the generating family $K_Q(\{\omega\})$ has the full piece $\{\omega\}$ among its members (and empties), so its sieve is maximal, and every covering sieve on $\{\omega\}$—being a superset of a sieve containing id, or reached by (J3) from such—contains $\mathrm{id}_{\{\omega\}}$; the null cover does not apply at nonempty

contexts. Membership of $\mathrm{id}_{\{\omega\}}$ in the pullback says exactly $\{\omega\} \in \sigma$. Hence σ contains every singleton of V. Finally, for a down-closed σ, containing every singleton of V is equivalent to $\bigcup \sigma = V$: one direction takes unions; the other picks, for $\omega \in V = \bigcup \sigma$, a member $S \ni \omega$ and applies down-closure to $\{\omega\} \subseteq S$. At $V = \varnothing$ both the empty sieve (generated by the null cover) and the maximal sieve $\{\mathrm{id}_\varnothing\}$ cover, and both have union $\varnothing$, consistently. □

The certificate corroborates the refinement identity as `transverse_refinement_is_singletons` and (in the marked layer) `joint_refinement_is_four_singletons`; the exact-topology verifier of Chapter 11 computes both presentations of the join and confirms their equality sieve-by-sieve.

Corollary 9.24 (Covering-sieve census for the join). *For a context V with $|V| = n$, the number of J_{atom}-covering sieves on V is: 2 for $n = 0$ (the empty and maximal sieves); and for $n \geq 1$, the number of down-closed families in $\mathcal{P}(V)$ containing all singletons, namely $1, 2, 9, 114$ for $n = 1, 2, 3, 4$. The total over the sixteen contexts is*

$$1 \cdot 2 \;+\; 4 \cdot 1 \;+\; 6 \cdot 2 \;+\; 4 \cdot 9 \;+\; 1 \cdot 114 \;=\; 2 + 4 + 12 + 36 + 114 \;=\; 168.$$

Proof. The characterization of covering sieves is Theorem 9.23. Counting: a down-closed family containing all singletons of V automatically contains $\varnothing$ (for $n \geq 1$) and is determined by its members of size ≥ 2, which form a family closed downward within sizes ≥ 2.

$n = 1$: no sets of size ≥ 2; exactly the maximal sieve: 1.

$n = 2$: the only candidate is the doubleton V itself, in or out: 2.

$n = 3$: any subfamily of the three doubletons ($2^3 = 8$ choices), plus the triple V only if all three doubletons are present (+1): 9.

$n = 4$: choose a set X_3 of triples, then doubletons X_2 containing every doubleton inside a chosen triple, then the top only if everything else is full. Summing over X_3 with free-doubleton counts as in the proof of Lemma 7.46 (the same intersection patterns, one level up): $X_3 = \varnothing$: $2^6 = 64$; $|X_3| = 1$ (4 ways, forcing 3 doubletons): $2^3 = 8$ each, 32; $|X_3| = 2$ (6 ways, forcing 5): 2 each, 12; $|X_3| = 3$ (4 ways, forcing all 6): 1 each, 4; $|X_3| = 4$: 1. Subtotal $64 + 32 + 12 + 4 + 1 = 113$; the full sieve (top included) adds 1: 114.

The weighted total over the class sizes $1, 4, 6, 4, 1$ is the displayed sum, 168. □

Remark 9.25 (Reconciling the two 168s). The exact-topology verifier reports 298 sieves in total (derived from first principles in Theorem 7.47) of which 168 are covering for the join—the number just derived. As flagged in Remark 7.48, this 168 is numerically equal to, but conceptually unrelated to, the count $\delta(4) = 168$ of *all* sieves on the single object Ω. The verifier computes the covering count twice, once from the join presentation $J_A \vee J_R$ and once from the singleton presentation J_{atom}, and confirms both the equality of the two topologies and the count; the derivation above shows the number is forced.

Remark 9.26 (Ledger record).

Ledger id	`THM-NT2-JOIN-DISCRETE-CONTEXT` (version 0.1)
Statement kind	`THEOREM`
Proof status	`INFORMAL_PROOF_PRESENT`
Review state	`IN_REVIEW`
Migration disposition	`IN_REVIEW`
Novelty status	`NOVELTY_UNASSESSED`
Admission ring	4

Claim boundary, verbatim: *"This is a structural gluing result for Set-valued context data, not a probabilistic marginal-extension theorem."* The boundary matters because the joined topology's "ordinary gluing over the four atomic contexts" invites a reading in terms of compatible marginals; Chapter 12's gluing-boundary discussion explains why that reading is blocked—pairwise-compatible probability marginals need not extend to a joint distribution, and no **Set**-valued sheaf statement licenses a probabilistic consistency claim.

Claim Boundary 9.27 (The `J_atom` naming boundary). The internal label J_{atom} (`J_atom` in the registered artifacts) is a name for the topology generated by the four-singleton partition. It does *not* assert that this topology is the standard category-theoretic *atomic topology*—the topology on a category satisfying the right Ore condition in which every nonempty sieve covers [**1**, **7**]—and on L it is not that: the sieve $\langle \varnothing \to \Omega \rangle = \{\varnothing \to \Omega\}$, i.e., the nonempty sieve generated by the empty piece, has union $\varnothing \neq \Omega$ and fails to cover, by Theorem 9.23. The preferred publication-layer name is $J_{\text{singleton}}$; this monograph uses J_{atom} where continuity with the registered artifacts demands it, always with this boundary in force.

8. Consolidated ledger view

Table 1 collects the status fields of every record in theorem ledger v0.1, including the three records whose mathematical content lives in other chapters: the resolved-identity theorem and the composition theorem (Chapter 10) and the backward-compatibility conjecture (Chapter 12). All statuses are reproduced verbatim from the ledger as of 2026-08-04; the monograph promotes none of them.

Table 1. Theorem ledger v0.1: status fields, verbatim. All records carry admission ring 4. Abbreviations: kind T = `THEOREM`, C = `CONJECTURE`; proof IPP = `INFORMAL_PROOF_PRESENT`, PR = `PROOF_REQUIRED`; review IR = `IN_REVIEW`, UA = `UNASSESSED`; novelty NU = `NOVELTY_UNASSESSED`.

Ledger id (`THM-NT2-...`)	Ver.	Kind	Proof	Review	Novelty	Ch.
`MARKED-COVERAGE`	0.1	T	IPP	IR	NU	9
`NONTRIVIAL-GLOBAL-COVERS`	0.1	T	IPP	IR	NU	9
`SUBCANONICAL-CONTEXT-FACTOR`	0.1	T	IPP	IR	NU	9
`DISTINCT-MODALITIES`	0.1	T	IPP	IR	NU	9
`RESOLVED-MODALITIES-IDENTITY`	0.1	T	IPP	IR	NU	10
`JOIN-DISCRETE-CONTEXT`	0.1	T	IPP	IR	NU	9
`PRODUCT-SITE-BACKWARD-COMPATIBILITY`	0.1	C	PR	UA	NU	12
`MODALITY-COMPOSITION`	0.2	T	IPP	IR	NU	10

Three remarks on the table. First, the dependency structure recorded in the ledger runs exactly as this monograph's chapter order suggests: `MARKED-COVERAGE` has no dependencies; `NONTRIVIAL-GLOBAL-COVERS`, `SUBCANONICAL-CONTEXT-FACTOR`, and `JOIN-DISCRETE-CONTEXT` depend on it; `DISTINCT-MODALITIES` depends on the first two; `RESOLVED-MODALITIES-IDENTITY` and `MODALITY-COMPOSITION` depend on `DISTINCT-MODALITIES`; and the conjecture depends on `JOIN-DISCRETE-CONTEXT`. Second, the sole version-0.2 record is the composition theorem, whose v0.2 statement carries the commutation result proved in Chapter 10; its claim boundary scopes it to $\mathrm{PSh}(L, \mathbf{Set})$ and lists the lift to $\mathrm{PSh}(\mathcal{C} \times L)$, coherence in the $\mathcal{C}$ coordinate, interaction with observational covers, proof-assistant verification, and independent review as open. Third, the sole conjecture—backward compatibility—carries proof status

`PROOF_REQUIRED` and review state `UNASSESSED`; its claim boundary states that backward compatibility "remains a Gate 1 proof and implementation obligation, not a completed theorem," and that even if proved, the resolved context factor is four independent legacy channels without cross-channel semantics. Chapter 12 develops both.

9. What the finite theorems establish, and what they do not

Assembled, the finite-context theory says: the transverse partition bases are genuine coverage bases (Lemma 9.2) generating genuine Grothendieck topologies (Theorem 9.5) whose global covers are proper—escaping, by construction, both legacy collapse mechanisms (Proposition 9.8); both topologies are subcanonical on the context factor (Theorem 9.11); they are distinct, with distinct non-identity sheafification modalities, witnessed by explicit finite presheaves whose gluing behavior is computed exactly—bijective ACBE gluing everywhere for F_A, a four-versus-one RPM failure at the global context, and the mirror facts for F_R (Theorem 9.17); and their join is exactly the singleton topology, the ordinary union coverage on the finite powerset lattice, with 168 covering sieves among the 298 (Theorem 9.23 and Corollary 9.24).

What they do not establish, in the ledger's own language and this chapter's boundaries: no empirical meaning for the ACBE or RPM marks (every record's claim boundary); no repair of legacy subcanonicity (Boundary 9.13); no ambient distinctness on $\mathrm{PSh}(\mathcal{C} \times L)$ (Boundary 9.20); no commutation claim *at this record version* (Remark 9.19); no standard-atomic-topology claim (Boundary 9.27); no probabilistic marginal-extension reading of the join (Remark 9.26); and no promotion of any status: every proof above is an `INFORMAL_PROOF_PRESENT` record's proof written at length, corroborated exhaustively by an executable certificate that is itself, per its own boundary, neither proof-assistant verification nor peer review nor a license for any claim beyond the mathematics on this one sixteen-object site.

Chapter 10 constructs the sheafification reflectors explicitly and proves the composition law $M_A M_R \cong M_{\mathrm{atom}} \cong M_R M_A$; Chapter 11 opens the certificate.

CHAPTER 10

Explicit Reflectors and the Composition Theorem

1. Purpose and status of this chapter

The preceding chapters established four finite-context facts about the marked context factor: the transverse partition families generate genuine Grothendieck topologies J_A and J_R on the sixteen-object lattice L; both topologies are subcanonical; the associated-sheaf modalities M_A and M_R are distinct and neither is trivial, witnessed by the explicit presheaves F_A and F_R; and the common refinement of the two global covers is the four-singleton cover, whose topology is the join $J_A \vee J_R = J_{\mathrm{atom}}$. Each of those results concerns one topology at a time, or the raw combinatorics of the covers. None of them says what happens when the two modalities are *composed*.

This chapter closes that question for the finite context factor. The answer is exact and, from the standpoint of the program's earlier informal hopes, corrective: the two modalities commute up to unit-compatible canonical natural isomorphism, and either composite is naturally isomorphic to sheafification for the joined topology $J_A \vee J_R$, which is itself the topology generated by the four-singleton partition. The proof is fully constructive. Rather than invoking sheafification as an abstract left adjoint and reasoning about it indirectly, we build the associated-sheaf reflector for each partition topology as an explicit product-over-blocks formula, verify the sheaf condition and the universal property directly against the definitions, identify the join of the two topologies by a two-sided sieve argument, and then exhibit the composite of the two explicit reflectors as a flattening of nested products onto atomic products. Every step is elementary; the value of the chapter is that no step is skipped.

Claim Boundary 10.1. Everything in this chapter transcribes and expands the registered artifact `MODALITY_COMPOSITION_PROOF_v0.1.md`, whose status is INTERNAL PROOF CANDIDATE / REVIEW REQUIRED, admission Ring 4, NOT LICENSED, disclosure internal only. Drafting this chapter promotes nothing: the composition theorem has not been reviewed by an independent category theorist, has not been formalized in a proof assistant, and licenses no empirical, probabilistic, product, or safety claim. The exhaustive executable corroboration described in Chapter 11 is corroboration, not proof. The result is scoped to $\mathrm{PSh}(L, \mathbf{Set})$; no statement in this chapter lifts it to $\mathrm{PSh}(\mathcal{C} \times L)$, includes the observational topology on $\mathcal{C}$, or interacts with the legacy comparison of Chapter 12.

2. Standing data

We fix the data used throughout, in the exact form assumed by the registered proof.

Notation 10.2. $\Omega = \{(0,0),(0,1),(1,0),(1,1)\}$ is the four-atom set; its elements are written $\omega_{ij} = (i,j)$. The context lattice is $L = \mathcal{P}(\Omega)$, regarded as a category with one arrow $U \to V$ exactly when $U \subseteq V$. A presheaf is a functor $F : L^{\mathrm{op}} \to \mathbf{Set}$, small in a fixed universe. The two transverse partitions of Ω are

$$A_i = \{(i,0),(i,1)\}, \qquad R_j = \{(0,j),(1,j)\}, \qquad i,j \in \{0,1\},$$

and Q is the partition of Ω into its four singletons. For a partition $P \in \{A, R, Q\}$ and a nonempty context V, the generating family is

$$K_P(V) = \{\, V \cap B \to V \mid B \text{ a block of } P \,\},$$

retaining an empty piece $\varnothing \to V$ whenever an intersection is empty. At the empty context the generator is the null (empty) family, which covers. J_A, J_R, J_{atom} are the topologies these bases generate; the publication-layer name for J_{atom} is $J_{\text{singleton}}$, since the label does not assert the standard category-theoretic atomic topology.

The registered proof assumes exactly five things: ordinary small **Set**-valued presheaves in a fixed universe; the powerset inclusion category L; the stated generating families with the null-cover convention; the ordinary Grothendieck-topology and associated-sheaf definitions [**1**, **4**]; and the transversality identity

$$A_i \cap R_j = \{(i, j)\}, \tag{1}$$

which is immediate from the definitions of the blocks. No legacy theorem, no semantic interpretation of the ACBE or RPM marks, and no empirical premise is used anywhere below. Pullbacks in L are intersections: the pullback of $U \to V$ along $W \to V$ is $U \cap W$.

Two elementary facts about sieves are used repeatedly and are recorded here once.

LEMMA 10.3. *Let J be a Grothendieck topology on a category, $S \in J(C)$ a covering sieve, and $T \supseteq S$ any sieve on C. Then $T \in J(C)$.*

PROOF. Apply the local-character axiom (J3) to T with respect to the covering sieve S: for each $f : D \to C$ in S we have $f \in T$, hence $f^*T \ni \mathrm{id}_D$, so f^*T is the maximal sieve on D and is covering by (J1). Therefore $T \in J(C)$. □

LEMMA 10.4. *If $J \leq J'$ (every J-covering sieve is J'-covering), then every J'-sheaf is a J-sheaf.*

PROOF. The sheaf condition for a presheaf quantifies over covering sieves; a J'-sheaf satisfies the gluing bijection for every sieve in J', in particular for every sieve in $J \subseteq J'$. □

Finally, we recall the basis criterion used throughout: the generating families above form a pullback-stable coverage basis (pullback along $U \subseteq V$ carries $K_P(V)$ to $K_P(U)$ as an indexed family, and to identity arrows on the empty object when $U = \varnothing$; refinement of a member by the same partition regenerates the same sieve because distinct blocks are disjoint), and a presheaf is a sheaf for the generated topology if and only if it satisfies the gluing condition for the basis families [**4**]. This was proved in full when the topologies were first constructed; we use it without further comment.

3. The one-partition reflector

The composition theorem requires knowing sheafification *exactly*, not just abstractly as a left adjoint. On the finite lattice L with a partition coverage, sheafification admits a closed form: a product over the blocks the context meets.

CONSTRUCTION 10.5 (One-partition reflector). Fix a partition $P \in \{A, R, Q\}$. For a context V, let

$$I_P(V) = \{\, B \in P \mid V \cap B \neq \varnothing \,\}$$

be the set of blocks that V meets. For every presheaf F, define a presheaf $R_P F$ by

$$(R_P F)(V) \;=\; \prod_{B \in I_P(V)} F(V \cap B), \tag{2}$$

where the empty product is the singleton $\mathbf{1}$. In particular

$$(R_P F)(\varnothing) = \mathbf{1} \tag{3}$$

even when $F(\varnothing)$ has zero, two, or more elements. For an inclusion $U \subseteq V$, the restriction map $(R_P F)(V) \to (R_P F)(U)$ *drops* the coordinates B with $U \cap B = \varnothing$ and applies the restriction map $F(V \cap B) \to F(U \cap B)$ of F on each retained coordinate. The unit is the natural transformation

$$\eta_{P,F,V} : F(V) \to (R_P F)(V), \qquad x \mapsto \big(x|_{V\cap B}\big)_{B \in I_P(V)}, \tag{4}$$

where $x|_{V\cap B}$ denotes the image of x under the restriction map of F along $V \cap B \subseteq V$.

Proposition 10.6 (Well-definedness). *$R_P F$ is a presheaf, R_P is a functor on* $\mathrm{PSh}(L, \mathbf{Set})$*, and η_P is a natural transformation* $\mathrm{id} \Rightarrow R_P$.

Proof. *Functor laws for $R_P F$.* Restriction along id_V retains every coordinate of $I_P(V)$ and applies F-restriction along identities, which is the identity by functoriality of F. For $W \subseteq U \subseteq V$: a coordinate B survives the composite $V \to U \to W$ restriction exactly when $W \cap B \neq \varnothing$, which (since $W \subseteq U$ forces $U \cap B \neq \varnothing$ as well) is the same survival condition as for the one-step restriction $V \to W$; on each surviving coordinate the two routes apply $F(V \cap B) \to F(U \cap B) \to F(W \cap B)$ and $F(V \cap B) \to F(W \cap B)$, equal by functoriality of F. Coordinates dropped at the first stage stay dropped, coordinatewise agreement gives equality of the maps.

Functoriality of R_P. A presheaf map $\tau : F \to F'$ induces $(R_P\tau)_V = \prod_{B \in I_P(V)} \tau_{V\cap B}$; this commutes with the restriction maps of $R_P F$ and $R_P F'$ because τ is natural at each $V \cap B \supseteq U \cap B$, and it preserves identities and composites coordinatewise.

Naturality of the unit. For $U \subseteq V$ and $x \in F(V)$, both routes around the naturality square send x to the tuple $\big(x|_{U\cap B}\big)_{B \in I_P(U)}$: going right then down restricts each coordinate $x|_{V\cap B}$ further to $U \cap B$ and drops the coordinates U misses; going down then right restricts x to U first and then forms the tuple. Both computations are composites of restriction maps of F along the same inclusions, hence equal. Naturality of η_P in F (that is, $\eta_{P,F'} \circ \tau = (R_P\tau) \circ \eta_{P,F}$ for $\tau : F \to F'$) is the componentwise naturality of τ along the inclusions $V \cap B \subseteq V$. □

Remark 10.7 (Two structural observations). Two features of formula (2) do all the later work. First, the value at a context that meets exactly one block is canonically the value of F at the intersection: if $V \neq \varnothing$ and $V \subseteq B_0$ for a block B_0, then $I_P(V) = \{B_0\}$ and $(R_P F)(V) = F(V \cap B_0) = F(V)$ via the one-factor product, and under this identification the unit $\eta_{P,F,V}$ is the identity map of $F(V)$. We call this the *singleton-factor bijection*; it is the engine of the uniqueness argument in Lemma 10.13. Second, the value at the empty context is forcibly the singleton, clause (3), regardless of $F(\varnothing)$. This is not an arbitrary convention: the null cover of the empty context has exactly one matching family (the empty one), so every sheaf for any of the three topologies must have a one-element value at $\varnothing$ (Lemma 10.8 below), and the reflector must therefore correct a nonterminal empty-context value. The executable fixture of Chapter 11 that exercises exactly this clause is a constant two-section presheaf whose empty-context value is deliberately nonterminal.

Lemma 10.8. *Let $J \in \{J_A, J_R, J_{\mathrm{atom}}\}$. Every J-sheaf G has $G(\varnothing)$ a one-element set.*

Proof. The null family covers $\varnothing$ in each basis, hence the sieve it generates (the empty sieve) is covering. A matching family for the empty family is a family of sections indexed by the empty set; there is exactly one such family. The sheaf condition demands that the

canonical map from $G(\varnothing)$ to this one-element set of matching families be a bijection, so $G(\varnothing)$ is a singleton. □

Example 10.9 (The reflector in action). Three computations calibrate the construction; the appendices tabulate the witness values used here.

(1) *Representables are fixed.* For yW with $W \in L$, the subcanonicity theorem (Theorem NT2-2, proved earlier in the monograph) says yW is already a J_P-sheaf; by Corollary 10.16 below, the unit $yW \to R_P\, yW$ is an isomorphism. Concretely: a matching tuple of compatible sections over the pieces of V exists exactly when every piece lies in W, which by joint exhaustiveness of the partition happens exactly when $V \subseteq W$.

(2) *The ACBE reflector annihilates the RPM witness.* The RPM witness F_R assigns to V the bit assignments on the complete R-blocks contained in V. For every context V and every i, the piece $V \cap A_i$ is a subset of A_i and therefore contains no complete R-block, since $R_j \subseteq A_i$ is impossible by transversality (1). Hence $F_R(V \cap A_i) = \mathbf{1}$ for every piece, and $(R_A F_R)(V) = \mathbf{1}$ at every context: $M_A F_R$ is the terminal presheaf. Symmetrically $M_R F_A$ is terminal. The modalities are not merely non-identity; each one can destroy all the information carried by the other's witness.

(3) *The empty-context correction.* Let C be the constant presheaf with $C(V) = \{0, 1\}$ for all V, all restrictions the identity. Then $(R_P C)(\varnothing) = \mathbf{1} \neq C(\varnothing)$: the unit is not injective at the empty context, and C is not a J_P-sheaf precisely because it fails the null-cover condition. At nonempty V, $(R_P C)(V)$ is a product of two-element sets, one per block V meets; for $V = \Omega$ and $P = A$ this is a four-element set, so the reflector can also *enlarge* values.

Proposition 10.10 (Limit preservation). *R_P preserves all small limits of presheaves; in particular it is left exact.*

Proof. Limits in $\mathrm{PSh}(L, \mathbf{Set})$ are computed pointwise. At each context V, formula (2) exhibits $(R_P F)(V)$ as a finite product of the evaluation functors $F \mapsto F(V \cap B)$, each of which preserves limits (evaluation is limit-preserving pointwise), and finite products of limit-preserving functors preserve limits. Left exactness is the finite-limit case. This is consistent with, and here independent of, the general fact that sheafification is left exact [5]; we will only conclude that R_P *is* sheafification in Lemma 10.13. □

4. The sheaf lemma

Lemma 10.11 ($R_P F$ is a J_P-sheaf). *For every presheaf F and every $P \in \{A, R, Q\}$, the presheaf $R_P F$ is a sheaf for J_P.*

Proof. Because the generating families form a coverage basis, it suffices to verify the gluing bijection for the basis families [4]. There are two cases.

The empty context. The null cover requires exactly one matching family and exactly one section. The empty family has exactly one matching family (the empty one), and $(R_P F)(\varnothing) = \mathbf{1}$ by clause (3), so the canonical map is a bijection between one-element sets.

A nonempty context V with cover $K_P(V)$. We first compute the value of $R_P F$ on each piece, then dispose of the compatibility constraints, then exhibit the gluing bijection.

Values on pieces. If $V \cap B \neq \varnothing$, then $I_P(V \cap B) = \{B\}$: the piece is contained in the single block B, and blocks are pairwise disjoint, so it meets no other block. Hence

$$(R_P F)(V \cap B) = F\big((V \cap B) \cap B\big) = F(V \cap B),$$

the singleton-factor bijection of Remark 10.7. If $V \cap B = \varnothing$, the piece is the empty context, where $(R_P F)(\varnothing) = \mathbf{1}$.

The disjoint-overlap argument. Let $B \neq B'$ be distinct blocks. The overlap of the corresponding pieces is the pullback

$$(V \cap B) \cap (V \cap B') = V \cap (B \cap B') = V \cap \varnothing = \varnothing,$$

since distinct blocks of a partition are disjoint. Compatibility of a matching family on this overlap is the requirement that the two restricted sections agree in $(R_P F)(\varnothing) = \mathbf{1}$ — a one-element set, in which any two elements are equal. Compatibility is therefore *automatic* for every pair of distinct pieces. More generally, any arrow $U \to V$ that factors through two distinct pieces satisfies $U \subseteq (V \cap B) \cap (V \cap B') = \varnothing$, so all overlap conditions in the sieve generated by $K_P(V)$ reduce to conditions in the one-element set. Compatibility of a piece with itself is trivially the requirement that a section agree with itself. Empty pieces impose no constraint at all: a matching family assigns to the arrow $\varnothing \to V$ the unique element of $\mathbf{1}$.

The gluing bijection. By the two preceding paragraphs, a matching family for $K_P(V)$ is precisely a free choice of one element of $F(V \cap B)$ for each block $B \in I_P(V)$ (the nonempty pieces), together with the forced unique elements on empty pieces. The set of matching families is therefore canonically

$$\prod_{B \in I_P(V)} F(V \cap B) \;=\; (R_P F)(V).$$

The canonical gluing map sends a section $s \in (R_P F)(V)$ to the family of its restrictions to the pieces. On the nonempty piece $V \cap B$, the restriction map of $R_P F$ retains exactly the coordinate B (all other blocks meet $V \cap B$ in $\varnothing$) and applies the identity $F(V \cap B) \to F(V \cap B)$; that is, restriction to the piece $V \cap B$ is projection onto the B-coordinate. The gluing map is therefore the canonical reindexing of the product by its own coordinates, which is a bijection. □

Remark 10.12. The proof shows why clause (3) is load-bearing twice over. If one had instead set $(R_P F)(\varnothing) = F(\varnothing)$, the null-cover case would fail whenever $F(\varnothing)$ is not a singleton, and the disjoint-overlap argument would fail as well: compatibility on empty overlaps would become a genuine constraint (agreement in $F(\varnothing)$), and the matching-family set would no longer be the full product. Both failure modes are exercised by the constant two-section fixture of Example 10.9(3), and both are checked executably in Chapter 11.

5. The universal property

Lemma 10.11 shows $R_P F$ is *a* sheaf receiving a map from F. The next lemma shows it is *the* associated sheaf: the unit is universal among maps from F to J_P-sheaves. This is the step that identifies the explicit formula with the abstract modality, and it is proved directly rather than by appeal to a general sheafification construction.

Lemma 10.13 (Universal property). *For every presheaf F and every J_P-sheaf G, composition with the unit (4) is a bijection*

$$\mathrm{Hom}_{\mathrm{Sh}(L,J_P)}(R_P F, G) \xrightarrow{\;\cong\;} \mathrm{Hom}_{\mathrm{PSh}(L)}(F, G), \qquad \beta \mapsto \beta \circ \eta_{P,F},$$

natural in F and G. Consequently $\eta_P : \mathrm{id} \Rightarrow R_P$ exhibits R_P as the associated-sheaf reflector for J_P, left adjoint to the inclusion $\mathrm{Sh}(L, J_P) \hookrightarrow \mathrm{PSh}(L, \mathbf{Set})$.

PROOF. Fix $\alpha : F \to G$ with G a J_P-sheaf. We construct a factorization, prove it is a presheaf map, prove it factors α, prove it is unique, and then address naturality.

Existence. Let V be a context and $(x_B)_{B \in I_P(V)} \in (R_P F)(V)$ a tuple. Apply α componentwise: for each $B \in I_P(V)$, the element $\alpha_{V \cap B}(x_B) \in G(V \cap B)$ is a section of G over the piece $V \cap B$; assign to each empty piece the unique element of $G(\varnothing) = \mathbf{1}$ (Lemma 10.8). These local sections form a matching family for $K_P(V)$: every overlap of distinct pieces is the empty context by the disjoint-overlap computation of Lemma 10.11, and $G(\varnothing)$ is a one-element set, so all compatibility conditions hold automatically. Since G is a J_P-sheaf, this matching family glues to a *unique* section of $G(V)$; define

$$\bar{\alpha}_V\big((x_B)_B\big) \in G(V)$$

to be that glued section, characterized by $\bar{\alpha}_V((x_B))|_{V \cap B} = \alpha_{V \cap B}(x_B)$ for every $B \in I_P(V)$. At $V = \varnothing$, define $\bar{\alpha}_\varnothing$ to be the unique map $\mathbf{1} \to G(\varnothing)$ between one-element sets.

Naturality of $\bar{\alpha}$ in V. Let $U \subseteq V$ and $(x_B) \in (R_P F)(V)$; write $s = \bar{\alpha}_V((x_B))$. We must show $s|_U = \bar{\alpha}_U\big((x_B)|_U\big)$, where $(x_B)|_U$ is the restricted tuple $\big(x_B|_{U \cap B}\big)_{B \in I_P(U)}$. Both sides are sections of $G(U)$; by the uniqueness half of the sheaf condition for the cover $K_P(U)$ of U (separatedness), it suffices to show they agree on each piece $U \cap B$ with $B \in I_P(U)$. The right side restricts to $\alpha_{U \cap B}(x_B|_{U \cap B})$ by its defining property. The left side restricts to

$$(s|_U)|_{U \cap B} = s|_{U \cap B} = \big(s|_{V \cap B}\big)|_{U \cap B} = \big(\alpha_{V \cap B}(x_B)\big)|_{U \cap B} = \alpha_{U \cap B}\big(x_B|_{U \cap B}\big),$$

using functoriality of restriction in G along $U \cap B \subseteq V \cap B \subseteq V$, the defining property of s, and naturality of α along $U \cap B \subseteq V \cap B$. The two sections agree on every piece of a cover of U, hence are equal. (When $U = \varnothing$ both sides lie in the one-element set $G(\varnothing)$.)

Factorization. For $x \in F(V)$, $\bar{\alpha}_V(\eta_{P,F,V}(x)) = \bar{\alpha}_V\big((x|_{V \cap B})_B\big)$ is the unique section of $G(V)$ restricting on each piece to $\alpha_{V \cap B}(x|_{V \cap B})$. By naturality of α, the section $\alpha_V(x)$ has exactly these restrictions: $\alpha_V(x)|_{V \cap B} = \alpha_{V \cap B}(x|_{V \cap B})$. Separatedness of G for $K_P(V)$ forces $\bar{\alpha}_V(\eta_{P,F,V}(x)) = \alpha_V(x)$, i.e. $\bar{\alpha} \circ \eta_{P,F} = \alpha$.

Uniqueness via the singleton-factor bijection. Let $\beta : R_P F \to G$ be any presheaf map with $\beta \circ \eta_{P,F} = \alpha$. First consider a context W that is a nonempty subset of a single block B. There the unit $\eta_{P,F,W} : F(W) \to (R_P F)(W)$ is the singleton-factor bijection of Remark 10.7 (one coordinate, identity map). Hence on such contexts β is completely determined: $\beta_W = \alpha_W \circ \eta^{-1}_{P,F,W}$. Now let V be arbitrary nonempty and $(x_B) \in (R_P F)(V)$. For each $B \in I_P(V)$, naturality of β along $V \cap B \subseteq V$ gives

$$\beta_V\big((x_B)\big)\big|_{V \cap B} = \beta_{V \cap B}\big((x_B)|_{V \cap B}\big) = \beta_{V \cap B}\big(\eta_{P,F,V \cap B}(x_B)\big) = \alpha_{V \cap B}(x_B),$$

where the middle equality uses that restriction of $R_P F$ to the piece $V \cap B$ is projection onto the B-coordinate, which under the singleton-factor bijection is exactly the unit applied to x_B. So $\beta_V((x_B))$ is a section of $G(V)$ whose restriction to every piece of $K_P(V)$ agrees with that of $\bar{\alpha}_V((x_B))$; separatedness forces $\beta_V = \bar{\alpha}_V$. At the empty context both maps land in the one-element set $G(\varnothing)$ and are equal. Hence $\beta = \bar{\alpha}$, and the assignment $\alpha \mapsto \bar{\alpha}$ is a two-sided inverse to composition with the unit.

Naturality of the bijection. In G: for a sheaf map $g : G \to G'$, we must check $\overline{g \circ \alpha} = g \circ \bar{\alpha}$; both are factorizations of $g \circ \alpha$ through the unit, and uniqueness (already proved, applied with target G') forces equality. In F: for a presheaf map $\tau : F' \to F$, we must check $\overline{\alpha \circ \tau} = \bar{\alpha} \circ R_P\tau$; the right side is a sheaf map $R_P F' \to G$ satisfying $(\bar{\alpha} \circ R_P\tau) \circ \eta_{P,F'} = \bar{\alpha} \circ \eta_{P,F} \circ \tau = \alpha \circ \tau$ (using naturality of the unit in the presheaf argument, Proposition 10.6), and uniqueness applied to $\alpha \circ \tau$ forces equality. Thus the bijection is natural in both variables, and R_P is left adjoint to the inclusion with unit η_P. □

Remark 10.14 (On the method, with the Stacks citation). The proof deliberately avoids assuming that a single application of the plus construction sheafifies an arbitrary presheaf, or that sheafification has any particular closed form; it verifies the universal property from scratch against the definitions. The general background — existence of the associated sheaf, its universal property, and its exactness properties — is recorded in the Stacks Project, Section 7.10, especially Theorem 7.10.10 and Proposition 7.10.12 [5]; the textbook treatment is [1, Chapter III]. The point of the direct proof is auditability: once F and G are fixed, each step reduces to a finite computation on the sixteen contexts. The quantifiers over presheaves F and sheaves G are not themselves finite, and no executable layer discharges them: Chapter 11 exercises the reflector, its unit, and the comparison maps on eighteen registered fixtures only. The universal property of Lemma 10.13 and the sheaf lemma (Lemma 10.11) are informally proved here and are *not* executably corroborated; their machine verification falls under admission gate 2 of Chapter 12.

Corollary 10.15 (R_P is THE reflector). *Any left adjoint to the inclusion* $\mathrm{Sh}(L, J_P) \hookrightarrow \mathrm{PSh}(L, \mathbf{Set})$ *is naturally isomorphic to* R_P*, compatibly with units. Writing* M_P *for the composite of any chosen sheafification with the inclusion, there are canonical natural isomorphisms* $M_A \cong R_A$*,* $M_R \cong R_R$*,* $M_{\mathrm{atom}} \cong R_Q$*.*

Proof. Uniqueness of left adjoints up to canonical natural isomorphism compatible with units [2, Ch. IV], applied to the adjunction established in Lemma 10.13. □

Corollary 10.16 (Fixed objects). *The unit* $\eta_{P,F} : F \to R_P F$ *is an isomorphism if and only if* F *is a* J_P*-sheaf. In particular* R_P *is idempotent up to canonical isomorphism:* $\eta_{P,R_P F}$ *and* $R_P(\eta_{P,F})$ *are isomorphisms* $R_P F \to R_P R_P F$*.*

Proof. If $\eta_{P,F}$ is an isomorphism, F is isomorphic to the sheaf $R_P F$ and is a sheaf. Conversely, if F is a J_P-sheaf, then $\mathrm{id}_F : F \to F$ factors through the unit by Lemma 10.13, giving $r : R_P F \to F$ with $r \circ \eta = \mathrm{id}$; and both $\eta \circ r$ and $\mathrm{id}_{R_P F}$ factor η through itself, so uniqueness gives $\eta \circ r = \mathrm{id}$. Idempotency follows since $R_P F$ is a sheaf by Lemma 10.11. □

Remark 10.17 (Modalities and local operators). In topos-theoretic language, each M_P is a left-exact idempotent monad on the presheaf topos — what this monograph calls a *modality* — and such monads correspond to local operators (Lawvere–Tierney topologies) on $\mathrm{PSh}(L, \mathbf{Set})$; for a presheaf topos over a small site these in turn correspond to Grothendieck topologies on the base. The general theory is in [7] and [1, Chapter V]. We use this correspondence only as orientation; every claim below is proved at the level of the explicit formulas.

6. The join of the two topologies

The composite of the two reflectors will turn out to be sheafification for a single topology: the join. Chapter 9 already identified it, proving $J_A \vee J_R = J_{\mathrm{atom}}$ as Theorem NT2-4; the argument is repeated here in reflector-facing form, so that this chapter is self-contained and so that the covering characterisation is available in the shape the composition proof consumes.

Definition 10.18. Order Grothendieck topologies on L by $J \leq J'$ iff every J-covering sieve is J'-covering. The collection of topologies containing two given topologies J_A, J_R is nonempty (the maximal topology contains both) and closed under intersection, so it has a least element: the *join* $J_A \vee J_R$, the smallest topology containing every J_A-cover and every J_R-cover.

LEMMA 10.19 (Join identification; second proof). *$J_A \vee J_R = J_{\text{atom}}$, where J_{atom} is the topology generated by the singleton partition Q with the null cover at the empty context.*

PROOF. We prove the two inequalities separately.

Step 1: $J_A \vee J_R \leq J_{\text{atom}}$ (singletons lie inside pieces). It suffices to show that J_{atom} is a topology containing J_A and J_R; the join, being least among such, is then contained in it. Fix a nonempty V and consider the sieve S_A generated by $K_A(V)$. Every singleton $\{\omega\} \subseteq V$ with $\omega = (i, j)$ satisfies $\{\omega\} \subseteq V \cap A_i$: the singleton lies inside exactly one ACBE piece. Hence the arrow $\{\omega\} \to V$ factors through a member of $K_A(V)$ and belongs to S_A. The sieve S_Q generated by $K_Q(V)$ consists precisely of the arrows factoring through some singleton piece — that is, all $U \subseteq V$ with $|U| \leq 1$ — and we have just shown every generator of S_Q lies in S_A, so $S_Q \subseteq S_A$. Since S_Q is J_{atom}-covering, Lemma 10.3 makes S_A covering in J_{atom} as well. The same argument with R_j in place of A_i handles $K_R(V)$. At the empty context all three bases prescribe the null cover, so the covering sieves agree there. Every generating cover of J_A and of J_R is therefore J_{atom}-covering; since J_A (resp. J_R) is the least topology containing its generating sieves, $J_A \leq J_{\text{atom}}$ and $J_R \leq J_{\text{atom}}$, hence $J_A \vee J_R \leq J_{\text{atom}}$.

Step 2: $J_{\text{atom}} \leq J_A \vee J_R$ (composite ACBE-then-RPM refinement via transversality). Let J' be any topology containing J_A and J_R; we show every J_{atom}-generating sieve is J'-covering, which gives $J_{\text{atom}} \leq J'$ and in particular $J_{\text{atom}} \leq J_A \vee J_R$. Fix nonempty V and let $T = S_Q$ be the sieve on V generated by the singleton family; as noted, T is the set of subsets of V of size at most one. We verify $T \in J'(V)$ by the local-character axiom (J3) applied along the J'-covering sieve S_A generated by $K_A(V)$: it suffices that $g^*T \in J'(W)$ for every arrow $g : W \to V$ in S_A.

So let $g : W \to V$ lie in S_A; then $W \subseteq V \cap A_i$ for some i. If $W = \varnothing$, then $\varnothing \to V$ lies in T (the empty set has size zero), so g^*T contains $\mathrm{id}_\varnothing$ and is maximal, hence covering. If $W \neq \varnothing$, compute the pullback sieve: $g^*T = \{\, U \subseteq W \mid U \in T \,\} = \{\, U \subseteq W \mid |U| \leq 1 \,\}$. Now consider the RPM cover $K_R(W)$ of W, which is J_R-covering hence J'-covering. Its pieces are $W \cap R_j$; since $W \subseteq A_i$, transversality (1) gives

$$W \cap R_j \;\subseteq\; A_i \cap R_j \;=\; \{(i, j)\},$$

so every RPM piece of W is empty or a singleton — this is the composite ACBE-then-RPM refinement collapsing to singletons. The sieve generated by $K_R(W)$ is the set of subsets of W contained in some piece, i.e. exactly the subsets of size at most one: every singleton $\{(i, j)\} \subseteq W$ is contained in the piece $W \cap R_j$, and $\varnothing$ is contained in every piece. Hence the sieve generated by $K_R(W)$ *equals* g^*T, which is therefore J'-covering. (Null-cover handling: if some ACBE piece $V \cap A_i$ is empty, the corresponding arrows contribute the $W = \varnothing$ case already treated; and at $V = \varnothing$ itself the J_{atom} generator is the null family, covering in J' because $J_A \leq J'$ prescribes the same null cover.)

By (J3), $T \in J'(V)$. Every generating sieve of J_{atom} is thus J'-covering, and since J_{atom} is the least topology containing its generators, $J_{\text{atom}} \leq J'$. Taking $J' = J_A \vee J_R$ completes Step 2, and the two steps give equality. □

COROLLARY 10.20. *A sieve on V is $(J_A \vee J_R)$-covering if and only if it contains every singleton of V; equivalently, the joined context coverage is the ordinary union coverage on the finite powerset lattice $\mathcal{P}(\Omega)$.*

PROOF. By Lemma 10.19 the join is generated by the singleton families. A sieve containing every singleton of V contains the generated sieve S_Q and is covering by Lemma 10.3. Conversely, the collection of sieves containing all singletons of their target is itself a Grothendieck

topology (the verification is the standard one for the union coverage on a powerset: maximality is clear; pullback stability holds because singletons of $U \subseteq V$ are singletons of V; local character holds because a singleton missing from T would pull back along its own inclusion in the covering sieve to a non-covering sieve on a one-point context, where covering sieves must contain the identity), and it contains the generators, hence contains J_{atom}. □

Claim Boundary 10.21. The internal label J_{atom} does not assert the standard category-theoretic *atomic topology* (the topology in which every nonempty sieve covers); $J_{\mathrm{singleton}}$ is the preferred publication-layer name. Chapter 11 records the exact-topology enumeration that corroborates Lemma 10.19: all 298 sieves over the sixteen contexts are enumerated, both presentations (join and singleton) are saturated to a fixed point, the Grothendieck axioms are checked independently on both computed topologies, and both report the same 168 covering sieves.

7. The composition theorem

We can now compute the composite of the two explicit reflectors exactly. The composite $M_A M_R$ of the modalities means: sheafify for J_R, include back into presheaves, sheafify for J_A, include back. By Corollary 10.15 this composite is canonically naturally isomorphic to $R_A R_R$, so it suffices to compute the latter.

Theorem 10.22 (Modality composition). *On* $\mathrm{PSh}(L, \mathbf{Set})$ *there are unit-compatible canonical natural isomorphisms*

$$M_A M_R \;\cong\; M_{\mathrm{atom}} \;\cong\; M_R M_A,$$

and, with Lemma 10.19, either composite is naturally isomorphic to sheafification for the joined topology $J_A \vee J_R$. *These are natural isomorphisms, not strict equalities.*

Proof. *The objectwise computation.* Fix a presheaf F and a context V. Expand formula (2) twice:

$$(R_A R_R F)(V) = \prod_{B \in I_A(V)} (R_R F)(V \cap B) = \prod_{\substack{i \\ V \cap A_i \neq \varnothing}} \; \prod_{\substack{j \\ V \cap A_i \cap R_j \neq \varnothing}} F(V \cap A_i \cap R_j).$$

By transversality (1),

$$V \cap A_i \cap R_j = V \cap \{(i,j)\} = \begin{cases} \{(i,j)\} & \text{if } (i,j) \in V, \\ \varnothing & \text{otherwise,} \end{cases}$$

so the inner index condition $V \cap A_i \cap R_j \neq \varnothing$ says exactly $(i,j) \in V$, and the outer condition $V \cap A_i \neq \varnothing$ says exactly that some j has $(i,j) \in V$ — which is implied by the inner condition and therefore imposes no further restriction on the pairs that occur. The set of index pairs (i,j) appearing in the nested product is thus in canonical bijection with the set of atoms $\omega \in V$, and each factor is $F(\{\omega\})$.

The flattening and regrouping bijections. Define

$$\varphi_{F,V} : (R_A R_R F)(V) \longrightarrow \prod_{\omega \in V} F(\{\omega\}) = (R_Q F)(V), \qquad \varphi_{F,V}\Big(\big((x_{ij})_j\big)_i\Big) = \big(x_{ij}\big)_{(i,j)\in V},$$

which forgets the nesting of a tuple of tuples and reindexes the entries by the atoms of V; and define

$$\psi_{F,V} : (R_Q F)(V) \longrightarrow (R_A R_R F)(V), \qquad \psi_{F,V}\big((y_\omega)_{\omega\in V}\big) = \Big(\big(y_{(i,j)}\big)_{j:(i,j)\in V}\Big)_{i:V\cap A_i\neq\varnothing},$$

which groups the atomic coordinates first by ACBE block and then by RPM block. Since the index sets on the two sides are in the canonical bijection just described and both maps merely transport entries along that bijection, $\varphi_{F,V}$ and $\psi_{F,V}$ are mutually inverse. At the empty context both sides are the empty product $\mathbf{1}$ ($I_A(\varnothing) = \varnothing$ on the nested side; no atoms on the flat side), and $\varphi_{F,\varnothing}$ is the unique map between one-element sets: the null cover requires no exception to the theorem.

Naturality in V. Let $U \subseteq V$. On the flat side, the restriction map of $R_Q F$ drops the coordinates of atoms $\omega \in V \setminus U$ and applies the F-restriction $F(\{\omega\}) \to F(\{\omega\})$ along the identity inclusion $\{\omega\} \subseteq \{\omega\}$ — which is the identity map, by functoriality of F — on the retained atoms. On the nested side, the outer restriction drops the i-coordinates with $U \cap A_i = \varnothing$ and applies the restriction of $R_R F$ from $V \cap A_i$ to $U \cap A_i$ on the rest; that inner restriction in turn drops the j-coordinates with $U \cap A_i \cap R_j = \varnothing$ and applies the identity $F(\{(i,j)\}) \to F(\{(i,j)\})$ on the rest. Both routes therefore implement the same operation on entries: delete the entries indexed by atoms outside U, keep the remaining entries unchanged. Hence $\varphi_{F,U} \circ \mathrm{res}^{R_A R_R F}_{V,U} = \mathrm{res}^{R_Q F}_{V,U} \circ \varphi_{F,V}$: flattening commutes with restriction, and φ_F is a map of presheaves (with inverse ψ_F).

Naturality in F. For a presheaf map $\tau : F \to F'$, the maps $(R_A R_R \tau)_V$ and $(R_Q \tau)_V$ both act by applying $\tau_{\{\omega\}}$ independently to each atomic coordinate — on the nested side after unwinding the two componentwise definitions, on the flat side directly. Transporting entries along the index bijection commutes with applying the same map to each entry, so $\varphi_{F',V} \circ (R_A R_R \tau)_V = (R_Q \tau)_V \circ \varphi_{F,V}$: the isomorphism is natural in F.

Unit compatibility. The unit of the composite reflector at F is the composite $\eta_{A,R_R F} \circ \eta_{R,F} : F \to R_R F \to R_A R_R F$. Chasing $x \in F(V)$: first $\eta_{R,F,V}(x) = (x|_{V \cap R_j})_j$; then applying η_A at $R_R F$ restricts this tuple to each ACBE piece, and the restriction of $R_R F$ from V to $V \cap A_i$ sends $(x|_{V\cap R_j})_j$ to $\big(x|_{V\cap A_i \cap R_j}\big)_{j : V \cap A_i \cap R_j \neq \varnothing}$. The resulting nested tuple has (i,j)-entry $x|_{\{(i,j)\}}$, so

$$\varphi_{F,V}\big((\eta_{A,R_R F} \circ \eta_{R,F})_V(x)\big) = \big(x|_{\{\omega\}}\big)_{\omega \in V} = \eta_{Q,F,V}(x) :$$

the flattening carries the composite unit to the unit of R_Q exactly.

Symmetry, commutation, and identification with the join. Exchanging the roles of A and R throughout gives a unit-compatible natural isomorphism $R_R R_A \cong R_Q$ by the same three checks (the objectwise computation is symmetric in the two partitions because transversality (1) is). Composing the first comparison with the inverse of the second yields a canonical natural isomorphism $R_A R_R \cong R_R R_A$. By Corollary 10.15, $M_A M_R \cong R_A R_R$ and $M_R M_A \cong R_R R_A$ and $M_{\mathrm{atom}} \cong R_Q$; and by Lemmas 10.13 and 10.19, R_Q is the associated-sheaf reflector for $J_{\mathrm{atom}} = J_A \vee J_R$, so either composite is naturally isomorphic to sheafification for the joined topology.

Finally, the isomorphisms are canonical — they are determined by the index bijection induced by transversality, with no choices — but they are not strict equalities: nested and flat **Set**-products need not be literally equal sets in a chosen implementation, and the theorem asserts nothing stronger than natural isomorphism. □

8. Corollaries

Corollary 10.23 (Commutation). *The finite-context modalities commute up to canonical natural isomorphism: $M_A M_R \cong M_R M_A$, compatibly with the composite units.*

PROOF. Immediate from Theorem 10.22, composing the two comparisons through M_{atom}; unit compatibility holds because both comparisons carry their composite units to the unit of M_{atom}. □

COROLLARY 10.24 (Composites land in the intersection subcategory). *For every presheaf F, the composite value $M_A M_R F$ (equivalently $M_R M_A F$, equivalently $M_{\mathrm{atom}} F$) is both a J_A-sheaf and a J_R-sheaf: either composite lands in* $\mathrm{Sh}(L, J_A) \cap \mathrm{Sh}(L, J_R) \subseteq \mathrm{PSh}(L, \mathbf{Set})$.

PROOF. By Theorem 10.22 the composite is naturally isomorphic to $R_Q F$, a sheaf for $J_{\mathrm{atom}} = J_A \vee J_R$ by Lemma 10.11. Since $J_A \leq J_A \vee J_R$ and $J_R \leq J_A \vee J_R$, Lemma 10.4 makes every $(J_A \vee J_R)$-sheaf a sheaf for each of the coarser topologies. □

COROLLARY 10.25 (Identity on atomic sheaves). *On an already J_{atom}-sheaf, both modalities and both composites are naturally isomorphic to the identity. This does* not *make either modality the identity on the ambient presheaf category: the witnesses F_A and F_R are moved (indeed, by Example 10.9(2), each is collapsed to the terminal presheaf by the opposite reflector).*

PROOF. Let F be a J_{atom}-sheaf. By Lemma 10.4, F is also a J_A-sheaf and a J_R-sheaf, so by Corollary 10.16 each unit $\eta_{A,F}$, $\eta_{R,F}$, $\eta_{Q,F}$ is an isomorphism; the composite units are then isomorphisms as well, and unit compatibility (Theorem 10.22) transports this to either composite. The non-triviality clause is the distinctness theorem (Theorem NT2-3, proved earlier in the monograph) together with Example 10.9(2). □

THEOREM 10.26 (Identity on the resolved subcategory). *Every sheaf for $J_A \vee J_R$ is separately a J_A-sheaf and a J_R-sheaf. Consequently both reflectors M_A and M_R restrict, up to their canonical reflector isomorphisms, to the identity on the resolved subcategory* $\mathrm{Sh}(L, J_A \vee J_R)$.

PROOF. The first sentence is Lemma 10.4 applied to $J_A \leq J_A \vee J_R$ and $J_R \leq J_A \vee J_R$: a sheaf for the finer topology satisfies the gluing bijection for every covering sieve of any coarser topology contained in it. For the second sentence: a reflector is naturally isomorphic to the identity exactly on its subcategory of local objects (Corollary 10.16), and the first sentence places every resolved object in both subcategories of local objects simultaneously. □

Design argument. The nontrivial modal action of M_A and M_R exists only on objects not yet resolved, which is precisely why the successor package of Chapter 12 keeps the ambient presheaf topos, and not the resolved subtopos, as the carrier of the modalities. Accordingly no text of the program may describe M_A or M_R as nontrivial endomodalities of $\mathcal{E}_{\mathrm{res}}$. This is a design consequence of Theorem 10.26, not an additional theorem.

PROPOSITION 10.27 (The modality algebra). *Up to canonical natural isomorphism, the set* $\{\mathrm{id}, M_A, M_R, M_{\mathrm{atom}}\}$ *is closed under composition, and composition realizes the join of the corresponding topologies* $\{J_{\mathrm{triv}}, J_A, J_R, J_{\mathrm{atom}}\}$ *(with J_{triv} the minimal topology, whose modality is the identity): for topologies J, J' in this set with modalities M, M',*

$$M M' \;\cong\; M_{J \vee J'}.$$

In particular $M_A M_A \cong M_A$, $M_R M_R \cong M_R$, $M_A M_{\mathrm{atom}} \cong M_{\mathrm{atom}} M_A \cong M_{\mathrm{atom}}$, and symmetrically for M_R; the only composite producing something new from distinct nontrivial inputs is $M_A M_R \cong M_R M_A \cong M_{\mathrm{atom}}$.

PROOF. The mixed case $M_A M_R$ is Theorem 10.22, and composites with the identity are trivial. Idempotency $M_P M_P \cong M_P$ is Corollary 10.16. For $M_{\text{atom}} M_A \cong M_{\text{atom}}$, compute with the explicit formulas:

$$(R_Q R_A F)(V) = \prod_{\omega \in V} (R_A F)(\{\omega\}) = \prod_{\omega \in V} F(\{\omega\}) = (R_Q F)(V),$$

since a singleton context meets exactly one ACBE block and the singleton-factor bijection applies; the identification is natural in V and F and unit-compatible by the same coordinate bookkeeping as in Theorem 10.22. For $M_A M_{\text{atom}} \cong M_{\text{atom}}$:

$$(R_A R_Q F)(V) = \prod_{i \,:\, V \cap A_i \neq \varnothing} (R_Q F)(V \cap A_i) = \prod_i \prod_{\omega \in V \cap A_i} F(\{\omega\}) \;\cong\; \prod_{\omega \in V} F(\{\omega\}),$$

regrouping the atoms of V by the ACBE block containing each — a special case of the flattening argument. The RPM cases are mirrors. Finally, the composition table matches the join table of the four topologies: $J_A \vee J_A = J_A$, $J_A \vee J_{\text{atom}} = J_{\text{atom}}$ (by $J_A \leq J_{\text{atom}}$, Lemma 10.19 Step 1), and $J_A \vee J_R = J_{\text{atom}}$ (Lemma 10.19). □

REMARK 10.28. Proposition 10.27 packages the chapter's results as a single algebraic statement: the finite-context modalities form a commutative idempotent monoid canonically isomorphic to the join-semilattice generated by J_A and J_R. That semilattice has exactly four elements — the trivial topology, the two factor topologies, and their join — so the modal landscape of the finite site is completely mapped: there are no hidden composites, no order-dependent operators, and no further modalities generable from these by composition. Any richer modal algebra requires a richer site.

EXAMPLE 10.29 (Both routes, on concrete fixtures). Two spot computations, both reproduced by the executable layer of Chapter 11, illustrate the composition theorem numerically.

On the ACBE witness. At the global context, $(R_R F_A)(\Omega) = F_A(R_0) \times F_A(R_1) = \mathbf{1} \times \mathbf{1}$, and more generally $R_R F_A$ is terminal (Example 10.9(2), with roles exchanged); applying R_A to a terminal presheaf returns a terminal presheaf. Along the other route, $(R_Q F_A)(V) = \prod_{\omega \in V} F_A(\{\omega\})$ is a product of one-element sets. Both composites are terminal, as the theorem requires — and the example shows the composite destroying strictly more than either factor preserves: F_A is fixed by M_A and annihilated by M_R, so the composite in either order annihilates it.

On the constant two-section presheaf. Let C be constant at $\{0, 1\}$. Then $(R_R C)(A_i) = C(\{(i, 0)\}) \times C(\{(i, 1)\})$ has 4 elements, so $(R_A R_R C)(\Omega) = 4 \times 4 = 16$; and directly $(R_Q C)(\Omega) = \prod_{\omega \in \Omega} C(\{\omega\}) = 2^4 = 16$. At a triple T_{ij} both routes give $2^3 = 8$; at the empty context both give 1 (against $C(\varnothing) = \{0, 1\}$, exhibiting the empty-context correction). The flattening map matches the nested 4×4 indexing with the flat 2^4 indexing atom by atom, which is exactly the bijection whose naturality the verifier checks on this fixture.

9. The narrative correction: distinct but commuting

ADVERSE FINDING 10.30. The program's earlier informal materials hoped that the ACBE and RPM lenses would eventually support *interacting, order-sensitive* modal structure — that resolving evidence through the ACBE lens first and the RPM lens second might differ from the opposite order, and that this order sensitivity might itself carry diagnostic content. Theorem 10.22 closes that hope for the present finite site, in both directions at once. The modalities are *distinct* and *non-identity*: Theorem NT2-3 exhibits explicit witnesses separated by them, and Example 10.9(2) shows each can annihilate the other's witness. Yet

they *commute*: both composites are canonically the joined sheafification, and the order of application is invisible up to canonical natural isomorphism.

CLAIM BOUNDARY 10.31. Distinctness does not imply noncommutation, and no current artifact licenses a noncommutation claim. Any text, in any register of the program, that describes the ACBE and RPM modalities of the present finite site as noncommuting, order-sensitive, or "interacting" misstates the registered mathematics. A successor construction seeking genuinely order-sensitive modalities must *change the site again* — different generators, a different context object, or added structure — rather than reinterpret this one; the present chapter proves that on this site the order sensitivity is not there to be found. Symmetrically, the commutation proved here is a finite-context fact about $\mathrm{PSh}(L, \mathbf{Set})$ only: the product-site lift to $\mathrm{PSh}(\mathcal{C} \times L)$, its coherence in the $\mathcal{C}$ coordinate, and its interaction with the observational topology remain OPEN (Chapter 12), so no artifact may assert ambient commutation either.

REMARK 10.32 (What the correction changes downstream). The correction reaches three things downstream of it. First, any pipeline design that schedules ACBE resolution before RPM resolution (or vice versa) for *mathematical* reasons is unjustified at the finite-context level: the results agree canonically, and a scheduling choice can be defended only on implementation or audit grounds — for example, because the marked provenance records of the two orders differ even though the sheaf-theoretic outputs do not. Second, the joined resolution is maximally destructive among the three modalities: by Corollary 10.24 it factors through both sheaf conditions, and on the witnesses it collapses everything (both $R_Q F_A$ and $R_Q F_R$ are terminal, since no complete block survives inside a singleton). Whatever information a workflow wishes to retain must be retained *before* joint resolution or outside the topology, in the marked records. Third, the resolved subcategory is modally inert (Theorem 10.26); the ambient category is the only home for the modal dynamics, which is the formal content of the package definition $\mathrm{NoeTop}_2 := \mathcal{E}_{\mathrm{amb}}$ discussed in Chapter 12.

10. Status recapitulation

CLAIM BOUNDARY 10.33. The results of this chapter — Lemmas 10.11, 10.13, 10.19 and Theorem 10.22 with its corollaries — constitute the registered finite-context modality-composition record. Its status is INTERNAL PROOF CANDIDATE / REVIEW REQUIRED: an informal proof is present in full (this chapter), exhaustive executable corroboration exists (Chapter 11), and nothing else. The record is Ring 4 and NOT LICENSED; it has not been independently reviewed, not formalized beyond the bounded Lean scope stated in Chapter 11, and carries NOVELTY UNASSESSED status — no literature search has established whether the explicit product-over-blocks reflector for transverse partition coverages on a powerset lattice, elementary as it is, is or is not standard. The chapter licenses no claim about $\mathrm{PSh}(\mathcal{C} \times L)$, no probabilistic or enriched gluing statement, no marked-object semantics (the proof concerns ordinary **Set**-valued presheaves; external provenance records neither enter nor are affected), and no empirical, causal, safety, or product claim of any kind.

CHAPTER 11

The Executable Certificate

1. Why an executable layer exists

The finite-context theory of the preceding chapters has an unusual property for a body of category-theoretic results: every quantifier in every statement ranges over finite, small, explicitly enumerable data. There are 4 atoms, 16 contexts, 81 inclusion arrows, and — as we derive below — exactly 298 sieves in total across all contexts. The witness presheaves have at most four sections at any context. Nothing is infinite, nothing is chosen, nothing depends on a universe beyond bookkeeping. A theory this small can be checked *exhaustively*: not sampled, not spot-checked, but walked in its entirety by a machine.

The Noetic Topos 2.0 candidate does exactly that, with two independent executable layers and a bounded proof-assistant layer. The first layer is a dependency-free finite-site checker whose output is the registered artifact `FINITE_MODEL_CERTIFICATE_v0.1.json`, covering the site combinatorics, subcanonicity, the witnesses, the marked-cover integrity rules, and both legacy collapse mechanisms. The second is the modality-composition verifier, which implements the explicit reflector of Chapter 10 and computes both relevant Grothendieck topologies *exactly*, by enumeration and saturation, rather than trusting the pen-and-paper join argument. The third is a deliberately narrow Lean scope. This chapter walks through all three, check by check, and closes with the certificate's own statement of what all this execution does and does not establish.

Claim Boundary 11.1. The relationship between the proofs and the programs must be stated with care, because it is easy to overclaim in either direction. The chapters' informal proofs are general arguments over the fixed finite site; the programs are exhaustive executions over the same site. Execution *corroborates* the proofs — a bug in a proof about a 16-object site would almost certainly produce a counterexample the exhaustive run would find — but it is not itself a proof: the checker could share a misformalization with the prose, and passing tests are evidence about the implemented model, not about the mathematical English. Conversely the proofs do not make the programs redundant: the programs pin down every convention (indexing, empty pieces, null covers, serialization) at a level of concreteness prose cannot reach, and they are what a future reviewer can rerun. Neither layer is licensed to substitute for the other, and neither is licensed to substitute for independent review.

2. Methodology of exhaustive checking

Before walking the checks, we state four methodological commitments, because they distinguish this executable layer from ordinary software testing and because each is auditable in its own right.

Determinism. Both verifiers are deterministic programs over fixed finite data. There is no randomness, no sampling, no property-based generation with a seed: a run either passes identically every time at a given commit or fails identically every time. This is what allows the certificate to be a *record* rather than a statistic — the expected-result block asserts exact

values, not distributions, and a reviewer who reruns a verifier at the bound commit either reproduces the block bit-for-bit or has found a discrepancy to file.

Dependency-freedom. The finite-site checker is deliberately free of third-party dependencies: contexts are frozen sets, presheaves are dictionaries, sieves are sets of subsets, and every categorical notion is implemented from its definition rather than imported from a library. The cost is verbosity; the benefit is that the trusted computing base for the certificate is the checker file itself plus the language runtime, and a reviewer auditing the formalization has one file to read against the definitions of Chapters 7–9.

Exhaustive quantification. Every "for all" in the checked properties is discharged by enumeration over its entire domain: all contexts, all inclusion arrows, all generating covers, all members, all sieves where sieves are quantified. Section 3.7 tabulates the domain of each check explicitly, because "exhaustive" is only meaningful relative to a stated domain.

Negative expectations as first-class results. The certificate registers expected *failures* — the two false sheaf verdicts, the rejection branches of the resolver, the two legacy collapse mechanisms — with the same standing as expected successes. A model in which F_A accidentally satisfied RPM gluing would *fail* the certificate, exactly as a model in which a representable failed the sheaf condition would. This matters for evidential value: a check suite that can only confirm positives cannot catch a construction that is degenerately too strong, and the historical failure being repaired — the legacy collapse — was precisely a construction whose covers were degenerately too strong (maximal everywhere).

3. The finite-site checker and its twenty-three checks

The finite-site layer is the module and test suite

`src/noetic_topos2/finite_site.py`
`tests/noetic_topos2/test_finite_site.py`

Its certificate declares schema version

`noetic_topos2.finite_model_certificate.v0.1`,

document status INTERNAL FINITE CHECK COMPLETE / INDEPENDENT PROOF REVIEW OPEN, and model identifier

`NT2-CONTEXT-2X2-TRANSVERSE-PARTITIONS`,

naming exactly the model of the monograph: the 2×2 atom grid, its powerset context lattice, and the two transverse coordinate partitions. The certificate's expected-result block registers two cardinality checks and twenty-one named verdicts (counting each paired ACBE/RPM verdict as a single named check), twenty-three registered checks in all, together with an aggregate flag `all_required_checks_pass`. All pass at the bound commit

`a1cd990109e7561dfd03949bf0e1653461c7ae44`

(the base commit registered in Chapter 1).

One gap in this layer's audit surface belongs here rather than in a footnote to it: the monograph reproduces no invocation for the finite-site checker. The module and suite paths above are the only locators supplied, and Section 4.4 registers commands for the composition layer alone, so — as with the Lean layer of Section 5 — this layer cannot be rerun from the monograph by itself. What the checks below report is accordingly a reading of the certificate's expected-result block; a reviewer who wants to reproduce that block must take the invocation from the registered artifact set rather than from these pages.

We take the checks in order, stating for each what is computed and which result of the monograph it corroborates.

3.1. Cardinalities and cover geometry.

CHECK CLASS 11.2. `atom_count` = 4, `context_count` = 16. The checker constructs Ω and $L = \mathcal{P}(\Omega)$ from scratch and counts. These two numbers anchor every later enumeration: a model that silently dropped an atom or a context would fail here before any sheaf-theoretic check ran. Corroborates the standing data of the context lattice definition (the sixteen-object poset with 81 inclusion arrows; the arrow count $\sum_V 2^{|V|} = 1 + 4 \cdot 2 + 6 \cdot 4 + 4 \cdot 8 + 16 = 81$ is exercised by the composition verifier below).

CHECK CLASS 11.3. `proper_global_covers` **(ACBE and RPM).** Verifies that at the global context Ω, every member of $K_A(\Omega) = \{A_0 \to \Omega, A_1 \to \Omega\}$ and of $K_R(\Omega)$ is a *proper* subcontext. This is the single most load-bearing geometric fact in the repair: the legacy clauses collapsed precisely because their generating sieves contained identities, and a sieve on a poset generated by arrows from proper subobjects cannot contain the identity. Corroborates the properness proposition of the finite-context theory (ledger THM-NT2-NONTRIVIAL-GLOBAL-COVERS) and, by contrast, the collapse diagnosis for the legacy site.

CHECK CLASS 11.4. `jointly_exhaustive` **(ACBE and RPM).** Verifies $A_0 \cup A_1 = \Omega = R_0 \cup R_1$ and, contextwise, that the pieces of $K_P(V)$ union to V. Joint exhaustiveness is what makes the generating families *covers* in the intuitive sense and is used in the subcanonicity proof (a compatible family over all pieces determines behavior on all of V). Corroborates the coverage-basis lemma behind Theorem NT2-1.

CHECK CLASS 11.5. `pullback_stable` **(ACBE and RPM).** For every pair $U \subseteq V$ and both partitions, the checker pulls back every member of $K_P(V)$ along the inclusion and verifies the resulting family is $K_P(U)$ (as an indexed family; identity arrows on the empty object when $U = \varnothing$, where the registered null cover covers as well). Pullback stability is axiom (J2) at the level of the basis and is the hypothesis under which basis-level gluing suffices for the generated topology [4]. Corroborates the coverage-basis lemma and hence Theorem NT2-1 (ledger THM-NT2-MARKED-COVERAGE).

3.2. Subcanonicity.

CHECK CLASS 11.6. `representables_are_sheaves`. For each of the 16 objects $W \in L$, the checker builds the representable presheaf yW (a singleton at V when $V \subseteq W$, empty otherwise), and verifies the gluing bijection at every context against every generating cover of both topologies, including the null cover at the empty context. This is the full finite content of Theorem NT2-2 (ledger THM-NT2-SUBCANONICAL-CONTEXT-FACTOR): both J_A and J_R are subcanonical on L. Note what a failure here would mean: a representable failing the sheaf condition would put the finite site in the same position as the legacy site, whose jointly-surjective coverage is provably not derivably subcanonical; the check certifies that the repair does not reintroduce that gap on the context factor.

3.3. Witness functoriality and the four verdicts.

CHECK CLASS 11.7. `acbe_witness_functorial`, `rpm_witness_functorial`. Before any sheaf verdict is computed, the checker validates that the proposed witness data F_A and F_R — bit assignments on complete contained blocks, with restriction forgetting coordinates — actually constitute presheaves: identity restrictions are identities and all composable restriction pairs commute. This ordering is deliberate methodology: a nonfunctorial section assignment could otherwise pass or fail sheaf conditions meaninglessly. The check corroborates the well-definedness half of the distinctness theorem's witness construction (Theorem NT2-3), and it

enforces the program-wide discipline that a "witness" must be validated as an object before being scored as evidence.

CHECK CLASS 11.8. **The four sheaf/non-sheaf verdicts:** `acbe_witness_is_acbe_sheaf` **(true),** `acbe_witness_is_rpm_sheaf` **(false),** `rpm_witness_is_rpm_sheaf` **(true),** `rpm_witness_is_acbe_sheaf` **(false).** The checker computes, for each witness and each topology, the matching-family set of every generating cover at every context and compares it with the value set. The registered expectations are asymmetric on purpose: two verdicts are expected *true* and two expected *false*, and the certificate records the false ones as first-class expectations rather than as absences. Together the four verdicts are exactly the content of Theorem NT2-3 (ledger THM-NT2-DISTINCT-MODALITIES) as scoped to $\mathrm{PSh}(L, \mathbf{Set})$: each subcategory $\mathrm{Sh}(L, J_A)$, $\mathrm{Sh}(L, J_R)$ contains an explicit object the other excludes, so the subcategories are distinct and neither modality is trivial. The quantitative core — $F_A(\Omega)$ has 4 sections while the RPM matching-family set at Ω is a singleton, so the gluing map $4 \to 1$ fails injectivity — is tabulated in full in Appendix C.

3.4. Transverse refinement and singleton contexts.

CHECK CLASS 11.9. `transverse_refinement_is_singletons.` Computes all four intersections $A_i \cap R_j$, one per index pair, and verifies each equals the singleton $\{(i, j)\}$; contextwise, verifies that the common refinement of $K_A(V)$ and $K_R(V)$ has as nonempty members exactly the singletons of V. This is the transversality identity, the one equation on which the join lemma, the composition theorem's flattening bijection, and the four-channel expressivity boundary all rest. Corroborates Theorem NT2-4 (ledger THM-NT2-JOIN-DISCRETE-CONTEXT) and the input identity of Chapter 10's Theorem on modality composition.

CHECK CLASS 11.10. `singleton_unmarked_families_coincide, singleton_marked_covers_distinct.`

At each singleton context $\{(i, j)\}$, the checker verifies two facts in deliberate tension. First, the bare arrow families coincide: $K_A(\{(i, j)\})$ and $K_R(\{(i, j)\})$ are the same unmarked list (the identity piece and an empty piece), so the generated sieves are equal and *the topology cannot distinguish the two covers there.* Second, the marked records of those same covers remain distinct: they carry different cover kinds (ACBE versus RPM) and therefore different canonical digests. The pair corroborates the central design argument of the marks chapter: provenance survives exactly where the topology forgets, and an audit trail that lived only in the topology would be lost at precisely these contexts. (Appendix A additionally derives, from the definitions, the full list of contexts where the unmarked families coincide; the certificate registers the singleton instances, which are the load-bearing ones.)

3.5. Marked-cover integrity.

CHECK CLASS 11.11. `marked_cover_roundtrip_stable.` Serializes marked covers to canonical JSON, rehashes, deserializes, and verifies that the reconstructed record and its SHA-256 digest are stable across the round trip. This is the base integrity property of the provenance layer: a digest that did not survive its own serialization could not chain.

CHECK CLASS 11.12. `marked_pullbacks_bind_parent_provenance, marked_pullbacks_preserve_topology_family.` Pulling a marked cover back along an inclusion creates a new record of derivation kind PULLBACK; the first check verifies the child's provenance chain contains its exact parent digest, so derived records are cryptographically bound to their origins. The second verifies that pullback does not mutate the cover family: the child's members are exactly the pullback of the parent's members, i.e. the marked layer and the

topological layer move in lockstep under (J2). Together they corroborate the marked-cover construction of the marks chapter and the pullback-stability geometry of Check 11.5, now at the serialization level.

CHECK CLASS 11.13. `joint_refinement_is_four_singletons`, `joint_refinement_retains_both_parent_digests`.
The ACBE/RPM common refinement at Ω is built as a derived record of kind COMMON REFINEMENT. The first check verifies its members are exactly the four singleton covers — the marked-layer image of the transversality identity (Check 11.9). The second verifies the derived record carries *both* generating-cover digests as parents: the joint refinement remembers that it came from one ACBE and one RPM cover, which is the provenance fact the joined topology itself provably cannot retain. Corroborates Theorem NT2-4 on the topological side and the derivation-node schema on the audit side.

CHECK CLASS 11.14. `marked_derived_records_resolve`. Exercises the fail-closed deserialization contract for derived records: resolution of a derived record requires a parent resolver; every parent digest is rehashed, the complete acyclic parent derivation is checked, and the child's members and mark are recomputed and compared. The check also covers the rejection branches: a common refinement without exactly one ACBE and one RPM generating parent, or a pullback without its exact parent, is rejected rather than treated as evidence, as are unknown or duplicate JSON keys and uncontrolled review states. This corroborates no theorem; it corroborates an *invariant* — the integrity discipline that makes the marked layer usable as audit data at all.

3.6. The legacy collapse checks.

CHECK CLASS 11.15. `legacy_acbe_identity_forces_maximal_sieve`, `legacy_rpm_split_epi_forces_identity`.
The checker builds small executable models of the two legacy failure mechanisms and confirms each. For the ACBE clause: a sieve containing an identity is closed under precomposition, hence contains every arrow into its target and is maximal. For the RPM clause: a sieve containing a split epimorphism ε with section η contains $\varepsilon \circ \eta = \mathrm{id}$ by precomposition closure, hence is maximal by the first mechanism. These are the two elementary facts that make the legacy ACBE/RPM cover clauses contribute only maximal sieves, collapsing the legacy topology to its jointly-surjective clause and erasing all intended modal structure. Corroborates the legacy-collapse proposition of the baseline chapter — the negative result the entire successor construction exists to repair — and does so inside the same certificate that verifies the repair, so that the diagnosis and the cure are checked by one artifact.

ADVERSE FINDING 11.16. At the bound commit, the finite-site suite comprises 8 tests and passes in full; the certificate's aggregate flag `all_required_checks_pass` is *true* over the twenty-three registered checks above. The distinction between the 8 tests and the 23 checks is granularity, not content: a test is a unit-test method that may assert many registered checks (for example, one test sweeps all pullback verifications for both partitions), while a check is a named line of the certificate's expected-result block. The certificate, not the test count, is the audit surface.

3.7. Quantifier domains and counterfactual failures. Exhaustiveness claims are only as strong as their stated domains. The following table records, for each registered check group, the exact domain enumerated and what a failure would have falsified. Domain sizes use the site constants: 16 contexts, 81 inclusion arrows, 2 partitions with 2 members each, 16 representables, 2 witnesses, 4 singleton contexts.

Check group	Domain exhausted	A failure would falsify
cardinalities	the constructed Ω and L	the standing data themselves
proper global covers	2 partitions $\times$ 2 members at Ω	nonmaximality of the generated sieves
jointly exhaustive	2 partitions $\times$ 16 contexts	the coverage-basis lemma (union of pieces)
pullback stable	2 partitions $\times$ 81 arrows $\times$ 2 members	basis axiom (J2); Theorem NT2-1
representables are sheaves	16 representables $\times$ 2 topologies $\times$ all generating covers at all 16 contexts	Theorem NT2-2 (subcanonicity)
witness functoriality	2 witnesses $\times$ (81 restriction maps + 256 composable pairs)	well-definedness of the witnesses
four sheaf verdicts	2 witnesses $\times$ 2 topologies $\times$ all generating covers at all contexts	Theorem NT2-3 (distinctness), in either direction
transverse refinement	4 index pairs; refinements at all 16 contexts	Theorem NT2-4; the composition theorem's input identity
singleton coincidence pair	4 singleton contexts $\times$ both families, unmarked and marked	the marks-necessity argument
marked integrity trio	serialization round trips, pullback derivations along arrows, family preservation	the provenance-layer invariants
joint refinement pair	the derived record at Ω and its parent set	the derivation-node schema; NT2-4 at the marked layer
derived-record resolution	resolution and all rejection branches	the fail-closed contract
legacy collapse pair	both executable collapse mechanisms	the collapse diagnosis motivating the successor

Reading the third column as a whole makes the certificate's structure visible: the checks are not a grab-bag but a covering of the monograph's finite claims — standing data, coverage basis, subcanonicity, distinctness, transversality, marks, and the legacy diagnosis — such that each finite theorem has at least one check whose failure would falsify it in the implemented model.

4. The modality-composition verifier

The second executable layer, `verify_modality_composition.py`, is independent of the finite-site checker and targets Chapter 10 specifically: the explicit reflector formula, its unit, both flattening/regrouping maps, and — crucially — the identification of the joined topology. Its registered invocation is

```
python docs/working/project_coherence/gate1/
    noetic_topos2/modality_composition/
      verify_modality_composition.py
```

and its scoped unit-test discovery command runs over the suite

`tests/noetic_topos2_modality_composition`.

Both are reproduced as copy-pasteable displays in Section 4.4.

4.1. Exact topology closure. The verifier does not sample sieves and does not trust the pen-and-paper join argument. It computes both relevant topologies *exactly*: it enumerates every sieve on every context, seeds the covering collection with the generated basis sieves of each presentation, and then saturates three closure rules — maximality (J1), pullback stability (J2), and local character (J3) — to a fixed point. The two presentations are computed independently: once seeded from the ACBE and RPM bases jointly (the join presentation), and once seeded from the singleton basis (the singleton presentation). The run then verifies that the two fixed points are *equal* as sets of covering sieves, and checks the three Grothendieck axioms once more on each computed result, independently, so that agreement cannot be an artifact of a shared saturation bug reaching the same wrong fixed point from a shared seed.

The enumeration sizes are derived below; they are checkable by hand and pin the claim that the run is exhaustive rather than sampled.

Remark 11.17 (Deriving the registered counts). A sieve on a context V in the poset L is a downward-closed set of subcontexts of V, i.e. a down-set in the Boolean lattice $\mathcal{P}(V)$. The number of down-sets of the Boolean lattice on n points is the Dedekind number $M(n)$, with $M(0) = 2$, $M(1) = 3$, $M(2) = 6$, $M(3) = 20$, $M(4) = 168$. Summing over the sixteen contexts by size:

$$1 \cdot M(4) + 4 \cdot M(3) + 6 \cdot M(2) + 4 \cdot M(1) + 1 \cdot M(0) = 168 + 80 + 36 + 12 + 2 = 298$$

sieves in all — the registered total. For the joined (equivalently singleton) topology, a sieve on V covers iff it contains every singleton of V (Chapter 10, join corollary). Such sieves automatically contain $\varnothing$ and are determined by a down-set of the subcontexts of size at least two. Counting per context size: at $|V| = 0$ both sieves on $\varnothing$ cover (the null cover generates the empty sieve, and its supersieve is maximal), giving 2; at $|V| = 1$ only the maximal sieve, giving 1 each; at $|V| = 2$ the sieve of proper subcontexts and the maximal sieve, giving 2 each; at $|V| = 3$ a free down-set choice over the three 2-subsets and the top, giving 9 each; at $|V| = 4$ a count over down-sets of the size-≥ 2 subsets of Ω gives 114. The total is

$$1 \cdot 2 + 4 \cdot 1 + 6 \cdot 2 + 4 \cdot 9 + 1 \cdot 114 = 2 + 4 + 12 + 36 + 114 = 168$$

covering sieves — the registered count, reported identically for both presentations. The arrow count $81 = \sum_V 2^{|V|}$ was derived in Check 11.2. These derivations are corroboration arithmetic for this monograph; the registered numbers themselves come from the verifier's enumeration. Appendix A tabulates the per-context values.

4.2. What the composition run checks. The deterministic run verifies, in order:

(1) **Base combinatorics.** All 4 atoms, 16 contexts, and 81 inclusion arrows are constructed and counted.
(2) **The sieve universe.** All 298 sieves across the sixteen contexts are enumerated.
(3) **Topology equality.** The join presentation and the singleton presentation saturate to the same 168 covering sieves — the executable content of the join lemma $J_A \vee J_R = J_{\text{atom}}$.
(4) **Axioms on both results.** The Grothendieck axioms (J1)–(J3) are verified independently on each of the two computed topologies.

(5) **Composite decompositions.** At every context, both index-level decompositions of the nested product — the ACBE-then-RPM expansion and the RPM-then-ACBE expansion — are computed and compared with the atomic product, corroborating the objectwise computation in the composition theorem's proof.
(6) **Natural isomorphisms on an adversarial fixture.** Both canonical natural isomorphisms are checked on a constant two-section presheaf whose empty-context value is *deliberately nonterminal*. This fixture exists to exercise the $(R_P F)(\varnothing) = \mathbf{1}$ clause: a wrong implementation that carried $F(\varnothing)$ through the reflector would satisfy most checks on well-behaved presheaves and fail exactly here, at the null cover and at every empty overlap.
(7) **Structured fixtures.** A bit-assignment presheaf (the witness shape) and all 16 representables are run through the same comparisons.
(8) **The full battery.** Inverse laws for the flattening and regrouping maps, restriction naturality (naturality in V), commutation naturality (naturality in F of the composed comparison), and unit compatibility are verified across 18 executable fixtures and 3,012 restriction squares.

Each item corroborates a specific labeled step of Chapter 10: items (3) and (4) the join lemma; item (5) the objectwise flattening computation; items (6)–(8) the naturality, inverse, and unit-compatibility paragraphs of the composition theorem and the well-definedness proposition of the reflector. Two steps of Chapter 10 have no counterpart here: the universal property of the reflector and the sheafhood of $R_P F$ carry no registered check in either executable layer. The covering claim of Section 3.7 is accordingly scoped to the finite theorems listed there and does not extend to those two lemmas, whose machine verification is admission gate 2.

4.3. The fixture battery. The 18 executable fixtures decompose as $16 + 1 + 1$: the sixteen representable presheaves, the constant two-section presheaf with nonterminal empty-context value, and the bit-assignment presheaf. The three families are chosen to stress different failure surfaces; each earns its place as follows.

The representables are the fixtures on which everything should be *inert*: they are sheaves for both factor topologies (subcanonicity), so the units and both composites should act by isomorphisms, and any fixture-dependent bug that breaks canonicity — an indexing error that permutes coordinates, say — shows up as a non-isomorphism or a failed naturality square on some yW. Sixteen of them cover every shape of context, including the empty representable $y\varnothing$ (a singleton exactly at $\varnothing$), which probes the null-cover conventions from the sheaf side.

The constant two-section presheaf is the designed adversary for the empty-product clause, as discussed in Chapter 10: its empty-context value has two elements, so any implementation that carries $F(\varnothing)$ through the reflector, or that treats empty overlaps as imposing real constraints, fails on it and on nothing else so cheaply. It also exercises value *shrinkage* at $\varnothing$ simultaneously with value *growth* at Ω (the reflector value there is a product of two-element sets), so restriction squares crossing the empty context are genuinely two-sided.

The bit-assignment presheaf has the witnesses' shape: values of different cardinalities at different contexts, coordinate-forgetting restrictions, and sensitivity to which blocks are complete. On this fixture the composite decompositions are nondegenerate — nested and flat products differ as concrete data structures — so the flattening and regrouping maps are tested where they actually move something.

Across the battery, the run verifies the inverse laws ($\psi \circ \varphi = \mathrm{id}$ and $\varphi \circ \psi = \mathrm{id}$ objectwise), restriction naturality, commutation naturality, and unit compatibility, totaling 3,012 restriction squares — the registered count of commuting-square instances discharged; the per-fixture breakdown is an implementation detail of the verifier and the monograph cites only the registered total.

4.4. Rerunning the layer. The composition layer's registered commands are reproduced in the artifact and in Chapter 10's source document; the deterministic verifier is invoked as a single script with no arguments, and the scoped suite by standard unit-test discovery over `tests/noetic_topos2_modality_composition`. A reviewer's workflow is: check out commit `a1cd990109e7561dfd03949bf0e1653461c7ae44` (the base commit registered in Chapter 1); run the two registered commands

```
python docs/working/project_coherence/gate1/
  noetic_topos2/modality_composition/
  verify_modality_composition.py
   python -m unittest discover
  -s tests/noetic_topos2_modality_composition
  -p "test_*.py" -v
```

and diff the verifier's report against the `expected_execution_summary` block of

`MODALITY_COMPOSITION_CERTIFICATE_v0.1.json`.

Any discrepancy is, by the fail-closed discipline, a finding about the artifact set — either a regression or an environment sensitivity — and not something to be waved through.

Adverse Finding 11.18. At the bound commit, the modality-composition suite comprises 6 tests and passes in full, and the deterministic verifier run completes with every check above reporting success.

5. The Lean layer: scope and explicit non-scope

The bounded proof-assistant layer is a Lean Gate 1 verifier reported as passing at the bound commit; v0.1 registers no Lean project path, toolchain pin, or build command, so this layer alone is not independently rerunnable from this monograph. Its scope is deliberately narrow, and the narrowness is registered rather than incidental.

Check Class 11.19. **Formalized (in scope).** Four-atom exhaustiveness (the atoms enumerate Ω); properness of the global cover pieces; the transverse refinement identity $A_i \cap R_j = \{(i, j)\}$; and the witness-map lemmas (the elementary combinatorial facts about complete-block containment that feed the witness constructions).

Claim Boundary 11.20. **Not formalized (explicit non-scope).** The Lean layer does *not* formalize: the sixteen presheaves; any sheaf condition; topology generation or the saturation of (J1)–(J3); the subcanonicity theorem; the witness sheaf/non-sheaf verdicts; the reflector construction or its universal property; the join lemma; the composition theorem; the product-site comparison; or the legacy round trip. Any sentence of the form "the Noetic Topos 2.0 results are formally verified" is therefore false. The accurate sentence is: a small set of finite combinatorial lemmas at the base of the construction is machine-checked in Lean; the sheaf-theoretic theory above them is informally proved and exhaustively executed, and its proof-assistant formalization is a registered open obligation (Chapter 12).

6. The certificate's own claim boundary

The certificate closes with a claim boundary of its own, and it governs every number in this chapter. Quoting the registered artifact `FINITE_MODEL_CERTIFICATE_v0.1.json` verbatim:

> "This is exhaustive execution over one sixteen-object finite context site. It is not proof-assistant verification, external peer review, empirical validation, or a product-claim license."

Claim Boundary 11.21. Unpacking the four negations, each against the temptation it blocks:

(1) *Not proof-assistant verification.* The Python layers are ordinary programs; their correctness is not machine-checked, and the Lean scope above excludes the sheaf theory. Exhaustiveness over the model does not transfer trust to the model's formalization.

(2) *Not external peer review.* No independent category theorist has reviewed the proofs or the code. The composition result in particular remains an INTERNAL PROOF CANDIDATE / REVIEW REQUIRED artifact.

(3) *Not empirical validation.* Nothing here measures anything. The ACBE and RPM coordinates are formal marks; no execution result bears on whether either coordinate detects any cognitive, causal, safety, or alignment property of any system.

(4) *Not a product-claim license.* The certificate's passing state may not be cited, in any register of the program, as grounds for a capability, quality, compliance, or assessment claim about any product. The artifact set remains Ring 4 and NOT LICENSED.

What the execution *does* establish is narrower and still valuable: that the registered finite model, as implemented, satisfies every one of the enumerated properties without exception, over its entire state space; that the prose theorems and the executable model agree everywhere both speak; and that the negative expectations (the two *false* sheaf verdicts, the rejection branches of the marked-record resolver, the two legacy collapse mechanisms) are checked as first-class results rather than assumed.

CHAPTER 12

The Ambient Candidate, Boundaries, and the Research Program

1. From the finite factor to the ambient package

The finite-context theory of Chapters 7, 9 and 10 is complete on its own terms, but it is a theory of the context factor L alone. The Noetic Topos 2.0 candidate is a construction over the *product* of the registered legacy category with that factor, and the way the candidate is packaged is itself a mathematical decision with a theorem behind it. This closing chapter assembles the package, states every boundary that fences it, records the backward-compatibility conjecture in its exact conjectural form, sets out the expressivity obligation with its acceptance criteria, situates the work against the literature, and lists the admission gates that stand between the present artifact set and any change of status.

Construction 12.1 (The successor package). Let $\mathcal{C}$ denote the registered legacy category Noetica, assumed small in the chosen universe, with its proven jointly-surjective topology J_{obs} (the legacy topology, written J_{legacy} in the baseline chapters). The successor begins with the ambient presheaf topos

$$\mathcal{E}_{\text{amb}} = \text{PSh}(\mathcal{C} \times L),$$

and generates three topology refinements on it:

(1) legacy observational covers in the $\mathcal{C}$ coordinate;
(2) K_A covers in the L coordinate; and
(3) K_R covers in the L coordinate.

Writing sheafification-followed-by-inclusion for these refinements as M_{obs}, M_A, and M_R, the fully resolved topology and subtopos are

$$J_{\text{res}} = J_{\text{obs}} \vee J_A \vee J_R, \qquad \mathcal{E}_{\text{res}} = \text{Sh}(\mathcal{C} \times L,\, J_{\text{res}}).$$

The underlying topos candidate and the full research object are

$$\text{NoeTop}_2 := \mathcal{E}_{\text{amb}},$$

$$\text{NoeSystem}_2 = \big(\text{NoeTop}_2,\ M_{\text{obs}},\ M_A,\ M_R,\ \mathcal{E}_{\text{res}},\ \textsc{external mark record}\big).$$

NoeSystem_2 is a structured package — a tuple of a topos, three modalities, a subtopos, and an external audit layer. It is *not* itself a topos or a category, and the provenance record is external audit data, not an internal topos object.

The definitional choice $\text{NoeTop}_2 := \mathcal{E}_{\text{amb}}$ — the ambient topos, not the resolved subtopos — is forced by the resolved-identity theorem of Chapter 10: every sheaf for the joined context coverage is separately a sheaf for each factor topology, so on $\mathcal{E}_{\text{res}}$ both M_A and M_R restrict to the identity up to canonical isomorphism. Had the successor been defined as the already-joined sheaf category, the ACBE and RPM reflectors would be identity on every admitted object and the modal structure would be erased a second time — a repetition, one level up, of exactly the collapse the successor exists to repair. The intended discipline is therefore:

raw and partially resolved evidence lives in the ambient topos, acted on by the modalities; only jointly resolved objects enter $\mathcal{E}_{\mathrm{res}}$; and no text may describe M_A or M_R as nontrivial endomodalities of the resolved subtopos.

Claim Boundary 12.2 (The executable schema is narrower than the notation). Construction 12.1 is a research specification, not a completed implementation. The v0.1 executable schema is deliberately narrower: it represents ACBE/RPM context-factor families and their joint refinement only. *Typed observational covers* in the legacy $\mathcal{C}$ coordinate remain OPEN: no typed construction of observational cover records, general arrows of $\mathcal{C} \times L$, full product-site covers, or their serialization exists in v0.1. Consequently M_{obs}, J_{res}, and $\mathcal{E}_{\mathrm{res}}$ are, at present, notation with stated intent, awaiting the typed constructions and proofs listed in the admission gates below. Relatedly, the once-proposed lift of the distinctness theorem to $\mathrm{PSh}(\mathcal{C} \times L)$ by external product of each witness with a representable $y\mathcal{C}_0$ is *not* sound as stated — at the null cover over $(c, \varnothing)$ the product value need not be a singleton, and duplicated representable coordinates can obstruct gluing — so ambient distinctness of M_A and M_R also remains OPEN; the proved distinctness result is scoped to $\mathrm{PSh}(L, \mathbf{Set})$.

Remark 12.3 (Anatomy of the failed ambient lift, and the candidate repair). The failed lift is instructive enough to dissect. The proposal was to carry the witness F_A to the product site as an external product $y\mathcal{C}_0 \times F_A$ — representable in the legacy coordinate, witness in the context coordinate — and argue that at $(\mathcal{C}_0, \Omega)$ the four-versus-one counting argument survives. Two things go wrong. At a null cover over a pair $(c, \varnothing)$, the sheaf condition for the context-coordinate refinement requires a one-element value, but the product value there has as many elements as $y\mathcal{C}_0(c)$, which can be empty or large; the external product does not satisfy the empty-context discipline that Chapter 10's terminal-empty-value lemma forces on every sheaf. And along legacy arrows duplicating representable coordinates, matching families can fail to glue for reasons that have nothing to do with the context factor, contaminating the witness argument. The candidate repair replaces $y\mathcal{C}_0$ by the terminal $\mathcal{C}$-presheaf, trivializing the legacy coordinate — but that repair is only well-posed after the lifted topology is actually defined, requires $\mathcal{C}$ nonempty, and needs naturality checked in both coordinates. None of that has been done, and the lesson already banked is methodological: ambient statements do not follow from finite-context statements by decorating witnesses, and every future lift claim must be proved against the typed product-site definitions of admission gate 3.

Remark 12.4 (Coordinates, refinements, and the intended evidence flow). Some unpacking of Construction 12.1 is useful even while its product-site layer remains open. The three refinements are *coordinate-local* by design: an observational cover touches only the $\mathcal{C}$ coordinate (it is a legacy jointly-surjective family, paired with identities on the context coordinate), while an ACBE or RPM cover touches only the L coordinate (a K_A or K_R family paired with identities on the legacy coordinate). Generating a topology on the product from coordinate-local families is standard in outline, but the candidate treats none of it as automatic: which sieves the joint generation produces at mixed objects, whether sheafification for a context-coordinate refinement can be computed objectwise in $\mathcal{C}$ (pointwise application of Chapter 10's reflector formula), and how the three refinements interact under joins are precisely the contents of admission gates 3 and 4 below. Subject to those gates, the intended evidence flow through the package is a pipeline with typed stages: raw evidence enters as a presheaf on $\mathcal{C} \times L$ (an object of $\mathcal{E}_{\mathrm{amb}}$), carrying arbitrary context-coordinate structure and its marked cover records; partial resolution applies M_A or M_R — meaningfully, in either order, by the composition theorem — while audit code reads the marks that the reflectors provably do not see; and joint resolution lands the object in the modally inert resolved zone, after

which only observational structure remains active. The stage boundaries are mathematically enforced: Chapter 10's resolved-identity theorem makes "resolved but still ACBE-active" a contradiction in terms, which is exactly the kind of design error the package's typing is meant to make unwritable.

REMARK 12.5 (What the package buys, conditionally). Even in its partly-open state, the package fixes three commitments that constrain all future work. First, the carrier of modal structure is ambient: any future theorem about modal dynamics must be stated on $\mathcal{E}_{\mathrm{amb}}$, and any statement on $\mathcal{E}_{\mathrm{res}}$ must be about resolved, modally inert objects. Second, provenance is external: the marks travel beside the topology, in hash-chained records with fail-closed resolution (Chapter 8; executable checks in Chapter 11), because the topology provably cannot carry them. Third, the package is honest about its own type: $\mathrm{NoeSystem}_2$ is a tuple, and calling it a topos would overclaim structure that only its first component has.

2. Gluing boundaries: what Set-valued descent does not license

The finite theory is a theory of **Set**-valued descent: gluing means bijectivity of the canonical map from global sections to matching families of local sections. Two adjacent temptations — reading probabilistic or linear-algebraic consequences out of **Set**-valued sheaf conditions — must be blocked explicitly, because both are natural-sounding and both are wrong.

2.1. The probabilistic boundary and the marginal problem. Probability distributions do not inherit **Set**-valued gluing by analogy. The obstruction is classical: a family of pairwise-compatible marginal distributions need not extend to any global joint distribution. This is the *marginal problem*, and nonextendable compatible families were exhibited already by Vorob'ev [10]; the sheaf-theoretic reformulation of such obstructions, in which empirical models are compatible families over a cover of measurement contexts and contextuality is precisely the failure of a global section, is developed by Abramsky and Brandenburger [11]. The relevance to this program is direct and cautionary: the entire motivation for a context site is to manage heterogeneous evidence, evidence in practice is often probabilistic, and it is therefore tempting to treat a passing **Set**-valued sheaf test as evidence that probabilistic summaries over the same cover cohere. They need not.

EXAMPLE 12.6 (The obstruction in miniature). The phenomenon fits in three binary variables. Let X, Y, Z each take values in $\{0,1\}$, and prescribe the three pairwise marginals to be perfect anticorrelation: each of (X,Y), (Y,Z), (X,Z) is uniform on $\{(0,1),(1,0)\}$. Every pair of these marginals is compatible — each single-variable marginal is uniform on $\{0,1\}$ however computed, so the family agrees on all overlaps. Yet no joint distribution has all three as marginals: a joint supported on triples (x,y,z) with $x \neq y$, $y \neq z$, and $x \neq z$ would need three pairwise-distinct values from a two-element set, so its support is empty, contradicting total mass one. Pairwise compatibility — the exact probabilistic analogue of a matching family — holds; the global section does not exist. In gate terms (Protocol 12.7): the cover $\{XY, YZ, XZ\}$ is cyclic, so gate 1 does not apply; the gate-2 linear program for this instance is infeasible, with the counting argument just given convertible into a Farkas witness; and the family has a nonvanishing obstruction in the sense of the contextuality literature [12, 13]. Any evidence-aggregation design over the Noetic context factor that reports probabilistic summaries per cover piece must therefore treat joint consistency as a per-instance obligation, not a structural gift of the site.

A probabilistic successor to the present construction must therefore pass through one of four explicit gates, each of which supplies what the analogy omits:

PROTOCOL 12.7 (The four probabilistic gates).

(1) **Junction-tree extension.** Restrict to covers whose intersection structure is acyclic (a junction tree), and supply a proved extension theorem that compatible marginals over such a cover always admit a global joint. Acyclicity is exactly the classical regime in which the marginal problem is solvable, and the proof obligation is the extension theorem itself, stated for the specific cover class used.

(2) **LP feasibility certificate.** For a fixed finite cover and finite outcome spaces, existence of a global joint is a finite linear program: variables the putative joint probabilities, constraints the marginalization equations and nonnegativity. A probabilistic consistency claim is then licensed by a feasibility certificate for that LP — an explicit feasible point or a solver artifact — and refuted by an infeasibility certificate (a Farkas witness). The certificate is per-instance: it licenses the instance, not the method.

(3) **Obstruction calculation with stated coefficients.** Compute a cohomological obstruction to extending the compatible family, in the style of the sheaf-theoretic contextuality literature [12], with the coefficient system stated explicitly. Vanishing of the obstruction in the stated coefficients is the licensed claim — no more; such obstructions can be one-directional, and the direction proved must be the direction cited.

(4) **Quantitative relaxation.** Where exact extension fails or is not decided, a quantitative claim is permitted only with the relaxation stated (e.g. nearest extendable family in a named metric), the residual reported, and the uncertainty of the residual reported. The sheaf-theoretic literature standardizes one such relaxation: the contextual fraction of [14], computed by the same linear program as gate 2.

No passing **Set**-valued sheaf test licenses a probabilistic consistency claim; only a gate artifact of one of these four kinds does.

2.2. The linear and cellular boundary. A second analogy must be blocked in the same motion. The applied-topology literature attaches spectral invariants — sheaf Laplacians, harmonic sections, diffusion dynamics — to *cellular sheaves* of vector spaces, and these tools are attractive for evidence aggregation. But a linear or cellular sheaf is additional structure, not a reinterpretation: it requires finite-dimensional stalks (or controlled Hilbert-space stalks), chosen inner products, and weighted coboundary or restriction data before a Laplacian is even defined [15]. Nothing in the present construction creates that structure. The context lattice L carries **Set**-valued presheaves; no stalk is a vector space, no inner product has been chosen, and no coboundary operator exists. Any future spectral or diffusion-style claim over the Noetic context factor requires a separate linear representation with its own registered definitions and proofs, and bounded Hilbert-valued cellular sheaves need not be finite-dimensional, so even the passage from finite to infinite dimensions requires care rather than analogy.

To make the missing structure concrete: a cellular sheaf in the sense of [15] assigns a vector space to each cell of a complex and a linear map to each incidence, and its Laplacian is built from a chosen coboundary operator and inner products on stalks — so that "harmonic" sections, spectral gaps, and consistency energies are all statements about *that chosen linear data*, and change when the inner products change. The Noetic context factor supplies only the incidence skeleton (the inclusion order on L) and **Set**-valued stalks. Between here and any Laplacian-style consistency score lie, at minimum: a linearization of section sets (itself a modeling decision with alternatives), a choice of inner products with a stated rationale, a

weighting of the coboundary reflecting cover semantics, and proofs that the resulting invariants respect the reflectors of Chapter 10 or an argument for why they need not. Each is a registered-definition-sized gap, none is closed by this monograph, and any program text that computes a "coherence energy" over the context factor without them is operating outside the licensed mathematics.

3. Backward compatibility as a conjectural comparison

The successor would be of limited use if it stranded the legacy theory. The intended relationship is a comparison, and its status is exactly CONJECTURE with PROOF REQUIRED — the only record in the finite theorem set that is not backed by a proof, and the monograph states it in conjectural form for that reason.

CONJECTURE 12.8 (Product-site round trip; ledger THM-NT2-PRODUCT-SITE-BACKWARD-COMPATIBILITY, PROOF REQUIRED). *Let $\mathcal{E}_{\text{legacy}} = \mathrm{Sh}(\mathcal{C}, J_{\text{legacy}})$ be the legacy sheaf topos, preserved verbatim. After atomic resolution of the context factor there is an equivalence*

$$\mathcal{E}_{\text{res}} \;\simeq\; \mathcal{E}_{\text{legacy}}^{4},$$

under which, for a registry-fixed neutral atom ω_{00}, the embedding

$$\iota : X \longmapsto (X, 1, 1, 1)$$

— the legacy object in the neutral channel, terminal objects in the remaining three channels — is fully faithful, and evaluation at the neutral channel satisfies a specified natural isomorphism

$$\mathrm{ev}_{00} \circ \iota \;\cong\; \mathrm{Id}_{\mathcal{E}_{\text{legacy}}}.$$

REMARK 12.9 (Why the shape is plausible and why that is not a proof). The finite-context theory makes the *shape* of the conjecture plausible: the joined context coverage resolves the four singleton atoms (Chapter 10), and sheaves on a discrete four-point context factor should decompose into four independent components. But the conjecture is a statement about the *product* site $\mathcal{C} \times L$ with the observational topology in play, and none of the required ingredients — the typed product-site covers (Boundary 12.2), the comparison functors, the full-faithfulness argument, the coherence of the equivalence with J_{obs} — currently exists. A proof would need, at minimum: a typed construction of J_{res} on $\mathcal{C} \times L$; a computation showing sheafification for the joined context coordinate is objectwise in $\mathcal{C}$ (the product-site lift of the composition theorem, itself open); and the equivalence with naturality in all data.

CLAIM BOUNDARY 12.10 (Two round trips, and neither is complete). The compatibility contract requires *two distinct* round trips, and they must not be conflated:

(1) **The categorical round trip.** The equivalence and natural isomorphism of Conjecture 12.8. This is mathematics, and its status is PROOF REQUIRED.
(2) **The serialization round trip.** The implementation requirement that registered identifiers and canonical payload hashes survive migration into and out of the new context factor *by strict equality*. This is engineering with a cryptographic acceptance test, and it is likewise incomplete: the v0.1 context-factor checker does not establish it.

A categorical natural isomorphism is not a bit-for-bit serialization receipt, and a passing serialization test would not prove the equivalence. Both are required; possession of one licenses nothing about the other. Additionally, because subcanonicity of the legacy site is not derivable from the registered axioms (the baseline countermodel), bare legacy site objects are represented in the sheaf topos by their sheafifications $a_{\text{legacy}}(yX)$ before any marking or

migration — never by silently treating yX as a sheaf [6]. This monograph claims no backward compatibility; no legacy theorem is promoted merely because it has been serialized into the new context factor.

REMARK 12.11 (What full faithfulness would and would not buy). We spell out the operational content the conjecture would have if proved, because that content is the reason the conjecture is registered at all. Full faithfulness of ι would say that legacy objects and *all their morphisms* survive the move into the neutral channel without collapse or spurious identification: maps between embedded legacy objects in the resolved successor would correspond exactly to legacy maps. The natural isomorphism $\mathrm{ev}_{00} \circ \iota \cong \mathrm{Id}$ would say the move is reversible at the object level by evaluation. Together they would license the migration discipline the program wants: analyze in the successor, project back, and lose nothing that the legacy semantics could see. What even a full proof would *not* buy: preservation of anything the legacy semantics could not see (marks are external on both sides and migrate by their own serialization contract); any statement about the three non-neutral channels, which the embedding deliberately fills with terminal objects; and any transfer of legacy theorem *statuses* — a legacy theorem cited in successor context still carries its own ledger record, including the legacy subcanonicity gap. The neutral channel being registry-fixed matters for the same discipline: if different artifacts chose different neutral atoms, their embeddings would disagree by a nontrivial symmetry of the context factor, and cross-artifact comparisons would silently break; fixing ω_{00} in the registry makes the choice auditable.

4. The expressivity boundary and the cross-context obligation

Suppose Conjecture 12.8 were proved tomorrow. A further boundary would immediately bind, and it is in some ways the deepest one in the monograph: $\mathcal{E}^4_{\mathrm{legacy}}$ is four *independent* legacy channels. The discrete resolved context factor contains no remaining overlap, no transition law, and no cross-atom coupling. The composition theorem lands every composite in the atomic subcategory, where the modalities freeze (Chapter 10, identity-on-atomic corollary); after resolution there is nothing left for the context factor to say about how the four channels relate. The sixteen-context construction is therefore a finite *separation and provenance reference model* — it proves that distinct, nontrivial, commuting context modalities are possible where the legacy clauses provably could not provide them — and not a substantively rich final semantics for contextual cognition.

The registered proof-obligation ledger carries this as a first-class record, and its content is reproduced here in full because it, rather than any theorem, determines the shape of the next stage of the program. The *ledger* is not, and the distinction matters to a reader auditing completeness: what follows is one obligation record, not the register. How many records the ledger holds in total is not among the figures registered in the source artifacts of Chapter 1, Section 1.9.2, so under Rule 1.3(iv) that count is declared unspecified here rather than guessed; the register itself lives in the proof-obligation ledger v0.1, cited there with its digest. What this chapter does reproduce whole is the ten admission gates of Section 6.

INVARIANT 12.12 (Obligation OBL-NT2-CROSS-CONTEXT-SEMANTICS, v0.1). *Kind:* design-and-proof obligation. *Statement:* choose and formalize one conservative successor contract: either (a) the structured ambient modal package is the principal Noetic Topos 2.0 semantics, or (b) the resolved semantics gains typed overlap or transition data with proved coherence. The finite discrete product $\mathcal{E}^4_{\mathrm{legacy}}$ alone supplies separation and provenance channels but no cross-atom coupling. *Assumptions:* the joined context coverage resolves the four singleton

atoms; the proposed product-site comparison is $\mathcal{E}_{\mathrm{res}} \simeq \mathcal{E}^4_{\mathrm{legacy}}$. *Dependencies:* the join theorem (ledger THM-NT2-JOIN-DISCRETE-CONTEXT) and the backward-compatibility conjecture (ledger THM-NT2-PRODUCT-SITE-BACKWARD-COMPATIBILITY). *Review state:* UNASSESSED; *resolution status:* OPEN; *ring:* 4.

Acceptance criteria (all four required):

(1) a single conservative successor contract is selected without asserting unproved equivalences;
(2) the selected contract has typed definitions and explicit coherence obligations;
(3) any claimed coherence theorem has a schema-valid theorem-ledger record and a reviewable proof artifact; and
(4) the four-independent-channel limitation remains explicit unless cross-context coupling is proved.

Claim boundary of the record itself: the current finite model is a reference construction; it is not licensed as the final substantive Noetic Topos 2.0 semantics.

REMARK 12.13 (The two branches, concretely). Branch (a) names the ambient package — topos plus modalities plus resolved subtopos plus marks — as the semantics, accepting that the interesting structure lives in the *unresolved* region and that resolution is an endpoint, not a home. Branch (b) enriches the resolved side: typed overlap objects, transition data between atoms, or a coarser-than-discrete resolved context structure, each with a coherence theorem. The branches are not exclusive forever, but the obligation demands that *one* be selected and formalized first, precisely to prevent the program from oscillating between an unformalized version of each. v0.1 licenses neither branch, and this monograph selects neither; selection is a gated decision with its own review.

5. Related work and novelty status

The mathematical background this monograph stands on is entirely standard, and no novelty is claimed for any of it: sites, sieves, and the passage from a coverage to a Grothendieck topology; sheafification as a left-exact left adjoint and its universal property; representable sheaves and subcanonicity; and the Grothendieck-topos framework generally [**1**, **4**, **5**, **6**]; with the local-operator perspective on modalities available in [**7**]. Two applied literatures motivate the explicit boundaries of Section 2 without being incorporated into the construction: the sheaf-theoretic treatment of nonlocality and contextuality, where global-section failure over a cover of measurement contexts is the formal content of contextuality [**11**], with the underlying marginal-problem obstructions classical [**10**]; and the spectral theory of cellular sheaves [**15**], which shows exactly how much added linear structure a Laplacian requires and therefore exactly what the present **Set**-valued theory lacks.

Against that background, the candidate's specific contributions are three: the collapse diagnosis as applied to the registered legacy ACBE/RPM clauses; the transverse two-partition finite context factor with its exact reflector calculus (Chapter 10); and the parallel hash-chained provenance layer with fail-closed resolution (Chapter 8, corroborated in Chapter 11). Their status must be stated exactly: all theorem-ledger records carry NOVELTY UNASSESSED. No dedicated literature search has been performed for the relationship of the marked-cover construction to colored sites, indexed categories, or stacks; nor for whether the explicit product-over-blocks reflector for partition coverages on a powerset lattice appears elsewhere (it is elementary enough that it plausibly does); nor for prior uses of transverse partition pairs to manufacture commuting distinct modalities. This monograph accordingly does not call any of its results new theorems in the scholarly sense. The claim made is weaker and

auditable: the results are *internally novel* to the program, correctly scoped, and fully proved or fully flagged.

6. The admission gates

Noetic Topos 2.0 may leave Ring 4 only after all ten of the following close. The list is registered; this section adds, for each gate, what closing it concretely requires.

PROTOCOL 12.14 (The ten admission gates).

(1) **Independent category-theory review** of the collapse and countermodel propositions and of the finite-context theorems through the composition theorem. Closing requires a named, independent reviewer; internal re-derivation, however careful, does not qualify.
(2) **Proof-assistant formalization** of the finite site, the generated topologies, the witness presheaves, the reflector universal properties, the join, and the natural isomorphisms — the full sheaf-theoretic layer excluded from the current Lean scope (Chapter 11).
(3) **Typed observational and product-site covers:** a typed construction of observational covers and general arrows in $\mathcal{C} \times L$, discharging Boundary 12.2.
(4) **The product-site lift** of the composition theorem: its coherence in the $\mathcal{C}$ coordinate and its interaction with J_{obs}. Until this closes, commutation is a finite-context fact only.
(5) **The backward-compatibility conjecture** (Conjecture 12.8) with *both* round trips of Boundary 12.10: the categorical equivalence with proofs, and the strict-equality serialization round trip with tests.
(6) **A canonical cross-context decision** under obligation OBL-NT2-CROSS-CONTEXT-SEMANTICS (Invariant 12.12), meeting all four acceptance criteria.
(7) **Probabilistic descent**: separate probability-gluing theorems or feasibility certificates through the four gates of Protocol 12.7, for any probabilistic use of the site.
(8) **A dedicated novelty search** for the marked/provenance construction and the reflector calculus, converting NOVELTY UNASSESSED into an assessed status in either direction.
(9) **Registry reconciliation and IP review**: reconciliation with the shared symbol and theorem registries, and authorship, disclosure, citation, and IP review of the whole artifact set.
(10) **External referee review**, with explicit confirmation that no empirical or product wording was licensed anywhere in the artifact set.

Gates 1, 2, and 10 concern trust in the mathematics; gates 3, 4, and 5 concern the ambient and legacy extensions; gates 6 and 7 concern semantics beyond the finite reference model; gates 8 and 9 concern scholarly and institutional hygiene. They are conjunctive: no subset suffices, and no gate is closed by the drafting of this monograph.

The gates also carry an implicit dependency order that any realistic program plan must respect. Gate 3 (typed product-site covers) is upstream of gates 4 and 5: neither the product-site lift nor the backward-compatibility equivalence can even be *stated* precisely until the typed constructions exist, which is why both are conjectural or open rather than merely unproved. Gate 5 is upstream of gate 6: the cross-context obligation's assumptions include the proposed comparison $\mathcal{E}_{\text{res}} \simeq \mathcal{E}^4_{\text{legacy}}$, so a canonical semantics decision made before the comparison is settled would rest on an unproved premise — permitted only under branch (a), which does not need the comparison, and then only with the four-channel limitation stated

as conditional. Gates 1 and 2 can proceed in parallel with everything else and de-risk it: independent review or formalization that found an error in the finite-context theory would invalidate downstream work cheaply rather than expensively. Gates 8, 9, and 10 are terminal: novelty search, registry and IP review, and external refereeing evaluate the finished artifact set, and doing them against a moving target would waste their one-shot evidential value. None of this ordering relaxes conjunctivity; it only says in which order the program can stop being wrong soonest.

7. Conclusion

The monograph's argument closes where it began, with the two collapse mechanisms. The legacy ACBE and RPM cover clauses could not carry modal structure because identities entered their sieves: the reflexive descent map directly, the RPM counit through its section by sieve closure. That is not a defect of topos machinery but a theorem about what a Grothendieck topology can remember, and it dictated both halves of the repair. The topological half changes the generators: a finite context lattice with two transverse partition coverages whose global pieces are proper, so that the generated sieves are nonmaximal and the modalities live. On that factor the theory is complete and exact — genuine subcanonical topologies; distinct, non-identity modalities separated by explicit witnesses; a computed join; explicit reflectors verified directly against the universal property; and the composition theorem, which is corrective as well as constructive: the modalities are distinct but *commuting*, either composite is sheafification for the join, and no artifact licenses a noncommutation claim. The provenance half moves what the topology forgets into hash-chained marked records that fail closed, checked at the serialization level by the same certificate that checks the theorems.

Around that completed core, every remaining surface is fenced. The ambient package is named honestly as a structured tuple whose executable schema is narrower than its notation; the probabilistic and linear analogies are blocked by explicit gates; backward compatibility is a conjecture with two incomplete round trips; the resolved semantics is four independent channels until coupling is proved, under a registered obligation with acceptance criteria; novelty is unassessed; and ten conjunctive gates stand between the artifact set and any exit from Ring 4. The monograph's contribution is the pairing: a small theory proved to the bottom, and a boundary discipline maintained with the same precision as the proofs. Preserving that pairing — promoting nothing that the ledger does not promote, while defending everything it does — is the standard against which any successor to this successor should be judged.

APPENDIX A

Context and Cover Tables

This appendix tabulates, exhaustively and with derivations, the raw combinatorial data of the finite context site: the sixteen contexts and their complete-block content, the generating cover families $K_A(V)$ and $K_R(V)$ at every context, the contexts at which the two families coincide as unmarked data, and the per-context sieve and covering-sieve counts that the exact-topology enumeration of Chapter 11 reports. Everything here is derivable by hand from the definitions in Chapter 7 (restated as standing data in Chapter 10); where a value is also registered in the executable certificate, the correspondence is noted. No entry in this appendix carries any status beyond finite combinatorics: the ACBE and RPM coordinate names remain formal marks with no empirical interpretation.

1. Naming conventions

NOTATION A.1. Atoms are $\omega_{ij} = (i, j) \in \Omega$ for $i, j \in \{0, 1\}$, where the first coordinate is the formal ACBE mark and the second the formal RPM mark. Blocks are $A_i = \{\omega_{i0}, \omega_{i1}\}$ (fixed first coordinate) and $R_j = \{\omega_{0j}, \omega_{1j}\}$ (fixed second coordinate). For the ten contexts that are neither $\varnothing$, a block, nor Ω, we use: the four singletons $\{\omega_{ij}\}$; the two diagonals

$$\Delta = D_{=} = \{\omega_{00}, \omega_{11}\}, \qquad \nabla = D_{\neq} = \{\omega_{01}, \omega_{10}\};$$

and the four triples $T_{ij} = \Omega \setminus \{\omega_{ij}\}$, each named by the atom it omits. The two diagonals are the diagonal doubletons $D_{=}$, $D_{\neq}$ of Chapter 7, which the appendices abbreviate Δ, ∇ (not to be confused with the constant presheaf ΔS of Chapter 2). A block B is *complete in* V when $B \subseteq V$; the witness presheaves of Appendix C are built on complete-block content, so the containment columns below feed directly into those tables.

The containment rules used throughout are immediate from the definitions: $A_0 \subseteq V$ iff $\omega_{00}, \omega_{01} \in V$; $A_1 \subseteq V$ iff $\omega_{10}, \omega_{11} \in V$; $R_0 \subseteq V$ iff $\omega_{00}, \omega_{10} \in V$; $R_1 \subseteq V$ iff $\omega_{01}, \omega_{11} \in V$. Note that a two-element context contains a complete A-block only if it *is* that block, and likewise for R-blocks; the diagonals Δ and ∇ contain no complete block of either partition, because each meets each block in exactly one atom. A triple T_{ij} omits one atom and therefore breaks exactly one A-block (A_i) and exactly one R-block (R_j), containing the other one of each.

2. The sixteen contexts and their complete blocks

Context V	Elements	$\lvert V\rvert$	Arrows into V	Complete A-blocks	Complete R-blocks
$\varnothing$	—	0	1	none	none
$\{\omega_{00}\}$	ω_{00}	1	2	none	none
$\{\omega_{01}\}$	ω_{01}	1	2	none	none
$\{\omega_{10}\}$	ω_{10}	1	2	none	none
$\{\omega_{11}\}$	ω_{11}	1	2	none	none
A_0	ω_{00}, ω_{01}	2	4	A_0	none

Context V	Elements	$\lvert V\rvert$	Arrows into V	Complete A-blocks	Complete R-blocks
A_1	ω_{10}, ω_{11}	2	4	A_1	none
R_0	ω_{00}, ω_{10}	2	4	none	R_0
R_1	ω_{01}, ω_{11}	2	4	none	R_1
Δ	ω_{00}, ω_{11}	2	4	none	none
∇	ω_{01}, ω_{10}	2	4	none	none
T_{00}	$\omega_{01}, \omega_{10}, \omega_{11}$	3	8	A_1	R_1
T_{01}	$\omega_{00}, \omega_{10}, \omega_{11}$	3	8	A_1	R_0
T_{10}	$\omega_{00}, \omega_{01}, \omega_{11}$	3	8	A_0	R_1
T_{11}	$\omega_{00}, \omega_{01}, \omega_{10}$	3	8	A_0	R_0
Ω	all four	4	16	A_0, A_1	R_0, R_1

The "arrows into V" column counts the inclusions $U \subseteq V$, i.e. $2^{|V|}$; summing the column gives $1 + 4 \cdot 2 + 6 \cdot 4 + 4 \cdot 8 + 16 = 81$, the registered arrow count of the site (Chapter 11). The complete-block columns support a consistency count on the distribution of complete A-blocks over contexts, written $\alpha(V)$ as in Chapter 7: $\alpha(V) = 2$ occurs only at Ω (one context); $\alpha(V) = 1$ occurs at A_0, A_1, and the four triples (six contexts); $\alpha(V) = 0$ occurs at the remaining nine contexts ($\varnothing$, four singletons, R_0, R_1, Δ, ∇). Totals: $1 + 6 + 9 = 16$. The same distribution holds for the complete-R-block count $\rho(V)$ by the evident symmetry exchanging the two coordinates. These counts give the value cardinalities $|F_A(V)| = 2^{\alpha(V)}$ tabulated in Appendix C: nine contexts with one section, six with two, one with four.

3. The generating family $K_A(V)$ at every context

$K_A(V) = \{V \cap A_0 \to V,\ V \cap A_1 \to V\}$ at nonempty V, retaining empty pieces; the null family at $\varnothing$. Each row lists the two pieces; "$= V$" flags a piece equal to the whole context (an identity arrow) and "$= A_i$" flags a piece that is a full block.

Context V	A_0-piece $V \cap A_0$	A_1-piece $V \cap A_1$	Remarks
$\varnothing$	null family (no members)		covers by convention
$\{\omega_{00}\}$	$\{\omega_{00}\} = V$	$\varnothing$	identity piece present
$\{\omega_{01}\}$	$\{\omega_{01}\} = V$	$\varnothing$	identity piece present
$\{\omega_{10}\}$	$\varnothing$	$\{\omega_{10}\} = V$	identity piece present
$\{\omega_{11}\}$	$\varnothing$	$\{\omega_{11}\} = V$	identity piece present
A_0	$A_0 = V$	$\varnothing$	identity piece present
A_1	$\varnothing$	$A_1 = V$	identity piece present
R_0	$\{\omega_{00}\}$	$\{\omega_{10}\}$	both proper singletons
R_1	$\{\omega_{01}\}$	$\{\omega_{11}\}$	both proper singletons
Δ	$\{\omega_{00}\}$	$\{\omega_{11}\}$	both proper singletons
∇	$\{\omega_{01}\}$	$\{\omega_{10}\}$	both proper singletons
T_{00}	$\{\omega_{01}\}$	$\{\omega_{10}, \omega_{11}\} = A_1$	both proper
T_{01}	$\{\omega_{00}\}$	$\{\omega_{10}, \omega_{11}\} = A_1$	both proper
T_{10}	$\{\omega_{00}, \omega_{01}\} = A_0$	$\{\omega_{11}\}$	both proper
T_{11}	$\{\omega_{00}, \omega_{01}\} = A_0$	$\{\omega_{10}\}$	both proper
Ω	A_0	A_1	both proper: the global cover

Two structural observations, both used in the body of the monograph. First, the generated sieve is nonmaximal exactly when *no* piece is an identity: this happens at $\varnothing$, R_0, R_1, Δ, ∇, the four triples, and Ω (ten contexts). At the four singletons and at A_0, A_1, one piece is the identity, the generated sieve is maximal, and the cover imposes no sheaf constraint beyond functoriality; at $\varnothing$ the nonmaximality takes the degenerate form of the empty sieve, generated by the null family, which imposes the one-section constraint. The certificate check `proper_global_covers` registers the Ω row. Second, at every context each nonempty piece lies inside a single block, so the singleton-factor bijection of Chapter 10 applies to it; this is visible in the table as the absence of any piece meeting both blocks.

4. The generating family $K_R(V)$ at every context

$K_R(V) = \{V \cap R_0 \to V,\ V \cap R_1 \to V\}$, same conventions.

Context V	R_0-piece $V \cap R_0$	R_1-piece $V \cap R_1$	Remarks
$\varnothing$	null family (no members)		covers by convention
$\{\omega_{00}\}$	$\{\omega_{00}\} = V$	$\varnothing$	identity piece present
$\{\omega_{01}\}$	$\varnothing$	$\{\omega_{01}\} = V$	identity piece present
$\{\omega_{10}\}$	$\{\omega_{10}\} = V$	$\varnothing$	identity piece present
$\{\omega_{11}\}$	$\varnothing$	$\{\omega_{11}\} = V$	identity piece present
A_0	$\{\omega_{00}\}$	$\{\omega_{01}\}$	both proper singletons
A_1	$\{\omega_{10}\}$	$\{\omega_{11}\}$	both proper singletons
R_0	$R_0 = V$	$\varnothing$	identity piece present
R_1	$\varnothing$	$R_1 = V$	identity piece present
Δ	$\{\omega_{00}\}$	$\{\omega_{11}\}$	both proper singletons
∇	$\{\omega_{10}\}$	$\{\omega_{01}\}$	both proper singletons
T_{00}	$\{\omega_{10}\}$	$\{\omega_{01}, \omega_{11}\} = R_1$	both proper
T_{01}	$\{\omega_{00}, \omega_{10}\} = R_0$	$\{\omega_{11}\}$	both proper
T_{10}	$\{\omega_{00}\}$	$\{\omega_{01}, \omega_{11}\} = R_1$	both proper
T_{11}	$\{\omega_{00}, \omega_{10}\} = R_0$	$\{\omega_{01}\}$	both proper
Ω	R_0	R_1	both proper: the global cover

The table is the image of the K_A table under the coordinate exchange $(i, j) \mapsto (j, i)$, which swaps A-blocks with R-blocks, fixes $\varnothing$, Δ, and Ω, and permutes the remaining contexts accordingly ($A_i \leftrightarrow R_i$, ∇ fixed as a set with its atoms swapped, $T_{01} \leftrightarrow T_{10}$). This symmetry is the tabular form of the coordinate-swap automorphism of Chapter 7, by which every ACBE statement has an exact RPM mirror (used throughout Chapter 9).

5. Coincidence of the unmarked families

The marks chapter and the certificate checks `singleton_unmarked_families_coincide` / `singleton_marked_covers_distinct` turn on the question: at which contexts do $K_A(V)$ and $K_R(V)$ carry the same information once the block labels are stripped? Comparing the two tables row by row, as *unordered sets of member arrows* (the exact list the topology consumes, forgetting which block indexed which piece):

Context V	$K_A(V)$ as arrow set	$K_R(V)$ as arrow set	Coincide?
$\varnothing$	null	null	yes

Context V	$K_A(V)$ as arrow set	$K_R(V)$ as arrow set	Coincide?
$\{\omega_{ij}\}$ (each of 4)	$\{V, \varnothing\}$	$\{V, \varnothing\}$	yes
A_0	$\{A_0, \varnothing\}$	$\{\{\omega_{00}\}, \{\omega_{01}\}\}$	no
A_1	$\{A_1, \varnothing\}$	$\{\{\omega_{10}\}, \{\omega_{11}\}\}$	no
R_0	$\{\{\omega_{00}\}, \{\omega_{10}\}\}$	$\{R_0, \varnothing\}$	no
R_1	$\{\{\omega_{01}\}, \{\omega_{11}\}\}$	$\{R_1, \varnothing\}$	no
Δ	$\{\{\omega_{00}\}, \{\omega_{11}\}\}$	$\{\{\omega_{00}\}, \{\omega_{11}\}\}$	yes
∇	$\{\{\omega_{01}\}, \{\omega_{10}\}\}$	$\{\{\omega_{01}\}, \{\omega_{10}\}\}$	yes
T_{00}	$\{\{\omega_{01}\}, A_1\}$	$\{\{\omega_{10}\}, R_1\}$	no
T_{01}	$\{\{\omega_{00}\}, A_1\}$	$\{R_0, \{\omega_{11}\}\}$	no
T_{10}	$\{A_0, \{\omega_{11}\}\}$	$\{\{\omega_{00}\}, R_1\}$	no
T_{11}	$\{A_0, \{\omega_{10}\}\}$	$\{R_0, \{\omega_{01}\}\}$	no
Ω	$\{A_0, A_1\}$	$\{R_0, R_1\}$	no

Adverse Finding A.2. The unmarked families coincide at exactly seven of the sixteen contexts: the empty context, the four singletons, and the two diagonals Δ and ∇. At these contexts the generated sieves are equal and the topology retains no trace of which partition produced the cover; only the marked records (different cover kinds, different digests) distinguish them. At the remaining nine contexts the arrow sets differ and the topology itself separates the covers. Three refinements sharpen this statement. (i) The singleton case is the one registered as a certificate check (`singleton_unmarked_families_coincide`, with its partner `singleton_marked_covers_distinct`); the empty-context and diagonal cases are derived here directly from the tables and are not separately registered in v0.1. (ii) At Δ the coincidence is aligned block-by-block: the A_0-piece equals the R_0-piece ($\{\omega_{00}\}$) and the A_1-piece equals the R_1-piece ($\{\omega_{11}\}$). At ∇ the coincidence is *crossed*: the A_0-piece equals the R_1-piece ($\{\omega_{01}\}$) and the A_1-piece equals the R_0-piece ($\{\omega_{10}\}$). So even among coinciding contexts, the indexed-by-block families agree at Δ-type contexts only up to relabeling and at ∇ only up to a label crossing — a second, finer layer of information the unmarked arrow set forgets. (iii) The coincidence pattern is forced by block geometry: $K_A(V) = K_R(V)$ as arrow sets can happen only when every piece of both covers has at most one atom, i.e. when V meets each block in at most one atom, which characterizes exactly the seven listed contexts (the subsets of Ω containing no complete block and no two atoms sharing a block occur precisely at sizes ≤ 1 and at the two diagonals).

6. The common refinement at every context

The common refinement of $K_A(V)$ and $K_R(V)$ has as members the four pairwise intersections $V \cap A_i \cap R_j$, $(i, j) \in \{0, 1\}^2$. By transversality, $V \cap A_i \cap R_j = V \cap \{\omega_{ij}\}$: the member indexed by (i, j) is $\{\omega_{ij}\}$ when $\omega_{ij} \in V$ and $\varnothing$ otherwise. The refinement is thus the singleton family of V padded with empty pieces — the context-by-context content of the certificate check `transverse_refinement_is_singletons` and, at Ω, of `joint_refinement_is_four_singletons`.

Context V	Nonempty refinement members	Empty members
$\varnothing$	none	4
$\{\omega_{ij}\}$ (each of 4)	$\{\omega_{ij}\}$	3
A_0	$\{\omega_{00}\}, \{\omega_{01}\}$	2
A_1	$\{\omega_{10}\}, \{\omega_{11}\}$	2

Context V	Nonempty refinement members	Empty members
R_0	$\{\omega_{00}\}, \{\omega_{10}\}$	2
R_1	$\{\omega_{01}\}, \{\omega_{11}\}$	2
Δ	$\{\omega_{00}\}, \{\omega_{11}\}$	2
∇	$\{\omega_{01}\}, \{\omega_{10}\}$	2
T_{ij} (each of 4)	the three singletons of T_{ij}	1
Ω	all four singletons	0

Empty members and their repetitions do not change the generated sieve, so at every context the refinement generates the sieve of size-≤ 1 subcontexts — the generating sieve of J_{atom}. This is the per-context form of the join computation in Chapter 10, and it also explains the coincidence pattern of Finding A.2 from another angle: at the seven coinciding contexts, $K_A(V)$ and $K_R(V)$ each *already equal* their own common refinement up to empty members, so nothing remains to distinguish them at the arrow level.

7. Per-context sieve and covering-sieve counts

For completeness we tabulate the sieve universe and the joined-topology covering counts underlying the registered totals 298 (derived in Chapter 7) and 168 (derived in Chapter 9), both reported by the certificate of Chapter 11. A sieve on V is a down-set of $\mathcal{P}(V)$, counted by the Dedekind number $M(|V|)$; a sieve is covering for $J_A \vee J_R = J_{\text{atom}}$ iff it contains every singleton of V (Theorem NT2-4, Chapter 9; re-derived in reflector-facing form in Chapter 10).

Context class	Count of contexts	Sieves on each	Covering (join) on each	Subtotals (sieves / covering)
$\lvert V\rvert = 0$	1	$M(0) = 2$	2	2 / 2
$\lvert V\rvert = 1$	4	$M(1) = 3$	1	12 / 4
$\lvert V\rvert = 2$	6	$M(2) = 6$	2	36 / 12
$\lvert V\rvert = 3$	4	$M(3) = 20$	9	80 / 36
$\lvert V\rvert = 4$	1	$M(4) = 168$	114	168 / 114
Totals	16	—	—	298 / 168

Derivation notes for the covering column. At $\varnothing$ both sieves cover: the null cover generates the empty sieve, and its unique supersieve is maximal. At a singleton the only covering sieve is maximal (the sole generating family contains the identity piece). At $|V| = 2$ the covering sieves are the sieve of all proper subcontexts and the maximal sieve. At $|V| = 3$, a covering sieve contains all four subcontexts of size ≤ 1; the free data is a down-set over the three 2-subsets and the top, giving $2^3 = 8$ choices without the top plus 1 with it (the top forces all 2-subsets), total 9. At Ω, the free data is a down-set of the eleven subcontexts of size ≥ 2; counting by the set of included 3-subsets (each forcing its three 2-subsets, with any two 3-subsets sharing one 2-subset and any three covering all six) gives $64 + 32 + 12 + 4 + 1 = 113$ down-sets without Ω itself plus 1 with it, total 114. These hand counts reproduce the enumeration totals that the modality-composition verifier computes independently and reports identically for both presentations of the joined topology.

APPENDIX B

The Sieve Census

This appendix records, for reference, the combinatorial data underlying the executable topology closures discussed in Chapter 11. Every count in this appendix is reproduced from the modality-composition verifier's registered output or is directly derivable from the definitions in Chapters 7 and 9; none is an independent claim. The governing status remains DRAFT / GATE 1 RESEARCH CANDIDATE / RING 4 / NOT LICENSED, and exhaustive enumeration over one sixteen-object site is corroboration of the finite proofs, not proof-assistant verification or independent review.

1. The underlying poset

Recall from Chapter 7 that $\Omega = \{(0,0),(0,1),(1,0),(1,1)\}$ and $L = \mathcal{P}(\Omega)$ ordered by inclusion. We abbreviate an atom (i,j) by the two-digit string ij, so the four atoms are $00, 01, 10, 11$ and a context is written as the concatenation of its atoms in lexicographic order (the empty context is written $\varnothing$).

Table 1: The sixteen contexts of L by cardinality.

Size	Count	Contexts
0	1	$\varnothing$
1	4	00, 01, 10, 11
2	6	00 01 ($= A_0$), 10 11 ($= A_1$), 00 10 ($= R_0$), 01 11 ($= R_1$), 00 11, 01 10
3	4	00 01 10, 00 01 11, 00 10 11, 01 10 11
4	1	00 01 10 11 ($= \Omega$)

The two size-two contexts 00 11 and 01 10 are the *diagonal* doubletons: each contains exactly one atom from each ACBE block and one from each RPM block, so it contains no complete block of either partition. The four block doubletons A_0, A_1, R_0, R_1 each contain exactly one complete block.

2. Arrow count

PROPOSITION B.1. *The inclusion poset L has exactly* 81 *arrows (ordered pairs $U \subseteq V$).*

PROOF. An ordered pair $U \subseteq V$ is determined by independently assigning each of the four atoms to one of three roles: in neither U nor V; in V but not U; or in both. These roles are mutually exclusive and exhaustive because $U \subseteq V$ forbids the fourth combination (in U but not V). Hence the count is $3^4 = 81$, matching the verifier's reported arrow total. □

3. Sieve and covering-sieve totals

The modality-composition verifier enumerates every sieve on every context and saturates the Grothendieck axioms to a fixed point. Its registered totals are reproduced in Table 2. The two presentations — the join $J_A \vee J_R$ and the singleton topology $J_{\text{singleton}}$ of Chapter 9 — yield identical covering-sieve counts, corroborating the join identification (NT2-4).

Table 2: Registered totals for the sixteen-context site: cardinalities, sieve and covering-sieve counts, and the composition verifier's fixture and restriction-square counts.

Quantity	Count
Atoms	4
Contexts	16
Inclusion arrows	81
Sieves (all contexts)	298
Covering sieves, $J_A \vee J_R$ presentation	168
Covering sieves, $J_{\text{singleton}}$ presentation	168
Executable fixtures (composition verifier)	18
Restriction squares checked	3012

Remark B.2. The count of 298 sieves is the total across all sixteen contexts, not the number on a single context: a context of size k carries a lattice of sieves whose size grows with k, and the census sums these. The equality of the two 168 covering-sieve counts is the executable face of the join identification (the join lemma of Chapter 10): the topology generated jointly by the ACBE and RPM families and the topology generated by the four singletons are literally the same set of covering sieves, not merely isomorphic ones.

4. Block-completeness table

Table 3 records, for each context, which complete ACBE and RPM blocks it contains. This is the data that drives the witness presheaves of Chapter 9: $F_A(V)$ has $2^{\alpha(V)}$ sections, where $\alpha(V)$ is the number of complete ACBE blocks in V (Definition of complete blocks, Chapter 7), and symmetrically $|F_R(V)| = 2^{\rho(V)}$.

Table 3: Complete blocks contained in each context.

Context	Complete ACBE blocks	Complete RPM blocks
$\varnothing$	none	none
00, 01, 10, 11	none	none
00 01 (A_0)	A_0	none
10 11 (A_1)	A_1	none
00 10 (R_0)	none	R_0
01 11 (R_1)	none	R_1
00 11	none	none
01 10	none	none
00 01 10	A_0	R_0

Context	Complete ACBE blocks	Complete RPM blocks
00 01 11	A_0	R_1
00 10 11	A_1	R_0
01 10 11	A_1	R_1
Ω	A_0, A_1	R_0, R_1

REMARK B.3. Only Ω contains both complete ACBE blocks, so only there does F_A take its full four sections. By transversality no RPM piece $V \cap R_j$ of any context contains a complete ACBE block, so the RPM matching-family set for F_A is a one-element set at *every* context (Appendix C, Finding C.2); the gluing map therefore fails injectivity wherever $|F_A(V)| \geq 2$, that is at the two ACBE blocks, the four triples, and Ω. The distinctness computation of Chapter 9 is carried out at Ω because that is the context with the widest gap — four sections against one matching family — not because it is the only failure.

APPENDIX C

Witness Presheaf Value Tables

This appendix tabulates the two witness presheaves F_A and F_R in full: their value sets at all sixteen contexts, as counts and as explicit bit-assignment sets; their restriction maps; and the gluing computations at the global context Ω for both topologies, which are the registered core of the distinctness theorem (Theorem NT2-3, Chapter 9) and of the certificate verdicts `acbe_witness_is_acbe_sheaf` / `acbe_witness_is_rpm_sheaf` and their RPM mirrors (Chapter 11). Every entry is derived directly from the definitions; the derivations are shown. Context names and complete-block containments are those of Appendix A.

1. Definitions and the counting rule

NOTATION C.1. $F_A(V)$ is the set of functions from $\{\, i \in \{0,1\} \mid A_i \subseteq V \,\}$ to $\{0,1\}$: bit assignments on the *complete* ACBE blocks contained in V. For $U \subseteq V$, the restriction map forgets the coordinates i with $A_i \subseteq V$ but $A_i \not\subseteq U$ and keeps the remaining bits unchanged. F_R is defined symmetrically on complete RPM blocks. We write a section of $F_A(V)$ as a tuple over its index set: at Ω the four sections are (b_0, b_1) with $b_i \in \{0,1\}$ the bit on A_i; at a context with one complete block the two sections are (0) and (1); at a context with no complete block the unique section is the empty assignment, written ().

The counting rule is immediate: $|F_A(V)| = 2^{\alpha(V)}$ where $\alpha(V)$ is the number of complete A-blocks contained in V (Definition of complete blocks, Chapter 7), and likewise $|F_R(V)| = 2^{\rho(V)}$. The containment table of Appendix A gives $\alpha(V)$ and $\rho(V)$ at every context. Two facts drive every verdict below:

ADVERSE FINDING C.2. No RPM piece ever contains a complete ACBE block: for every context V and every j, $V \cap R_j \subseteq R_j$, and $A_i \subseteq R_j$ is impossible since $A_i \cap R_j = \{(i,j)\}$ is a proper subset of A_i. Hence $F_A(V \cap R_j) = \{()\}$ is a one-element set for every V and j. Symmetrically, $F_R(V \cap A_i)$ is a one-element set for every V and i. This single observation determines the entire failure pattern of the cross gluing computations below.

2. Value tables

Context V	$\lvert F_A(V)\rvert$	$\lvert F_R(V)\rvert$	$F_A(V)$ explicitly	$F_R(V)$ explicitly
$\varnothing$	1	1	$\{()\}$	$\{()\}$
$\{\omega_{00}\}$	1	1	$\{()\}$	$\{()\}$
$\{\omega_{01}\}$	1	1	$\{()\}$	$\{()\}$
$\{\omega_{10}\}$	1	1	$\{()\}$	$\{()\}$
$\{\omega_{11}\}$	1	1	$\{()\}$	$\{()\}$
A_0	2	1	$\{(0),(1)\}$ on A_0	$\{()\}$
A_1	2	1	$\{(0),(1)\}$ on A_1	$\{()\}$
R_0	1	2	$\{()\}$	$\{(0),(1)\}$ on R_0

Context V	$\lvert F_A(V)\rvert$	$\lvert F_R(V)\rvert$	$F_A(V)$ explicitly	$F_R(V)$ explicitly
R_1	1	2	$\{()\}$	$\{(0),(1)\}$ on R_1
Δ	1	1	$\{()\}$	$\{()\}$
∇	1	1	$\{()\}$	$\{()\}$
T_{00}	2	2	$\{(0),(1)\}$ on A_1	$\{(0),(1)\}$ on R_1
T_{01}	2	2	$\{(0),(1)\}$ on A_1	$\{(0),(1)\}$ on R_0
T_{10}	2	2	$\{(0),(1)\}$ on A_0	$\{(0),(1)\}$ on R_1
T_{11}	2	2	$\{(0),(1)\}$ on A_0	$\{(0),(1)\}$ on R_0
Ω	4	4	$\{(b_0,b_1) : b_i \in \{0,1\}\}$	$\{(c_0,c_1) : c_j \in \{0,1\}\}$

Derivation: each row applies the counting rule to the complete-block columns of Appendix A. Nine contexts contain no complete A-block (value 1), six contain exactly one (A_0, A_1, and the four triples; value 2), and Ω contains both (value 4); mirrored for F_R. The section totals over all contexts are $9 \cdot 1 + 6 \cdot 2 + 4 = 25$ for each witness.

3. Restriction maps

Restriction in F_A is determined by which complete blocks survive the inclusion. The following table describes every qualitatively distinct class of inclusions; all others are composites or have singleton source or target value sets (where the map is the unique one). Bits are named as in the value table.

Inclusion class	Surviving A-blocks	F_A restriction map	Character
$\Omega \supseteq T_{0j}$	A_1	$(b_0,b_1) \mapsto (b_1)$	drops b_0; two-to-one
$\Omega \supseteq T_{1j}$	A_0	$(b_0,b_1) \mapsto (b_0)$	drops b_1; two-to-one
$\Omega \supseteq A_i$	A_i	$(b_0,b_1) \mapsto (b_i)$	drops the other bit; two-to-one
$\Omega \supseteq R_j, \Delta, \nabla$, singletons, $\varnothing$	none	$(b_0,b_1) \mapsto ()$	constant; four-to-one
$T_{ij} \supseteq A_{1-i}$	A_{1-i}	$(b) \mapsto (b)$	identity on the surviving bit
$T_{ij} \supseteq$ any U with $A_{1-i} \not\subseteq U$	none	$(b) \mapsto ()$	constant; two-to-one
$A_i \supseteq$ proper U	none	$(b) \mapsto ()$	constant; two-to-one
source U with $\alpha(U) = 0$	none	$() \mapsto ()$	identity of a singleton

Functoriality is visible in the table: a composite of inclusions survives a bit exactly when both steps do, and on surviving bits every map is the identity, so both functor laws hold coordinatewise. This is the content the certificate validates as `acbe_witness_functorial` before computing any verdict; the F_R table is the mirror image under the coordinate exchange (replace A-blocks by R-blocks, $T_{ij} \supseteq A_{1-i}$ by $T_{ij} \supseteq R_{1-j}$, and b_i by c_j), validated as `rpm_witness_functorial`.

4. Gluing at Ω: the four registered computations

The sheaf condition at Ω compares the value set with the set of matching families over each global cover. Since distinct pieces of either cover meet in $\varnothing$, where both witnesses have the one-element value $\{()\}$, compatibility is automatic and the matching-family set is the plain product of the piece values. The four computations, tabulated:

Witness / cover	Piece values	Matching families	Value at Ω	Gluing map and verdict
F_A over $K_A(\Omega)$	$F_A(A_0) = \{(0), (1)\}$, $F_A(A_1) = \{(0), (1)\}$	$2 \times 2 = 4$	4	$(b_0, b_1) \mapsto ((b_0), (b_1))$: bijection — sheaf condition holds
F_A over $K_R(\Omega)$	$F_A(R_0) = \{()\}$, $F_A(R_1) = \{()\}$	$1 \times 1 = 1$	4	all four of $(0,0)$, $(0,1)$, $(1,0)$, $(1,1) \mapsto ((), ())$: not injective — fails
F_R over $K_R(\Omega)$	$F_R(R_0) = \{(0), (1)\}$, $F_R(R_1) = \{(0), (1)\}$	$2 \times 2 = 4$	4	$(c_0, c_1) \mapsto ((c_0), (c_1))$: bijection — sheaf condition holds
F_R over $K_A(\Omega)$	$F_R(A_0) = \{()\}$, $F_R(A_1) = \{()\}$	$1 \times 1 = 1$	4	four-to-one onto $((), ())$: not injective — fails

Derivations. Row 1: a matching family for $K_A(\Omega)$ is a free pair (one section over each A_i; the overlap $A_0 \cap A_1 = \varnothing$ imposes agreement in a singleton), and the canonical map sends the global assignment (b_0, b_1) to its restrictions, which by the restriction table are (b_0) on A_0 and (b_1) on A_1; this is the evident bijection, so F_A satisfies the ACBE gluing condition at Ω. Row 2: by Finding C.2 each RPM half carries the one-element value, so exactly one matching family exists; the canonical map sends each of the four global sections to it, failing injectivity, so F_A is not a J_R-sheaf — this is the registered "$4 \to 1$" computation, exhibited here element by element. Rows 3 and 4 are the mirror computations.

5. Gluing at the remaining contexts

The certificate's verdicts quantify over *all* contexts, not only Ω; for completeness we tabulate the matching-family counts everywhere. For the own-partition cover the count is $|F_A(V \cap A_0)| \cdot |F_A(V \cap A_1)|$ (empty pieces contribute the factor 1); for the cross cover it is identically 1 by Finding C.2. At $\varnothing$ the null cover has exactly one matching family.

Context V	$\lvert F_A(V)\rvert$	Matching for $K_A(V)$	Matching for $K_R(V)$	Verdicts for F_A (own / cross)
$\varnothing$	1	1	1	holds / holds
$\{\omega_{ij}\}$ (each)	1	$1 \cdot 1 = 1$	1	holds / holds
A_0	2	$2 \cdot 1 = 2$	1	holds / fails ($2 \to 1$)
A_1	2	$1 \cdot 2 = 2$	1	holds / fails ($2 \to 1$)
R_0	1	$1 \cdot 1 = 1$	1	holds / holds
R_1	1	$1 \cdot 1 = 1$	1	holds / holds
Δ	1	$1 \cdot 1 = 1$	1	holds / holds
∇	1	$1 \cdot 1 = 1$	1	holds / holds
T_{00}	2	$1 \cdot 2 = 2$	1	holds / fails ($2 \to 1$)
T_{01}	2	$1 \cdot 2 = 2$	1	holds / fails ($2 \to 1$)
T_{10}	2	$2 \cdot 1 = 2$	1	holds / fails ($2 \to 1$)
T_{11}	2	$2 \cdot 1 = 2$	1	holds / fails ($2 \to 1$)
Ω	4	$2 \cdot 2 = 4$	1	holds / fails ($4 \to 1$)

Reading the table: in the own-partition column the matching count equals the value count at every context, and inspection of the canonical maps (as in Row 1 of the Ω table; at contexts

with an identity piece the condition is trivial) shows each is a bijection — F_A is a J_A-sheaf, and by the basis criterion the generating families suffice. In the cross column the matching count is constantly 1, so the gluing map fails injectivity at exactly the seven contexts where $|F_A(V)| \geq 2$: the two A-blocks, the four triples, and Ω. The registered Ω computation is thus one instance of a uniform failure pattern; any one of the seven already establishes that F_A is not a J_R-sheaf, and the certificate's exhaustive run confirms the pattern at all of them. The mirrored table for F_R (exchange the two partitions, the two block families, and the two cover columns) shows F_R is a J_R-sheaf failing J_A-gluing at the two R-blocks, the four triples, and Ω.

6. Reflector values on the witnesses

For completeness we tabulate what the three reflectors of Chapter 10 do to the witnesses of Chapter 9, since these values are quoted in the body (the annihilation example and the both-routes example). Each entry is computed from the reflector formula $(R_P F)(V) = \prod_{B \in I_P(V)} F(V \cap B)$ together with Finding C.2 and the value tables above.

Presheaf	R_A applied	R_R applied	R_Q applied
F_A	$\cong F_A$ (unit iso; F_A is a J_A-sheaf)	terminal (1 at every context)	terminal
F_R	terminal (1 at every context)	$\cong F_R$ (unit iso)	terminal

Derivations. $R_A F_A \cong F_A$: the unit of a reflector is an isomorphism exactly on sheaves (Chapter 10), and the own-partition column of Section 5 verifies the sheaf condition. $R_A F_R$ terminal: every piece $V \cap A_i$ lies in a single ACBE block, hence contains no complete RPM block, so every factor $F_R(V \cap A_i)$ is a one-element set and the product is a one-element set at every context. $R_Q F_A$ terminal: every factor $F_A(\{\omega\})$ is a one-element set since a singleton contains no complete block. The terminal presheaf is a sheaf for all three topologies (every matching-family set of a constant-singleton presheaf is a singleton, including at the null cover), so these values are consistent with the sheaf lemma. The table also exhibits, in minimal form, the content of the composition theorem on these fixtures: both composite routes applied to either witness give the terminal presheaf, so the two routes agree here for the strongest possible reason — they agree with each other by agreeing with R_Q.

7. Section counts at a glance

Summing the value tables: each witness has 25 sections in total over the sixteen contexts (9 contexts contributing one section, 6 contributing two, Ω contributing four); each reflected cross-image ($R_A F_R$, $R_R F_A$, $R_Q F_A$, $R_Q F_R$) has exactly 16 sections, one per context. The asymmetry 25 versus 16 is the quantitative footprint of nontriviality: a modality that were naturally isomorphic to the identity would preserve section counts at every context, and the drop from $(1, 2, 4)$-valued to constantly 1-valued is what the certificate's false verdicts measure context-by-context.

Claim Boundary C.3. These tables establish combinatorial facts about two specific finite presheaves on L and nothing more. They corroborate the distinctness theorem's witnesses (Theorem NT2-3, Chapter 9, scoped to $\mathrm{PSh}(L, \mathbf{Set})$); they attach no interpretation to the ACBE and RPM marks, and they license no ambient, probabilistic, empirical, or product statement.

APPENDIX D

Glossary and Notation Index

This appendix collects every technical term and symbol used in the monograph, with a short self-contained definition and a plain-text pointer to the chapter or appendix where the notion is developed. It is a reading aid, not a normative source: where this glossary and a definition in the body could be read differently, the body governs, and where the body and a registered ledger artifact could be read differently, the ledger governs. Chapter references are given as plain text (for example "Chapter 10") rather than as links.

1. Terms

ACBE mark. The first coordinate of an atom of Ω, a formal label preserved for controlled migration vocabulary from the legacy construction. It carries no empirical interpretation until a separate operationalization and validation record exists; no theorem in the monograph attaches meaning to it beyond its combinatorics. Chapter 7; restated as standing data in Chapter 10; provenance role in Chapter 8, with executable checks in Chapter 11.

Admission gates. The ten conjunctive conditions — independent review, formalization, typed product-site covers, the product-site lift, the backward-compatibility proofs, the cross-context decision, probabilistic gates, novelty search, registry and IP review, external referee review — that must all close before Noetic Topos 2.0 may leave Ring 4. Enumerated and discussed in Chapter 12.

Admission ring (Ring 4). The program's containment level for the entire artifact set of this monograph: internal research material, not licensed for public, empirical, or product claims. Defined in Chapter 1; restated in the status boundaries of Chapters 7, 8, 9, 10, 11, and 12.

Ambient topos. The presheaf topos $\mathcal{E}_{\mathrm{amb}} = \mathrm{PSh}(\mathcal{C} \times L)$ over the product of the legacy category with the context lattice; the carrier of the three modalities and the home of unresolved and partially resolved evidence. The candidate definition sets $\mathrm{NoeTop}_2 := \mathcal{E}_{\mathrm{amb}}$. Chapter 12.

Associated sheaf (sheafification). The universal sheaf receiving a map from a given presheaf: the left adjoint to the inclusion of sheaves into presheaves. On the finite context factor it is realized exactly by the one-partition reflector. Chapter 4; realized explicitly in Chapter 10; general background in the cited Stacks Project material.

Atom. One of the four elements $\omega_{ij} = (i, j)$ of Ω, carrying a formal ACBE mark i and RPM mark j. Chapter 7; tabulated in Appendix A.

Backward-compatibility conjecture. The unproved comparison $\mathcal{E}_{\mathrm{res}} \simeq \mathcal{E}^4_{\mathrm{legacy}}$ together with the neutral-channel embedding ι and the natural isomorphism $\mathrm{ev}_{00} \circ \iota \cong \mathrm{Id}$; ledger status PROOF REQUIRED. Two round trips (categorical and strict serialization) are required and neither is complete. Chapter 12.

Bit-assignment presheaf. The value shape of the witness presheaves: functions from a set of complete blocks to $\{0, 1\}$, restriction forgetting coordinates. Also the name of an executable fixture in the composition verifier. Chapter 9; Appendix C; fixture in Chapter 11.

Block. A member of one of the partitions of Ω: an ACBE block A_i, an RPM block R_j, or a singleton block of Q. Chapter 7; Appendix A.

Certificate (finite model). The registered artifact `FINITE_MODEL_CERTIFICATE_v0.1.json`: the expected-result record of the finite-site checker, comprising the twenty-three registered checks and the aggregate pass flag, under its own claim boundary. Chapter 11.

Claim boundary. A first-class statement, attached to an artifact, of what the artifact does not establish. The monograph's governing example is the certificate's boundary: exhaustive execution is not proof-assistant verification, peer review, empirical validation, or a product-claim license. Chapters 10, 11, 12 and the appendices each carry their own.

Coincidence (of unmarked families). The phenomenon that $K_A(V)$ and $K_R(V)$ can be equal as bare arrow sets — at the empty context, the four singletons, and the two diagonals — so that the topology retains no trace of which partition produced the cover, while the marked records remain distinct. Proved in Chapter 7; re-derived tabularly in Appendix A; singleton cases registered in the certificate, Chapter 11.

Common refinement. The family of pairwise intersections of two covers. For the two global covers it is the four-singleton cover, by transversality; as a marked record it is a derivation node carrying both parent digests. Chapter 7; marked-record form in Chapter 8; checks in Chapter 11.

Complete block. A block B with $B \subseteq V$ for the context V at hand; the witness presheaves assign bits only to complete blocks; the counts are written $\alpha(V)$ and $\rho(V)$. Chapter 7; Appendix A tabulates completeness at every context.

Composition theorem. The chapter-10 result $M_A M_R \cong M_{\text{atom}} \cong M_R M_A$, unit-compatibly and naturally, with either composite naturally isomorphic to sheafification for $J_A \vee J_R$. Status: internal proof candidate, review required. Chapter 10; executable corroboration in Chapter 11.

Context (context lattice). An element of $L = \mathcal{P}(\Omega)$, ordered by inclusion; the sixteen-object poset carrying the finite theory. Chapter 7; enumerated in Appendix A.

Covering sieve. A sieve belonging to the topology at its object; the sheaf condition quantifies over covering sieves. Chapter 3; counted exactly in Chapters 9 and 11 and in Appendix A.

Cover kind. The field of a mark record naming the admitting clause (OBSERVATIONAL, ACBE, or RPM); distinct cover kinds at coinciding unmarked families are what the marked layer preserves. Chapter 8; executable checks in Chapter 11.

Dedekind number. The number $M(n)$ of down-sets of the Boolean lattice on n points ($2, 3, 6, 20, 168$ for $n = 0, \ldots, 4$); used to derive the registered total of 298 sieves. Derived in Chapter 7 (written δ there); reported in Chapter 11; per-context table in Appendix A.

Derivation kind. The field of a marked cover declaring how it arose: GENERATOR, PULLBACK, or COMMON REFINEMENT; derived kinds require parent resolution. Chapter 8; executable checks in Chapter 11.

Digest. The SHA-256 hash of a marked cover's canonical JSON, binding target, members, mark, and sorted parent digests; the unit of provenance chaining. Chapter 8; executable checks in Chapter 11.

Distinctness theorem. Theorem NT2-3: F_A is a J_A-sheaf but not a J_R-sheaf and symmetrically, so the two sheaf subcategories differ and neither modality is trivial; scoped to $\mathrm{PSh}(L, \mathbf{Set})$. Proved in Chapter 9; witness tables in Appendix C; verdict checks in Chapter 11.

Empty-product convention. The clause $(R_P F)(\varnothing) = \mathbf{1}$: the reflector's value at the empty context is the singleton regardless of $F(\varnothing)$, forced by the null cover. Load-bearing in the sheaf lemma and exercised by a dedicated adversarial fixture. Chapter 10; Chapter 11.

Evidence Packet. The enclosing program artifact that must separately bind raw ingress bytes; the cover digest is the canonical semantic-record digest only, not a substitute for that binding. Chapter 8 (Boundary: the Evidence Packet binds raw bytes; the cover digest does not); program-level notion.

Expressivity boundary. The observation that, even if the backward-compatibility conjecture is proved, $\mathcal{E}^4_{\text{legacy}}$ is four independent channels with no cross-atom coupling, so the finite model is a separation and provenance reference model rather than a final semantics. Chapter 12.

Fail-closed. The integrity posture of the marked layer: unknown or duplicate JSON keys, uncontrolled review states, and derived records with wrong or unresolvable parents are rejected rather than treated as evidence. Chapter 8; executable checks in Chapter 11.

Flattening and regrouping. The mutually inverse bijections φ and ψ between the nested product $(R_A R_R F)(V)$ and the atomic product $(R_Q F)(V)$, induced by transversality; the computational core of the composition theorem. Chapter 10.

Gluing (matching family). A matching family for a cover assigns a compatible local section to each member; the sheaf condition asks that the canonical map from global sections to matching families be a bijection. Chapters 3 and 4; computed for the witnesses in Appendix C.

Grothendieck topology. An assignment of covering sieves to objects satisfying maximality (J1), pullback stability (J2), and local character (J3). Chapter 3; computed exactly by saturation in Chapter 11.

Join (of topologies). The least topology containing two given topologies. The monograph's central join identification is $J_A \vee J_R = J_{\text{atom}}$ (Theorem NT2-4, Chapter 9; re-derived in reflector-facing form in Chapter 10).

Lean layer. The bounded proof-assistant scope: four-atom exhaustiveness, proper global pieces, the transverse refinement identity, and the witness-map lemmas — and nothing else; the sheaf theory is explicitly outside it. Chapter 11.

Legacy collapse. The diagnosis that the legacy ACBE and RPM cover clauses generate only maximal sieves (identity via reflexivity; identity via split-epi closure), so the legacy topology reduces to its jointly-surjective clause and retains no modal provenance. Chapter 5; executable checks in Chapter 11.

Legacy site. The registered category $\mathcal{C}$ = Noetica with its jointly-surjective topology J_{legacy} (also written J_{obs} in the ambient package) and sheaf topos $\mathcal{E}_{\text{legacy}}$. Chapter 5; ambient package in Chapter 12.

LP feasibility certificate. The second probabilistic gate: existence of a global joint distribution over a fixed finite cover posed as a finite linear program, with a feasibility or infeasibility certificate as the licensed artifact. Chapter 12.

Marginal problem. The classical fact that pairwise-compatible marginal distributions need not extend to a joint distribution; the reason Set-valued gluing licenses no probabilistic claim. Chapter 12, with the cited classical and sheaf-theoretic literature.

Marked cover. The immutable record (derivation kind, target, exact member arrows, mark record, parent digests) that travels in parallel with the unmarked family; the topology consumes the family, governance consumes the record. Chapter 8; executable checks in Chapter 11.

Mark record. The provenance component of a marked cover: cover kind, source authority reference, assumptions, evidence references, review state, provenance chain. Chapter 8; executable checks in Chapter 11.

Maximal sieve. The sieve of all arrows into an object; covering in every topology. A sieve containing an identity, or a split epimorphism with its section, is maximal — the two collapse mechanisms. Chapter 3; the collapse mechanisms in Chapter 5; checks in Chapter 11.

Modality. A left-exact idempotent monad on a presheaf topos; here, sheafification-followed-by-inclusion for a topology refinement (M_A, M_R, M_{atom}, M_{obs}). Trivial means naturally isomorphic to the identity. Chapter 4; explicit formulas in Chapter 10; ambient roles in Chapter 12.

Modality-composition verifier. The second executable layer: implements the explicit reflector, its unit, and both comparison maps, and computes both topologies exactly by sieve enumeration and saturation. Chapter 11.

Model ID. The registered identifier `NT2-CONTEXT-2X2-TRANSVERSE-PARTITIONS` naming the finite model checked by the certificate. Chapter 11.

Neutral channel. The registry-fixed atom ω_{00} used by the conjectural legacy embedding $\iota : X \mapsto (X, 1, 1, 1)$; evaluation there is ev_{00}. Chapter 12.

NoeSystem (NoeSystem_2). The full research object — the tuple of NoeTop_2, the three modalities, the resolved subtopos, and the external mark record. A structured package, not a topos or category. Chapter 12.

NoeTop (NoeTop_2). The underlying topos candidate, defined as the ambient presheaf topos $\mathcal{E}_{\text{amb}}$ (not the resolved subtopos, on which the modalities restrict to identity). Chapter 12.

Novelty status. All theorem-ledger records carry NOVELTY UNASSESSED: no dedicated literature search has been performed, and the monograph does not call its results new theorems in the scholarly sense. Chapter 12.

Null cover. The empty family at the empty context, which covers it in all three bases; forces every sheaf to have a one-element value at $\varnothing$. Chapter 7; used throughout Chapters 9 and 10.

Obligation (cross-context semantics). The registered record OBL-NT2-CROSS-CONTEXT-SEMANTICS: choose and formalize either the ambient modal package as principal semantics or a resolved semantics with typed overlap or transition data and proved coherence, under four acceptance criteria. Chapter 12.

One-partition reflector. The explicit construction $(R_P F)(V) = \prod_{B \in I_P(V)} F(V \cap B)$ with the empty product a singleton, restriction dropping absent coordinates, and unit given by restrictions; proved to be the associated-sheaf reflector for J_P. Chapter 10.

Presheaf. A functor $L^{\text{op}} \to \mathbf{Set}$ (more generally $\mathcal{C}^{\text{op}} \to \mathbf{Set}$): sets of sections with compatible restriction maps. Chapter 2; on L, Chapter 10.

Proof-obligation ledger. The registered artifact recording open design-and-proof obligations with statements, dependencies, acceptance criteria, and claim boundaries. Chapter 12.

Provenance chain. The parent-digest lineage of a marked cover, checked acyclically and recomputed on resolution. Chapter 8; executable checks in Chapter 11.

Pullback sieve; pullback stability. The pullback of a sieve along an arrow ($g^* S$); axiom (J2) requires covering sieves to pull back to covering sieves, and the bases satisfy it

at the family level (in L, pullback is intersection). Chapter 3; computed on L in Chapter 7; checks in Chapter 11.

Reflector. A left adjoint to the inclusion of a reflective subcategory; here always the associated-sheaf functor for one of the topologies, realized explicitly by R_P. Chapter 4 (general notion); explicit construction in Chapter 10.

Representable presheaf. $yW = \mathrm{Hom}_L(-, W)$: a singleton at $V \subseteq W$, empty otherwise. All sixteen are sheaves for both factor topologies (subcanonicity). Chapter 2 (definition); Chapter 9 (subcanonicity); usage in Chapter 10; check in Chapter 11.

Resolved subtopos. $\mathcal{E}_{\mathrm{res}} = \mathrm{Sh}(\mathcal{C} \times L, J_{\mathrm{res}})$, the sheaf subcategory for the fully joined refinement; modally inert by the resolved-identity theorem. Chapters 10 and 12.

Restriction map. The action of a presheaf on an inclusion $U \subseteq V$, written $x \mapsto x|_U$; for the witnesses, coordinate forgetting (Appendix C).

RPM mark. The second coordinate of an atom; formal label only, as for the ACBE mark. Chapter 7; tabulated in Appendix A.

Section-retraction. A pair $\eta : X \to R(X)$, $\varepsilon : R(X) \to X$ with $\varepsilon \circ \eta = \mathrm{id}$; the legacy RPM data. Its split epimorphism is one of the two collapse mechanisms. Chapter 2 (split epimorphisms); Chapter 5 (the legacy RPM data and the collapse).

Separatedness. The injectivity half of the sheaf condition: sections agreeing on a cover are equal. Chapter 4; used repeatedly in the universal-property proof of Chapter 10.

Sheaf. A presheaf whose gluing map is a bijection for every covering sieve; equivalently, for a coverage basis, for every basis family. Chapter 4; explicit reflector in Chapter 10.

Sieve. A precomposition-closed set of arrows into a fixed object; in the poset L, a down-set of subcontexts. Chapter 3; counted in Chapter 7, in Chapter 11, and in Appendix A.

Singleton-factor bijection. The canonical identification $(R_P F)(W) = F(W)$ at a nonempty context W contained in a single block; the engine of the uniqueness half of the universal property. Chapter 10.

Site. A category with a Grothendieck topology. Chapter 3.

Split epimorphism. An arrow with a section; a sieve containing one is maximal. Chapter 2 (definition), Chapter 3 (maximality lemma), Chapter 5 (the legacy collapse).

Subcanonical. A topology for which every representable presheaf is a sheaf; both factor topologies are subcanonical, while legacy subcanonicity is not derivable. Chapter 3 (definition); Theorem NT2-2, Chapter 9; used in Chapter 10; consequences in Chapter 12.

Theorem ledger. The registered record set assigning each theorem its proof status; drafting prose promotes nothing recorded there. Status recaps in Chapters 10 and 12.

Transverse partitions. The two partitions $\{A_0, A_1\}$ and $\{R_0, R_1\}$ of Ω with $A_i \cap R_j = \{(i, j)\}$; transversality is the single identity powering the join and composition results. Chapter 7; used in Chapters 9 and 10.

Unit. The natural transformation $\eta_P : \mathrm{id} \Rightarrow R_P$ (restrictions to pieces), universal among maps to sheaves; unit compatibility is part of the composition theorem. Chapter 10.

Universal property. The bijection $\mathrm{Hom}_{\mathrm{Sh}}(R_P F, G) \cong \mathrm{Hom}_{\mathrm{PSh}}(F, G)$ natural in both variables, exhibiting R_P as THE associated-sheaf reflector. Chapter 10.

Witness presheaf. F_A or F_R: the explicit bit-assignment presheaves separating the two sheaf subcategories. Defined and used in Chapter 9; Appendix C tabulates them in full.

2. Symbols

Symbol	Meaning	Where
Ω	the four-atom set $\{0,1\} \times \{0,1\}$	Chapter 7
ω_{ij}	the atom (i,j); ω_{00} is the neutral atom	Chapter 7; neutral-channel role in Chapter 12
L	the context lattice $\mathcal{P}(\Omega)$	Chapter 7
A_i, R_j	ACBE and RPM blocks	Chapter 7, Appendix A
Q	the singleton partition of Ω	Chapter 7
$D_=$ (Δ), $D_\neq$ (∇)	the diagonal contexts $\{\omega_{00},\omega_{11}\}$, $\{\omega_{01},\omega_{10}\}$ (not the constant presheaf ΔS of Chapter 2)	Chapter 7; Appendix A
T_{ij}	the triple $\Omega \setminus \{\omega_{ij}\}$	Chapter 7; Appendix A
$K_P(V)$	generating family of partition P at V	Chapter 7; Appendix A
J_A, J_R	the two factor topologies	Chapter 9
J_{atom}	the singleton-generated topology; preferred name $J_{\text{singleton}}$	Chapter 9
J_{obs}, J_{legacy}	the legacy jointly-surjective topology	Chapter 12
J_{res}	the fully joined refinement $J_{\text{obs}} \vee J_A \vee J_R$	Chapter 12
$\vee$	join of topologies (least upper bound)	Chapter 9
$\leq$	comparison of topologies (containment of covering sieves)	Chapter 9
$I_P(V)$	blocks of P meeting V	Chapter 10
R_P	the one-partition reflector for P	Chapter 10
η_P	the unit of R_P	Chapter 10
φ, ψ	flattening and regrouping bijections	Chapter 10
M_A, M_R, M_{atom}, M_{obs}	sheafification-then-inclusion modalities	Chapters 10, 12
F_A, F_R	the witness presheaves	Chapter 9; Appendix C
$\alpha(V)$, $\rho(V)$	counts of complete ACBE and RPM blocks in V	Chapter 7; Appendices A, B, C
yW	the representable presheaf of W	Chapter 2 (definition); Chapter 9 (subcanonicity); Chapters 10, 11
$\mathbf{1}$	a one-element set; the empty product	Chapter 10
PSh, Sh	presheaf and sheaf categories	throughout
Set	the category of small sets	throughout

Symbol	Meaning	Where
$\mathcal{C}$	the registered legacy category Noetica	Chapter 5; ambient package in Chapter 12
$\mathcal{E}_{\text{amb}}$, $\mathcal{E}_{\text{res}}$, $\mathcal{E}_{\text{legacy}}$	ambient, resolved, and legacy topoi	Chapter 12
NoeTop_2, NoeSystem_2	topos candidate and full package	Chapter 12
ι, ev_{00}	conjectural neutral-channel embedding and evaluation	Chapter 12
$M(n)$	the Dedekind number (down-sets of the n-cube); written $\delta(n)$ in Chapter 7	Chapter 7; Chapter 11, Appendix A
m_C	the maximal sieve on C	Chapter 3
g^*S	the pullback sieve of S along g	Chapter 3
id, Hom, Ob	identity, hom-sets, objects	throughout

REMARK D.1. Status vocabulary, for completeness: RING 4 (containment level of the artifact set), NOT LICENSED (no public, empirical, or product claim), INTERNAL PROOF CANDIDATE / REVIEW REQUIRED (an informal proof present, no independent review), PROOF REQUIRED (a conjecture with no proof), NOVELTY UNASSESSED (no literature search performed), and OPEN (a registered obligation not yet discharged). These words are load-bearing wherever they occur; this monograph uses them in exactly the senses of the registered ledgers, and Chapter 12 collects the gates under which any of them may change.

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